

KU Leuven
Biomedical Sciences Group
Faculty of Pharmaceutical Sciences
Department of Pharmaceutical and Pharmacological Sciences
Clinical Pharmacology and Pharmacotherapy Unit



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PHARMACOMETRICS APPROACHES FOR THERAPY INDIVIDUALIZATION IN PATIENTS WITH GASTROINTESTINAL DISORDERS

Yannick HOFFERT

Supervisor: Prof. Dr. Erwin Dreesen
Co-supervisor: Prof. Dr. Tim Vanuytsel
Prof. Dr. Jef Verbeek
Prof. Dr. Bram Verstockt
Chair: Prof. Dr. Pieter Annaert
Jury members: Prof. Dr. Isabel Spriet
Prof. Dr. Thomas Neyens
Prof. Dr. Brenda de Winter
Prof. Dr. Brian Cicali

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Promotiezaal
Universiteitshal
Naamsestraat 22
3000 Leuven
Belgium

Clinical Pharmacology and Pharmacotherapy Unit
Department of Pharmaceutical and Pharmacological Sciences
O&N II Herestraat 49 – box 521
3000 Leuven
Belgium

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WORDS OF THANKS

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LIST OF ABBREVIATIONS

-2LL	Minus twice the log of the likelihood
ALP	Alkaline phosphatase
ALT	Alanine transferase
AST	Aspartate transferase
ASUC	Acute severe ulcerative colitis
AUC	Area under the curve
AUROC	Area under the receiver operating characteristics curve
BMI	Body mass index
CC	Concentration control
CD	Crohn's disease
CI	Confidence interval
CL	Clearance
C _{max}	Maximum concentration
CNI	Calcineurin inhibitor
COSSAC	Conditional Sampling use for Stepwise Approach based on Correlation tests
COV	Covariance
CRP	C-reactive protein
CV	Coefficient of variation
CYP3A4	Cytochrome P450 3A4
DDI	Drug-drug interaction
DOAC	Direct oral anticoagulant
EBE	Empirical Bayes Estimate
EHR	Electronic health record
EMA	European Medicine Agency
EPAR	European public assessment report
EU	European Union
F	Bioavailability
FDA	Food and Drug Administration
FFEM	Full fixed effect model
FKBP	FK506 binding protein
FO	First-order estimation
FOCE	First-order conditional estimation
FOCE+I	First-order conditional estimation with interaction
FREM	Full random effect model
GGT	Gamma glutamyl transferase
GI	Gastrointestinal
GLP-2	Glucagon-like peptide 2
GOF	Goodness-of-fit
HPC	High-performance computing
IATDMCT	International Association of Therapeutic Drug Monitoring and Clinical Toxicology
IBD	Inflammatory bowel diseases
IF	Intestinal failure

IFALD	Intestinal failure-associated liver disease
IIRTA	International Intestinal Rehabilitation and Transplant Association
IIV	Inter-individual variability
IOV	Interoccasion variability
IPV	Inpatient variability
IQR	Interquartile range
ITx	Intestinal transplantation
IV	Intravenous
k_a	Absorption rate constant
KWS	‘Klinisch Werkstation’
LASSO	Least Absolute Shrinkage and Selection Operator
LDH	Lactate dehydrogenase
LIFT	Leuven Intestinal Failure and Transplantation
LLOQ	Lower limit of quantification
mAb	Monoclonal antibody
MCMC	Markov Chain Monte Carlo
MDR	Medical device regulation
MIPD	Model-informed precision dosing
NLME	Nonlinear mixed-effects
OFV	Objective function value
P-gp	P-glycoprotein
pcVPC	Prediction-corrected visual predictive check
PD	Pharmacodynamics
PDMS	‘Patiënt Data Management System’
PK	Pharmacokinetic
PKPD	Pharmacokinetic–pharmacodynamic
PopPK	Population pharmacokinetics
PRO2	Patient-reported outcome
PS	Parenteral support
PTA	Probability of target attainment
RCT	Randomized controlled trials
RFS	Rejection free survival
RHS	Regularized Horseshoe
ROC	Receiver operating characteristics
RSE	Relative standard error
RUV	Residual unexplained variability
SAEM	Stochastic approximation expectation maximization
SBS	Short bowel syndrome
SC	Subcutaneous
SCM	Stepwise covariate modeling
SIR	Sampling importance resampling
TCI	Target concentration intervention
TDM	Therapeutic drug monitoring
T_{max}	Time to maximum concentration
TNF	Tumor necrosis factor

UC	Ulcerative colitis
ULOQ	Upper limit of quantification
V	Volume of distribution
VAR	Variances
VIF	Variance inflation factor
VPC	Visual predictive check
VSC	Vlaams Supercomputer Centrum (Flemish Supercomputer Center)
VTE	Venous thromboembolism

GENERAL INTRODUCTION

Parts of the introduction are based on

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Iris K Minichmayr, Erwin Dreesen, Maddalena Centanni, Zhigang Wang, **Yannick Hoffert**, Lena E Friberg, Sebastian G Wicha. Model-informed precision dosing: State of the art and future perspectives. *Adv Drug Deliv Rev*. 2024 Dec;215:115421.

1. Complexity of gastrointestinal diseases

Gastrointestinal (GI) disorders comprise a wide spectrum of conditions affecting the digestive tract and associated organs, representing a substantial and growing burden on global healthcare systems.¹ Beyond their clinical impact, GI disorders also impose considerable economic consequences due to recurrent hospitalizations, the need for advanced diagnostics, lifelong pharmacological and nutritional support, and, in severe cases, complex surgical interventions. As life expectancy increases and the global prevalence of chronic inflammatory and functional GI diseases continues to rise, the demand for optimized, individualized therapeutic strategies has become increasingly pressing.²

Inflammatory bowel diseases (IBD), including Crohn's disease (CD) and ulcerative colitis (UC), are chronic, immune-mediated disorders characterized by relapsing inflammation of the GI tract. Their global prevalence has risen substantially over the past decades, exceeding 0.3% in Western countries, with the highest burden observed in North America, Europe, and Oceania.³⁻⁵ Newly industrialized regions are now experiencing a rapid increase in incidence resembling earlier epidemiological trends in the West. IBD typically manifests in adolescence or early adulthood but may occur at any age, leading to lifelong morbidity, impaired quality of life, and considerable healthcare utilization. CD and UC differ substantially in their anatomical distribution, depth of inflammation, clinical expression, and extraintestinal manifestations, necessitating tailored therapeutic strategies (**Figure 1, Table 1**).

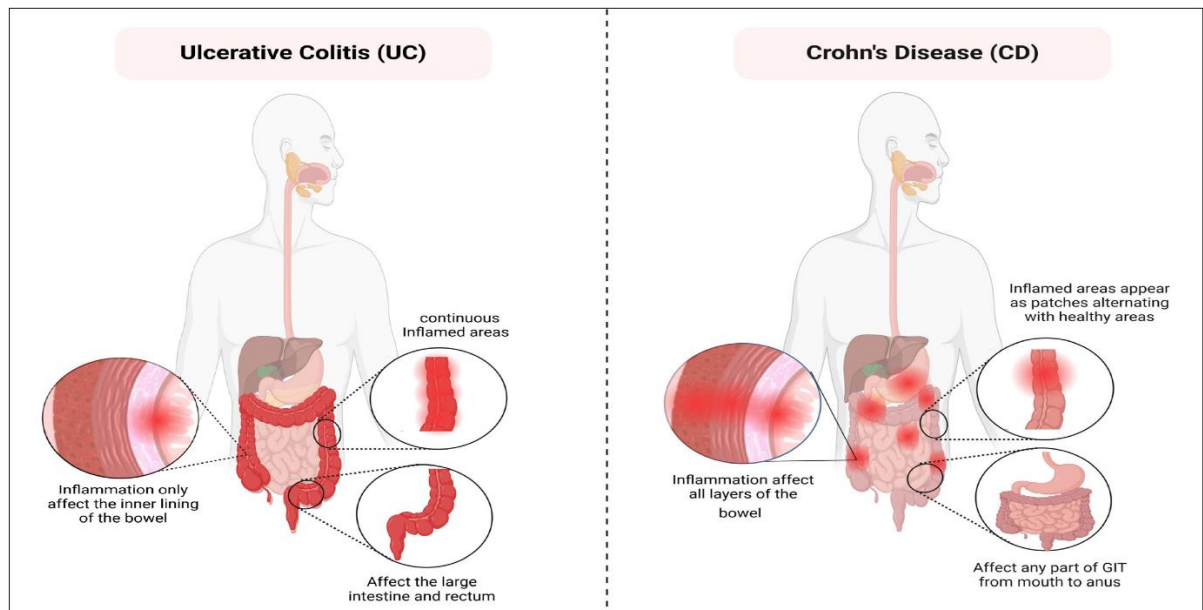


Figure 1. Difference between two forms of inflammatory bowel disease. Ulcerative colitis (UC) occurs only in the colon and rectum and is characterized by the continuous appearance of inflammation and an inner layer of the bowel that is involved in inflammation. Crohn's disease (CD) occurs in any part of gastrointestinal tract (GIT) characterized with the patchy appearance of inflammation and all layers of bowel involved in inflammation. Abbreviations. GIT, gastrointestinal tract.

Figure reproduced with permission from Hetta *et al.*⁶

Table 1. Comparison of characteristics for Crohn's disease and ulcerative colitis.

	Ulcerative colitis	Crohn's disease
<i>History</i>		
Onset	More indolent (wk/mo)	More abrupt (d/wk)
Location	Starts in rectum and moves proximally in colon	Anywhere in GI tract
Abdominal pain	Uncommon	Common
Non-GI symptoms	Uncommon	Common
<i>Physical examination</i>		
Fever	+	+++
Perianal disease	+	+++
Eye, joint, or skin changes	-	++
<i>Laboratory</i>		
Elevated CRP and/or ESR	+	+++
ASCA	-	++
<i>Rectal involvement</i>		
Continuous mucosal involvement	++	+
Transmural change on biopsy	-	+++

Abbreviations. ASCA, anti-*Saccharomyces cerevisiae*; CRP, C-reactive protein; d, days; ESR, erythrocyte sedimentation rate; GI, gastrointestinal; mo, month; wk, week.

Table reproduced with permission from Sairenji *et al.*⁷

Short bowel syndrome (SBS) represents one of the most severe downstream consequences of extensive intestinal disease or resection, and CD is among its leading causes in adults. Ischemia and CD were identified as the most common cause for SBS.^{8,9} It is defined as less than 180 to 200 cm of remaining small bowel length, leading to nutritional and fluid supplements. Consequently, patients with SBS present with signs of malabsorption including weight loss, diarrhea, steatorrhea, dehydration, malnutrition, and electrolyte imbalance.⁸ Based on the remaining intestinal anatomy, SBS can be classified in type 1 (end-jejunostomy), type 2 (jejunocolonic anastomosis), and type 3 (jejunoleal anastomosis) (**Figure 2**).¹⁰ Therapeutic options largely rely on dietary measures and off-label use of antisecretory therapy and antimotility agents.¹¹ More recently, the use of glucagon-like peptide 2 (GLP-2) analogues was introduced as a disease-modifying agent.¹² In its most severe forms, SBS can lead to intestinal failure, a highly disabling condition as a consequence of massive intestinal loss.¹³

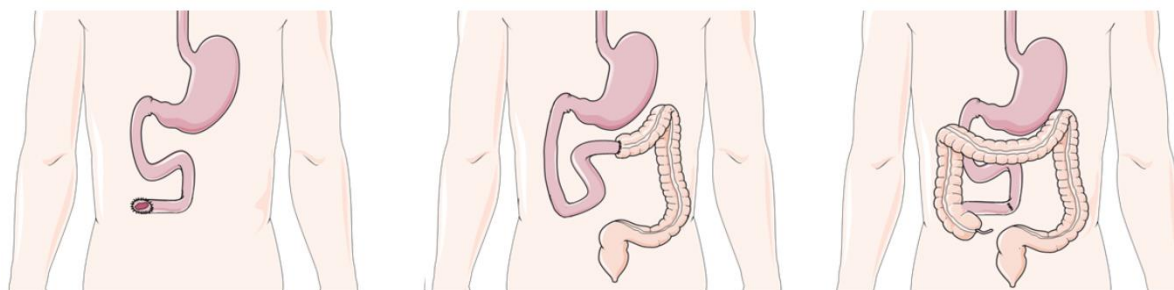


Figure 2. Short bowel syndrome (SBS) anatomy types. Classification of SBS according to anatomy of remnant bowel. Anatomy type 1 (end-jejunostomy; left), type 2 (jejunocolonic anastomosis; middle), and type 3 (jejunoleal anastomosis; right).

Figure reproduced with permission from Verbiest et al.¹⁴

Lastly, intestinal transplantation (ITx) has become a life-saving option for patients with irreversible intestinal failure, most commonly due to short bowel syndrome (SBS) following extensive surgical resections, or severe, treatment-refractory IBD leading to loss of functional small intestine. While parenteral nutrition can sustain many patients for years, long-term dependence is frequently complicated by catheter-related infections, loss of venous access, and progressive intestinal failure-associated liver disease (IFALD). Once

these complications develop, ITx is considered, either as an isolated bowel graft or as a combined liver–intestine transplant when significant hepatic dysfunction is present, or multivisceral transplant involving bowel, liver, pancreas and stomach (**Figure 3**). Combined transplantation is particularly indicated when portal hypertension, synthetic liver failure, or advanced cholestasis accompany chronic intestinal failure. Advances in surgical technique, immunosuppression, and postoperative care have steadily improved outcomes, making ITx a viable therapeutic strategy for select patients.

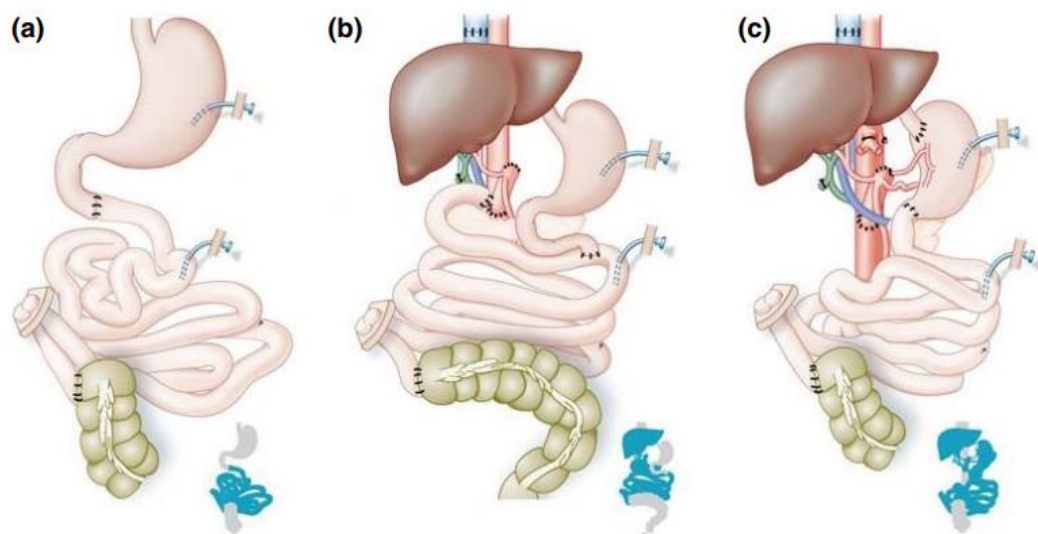


Figure 3. Schematic representation of the three intestinal transplantation types; (a) isolated intestinal, (b) combined liver–intestine, (c) multivisceral transplantation.
Figure reproduced with permission from Ceulemans *et al.*¹⁵

2. Moving beyond the *one-dose-fits-all* paradigm

The *one-dose-fits-all* paradigm is deeply rooted in clinical drug development. This simplistic approach aims to recommend a single dose level while providing a good balance between efficacy and safety for a patient population.¹⁶ This approach served the drug development process well in the last couple of decades, ranging from simplicity in industry to regulatory approval until clinical use of a drug. Furthermore, this approach is particularly useful for drugs with a wide therapeutic window where deviations in exposure do not lead to patient harm in any way. This is also reflected when evaluating the dosing regimens of new approved drugs. It has been shown that 78% of the approved drugs by the Food and Drug Administration (FDA) from 2013–2017 only had one approved dosing regimen, 12% had two, and 3% had three approved dosing regimens, reinforcing the paradigm.¹⁷ Another assumption of the *one-dose-fits-all* approach is that the same dose leads to the same exposure and therefore the same response. However, there are many known (and unknown) factors distorting this relationship. Any variability at any point in the exposure–response relationship will carry forward and finally superimpose on the therapeutic outcome. Besides difficulties in reaching therapeutic exposure and therefore therapeutic response, many drugs exhibit severe side effects and are characterized by a narrow therapeutic window (Figure 4).

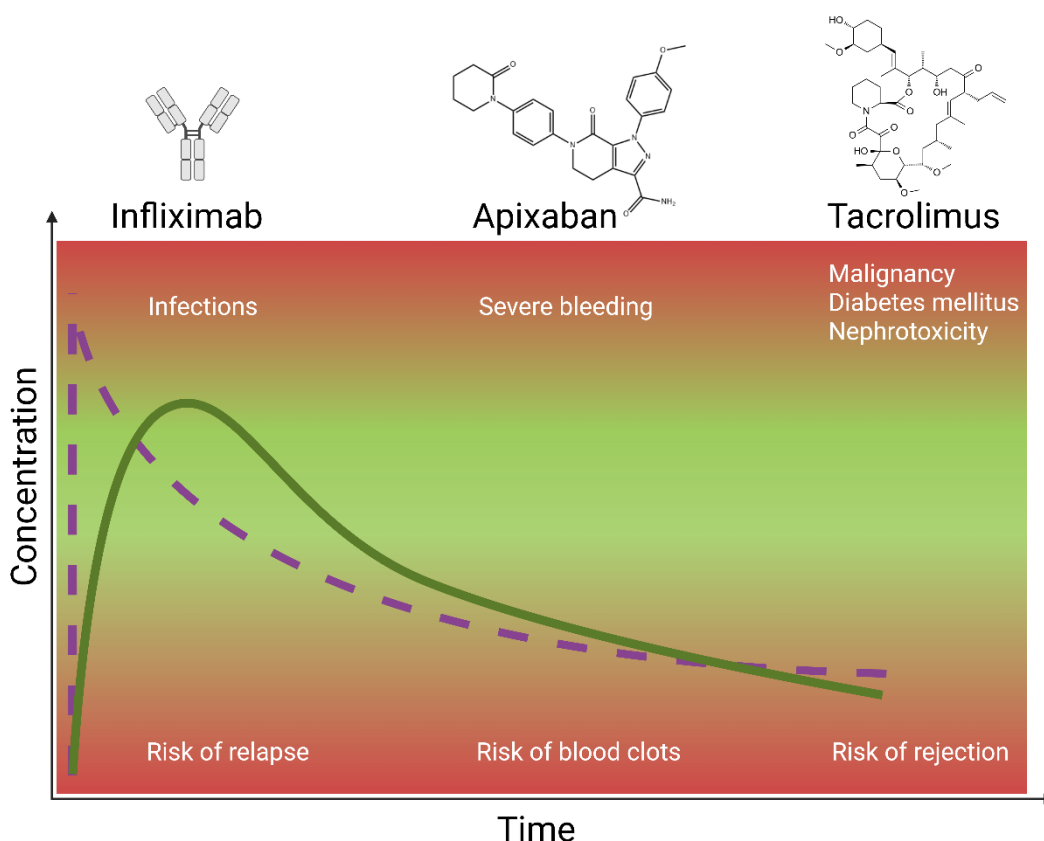


Figure 4. Schematic overview of therapeutic window for infliximab, apixaban and tacrolimus. The green background indicates therapeutic exposure, and the red background indicates sub- and supratherapeutic exposure. The purple dashed line depicts an intravenous and the green solid line an oral drug administration.

2.1 Infliximab

Infliximab is an anti-tumor necrosis factor (anti-TNF) monoclonal antibody (mAb) commonly used for the treatment of patients with IBD. Its effect is mediated by binding to the soluble and transmembrane form of TNF α and thereby neutralizing pro-inflammatory effects of TNF α on target cells.¹⁸ Despite progress in medical therapy, treatment of IBD remains challenging due to heterogeneous disease behaviour, variable response to therapy, and the persistent problem of primary non-response (10–30%) and secondary loss of response (up to 50%) to biologics including infliximab.^{19–22} These limitations partly illustrate the insufficiency of the historical *one-dose-fits-all* paradigm. Traditional step-up or accelerated top-down strategies continue to be employed, yet the emergence of biologics and small molecules has shifted therapeutic goals from symptom control toward a treat-to-target framework prioritizing endoscopic remission, and improvement in quality of life for

instance.^{23–25} Mechanisms defining the absorption, distribution, metabolism, and excretion of monoclonal antibodies like infliximab differ substantially from those of small molecules like tacrolimus or apixaban. Monoclonal antibodies are administered parenterally, for instance intravenously (IV) or subcutaneously (SC).²⁶ The IV route has the benefit of complete bioavailability ($F = 100\%$), a controlled rate, and in case of a bolus injection an immediate peak exposure, whereas the SC route has the benefits of convenient self-injection by the patient, and a depot effect of the drug in the administration compartment leading to a more sustained drug release and concentration. The primary route of antibody elimination is via intracellular degradation after receptor-mediated, target-mediated or fluid phase endocytosis in cells of the reticuloendothelial system.²⁷ One particular case is antidrug antibody-mediated metabolism, often occurring in lower exposure ranges, where therapeutic proteins may elicit a humoral immune response that can alter PK and thus the response to infliximab.²⁸ Antibody excretion via hepatic or renal elimination is less relevant for infliximab (IgG antibody). One clinically relevant drug excretion scenario is for patients with acute severe ulcerative colitis (ASUC), where the intestinal barrier function is damaged, leading to a loss of the drug via the stool and an increase in clearance.^{29,30} This phenomenon is often referred to as leaky gut or protein-losing enteropathy. Patients who are very sick lose drugs to a greater extent through the gut, leading to lower infliximab exposure. Yet, lower infliximab exposure leads to disease worsening, consequently enhancing the vicious circle.³¹

The complex dose–exposure–response interplay in IBD is also reflected in randomized controlled trials (RCT) comparing exposure-guided (therapeutic drug monitoring [TDM]-based) and standard dosing, showing in summary inconclusive results. The first infliximab RCT for patients with IBD by Steenholdt *et al.* showed that reactive TDM during maintenance reduced treatment costs, without improving clinical efficacy.³² The TAXIT and TAILORIX trials point towards a similar finding but failed to show benefits of reactive TDM. However, evidence for the merits of TDM emerged with positive results from NOR-DRUM B trial. The

distorted dose–exposure–response relationship is further complicated by the drug-titration paradox, which is inherently present when dosing is exposure-guided.³³ It has been demonstrated that infliximab clearance (derived from a population pharmacokinetics [popPK] model) remains a robust predictor of clinical efficacy when doses are optimized.³⁴

2.2 Apixaban

Apixaban is an oral, direct factor Xa inhibitor widely used for the prevention and treatment of venous thromboembolism and stroke prevention mainly in patients with atrial fibrillation. Its anticoagulant effect depends on systemic exposure following GI absorption, a process that may be profoundly altered in patients with SBS. Despite its favorable PK profile in the general population, the use of apixaban in SBS remains challenging because adequate absorption depends on sufficient functional intestinal surface area, intestinal transit time, and regional drug uptake.^{35,36} Apixaban is absorbed primarily in the proximal small intestine, with decreasing absorption distally. Typically, the maximum concentration after oral administration occur after 3–4 hours post-administration.³⁷ It undergoes limited first-pass metabolism via cytochrome P450 3A4 (CYP3A4) and P-glycoprotein (P-gp)-mediated transport, and has a half-life of approximately 12 hours.³⁷ In patients with SBS, loss of absorptive surface, accelerated transit, altered luminal pH, and adaptive changes in intestinal transporter expression can lead to highly variable and often reduced bioavailability, resulting in subtherapeutic drug exposure and an increased risk of thromboembolic events. This risk is particularly concerning in SBS patients with ischemic aetiology, who may exhibit ongoing prothrombotic tendencies. Traditional fixed-dose strategies for direct oral anticoagulants assume intact GI absorption and may therefore be insufficient in this population. These limitations highlight the need for an individualized treatment approach that considers the extent and location of bowel resection, residual

intestinal anatomy, and potential pharmacokinetic monitoring to ensure adequate apixaban exposure.

2.3 Tacrolimus

Tacrolimus is a calcineurin inhibitor (CNI) exerting its effect by impairing the immunophilin FK506 binding protein (FKBP), which inhibits the calcineurin phosphatase, a key mediator in T-cell proliferation.³⁸ Its potent immunosuppressive properties made it an integral part of contemporary immunosuppressive management after every solid organ transplantation.^{39–42} However, its use is complicated by a narrow therapeutic window and high intra- and interindividual variability, making it less suitable for the *one-dose-fits-all* approach. Exposure that is too low bears the risk of graft rejection, whereas exposure that is too high bears the risk of de novo malignancy, diabetes mellitus, and nephrotoxicity.^{43,44}

Tacrolimus is available as an oral and intravenous formulation. It undergoes rapid and incomplete oral gastrointestinal absorption, mainly in the small intestine (duodenum and ileum).⁴⁵ After absorption, it is extensively bound to red blood cells, albumin, and α_1 -acid glycoprotein. The mean elimination half-life is approximately 12 hours and blood clearance is approximately 0.06 L/h/kg.⁴⁶ The main metabolic pathway is CYP3A5- and CYP3A4-mediated, making it subject to genetic polymorphisms, inflammation-mediated suppression, and drug–drug interaction (DDI). Furthermore, it has been shown that poor liver function can decrease tacrolimus clearance and therefore increase the elimination half-life.^{47,48}

3. Pharmacometrics

Pharmacometrics is a discipline situated at the intersection between pharmacology, statistics, and medicine. Historically, pharmacometrics emerged from popPK and the identification of sources of variability in the dose–exposure relationship. However, nowadays, pharmacometrics has become an umbrella term for any type of top–down, data-driven modeling of pharmacological and biomedical data. Relationships of interest beyond dose-exposure (pharmacokinetics [PK]), extend to exposure-response (pharmacodynamics [PD]), or dose-exposure-response (PK-PD), and can be modelled accordingly.

3.1 Nonlinear mixed effects modeling

The foundation of modern pharmacometrics is a nonlinear mixed effects modeling approach. “Nonlinear” refers to the nonlinear relationship between the dependent variable (e.g., drug concentration, treatment response, survival probability), and independent variables (e.g., time). “Mixed” refers to the type of parametrization in the model. A mixed effects model combines fixed effects, which determine the typical patient response, with random effects, which model deviations from that typical patient response across individuals or groups.

Parameter estimation in a pharmacometrics model is based on minimizing an objective function value (OFV), often using maximum likelihood estimation. Maximum likelihood estimation relies on first-order estimation (FO), first order conditional estimation (FOCE), FOCE with interaction (FOCE+I), or Laplace estimation.^{49,50} However, there are additional approaches such as i) expectation maximization relying on importance sampling algorithms (IMP), or stochastic approximation expectation maximization (SAEM), ii) Bayesian estimation relying on Markov Chain Monte Carlo (MCMC), and iii) nonparametric estimation methods, the latter of which is beyond our workflow. In Maximum likelihood estimation, the OFV corresponds to minus twice the log of the likelihood (-2LL) and provides an overall

summary of how close the model estimation (i.e., a set of model parameters) describes the observed data (**Figure 5**). In general, the lower the OFV, the better the agreement between model predictions and observed data.

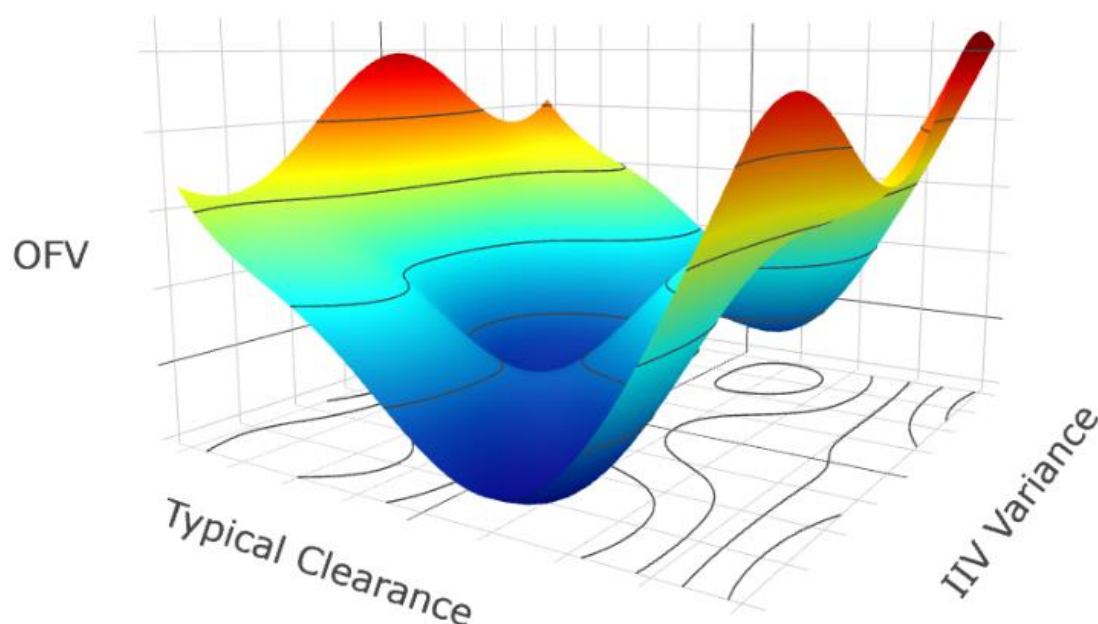


Figure 5. Objective function value landscape depicting a global minimum.
Abbreviations. IIV, Interindividual variability; OFV, objective function value.

In addition to OFV, other model evaluation tools are used during model development including graphical evaluations (e.g., goodness-of-fit plots), simulation-based methods (e.g., visual predictive check; VPC), the precision of parameter estimates (e.g., bootstrap, sampling importance resampling; SIR), and supplementary evaluation tools such as plausibility of model parameters and global and local diagnostics of multicollinearity such as condition number and variance inflation factor (VIF).

3.2 Model components

A traditional mixed effects model consists of a structural, statistical and covariate model. The structural model comprises key components of a pharmacokinetic concentration–time profile, such as clearance (CL), volume of distribution (V), and absorption rate constant (k_a), and may also incorporate key parameters of pharmacodynamic exposure–response models,

such as $E_{\max}$ or EC_{50} . Statistical models describe the variability in a population, whereas covariate models explain (part of) the identified variability.

Structural model

Structural models with shared parameters across the population can be algebraic or differential equations. For a typical pharmacokinetics model for instance, typical parameters represent volumes of compartments, absorption (zero order, first order, Erlang type absorption), and elimination processes (linear, nonlinear, target-mediated drug disposition).⁵⁰ Estimated structural parameters can be expressed as:

$$CL = \theta_1 \quad (\text{Eq. 1})$$

$$V = \theta_2 \quad (\text{Eq. 2})$$

Statistical model

Statistical models account for unexplained variability around the structural model and are typically categorized in two levels. Level 1 variability addresses the inter-individual variability (IIV) and intra-individual variability such as interoccasion variability (IOV) for instance.⁵⁰ The interindividual variability can be expressed:

$$CL_i = \theta_1 \times e^{\eta_{i,1}} \quad (\text{Eq. 3})$$

$$V_i = \theta_2 \times e^{\eta_{i,2}} \quad (\text{Eq. 4})$$

Where η describes the deviations from the typical value (θ) for an individual i . We assume normal random effects, with a mean of 0 and a variance of ω^2 , on the log-scale, which implies a log-normal distribution of the individual parameters, ensuring positivity. Variances (VAR) and covariances (COV) are organized in an omega (Ω) matrix:

$$\Omega = \begin{bmatrix} VAR(CL)[\omega^2_{1,1}] & COV(CL, V) \\ COV(CL, V) & VAR(V)[\omega^2_{2,2}] \end{bmatrix} \quad (\text{Eq. 5})$$

The IOV is included when structural parameters can change over time, particularly relevant in case of a crossover study design, or when relevant time-varying covariates are missing. The importance of including IOV in population PK/PD analyses, and the consequences if not handled properly, has been described before.^{51–53} The IOV can be taken into account using the following expression:

$$CL_{i,j} = \theta_1 \times e^{\eta_{i,1} + IOV} \quad (\text{Eq. 6})$$

$$V_{i,j} = \theta_2 \times e^{\eta_{i,2} + IOV} \quad (\text{Eq. 7})$$

Level 2 variability addresses the residual unexplained variability (RUV), which typically comes from assay variability, sampling errors including timing, and erroneous documentation. The error model can be an additive, proportional, or combined additive and proportional error model. In case of a combined proportional error model, the level 2 random-effects parameters can be expressed as:

$$Y_{i,j} = f(\vec{\theta}, \vec{\eta}_i, t_{i,j}) \times (1 + \varepsilon_{prop,i,j}) + \varepsilon_{add,i,j} \quad (\text{Eq. 8})$$

Where Y is the dependent variable (e.g., drug concentration in a PK model, or response in a PD model) for an individual *i* at time *j*, as a function of the independent variable (e.g., time) composed of a fixed-effects vector of the population ($\vec{\theta}$), the random effects vector of an individual ($\vec{\eta}_i$), and the time for the respective individual (**Figure 6**). Finally, $\varepsilon_{i,j}$ represents the residual unexplained variability for an individual *i* at time *j* and can be framed as a proportional or additive term. Similar to IIV, the RUV can now be compiled into a sigma (Σ) matrix that includes variance parameters for the combined additive (homoscedastic) and proportional (heteroscedastic) residual error model.

$$\Sigma = \begin{bmatrix} VAR(\varepsilon_1)[\sigma^2_{1,1}] & COV(\varepsilon_1, \varepsilon_2) \\ COV(\varepsilon_1, \varepsilon_2) & VAR(\varepsilon_2)[\sigma^2_{2,2}] \end{bmatrix} \quad (\text{Eq. 9})$$

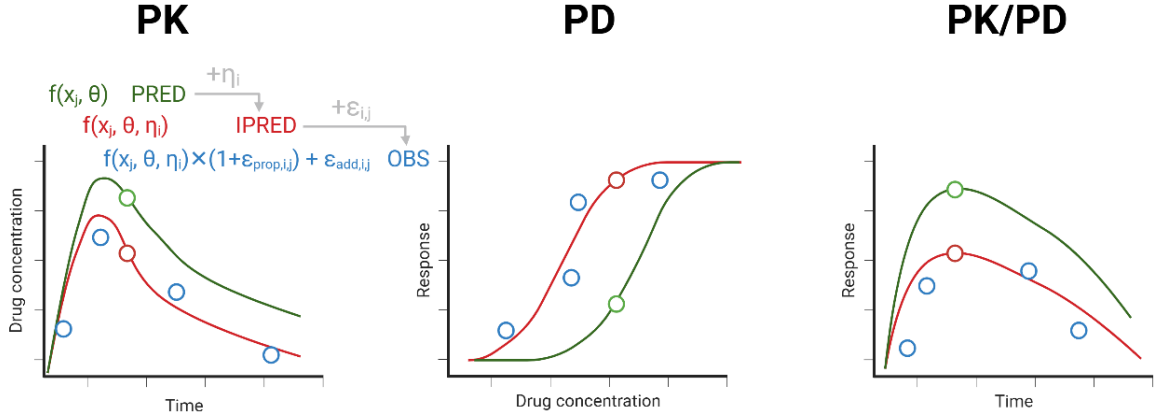


Figure 6. Summary representation of the hierarchical model parameter relationship of population predictions shown in green (PRED), individual predictions shown in red (IPRED) and observations shown in blue (OBS) for pharmacokinetics (PK), pharmacodynamics (PD), and PK/PD models.

Covariate model

A covariate model explains the identified variability in a model parameter across patient characteristics (so-called covariates), which can be for instance bodyweight, sex or disease progression markers. It is important to emphasize that in data-driven modeling, covariates explaining population-level variability usually represent proxies for underlying mechanisms that are mostly unknown or unreported. There are different ways to parameterize a covariate relationship depending on the covariate type (continuous [body weight] or discrete [sex]). The most widely used covariate selection method is the stepwise covariate modeling (SCM), where covariates are included and discarded in a forward inclusion and backward elimination manner, typically on a 5% and 1% significance level, respectively. In recent years, additional covariate modeling approaches emerged, including Least Absolute Shrinkage and Selection Operator (LASSO), Conditional Sampling use for Stepwise Approach based on Correlation tests (COSSAC), Regularized Horseshoe (RHS), Wald approximation method, full fixed effect model (FFEM), full random effect model (FREM), SCM+, and SCM+ with stage-wise filtering.^{54,55} These approaches differ in their underlying assumptions and performance

characteristics: stepwise methods (SCM, SCM+) are widely used and interpretable but can be computationally intensive and prone to selection bias, whereas full model approaches (FFEM, FREM) avoid selection bias but may suffer from imprecise estimates or increased model complexity. Penalized and Bayesian methods (e.g., LASSO, RHS) enable simultaneous evaluation of many covariates and may improve predictive performance, although they require more advanced implementation and can introduce shrinkage or dependence on prior assumptions. Finally, efficient search algorithms such as COSSAC and approximation-based methods (e.g., Wald approaches) aim to reduce computational burden while retaining similar selection performance.

3.3 Modeling software

Throughout the thesis, population modeling and simulation were carried out using NONMEM (ICON Development Solutions, Gaithersburg, MD, USA), with a GNU Fortran 95 compiler and the Pearl-speaks-NONMEM (PsN) toolkit on the Pirana interface software (Certara, Inc., Princeton, NJ, USA). Data handling, post-processing and visualization were performed in R within the RStudio (Posit) integrated development environment. For projects requiring higher computational power, the VSC (Flemish Supercomputer Center) infrastructure was used, funded by the Research Foundation Flanders and the Flemish Government. Deviations from this particular workflow were indicated in each chapter, where applicable.

4. Clinical applicability of pharmacometrics models

In recent years, the value of computer-based models in clinical decision-making has substantially increased. These models serve as powerful tools to simulate alternative dosing regimens, identify superior dosing strategies for instance covariate-based dosing, and support risk stratification by identifying patient subgroups at increased risk of unfavorable clinical outcomes. Particularly for special patient populations, such as patients with gastrointestinal disorders, model-based support is warranted. The importance of such model-informed approaches is increasingly recognized in regulatory guidance, including the ICH M15 guideline, which emphasizes the role of quantitative modeling in optimizing drug development and clinical use.⁵⁶

One particular relevant case of pharmacometrics in the clinic is the estimation of individual patients' PK (or PD) parameters using a model, individual patient data and a process called Bayesian forecasting. Bayesian statistics (and forecasting) is about updating prior beliefs once new data and information become available. In the pharmacometrics setting, a model (prior) is updated using observed individual patient data (likelihood) and model estimates are then sampled from a *posterior* distribution (Eq. 10);

$$OFV = \sum_{j=1}^n \frac{\{y_i - f(\theta, x_j)\}^2}{\sigma_{y_i}^2} + \sum_{p=1}^p \frac{\{\theta_{p,i} - \theta_p\}^2}{\omega_{\theta_p}^2} \quad (\text{Eq. 10})$$

with $\{y_i - f(\theta, x_j)\}$ depicting the difference between the j^{th} observation in the patient and its population model-predicted value using the vectors of individual model parameters θ and covariates x_j , and $\sigma_{y_i}^2$ depicting the residual variance around the model prediction within the patient. Furthermore, $\{\theta_{p,i} - \theta_p\}$ depicts the difference between the p^{th} individual model parameter and its population average value, and $\omega_{\theta_p}^2$ the variance of the IIV of the population parameter p .

Obtaining individual pharmacokinetics estimates is relevant to identify changes in drug PK over time, but also to obtain model-based exposure, which represents “one of the most valuable assets to the pharmacometrician”.⁵⁷ The estimation of individual patients' pharmacokinetic parameters always relies on a computational process and usually a software tool. In recent years, accessibility to suitable software tools has remarkably increased with sophisticated options and user-friendly interfaces.⁵⁸

Model-based TDM (commonly referred to as model-informed precision dosing [MIPD]) is increasingly acknowledged and employed for various drug classes, disease areas, and therapy monitoring approaches (**Figure 7**). Furthermore, there are tremendous efforts to improve the predictive performance of models to further enhance their clinical utility in an MIPD setting.⁵⁹⁻⁶¹

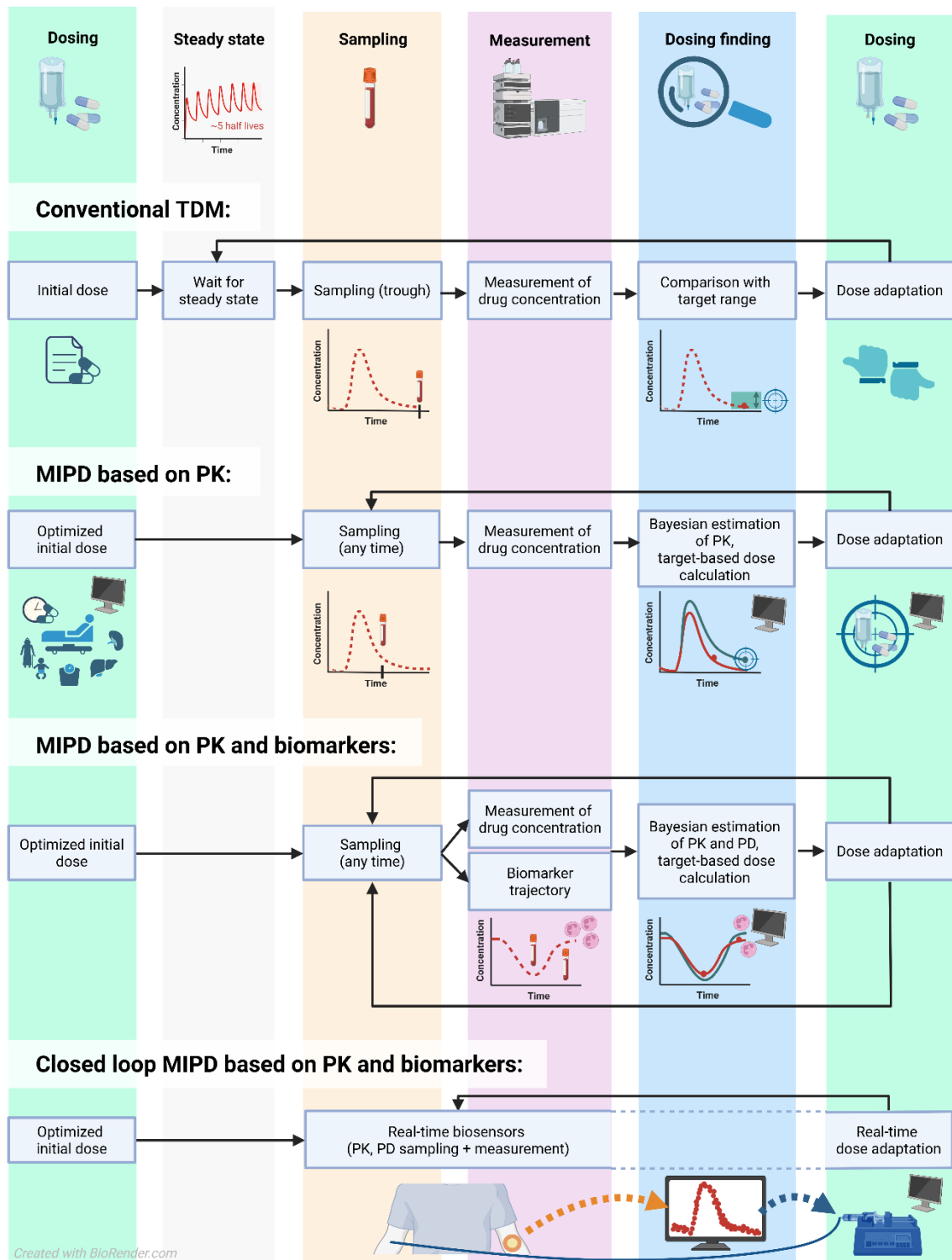


Figure 7. Different approaches of dosing and dosing individualization involving PK or biomarker measurements. Abbreviations. MIPD, model-informed precision dosing; PD: pharmacodynamics; PK, pharmacokinetics; TDM, therapeutic drug monitoring. Figure reproduced with permission from Minichmayr *et al.*⁶²

4.1 Pharmacokinetics–pharmacodynamics of infliximab in inflammatory bowel diseases

The concept of infliximab MIPD was previously proposed and validated in *in silico* simulations and retrospective studies.^{63–65} However, there is evidence accumulating and translating these results into RCTs. In pioneering work, Strik *et al.* evaluated and showed merits of MIPD of infliximab compared to standard dosing for maintaining remission during maintenance therapy.⁶⁶ In the MODIFI study (NCT04982172), a historically controlled interventional study, Wang *et al.* demonstrated opportunities for model-guided dosing interval extensions.⁶⁷ In pediatric patients, MIPD improved infliximab target attainment and reduced immunogenicity.⁶⁸ Additional results from ongoing RCTs like OPTIMIZE (NCT04835506), and REMODEL-CD (NCT05660746) are awaited.

In IBD, outcome measures range from non-invasive scores such as the patient-reported outcome (PRO2) score, to moderately invasive endoscopic assessments, and ultimately to hard clinical endpoints like colectomy. Capturing this diversity in real-world outcome data demands robust and versatile exposure–response modeling approaches.

4.2 Pharmacokinetics of apixaban in patients with short bowel syndrome

The European Medicine Agency (EMA) recognized the need for dose adaptation of apixaban and recommended a 50% dose reduction for patients with severely reduced kidney function or if two of the following apply; i) serum creatinine ≥ 133 $\mu\text{mol/L}$, ii) ≥ 80 years or iii) ≤ 60 kg.⁶⁹ Furthermore, a popPK modeling and simulation study in hospitalized patients demonstrated that patients with stage 3 or 4 chronic kidney disease require a two and four-fold longer preoperative apixaban withdrawal time to reach the recommended levels compared to patients with normal kidney function.⁷⁰ Furthermore, the authors conclude that model-based dosage individualization appears to be an elegant approach to predict the time to therapeutic and safe concentration.⁷⁰ Model-based dose individualization studies were

already performed for hospitalized patients and patients with nonvalvular atrial fibrillation.^{70,71} A recent study investigating the pharmacokinetics of apixaban in 13 patients with SBS (one dose-level per patient) concluded that “the absorption may not be significantly altered”.⁷² However, given the well-documented impairment of intestinal absorption in this population, model-based dose individualization may offer a promising strategy to optimize apixaban therapy.

4.3 Pharmacokinetics–pharmacodynamics of tacrolimus in patients undergoing solid organ transplantation

Recognizing the clinical importance of individualized dosing, EMA has approved exposure-guided dosing of tacrolimus,⁷³ and TDM is therefore routinely implemented in clinical practice. However, regulatory guidance does not specify how individualized dosing should be operationalized. In recent years, several prospective clinical studies have demonstrated the potential benefit of tacrolimus MIPD, as summarized by Minichmayr *et al.* For instance, it has been shown that model-guided dosing improved tacrolimus target concentration achievement in patients with kidney transplantation.⁷⁴ Similar results were shown by Fukudo *et al.* using a Bayesian approach for dose individualization in patients undergoing liver transplantation.⁷⁵ Despite these advances, there remains a notable lack of popPK models specifically developed for ITx, as well as a paucity of quantitative evidence linking tacrolimus exposure to long-term outcomes. Addressing these gaps is essential to further improve precision dosing strategies and long-term outcomes in intestinal transplant recipients.

5. Overall thesis aim and research question

The overall aim of this thesis is to develop and apply pharmacometrics approaches to advance individualized therapy in major gastrointestinal disorders.

Through quantitative modeling of drug exposure, therapeutic intervention, response, and

physiological changes, we aim to provide actionable insights that support model-informed precision dosing in IBD, SBS, and ITx.

OBJECTIVES

Over the last few decades, substantial advances in therapeutic options have been made for patients with gastrointestinal disorders. However, in this particular population, achieving adequate clinical efficacy remains a substantial challenge due to marked variability arising from (patho-)physiological changes. The heterogeneity and complexity of this special patient population remain underexplored, making the identification of patient subgroups at risk of therapeutic failure increasingly relevant.

We hypothesize that population modeling and **model-informed therapy individualization** of infliximab, apixaban, and tacrolimus may **improve exposure target attainment, reduce adverse events, and improve therapy outcomes** for patients with inflammatory bowel diseases, short bowel syndrome, and bowel transplantation.

The main objective of this doctoral research is to **explore the pharmacokinetics and pharmacodynamics** of these drugs and investigate the merits of **popPK/PD modeling and simulation** in therapy individualization.

In **chapter 1** we performed a modeling and simulation study to explore the pharmacokinetics–pharmacodynamics of subcutaneous infliximab.

In **chapter 2** we performed a modeling and simulation study to explore the pharmacokinetics–pharmacodynamics of infliximab in patients with acute severe ulcerative colitis.

In **chapter 3** we explored the pharmacokinetics of apixaban in patients with short bowel syndrome.

In **chapter 4** we developed a multistate model framework, and we explore long-term outcomes of patients undergoing enteral nutrition

In **chapter 5** we systematically reviewed available popPK models of tacrolimus.

In **chapter 6** we performed a modeling and simulation study to explore the pharmacokinetics-pharmacodynamics of tacrolimus in patients undergoing intestinal transplantation.

Chapter 1

Population pharmacokinetics–pharmacodynamics of subcutaneous infliximab and relapse risk stratification in inflammatory bowel disease

This chapter is based on

Yannick Hoffert, Zhigang Wang, Mathurin Fumery, Maria Nachury, Maëva Bazoge, Anthony Buisson, Erwin Dreesen. A Risk Stratification Tool for Relapse After Intravenous-To-Subcutaneous Switching of Infliximab in Patients With Inflammatory Bowel Diseases. *Am J Gastroenterol*. 2026 Jan 1;121(1):179-186.

Abstract

Objectives A subcutaneous formulation of infliximab was recently approved for maintenance therapy of inflammatory bowel disease. However, limited clinical experience, particularly with patients on escalated intravenous infliximab regimens, poses challenges for the transition to subcutaneous therapy. We investigated the pharmacokinetics and pharmacodynamics of subcutaneous infliximab to identify early predictors of relapse upon switching.

Methods We repurposed data from a prospective, multicenter trial involving patients with inflammatory bowel disease switching from intravenous to subcutaneous infliximab. We estimated each patient's infliximab clearance using Bayesian forecasting from a pre-switch sample and a population pharmacokinetics model. We performed pharmacodynamics modeling to evaluate pre-switch predictors of post-switch relapse. Relapse was defined as clinical recurrence (partial Mayo score >2 or Harvey–Bradshaw Index >4 leading to therapeutic escalation) or an increase in fecal calprotectin ≥ 150 $\mu\text{g/g}$ upon switching.

Results Using data from 98 patients with inflammatory bowel disease, we identified infliximab clearance and fecal calprotectin as independent predictors of relapse. A 2-item risk score stratified patients into the low-risk ($<19\%$ probability of relapse; 75/98; 77%) and high-risk ($\geq 19\%$ probability of relapse; 23/98; 23%) groups (sensitivity 0.75 [95% CI 0.48–0.93], specificity 0.87 [95% CI 0.77–0.93] positive predictive value 52% [95% CI 31–73%], negative predictive value 95% [95% CI 87–99%]). Our pharmacokinetics–pharmacodynamics model classified patients with and without relapse ($P < 0.0001$) with an area under the receiver operating characteristic curve of 0.83 (95% CI 0.71–0.93).

Conclusions Pre-switch infliximab clearance and fecal calprotectin are accurate predictors of relapse after switching to subcutaneous infliximab. An interactive risk stratification tool

facilitates confirmation of a stratified medicine approach to improve infliximab therapy in inflammatory bowel disease.

1. Introduction

Recently, a subcutaneous formulation of infliximab CT-P13 has been developed and added as a treatment option for patients with CD and UC.⁷⁶ It is well tolerated and offers maintained or improved efficacy, as observed in several pivotal trials.^{77,78} Additionally, it has been demonstrated that deep remission in patients with CD and UC was associated with higher infliximab concentrations following subcutaneous dosing.⁷⁷ Moreover, the subcutaneous formulation has the benefit of convenient self-injection and is preferred by the majority of patients.⁷⁹ Direct pharmacoeconomic benefits due to reduced demands on healthcare professionals and resources have been described as well.^{80,81}

A recent prospective trial investigated intravenous-to-subcutaneous switching, mainly focusing on patients on standard 8-weekly 5 mg/kg regimens.⁸² This poses questions regarding the suitability of the standard 2-weekly 120 mg subcutaneous regimen for patients on escalated regimens. Therefore, TDM was recommended prior to switching.⁸³ Furthermore, the need for reactive TDM in patients losing response to subcutaneous infliximab maintenance therapy was highlighted. However, the lack of an infliximab concentration threshold and a treatment decision strategy impedes the implementation of TDM. We previously proposed a subcutaneous dosing strategy aimed at avoiding a drop in infliximab trough concentrations upon switching.⁸⁴ Moreover, we developed a calculator to predict post-switch steady-state infliximab trough concentrations from the last pre-switch trough concentration, and we hypothesized that an overprediction may be explained by a sudden increase in infliximab clearance, thus being indicative of relapse. Confirmation of this hypothesis is still awaited.

We used data from the prospective, multicenter REMSWITCH trial to investigate the pharmacokinetics and pharmacodynamics of infliximab during intravenous-to-

subcutaneous switching and identify early predictors of relapse in patients with CD and UC.³²

2. Methods

2.1 Patients

We repurposed data from the prospective, multicenter REMSWITCH trial.⁸² Patients were treated with one of four intravenous infliximab dosing strategies; 5 mg/kg every 8 weeks, 10 mg/kg every 8 weeks, 10 mg/kg every 6 weeks, or 10 mg/kg every 4 weeks, followed by one of four subcutaneous dosing strategies; 120 mg every 2 weeks, 240 mg every 2 weeks, 120 mg every week, or 240 mg every week. Predose (trough) blood samples were drawn before the switch (visit 0 [v0]) and after the switch (v1). The time point right before the first SC administration was defined as baseline. The first visit after the switch was between weeks 4–8 and samples were drawn right before the next SC administration. Relapse was assessed at visit 1, visit 2 (8–16 weeks after the switch), and visit 3 (16–24 weeks after the switch). Patients who had a pre-switch infliximab concentration available were eligible for our analysis.

We collected patient information on age, bodyweight, sex assigned at birth (female/male), diagnosis (CD/UC), disease duration, antibodies towards infliximab (present/absent), C-reactive protein, fecal calprotectin, coadministration of methotrexate (yes/no), partial Mayo Score and Harvey–Bradshaw Index. Patients were eligible for our analysis only if variable missingness was less than 10% of the REMSWITCH dataset.

Buisson *et al.* previously provided a comprehensive description of the study protocol.⁸² The trial was approved by the local Ethics Committee (IRB00013412, “CHU de Clermont Ferrand IRB #1”, number 2022-CF001) in compliance with the French policy of individual data protection. We repurposed the trial data in this retrospective study, which was approved by the local Ethics Committee of CHU de Clermont Ferrand (2023-CF059) and KU Leuven (G-2024-8508). We performed the study in accordance with the Declaration of Helsinki, Good

Clinical Practice, and applicable regulatory requirements including individual data protection.

2.2 Bayesian forecasting with the population pharmacokinetics model

We coded the popPK model of infliximab from the information available in the European Public Assessment Report (EPAR) in R (version 4.4.1) (**Supplemental code**).²⁶ This model was built on data from studies CT-P13 1.5 (healthy subjects), CT-P13 3.5 Part 1 and Part 2 (subjects with rheumatoid arthritis), and CT-P13 1.6 Part 1 (subjects with CD). We used each patient's measured pre-switch (v0) infliximab trough concentration to estimate the individual pre-switch pharmacokinetics parameters (clearance and volume of distribution in the central compartment). This we did by performing maximum *a posteriori* Bayesian forecasting with the popPK model using the *mapbayr* package (version 0.10.0) in R.⁸⁵

2.3 Predictive performance assessment of the population pharmacokinetics model

We used the Bayesian forecasted maximum *a posteriori* pharmacokinetics parameters to simulate the individual patient's infliximab concentration–time course under the given dosing. Simulations were performed using the *mrgsolve* package (version 1.0.9) in R.

We assessed the predictive performance of the popPK model by comparing the observed post-switch (v1) infliximab trough concentration with the corresponding individual prediction from Bayesian forecasting and simulation. Bland–Altman analysis was performed using the *BlandAltmanLeh* package (version 0.3.1) in R to evaluate the agreement between observed and predicted infliximab concentrations.⁸⁶ We considered predictions unbiased when zero was included in the 95% confidence interval (CI) of the mean difference between observed and predicted infliximab concentrations. Individual infliximab predictions were compared with the predictions obtained in NONMEM (version 7.5.0).

2.4 Development of a relapse prediction model

We developed a logistic regression pharmacodynamics model to predict each patient's probability of relapse. Relapse was defined as clinical recurrence (partial Mayo score >2 or Harvey–Bradshaw index >4) leading to therapeutic escalation or an increase in fecal calprotectin of ≥ 150 $\mu\text{g/g}$ compared to baseline.⁸² We performed a covariate search to identify pre-switch (v0) predictors of relapse. The tested covariates included bodyweight, sex assigned at birth, diagnosis, presence/absence of antibodies towards infliximab, C-reactive protein, fecal calprotectin, coadministration of methotrexate, the intravenous administration regimen, the subcutaneous administration regimen, the measured infliximab concentration, and the Bayesian forecasted infliximab clearance. The covariate search was automated using the *glmulti* package in R (version 1.0.8), and the best model was defined based on the Akaike Information Criterion, a fitting parameter penalizing the number of parameters in a model,⁸⁷ and a VIF ≤ 5 was deemed acceptable to ensure low multicollinearity.⁸⁸

The predictive performance of the model was internally evaluated by assessing model discrimination (Receiver Operating Characteristics [ROC] analysis) and model calibration (VPC). We stratified patients into low-risk or high-risk groups based on the risk threshold with the highest Youden J index.

2.5 Implementation of the model into an interactive tool

We developed an interactive tool based on the integrated popPK and PD relapse prediction model. The *Shiny* package (version 1.7.4) in R was used for developing the web application, interfaced with RStudio (version 2023.09.1).

2.6 Statistics

Descriptive and inferential statistics were reported as mean and standard deviation or median and interquartile range (IQR), respectively, using a 2-tailed Student *t*-test and

Wilcoxon signed-rank test, as appropriate. For discrete variables a Fisher's exact test was used and values were reported as percentages. A 2-sided *P* value less than .050 was considered statistically significant. Statistical analyses and calculations were performed using R.

3. Results

3.1 Patients

Of 133 patients included in the REMSWITCH trial, 98 (73.6%) were eligible for inclusion in our analysis (**Figure S1**). The most common intravenous regimen was the label regimen of 5 mg/kg every eight weeks (45/98 patients, 45.9%). The most common subcutaneous regimen was also the label regimen of 120 mg every two weeks (93/98 patients, 94.9%). Sixteen patients (16.3%) experienced a relapse upon switching from intravenous to subcutaneous infliximab. A detailed overview of the relapse cohort criteria is provided in **Table S1**. The univariate analysis revealed a statistically significant difference in estimated clearance at baseline ($p<0.01$), fecal calprotectin at baseline ($p<0.001$), IV dosing regimen before the switch ($p<0.001$), and SC dosing regimen after the switch ($p<0.05$) between the no relapse and relapse cohort (**Table 2**).

Table 2. Baseline, regimen, and on-treatment characteristics of patients included in the analysis dataset

Characteristic	No relapse	Relapse	Significance level
Number of patients	82	16	
<i>Demographics at baseline</i>			
Age, years, mean $\pm$ SD*	38.6 $\pm$ 14.6	44.8 $\pm$ 12.8	ns
Bodyweight, kg, median [IQR]*	69.5 [59.3–80.0]	70.0 [65.0–78.5]	ns
Sex assigned at birth, female, n (%)	43 (52.4)	8 (50.0)	ns
Diagnosis, CD/UC, n (%)	65/17 (79.3/20.7)	11/5 (68.8/31.3)	ns
Disease duration, years, median [IQR]	10.0 [7.0–19.0]	15.5 [11.8–23.0]	ns
Estimated infliximab clearance, L/day, median [IQR]	0.31 [0.25–0.35]	0.40 [0.31–0.48]	<0.01
<i>Serology at baseline</i>			
Detectable antibodies towards infliximab, n (%)	1 (1.2)	1 (6.3)	ns
C-reactive protein, mg/L, median [IQR]	1.0 [0.0–3.0]	3.0 [1.4–3.3]	ns
Infliximab trough concentration, μ g/ml, median [IQR]	5.9 [4.1–7.7]	7.4 [5.7–14.4]	ns
<i>Disease activity at baseline</i>			
Harvey Bradshaw index for patients with CD, 0:1:2:3:4:5:6, n [†]	36:15:4:2:3:2:1	6:2:2:1:0:0:0	ns
Partial Mayo score for patients with UC, 0:1:2:3, n	14:2:1:0	4:1:0:0	ns
Fecal calprotectin, μ g/g, median [IQR]	30.0 [15.0–80.8]	218.0 [58.5–656.0]	<0.001
<i>Dosing regimens</i>			
Methotrexate co-medication, n (%)	2 (2.4)	1 (6.3)	ns
IV, 5 mg/kg, q8w, n (%)	40 (48.8)	5 (31.3)	
IV, 10 mg/kg, q8w, n (%)	29 (35.4)	3 (18.8)	<0.001
IV, 10 mg/kg, q6w, n (%)	11 (13.4)	2 (12.5)	
IV, 10 mg/kg, q4w, n (%)	2 (2.4)	6 (37.5)	
SC, 120, q2w, n (%)	80 (97.6)	13 (81.3)	
SC, 240, q2w, n (%)	1 (1.2)	1 (6.3)	<0.05
SC, 120, q1w, n (%)	1 (1.2)	2 (12.5)	

Abbreviations. CD, Crohn's disease; IQR, interquartile range; IV, intravenous; ns, not significant; qwx, every x week(s); SC, subcutaneous; SD, standard deviation; UC, ulcerative colitis. Significance levels were calculated using a 2-tailed Student t-test and Wilcoxon signed-rank test, as appropriate. For categorical covariates, Fisher's exact test was used.

*Age and bodyweight were not available for three study participants. [†]Harvey-Bradshaw Index was not available for two patients.

3.2 Predictive performance of the population pharmacokinetics model

From the 133 patients, 76 were eligible for inclusion in our analysis (**Figure S1**). The predicted post-switch infliximab trough concentrations showed acceptable agreement with the observed concentrations (mean bias -0.43 mg/L, 95% CI -1.64 mg/L to +0.78 mg/L) (**Figure 8**).

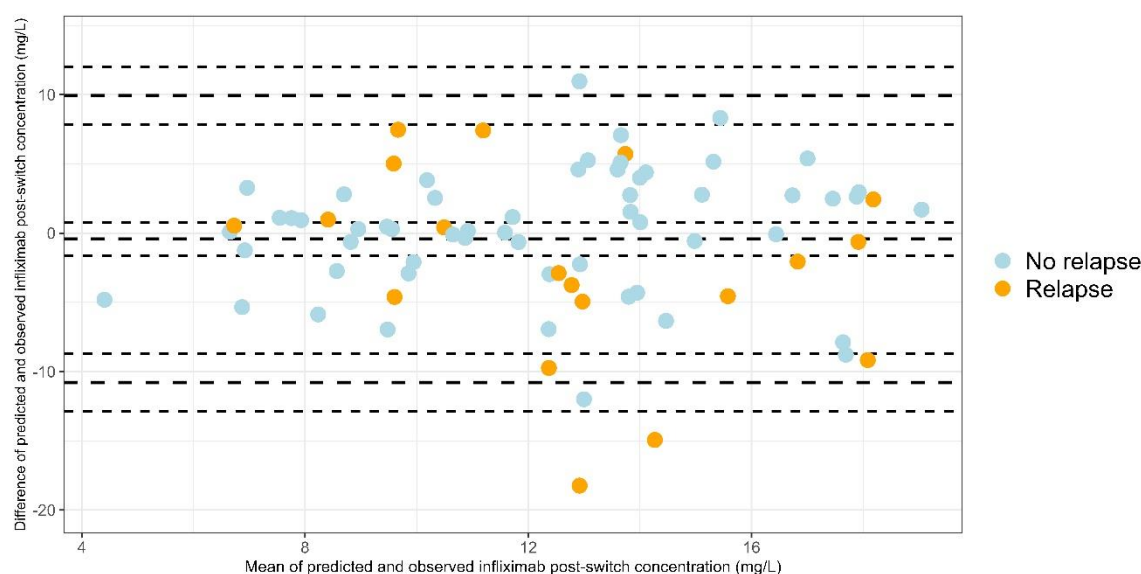


Figure 8. Bland–Altman plot showing the agreement between the observed and predicted post-switch infliximab trough concentrations. The horizontal long-dashed lines represent the mean and standard deviation. The short dashed lines represent the 95% confidence intervals for the mean and the standard deviation. From the 98 patients included in this study, 76 provided a pre-switch/post-switch infliximab trough concentration pair of which 19 experienced a relapse.

Predictions were unbiased, yet precision was poor, especially for patients who experienced a relapse (**Figure S2, Table S2**). We included visual representations of the individual patient’s simulated infliximab concentration–time courses in **Figure S3**.

3.3 Relapse prediction model

The pre-switch (v_0) infliximab trough concentration was not significantly different between patients with and without relapse ($p = 0.125$). Bayesian forecasted infliximab clearance was significantly lower in relapse-free patients (median 0.31 [IQR 0.25–0.35] L/day) than in patients who experienced a relapse (median 0.40 [IQR 0.31–0.48] L/day) ($p < 0.01$). The corresponding area under the ROC curve was 0.74 (95% CI 0.58–0.88). Fecal calprotectin was significantly lower in relapse-free patients (median 30.0 [IQR 15.0–80.8] $\mu\text{g/g}$) than in patients experiencing relapse (median 218.0 [IQR 58.5–656.0] $\mu\text{g/g}$) ($p < 0.001$). The corresponding area under the ROC curve was 0.74 (95% CI 0.58–0.89) (**Figure S4**).

Both pre-switch Bayesian forecasted infliximab clearance and fecal calprotectin were identified as predictors of relapse in the multivariable logistic regression pharmacodynamics model. The VIF of the two predictors was 1.02 indicating low

multicollinearity. An overview of the final model parameter estimates and their precision was appended in **Table S3**. The probability of relapse of patient i was calculated as follows:

$$P(\text{Relapse})_i = 1 - \frac{1}{1 + e^{-6.959 + 8.3953 \times \text{Clearance}_i + 0.5422 \times \text{Ln}(\text{Fecal calprotectin}_i)}} \quad (\text{Eq. 11})$$

The model-predicted probability of relapse was significantly higher in patients who experienced a relapse (median 32% [IQR 17–64%]) than in patients without a relapse (median 7.4% [IQR 4.7–13%]) ($p < 0.0001$). The corresponding area under the ROC curve was 0.83 (95% CI 0.71–0.93) (**Figure 9**). Calibration of the prediction model was good (**Figure 10**).

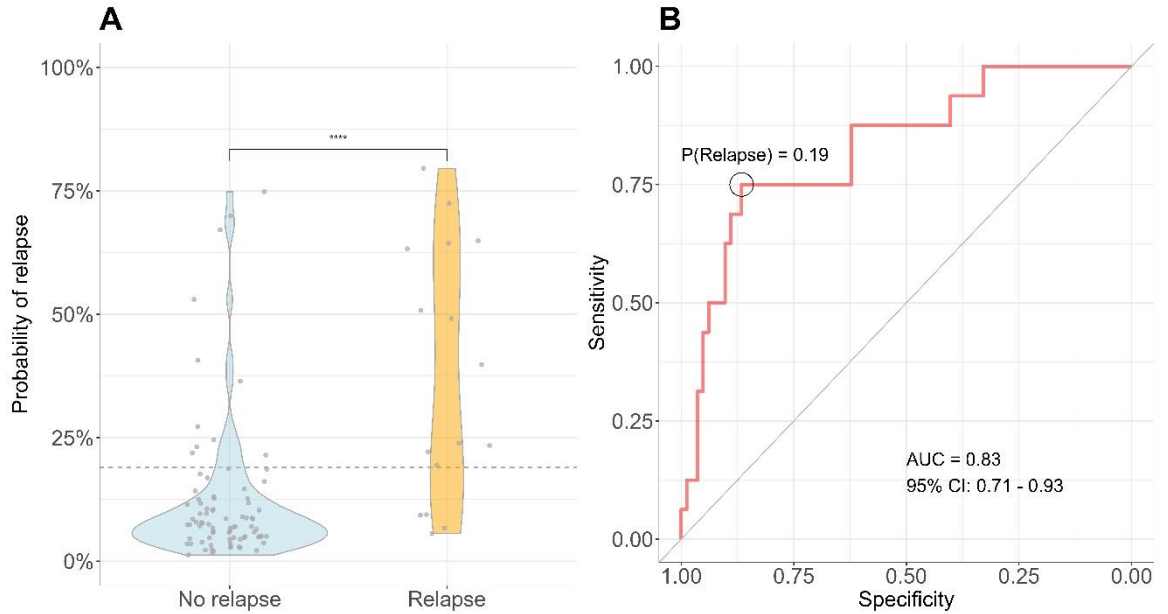


Figure 9. Model-predicted probability of relapse stratified for patients with and without relapse (A), and the corresponding receiver operating characteristics curve (B). The horizontal grey dashed line corresponds to the identified threshold of 19%. Confidence intervals are based on 2,000 non-parametric bootstraps. **** $p < 0.0001$ using Wilcoxon signed-rank test.

Abbreviations. AUC, area under the curve; CI, confidence interval; P, probability.

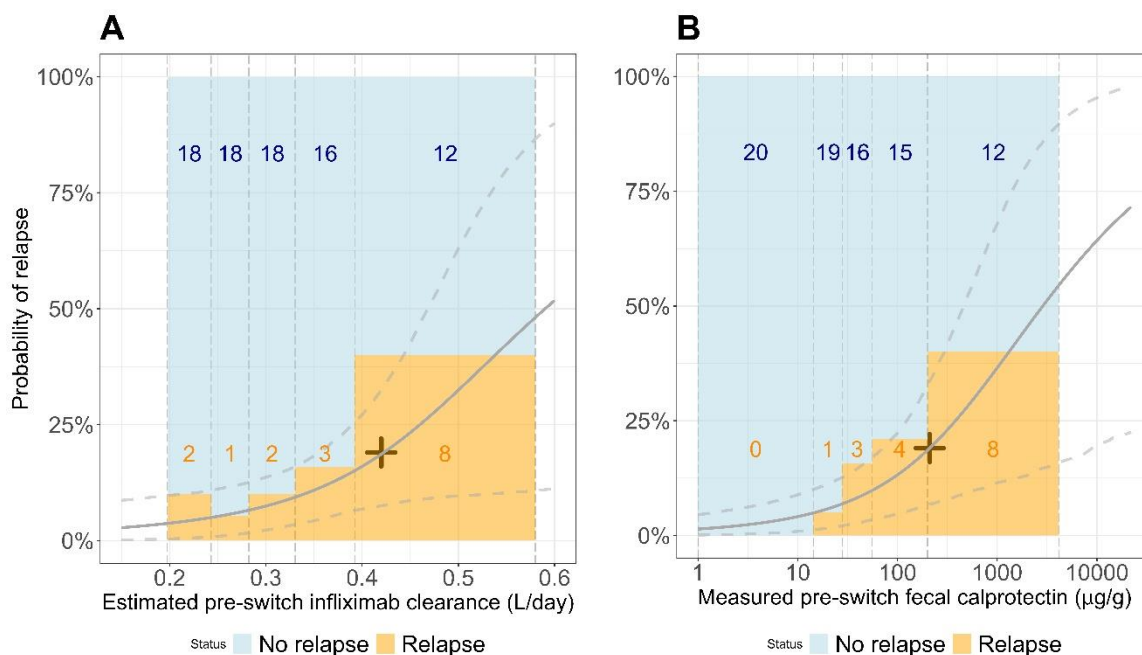


Figure 10. Visual predictive checks of the relapse prediction model. Numbers represent the number of patients in each quantile with (bottom) and without (top) relapse. The solid and dashed lines represent the model median and 95% confidence interval, respectively. The grey cross indicates the 19% probability of relapse cutoff for stratifying patients into low-risk and high-risk groups corresponding to an estimated pre-switch infliximab clearance, and measured pre-switch fecal calprotectin value of 0.42 L/day and 210.6 µg/g, respectively.

We identified a threshold probability of relapse of 19% (sensitivity 0.75 [95% CI 0.48–0.93], specificity 0.87 [95% CI 0.77–0.93], negative predictive value 95% [95% CI 87–99%], positive predictive value 52% [95% CI 31–73%]). Patients with a probability of relapse of 19% or higher (23/98, 23.5%) were defined as high-risk patients. Of these patients, twelve (52.1%) experienced a relapse. On the other hand, patients with less than 19% probability of relapse (75/98, 76.5%) were stratified into the low-risk group. Of these patients, only four (5.3%) experienced a relapse (**Figure S5**).

Figure 11 illustrates the distribution of patient risk factors (pre-switch Bayesian forecasted infliximab clearance and measured fecal calprotectin) in the model development dataset with respect to the simulated probability of relapse. Calibration of the final pharmacokinetics–pharmacodynamics relapse prediction model was good.

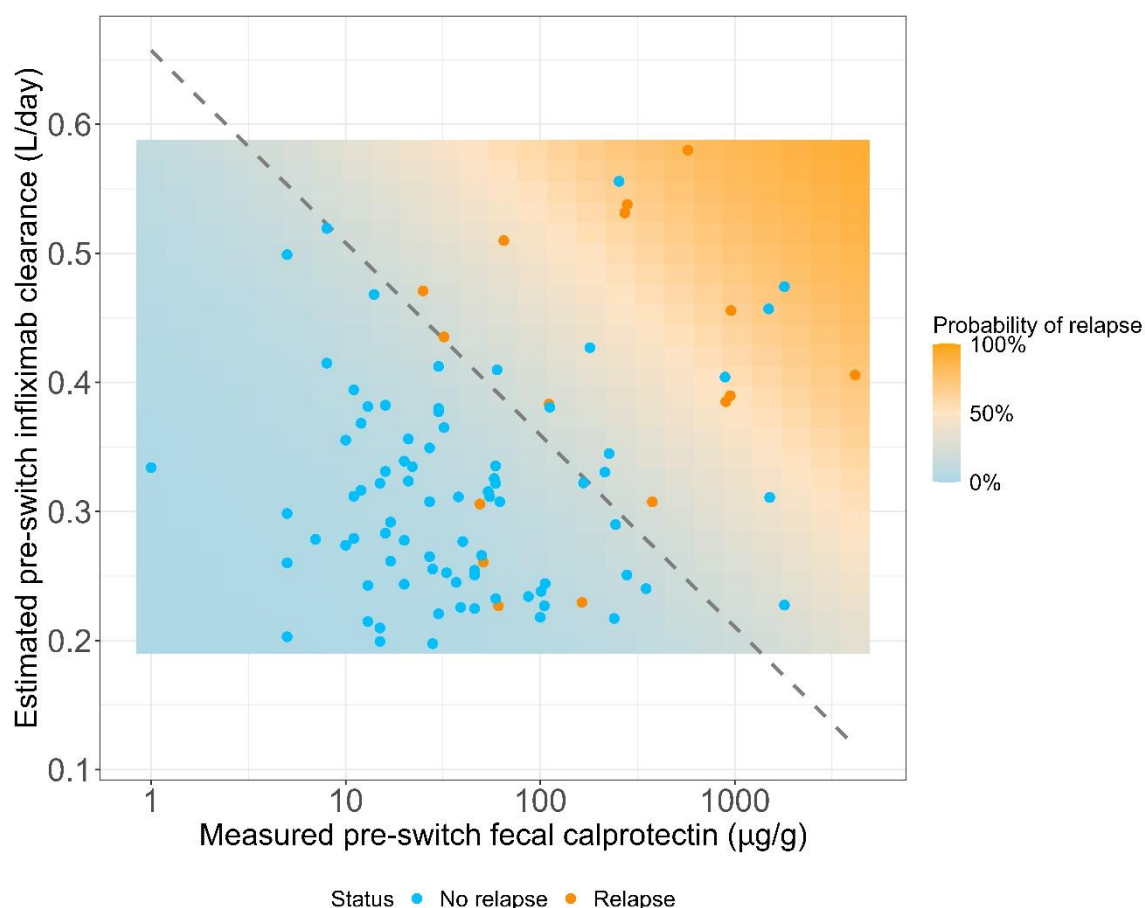


Figure 11. Heat map distribution of pre-switch Bayesian forecasted infliximab clearance and measured fecal calprotectin among low-risk and high-risk patients. The low-risk group was defined as patients with a probability of relapse below 19%, indicated by the dashed grey line. Model-predicted probabilities (background; color gradient) are overlaid with dots representing patients in the model development dataset.

3.4 Interactive prediction tool

We developed an interactive tool based on the integrated popPK and relapse prediction model. The performance of the tool (individual infliximab predictions obtained with the *mapbayr* package) was successfully cross-validated against NONMEM (**Figure S6**). Our pharmacokinetics–pharmacodynamics model offers an evidence-based approach to improve infliximab therapy in IBD, and its integration into a software tool can facilitate external confirmation of a stratified medicine approach to intravenous-to-subcutaneous switching of infliximab in patients with CD and UC. Based on user-provided patient information (bodyweight, coadministration of methotrexate, presence/absence of antibodies towards infliximab, a pre-switch infliximab trough concentration, pre-switch fecal calprotectin, and the intravenous infliximab dosing regimen), Bayesian forecasting will be

performed, the pre-switch infliximab clearance will be estimated, and the patient's probability of relapse will be calculated (**Figure 12**). A detailed description of the use is included in the **Supplementary Information**. The software tool can be accessed via: <https://lpmx.shinyapps.io/infliximab/>.

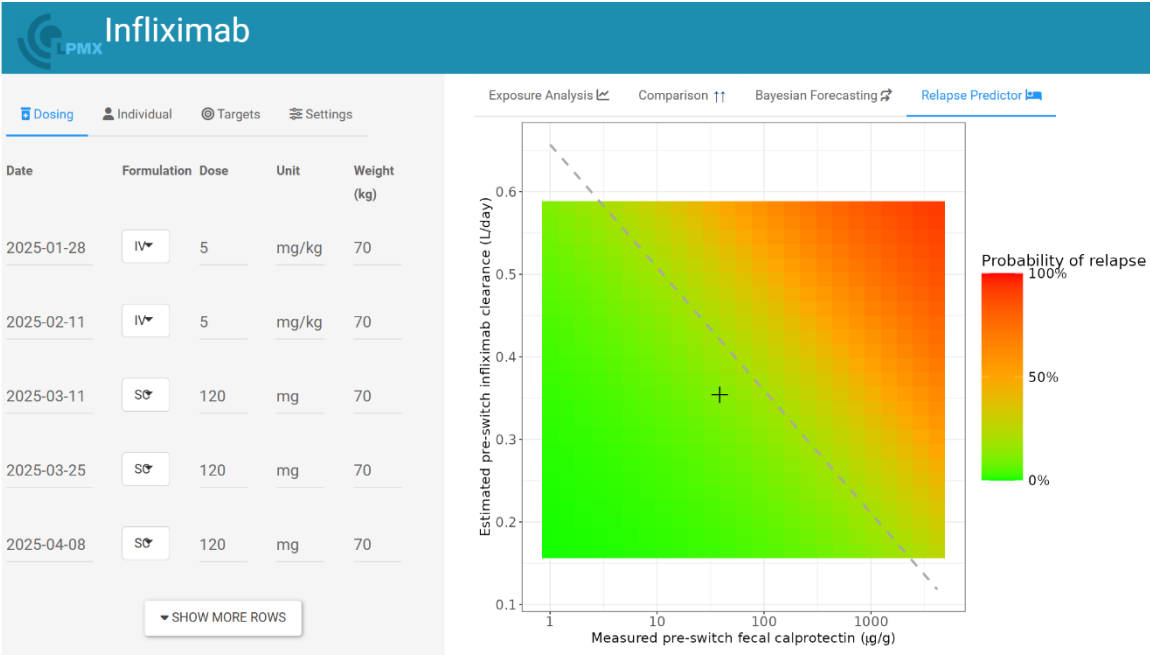


Figure 12. Snapshot of the web-based, interactive software tool.

4. Discussion

We developed a stratification algorithm that successfully assigns patients with CD and UC to low-risk and high-risk groups based on the predicted probability of relapse upon switching from intravenous to subcutaneous infliximab therapy. The risk calculation is based on measured pre-switch fecal calprotectin and the estimated pre-switch infliximab clearance, which can be calculated from the patient's bodyweight, coadministration of methotrexate, presence/absence of antibodies towards infliximab, a pre-switch infliximab trough concentration, and the intravenous infliximab dosing regimen. To allow the research community to externally validate our risk stratification algorithm, we integrated the popPK and logistic regression pharmacodynamics models into an interactive software tool that is freely available.

Estimated clearance and fecal calprotectin were identified as independent predictors for treatment outcome. The higher estimated clearance in the relapsed cohort is possibly driven by an increase in disease activity.^{89,90} The estimated clearance can, therefore, serve as an early predictor and alarm signal for clinical relapse. In our analysis, the infliximab trough concentrations were not associated with clinical relapse. While a clinician can modify the infliximab concentration by performing dosing optimizations, the infliximab clearance is an unmodifiable, intrinsic patient characteristic. The infliximab clearance is calculated by performing Bayesian forecasting with an infliximab concentration and the dosing regimen using a popPK model. While concentration-based TDM can be done using simple decision algorithms such as the TAXIT algorithm,⁹¹ clearance-based decision-making requires a software tool.⁸⁹

While TDM is traditionally based on infliximab (trough) concentrations, we now provide evidence for combined monitoring of infliximab exposure (pharmacokinetics) and fecal calprotectin (pharmacodynamics). While infliximab clearance alone had good

discriminative ability (area under the ROC curve of 0.74), also considering fecal calprotectin further increases the discriminatory power of the model (area under the ROC curve 0.83). The predictive performance of jointly monitoring infliximab and calprotectin should be weighed against the increased patient burden due to combined blood and stool sampling and increased healthcare costs. In case fecal calprotectin is not available, our software tool will use median imputation (38.5 µg/g), thereby allowing flexible patient stratification, still with good predictive performance.

We previously proposed a subcutaneous dosing strategy aimed at avoiding a drop in infliximab trough concentrations upon switching.⁸⁴ Thus, in order to select a subcutaneous dosing strategy that avoids a drop in infliximab trough concentrations upon switching, we need to be able to predict a patient's post-switch infliximab concentrations. Therefore, the initial objective of this study was to externally validate the predictive performance of the Celltrion infliximab popPK model published in the EPAR.²⁶ Using the REMSWITCH data, we showed that predictions are unbiased but imprecise. This may be explained by the highly variable and unpredictable rate and extent of absorption from the subcutaneous injection site.⁹² Further, documentation errors of the injection times may have affected the predictive performance of the pharmacokinetics model. In our dataset, the exact time points of injections were not available so we relied on the assumption of perfect therapy adherence at the prescribed time points. Lastly, when predicting post-switch infliximab exposure, we assumed pharmacokinetics parameters were constant over time. This assumption does not hold as an increase in disease activity will result in an increased infliximab clearance due to target-mediated drug disposition and protein-losing enteropathy.³⁰ These mechanisms may explain why we observed a (nonsignificant) trend towards overprediction of post-switch infliximab trough concentrations in patients who relapsed upon switching. In these patients, higher infliximab concentrations were predicted based on the assumption of unchanged infliximab clearance, but lower infliximab concentrations were eventually measured

because infliximab clearance increased due to increased disease activity. Larger datasets will be needed to validate this hypothesis.

We integrated the popPK and logistic regression pharmacodynamics models into a software tool, which can be used to (i) perform Bayesian forecasting to estimate infliximab clearance and predict individualized infliximab concentration–time profiles after switching, and (ii) predict the individual patient’s relapse risk based on estimated infliximab clearance and measured fecal calprotectin. Both functionalities are potentially clinically relevant, yet we would like to emphasize that the lack of external validation demands further research before clinical implementation can be considered. Therefore, we want to stress that our tool is not intended for clinical decision-making in routine IBD care. We welcome any suggestions to improve the quality and user experience of our tool, and we welcome research collaborations to expand validation of the tool.

Our pharmacokinetics–pharmacodynamics relapse prediction model has some limitations. First, the Bayesian forecasted maximum *a posteriori* clearance is deduced based on a popPK model mainly built on patients with CD and rheumatoid arthritis. Applying this model to our cohort of patients with CD and UC entails an additional source of uncertainty. A second available popPK model for SC infliximab from Hanžel et al was built on data from patients with IBD.⁹³ This model was difficult to reconstruct based on the available information in the publication. Furthermore, the visual predictive check of the final model showed biased predictions, especially for the standard SC dosing regimen of 120 mg infliximab. Second, the identified two-item risk score is based on non-modifiable predictors at baseline. Moreover, whether a patient received standard or escalated SC regimen was not retained in the final logistic regression model. Only five out of 98 patients were on escalated SC regimen, likely causing an underpowered analysis for this particular predictor. Results from dedicated trials like AMARETTO (NCT06113913) are awaited. We refrained from working with other relapse

definitions for instance based on clinical or biological relapse alone. The combined relapse definition allows us to retain a bigger cohort including 16 cases of relapse. Yet, we plan to update our model if more data becomes available to continuously increase the sample size. Lastly, our prediction model is not externally validated. Moreover, the data we used for model building was generated from a prospective multicenter trial. It remains crucial to externally validate the model, both retrospectively and prospectively, preferably with bigger datasets and more cases of relapse, prior to clinical implementation.

The present work introduces a pharmacokinetics–pharmacodynamics model of infliximab that can be used to predict individual patients’ probability of relapse upon switching from intravenous to subcutaneous infliximab. Our model supports treatment monitoring based on measured infliximab concentrations and fecal calprotectin. Our model discrimination and calibration were good. We integrated our model into a software tool to facilitate confirmation of a stratified medicine approach to improve infliximab therapy in IBD.

Chapter 1 Supplementary Material

Supplementary Figures

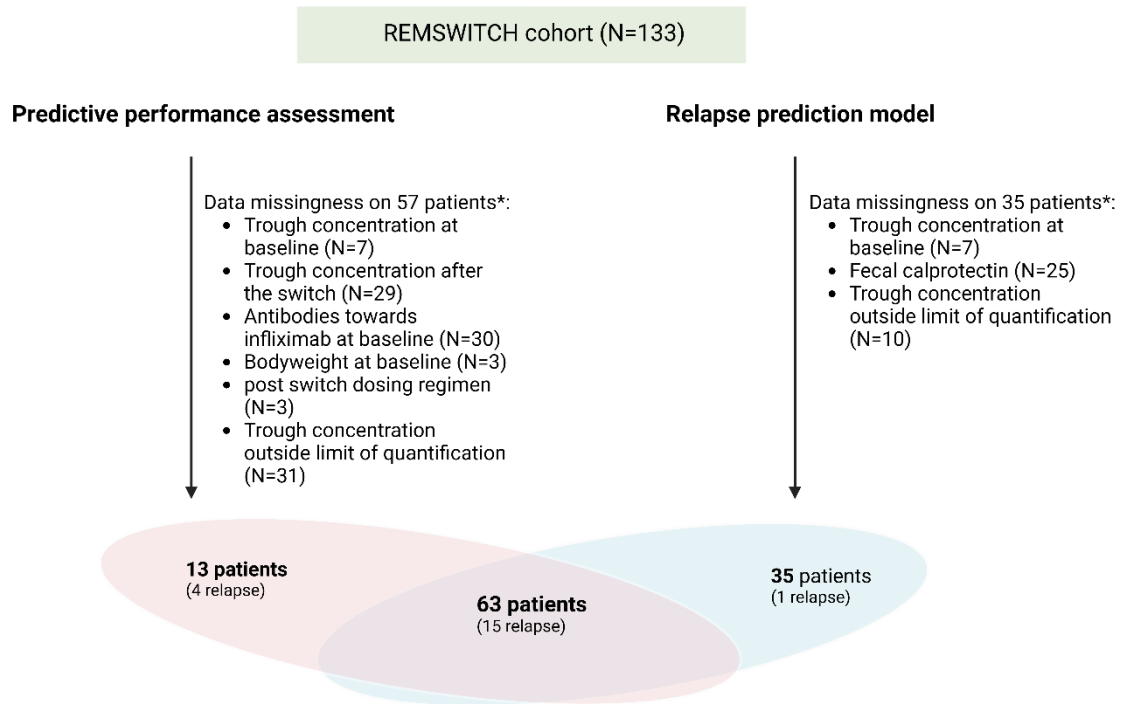


Figure S1. Inclusion and exclusion criteria for eligibility for the predictive performance assessment and the relapse prediction model. The two cohorts overlap in total of 63 patients (15 relapse). * Some patients fail on more than one criterion.

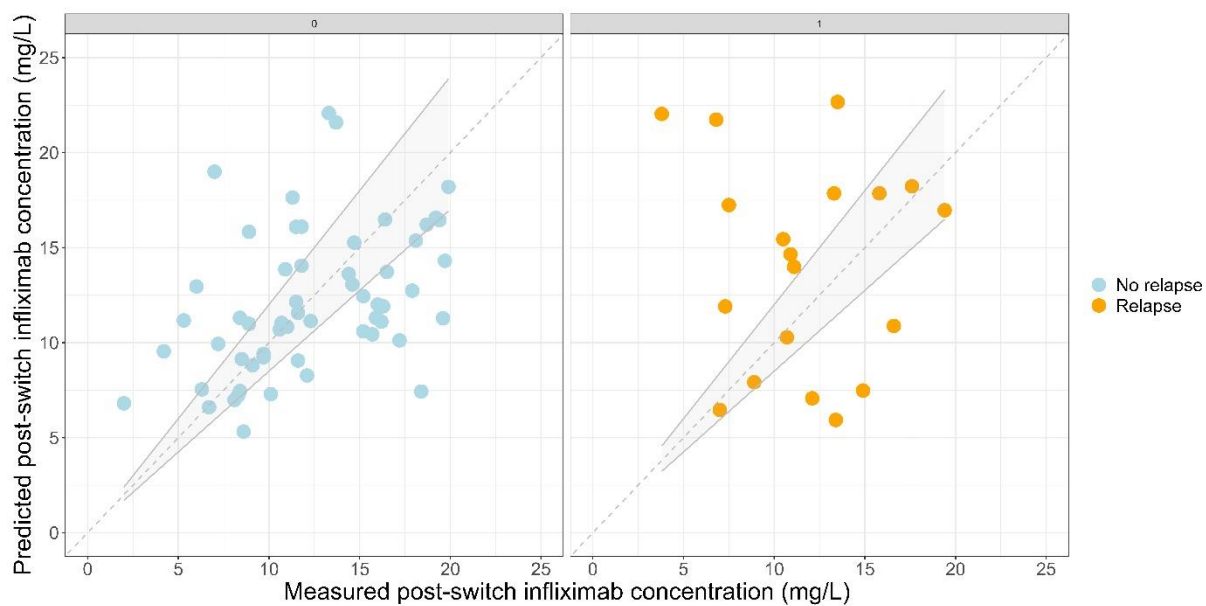
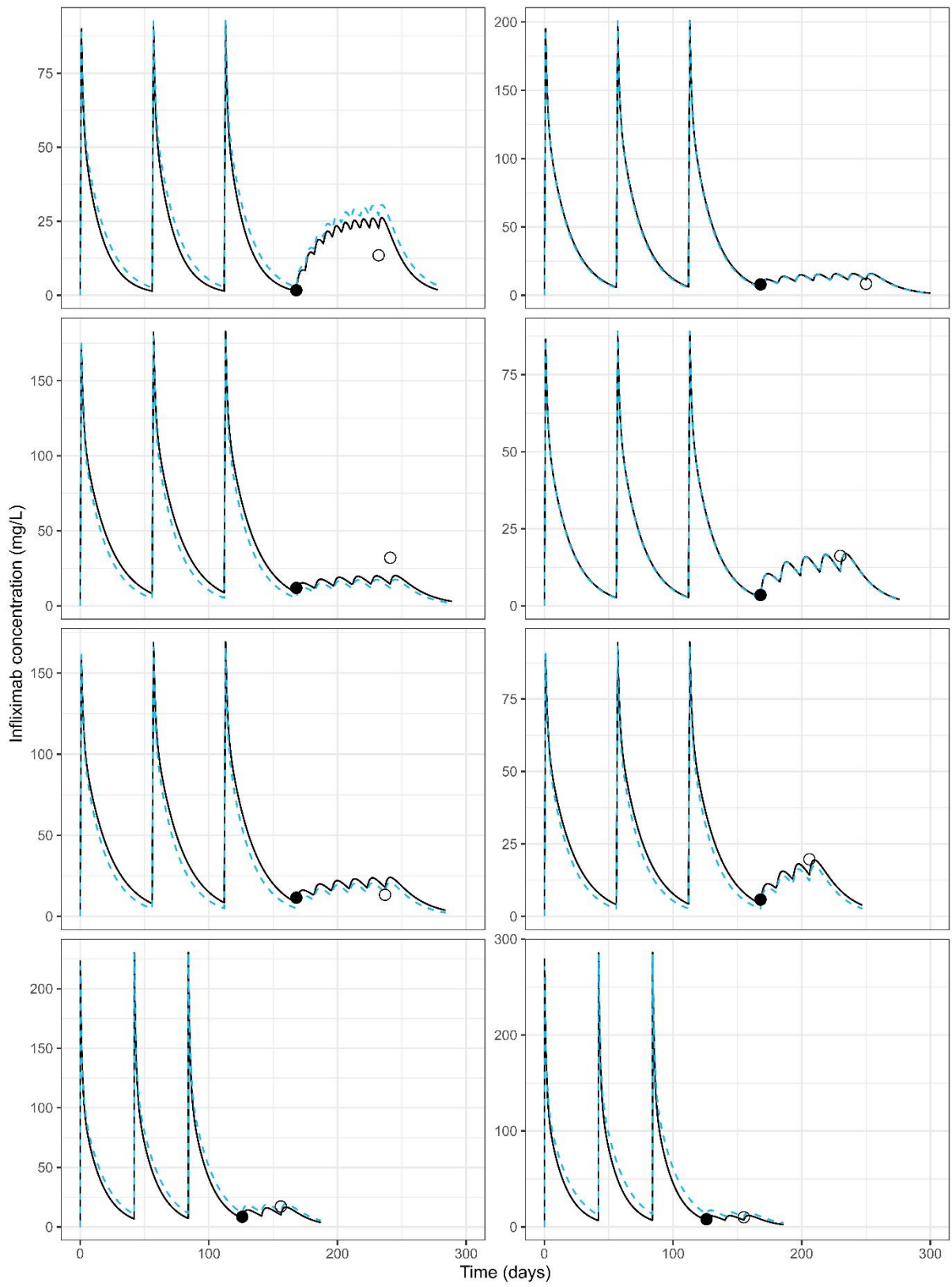
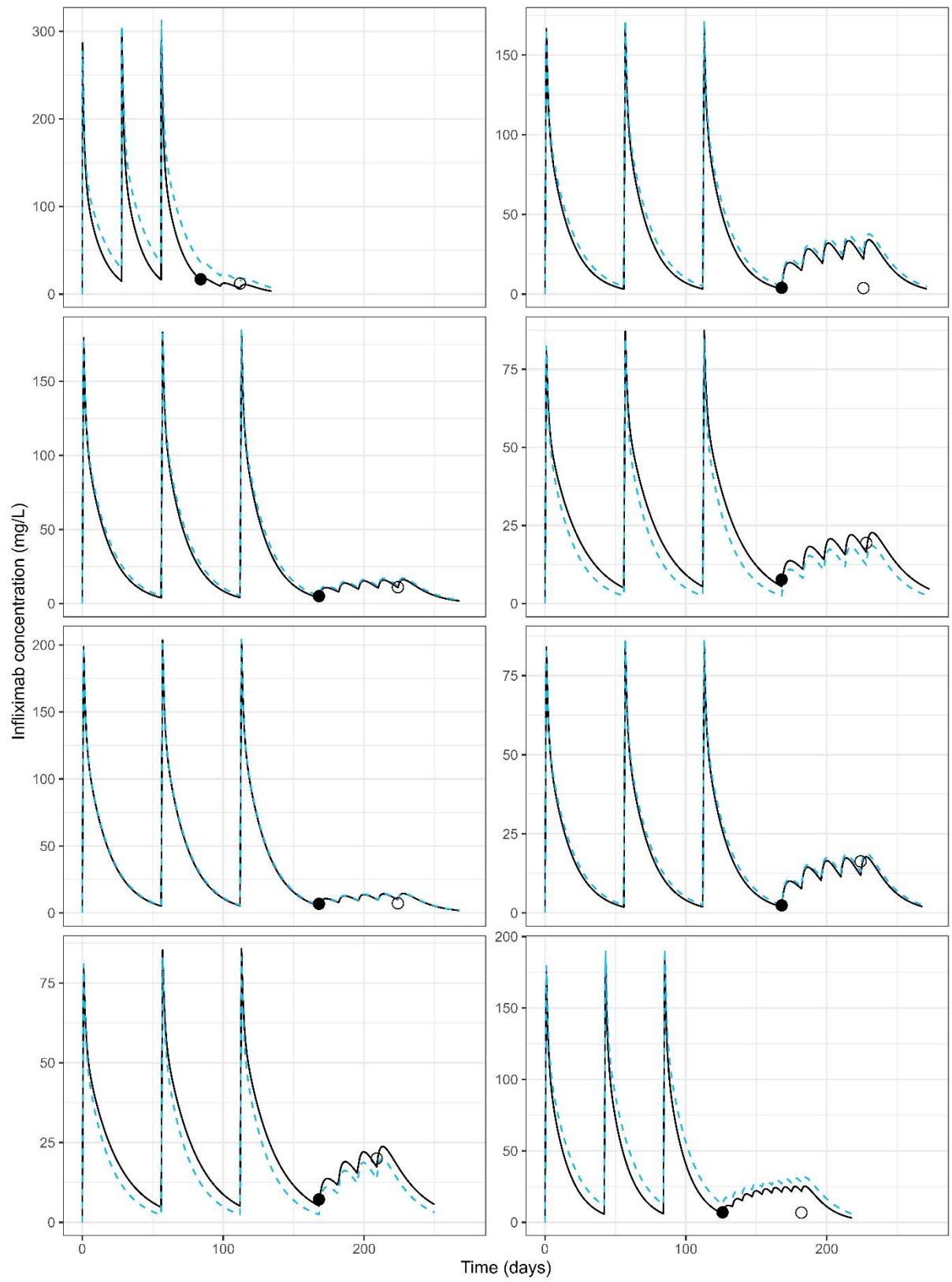
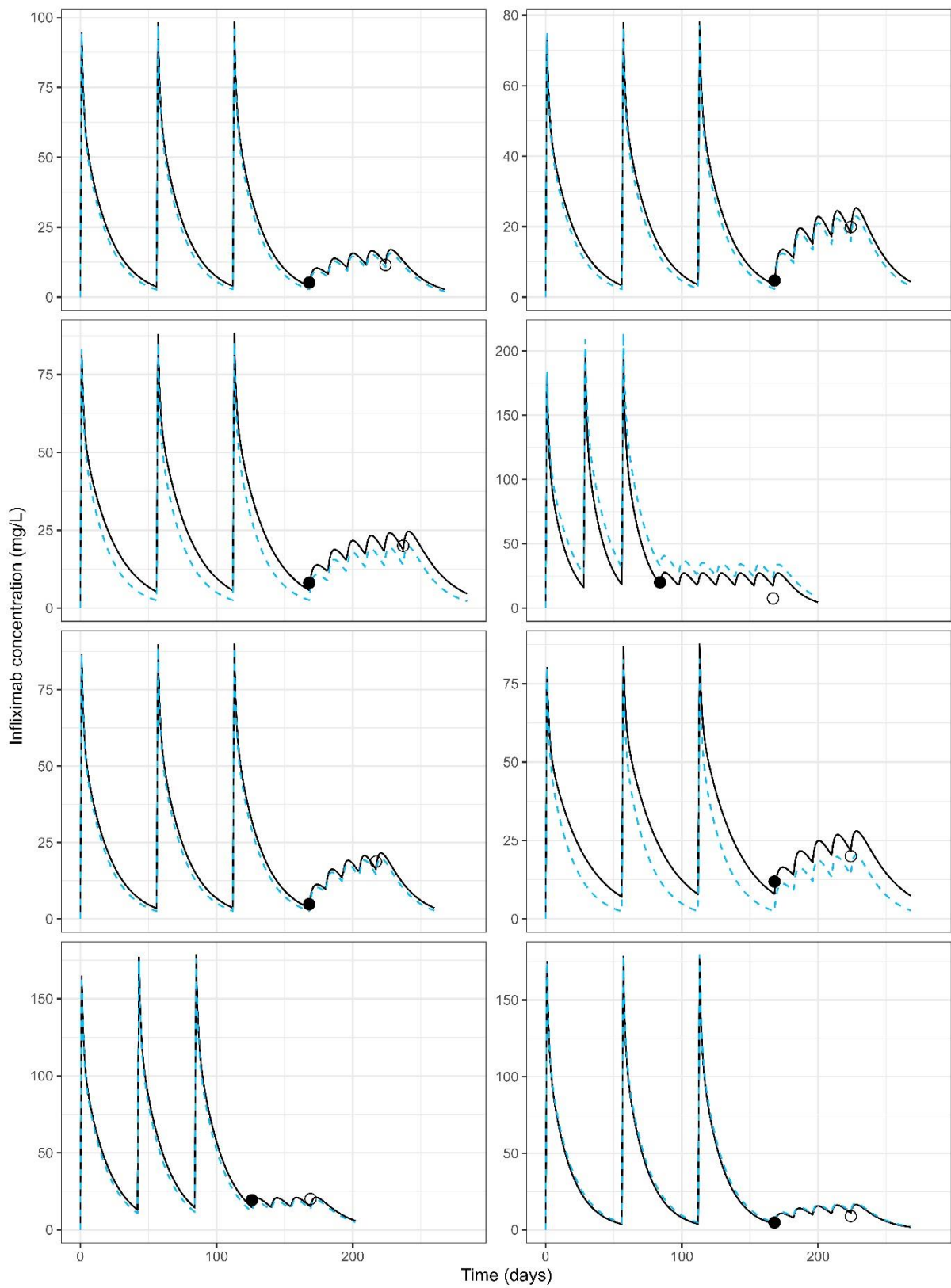
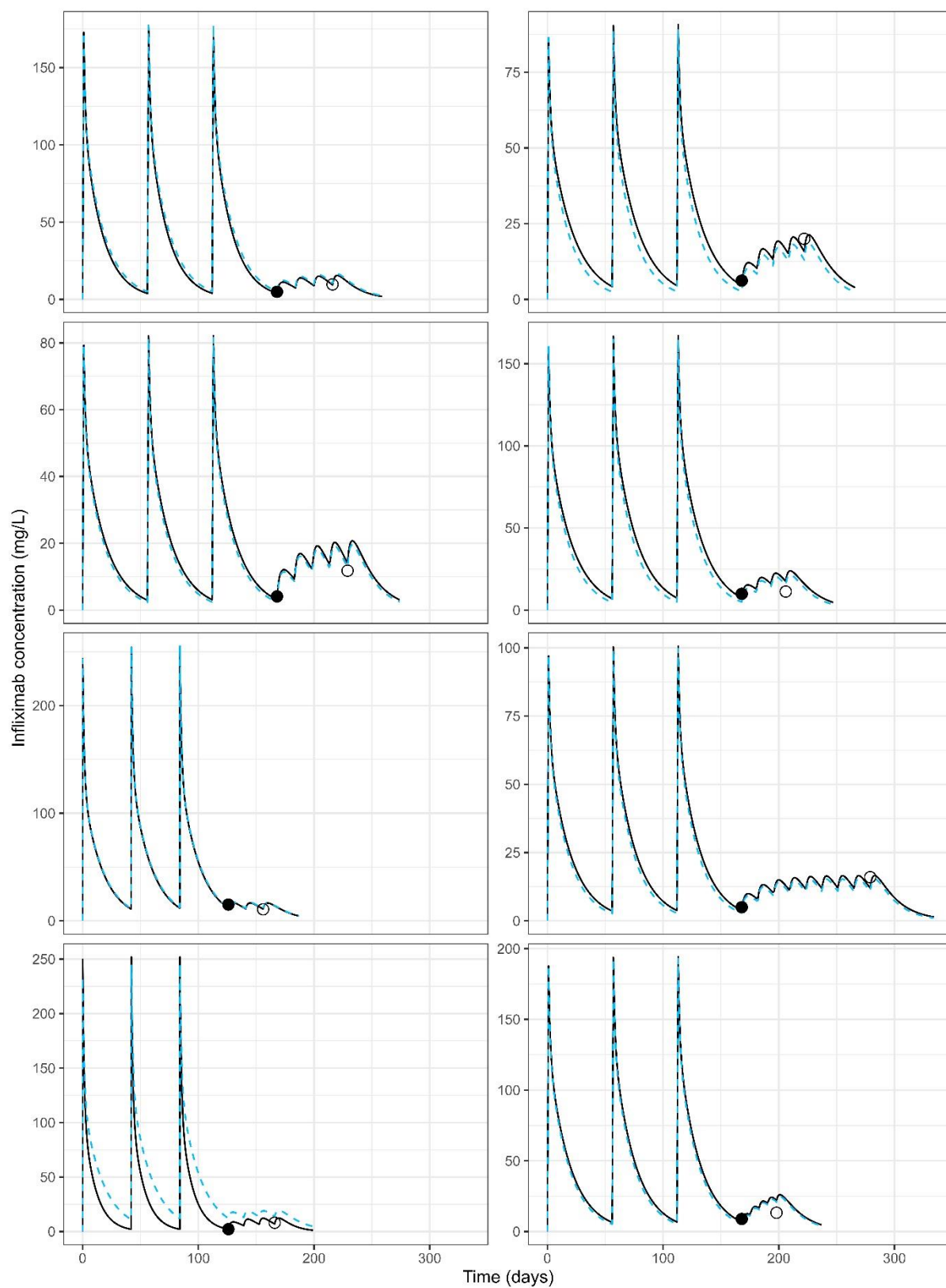


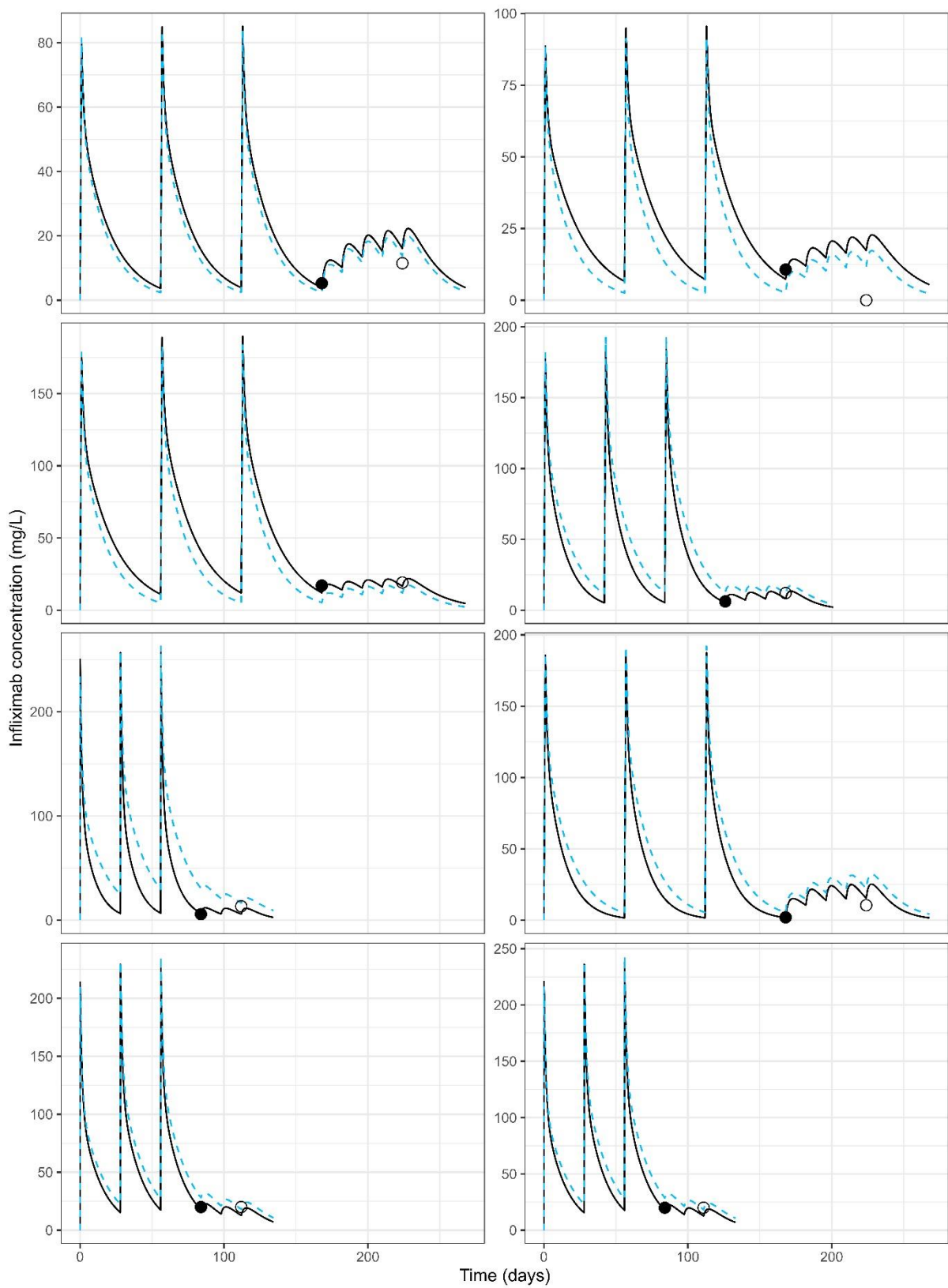
Figure S2. Scatterplot depicting predicted *versus* measured post-switch infliximab concentrations, stratified by relapse status. The grey dashed line indicates the unity line and the grey shaded area represents the uncertainty considering a $\pm 15\%$ assay variability. From the 133 patients included, only 76 provided an infliximab trough concentration after.

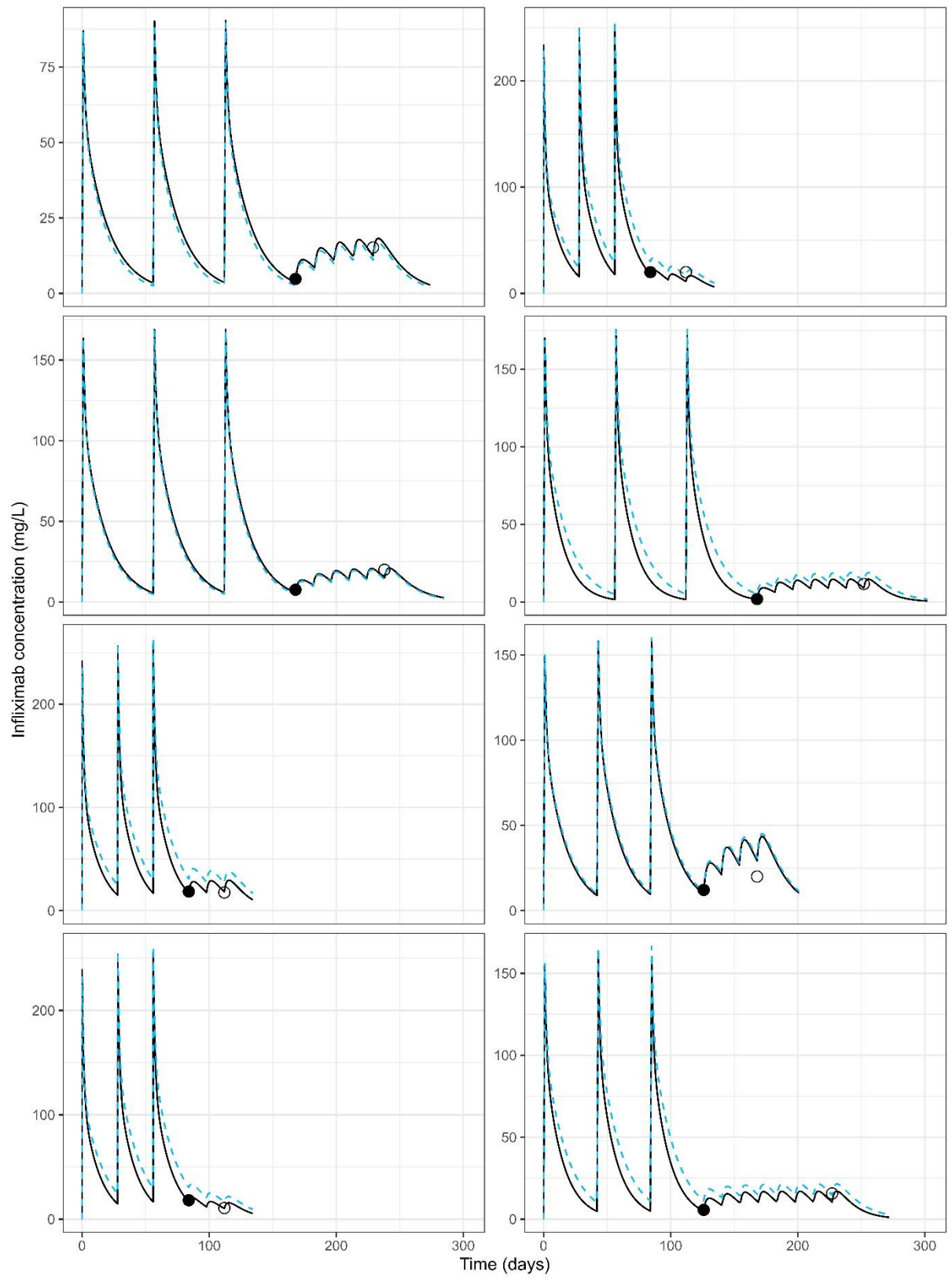


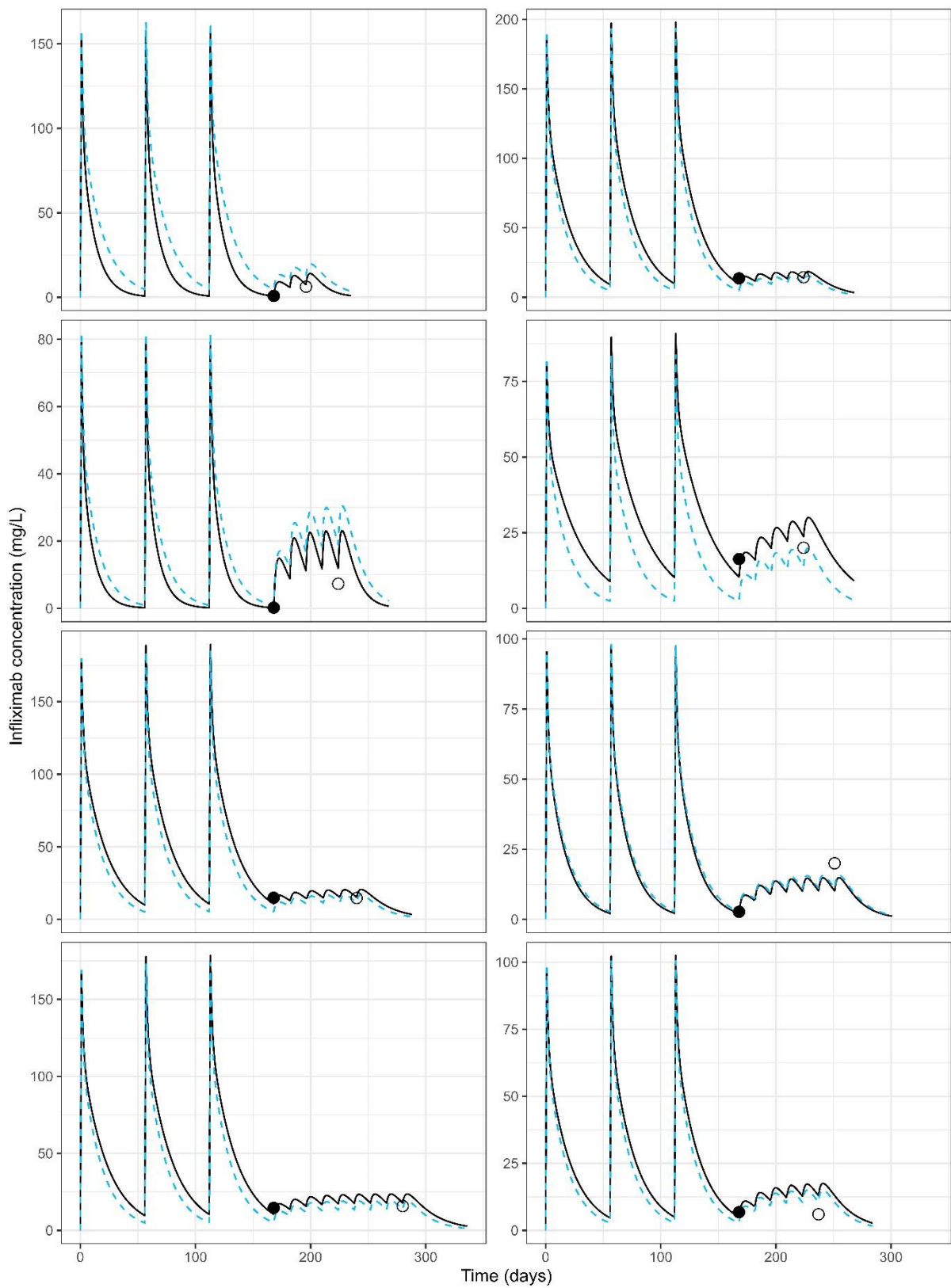


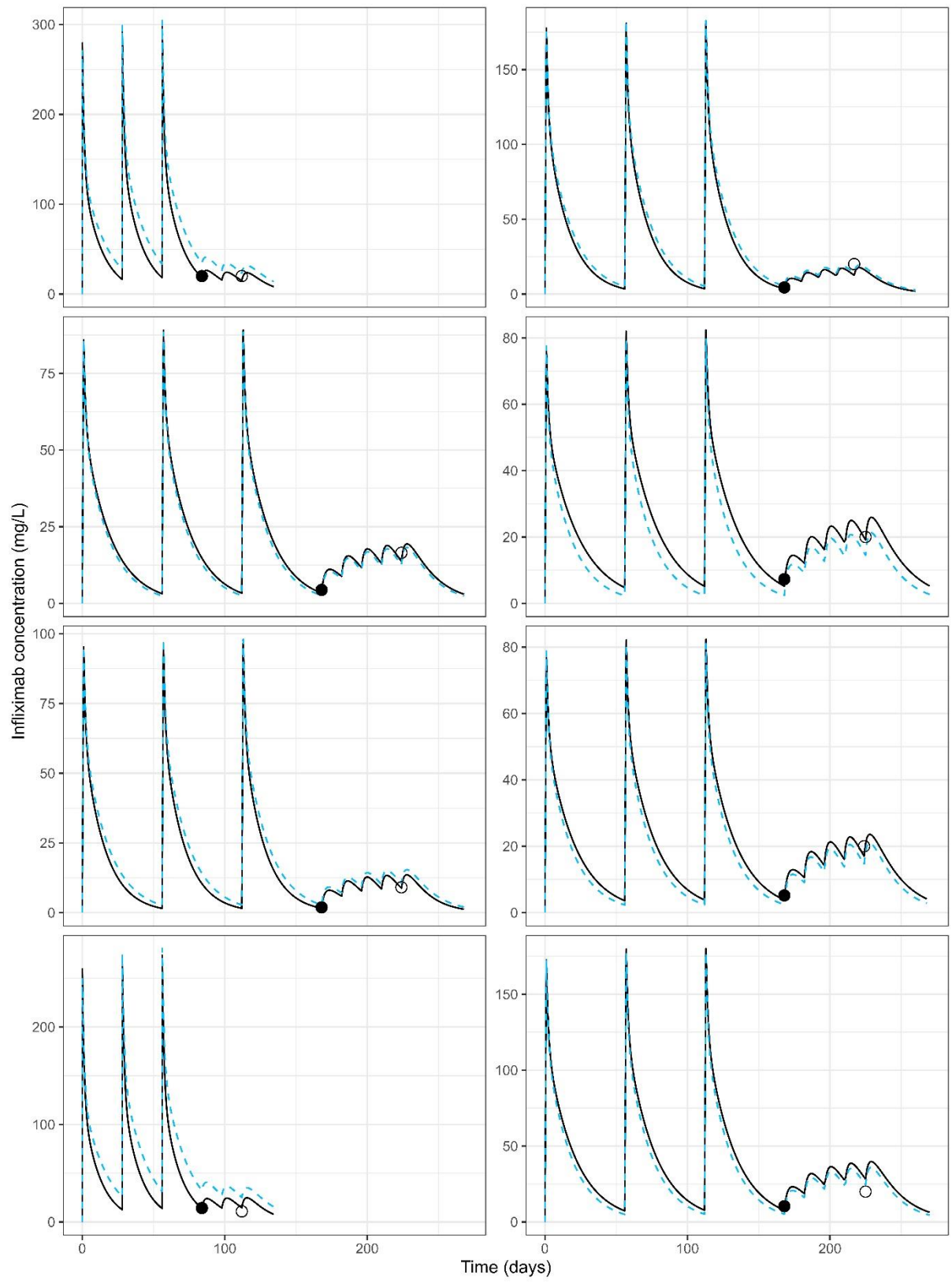


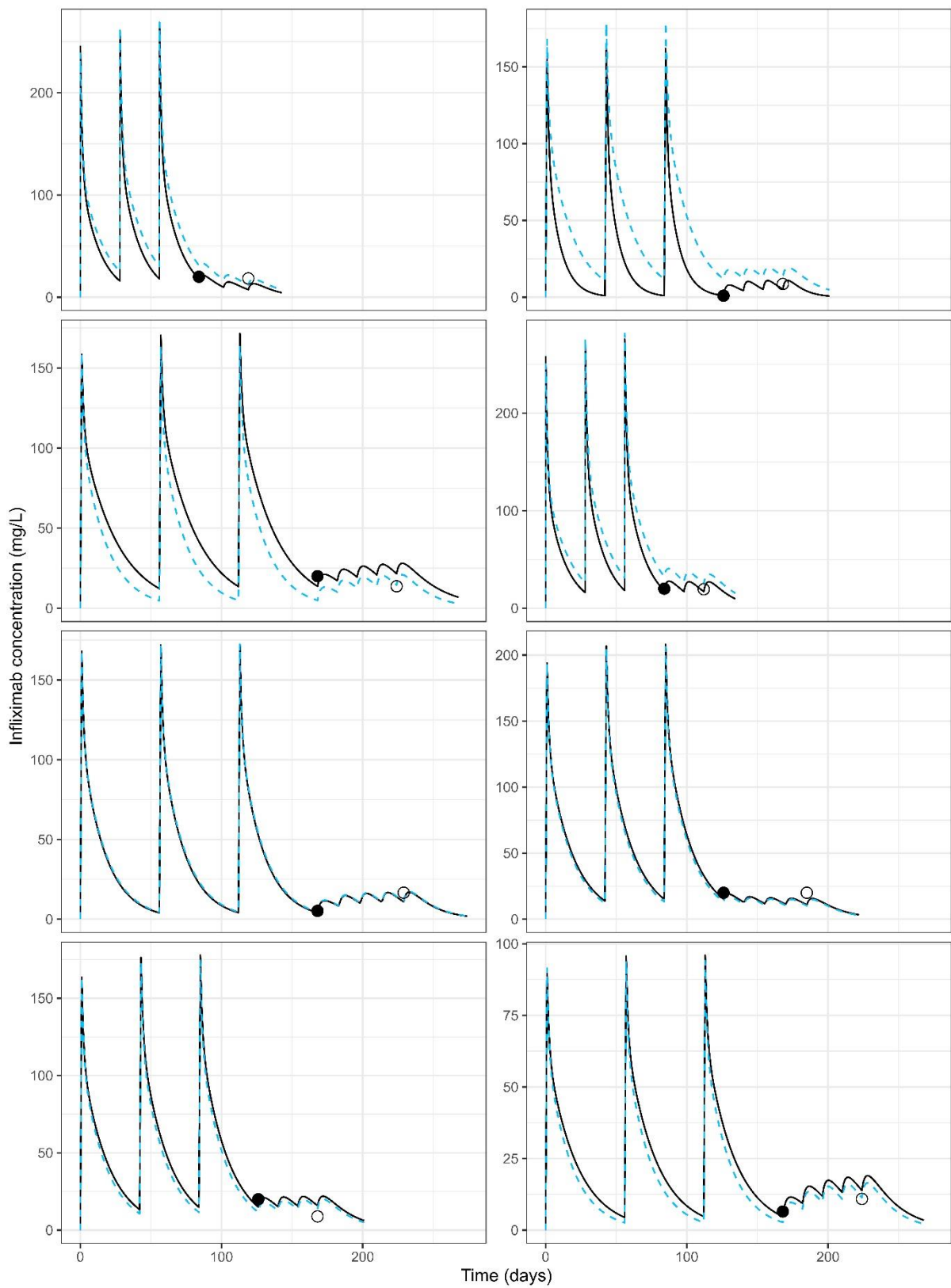












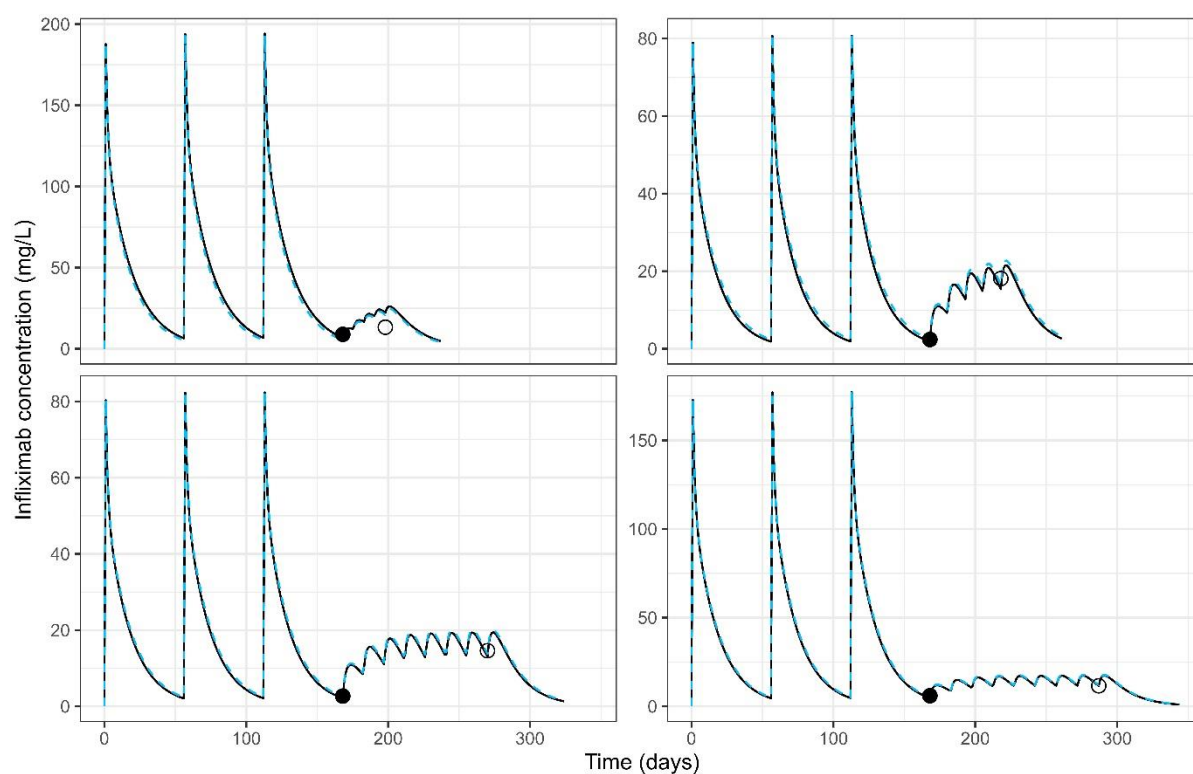


Figure S3. Individual infliximab concentration–time profile. The blue dashed line shows the population prediction (*a priori* prediction) and the black solid line shows the individual prediction (maximum *a posteriori* Bayesian forecasting). The black dot represents the measured pre-switch infliximab concentration (that was used for Bayesian forecasting) and the open circle represents the measured post-switch infliximab concentration.

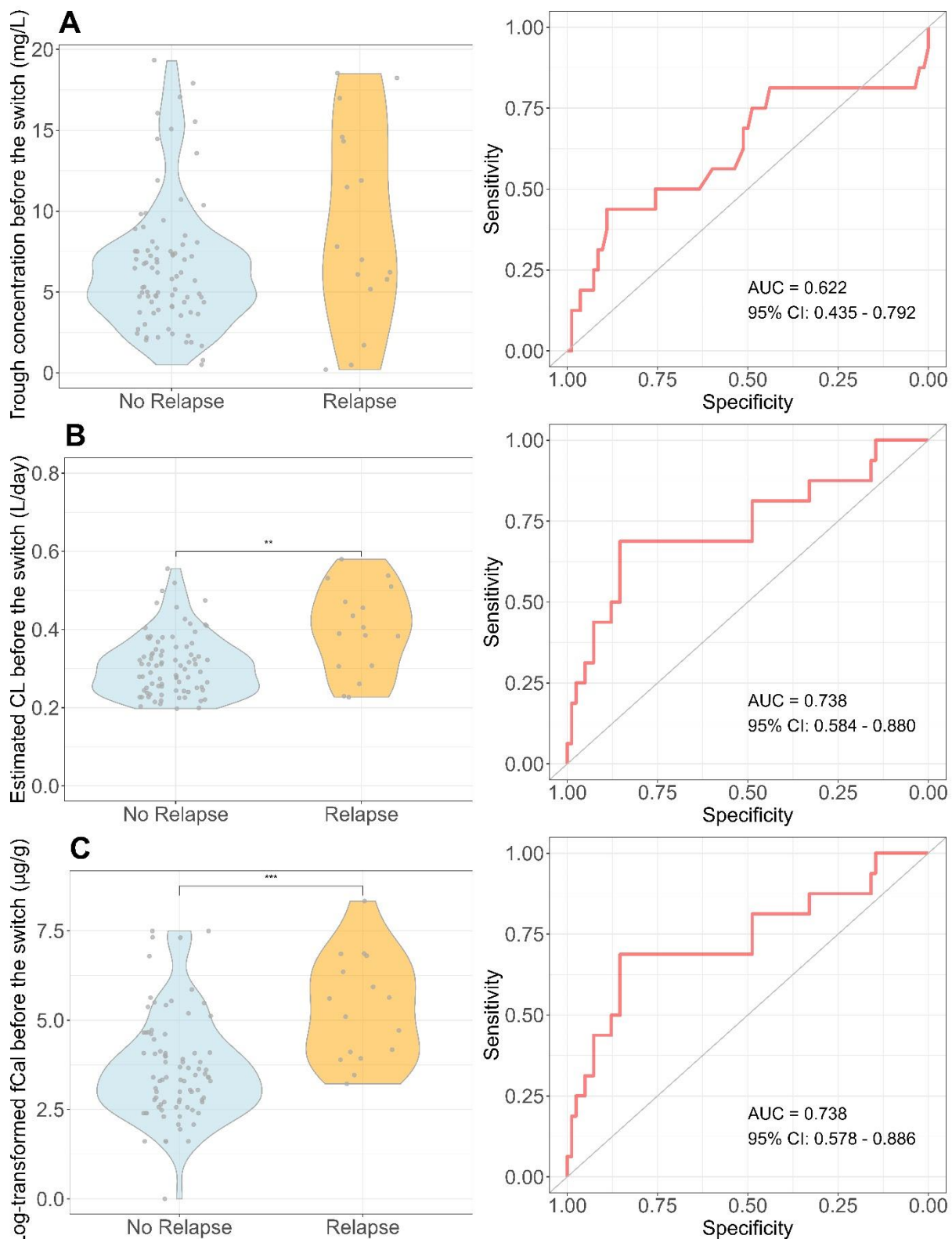


Figure S4. Pre-switch infliximab trough concentrations (A), Pre-switch estimated infliximab clearance (B), and pre-switch fecal calprotectin (C) stratified by relapse status and the corresponding receiver operating characteristic curve. Confidence intervals are based on 2,000 non-parametric bootstraps. *** $p < 0.001$, ** $p < 0.01$ using Wilcoxon signed-rank test. Abbreviations: AUC, area under the curve; CI, confidence interval; CL, clearance; fCal, fecal calprotectin.

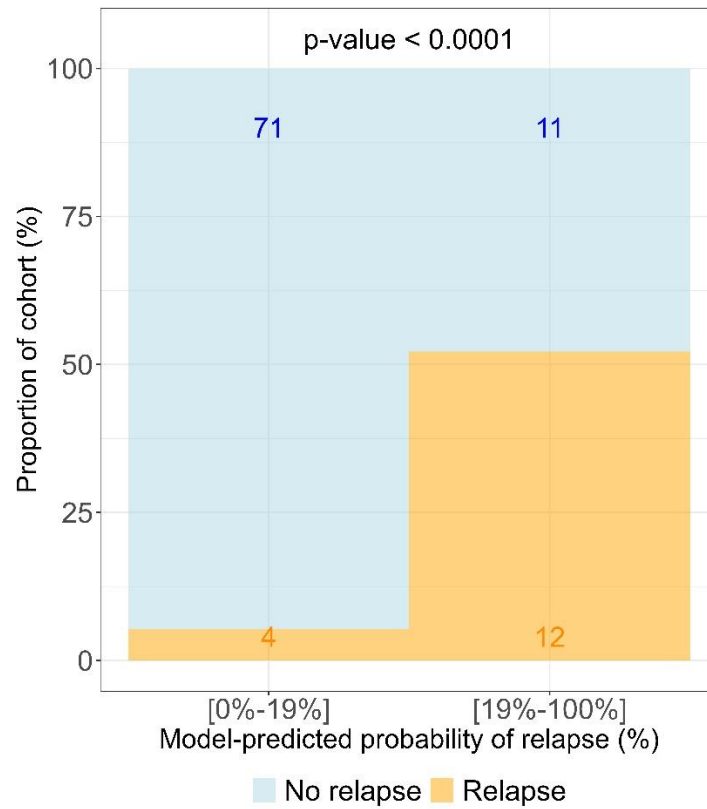


Figure S5. Proportion of cohort stratified by the identified probability threshold of 19% derived from the Youden J index. *P*-value is based on a Fisher-exact test.

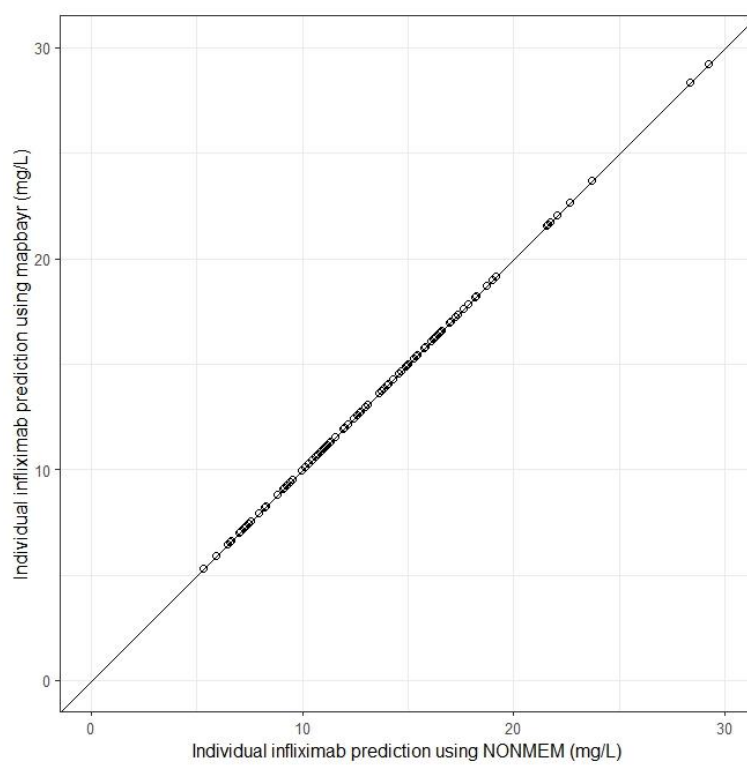


Figure S6. Comparison of the infliximab predictions obtained with the *mapbayr* package and NONMEM.

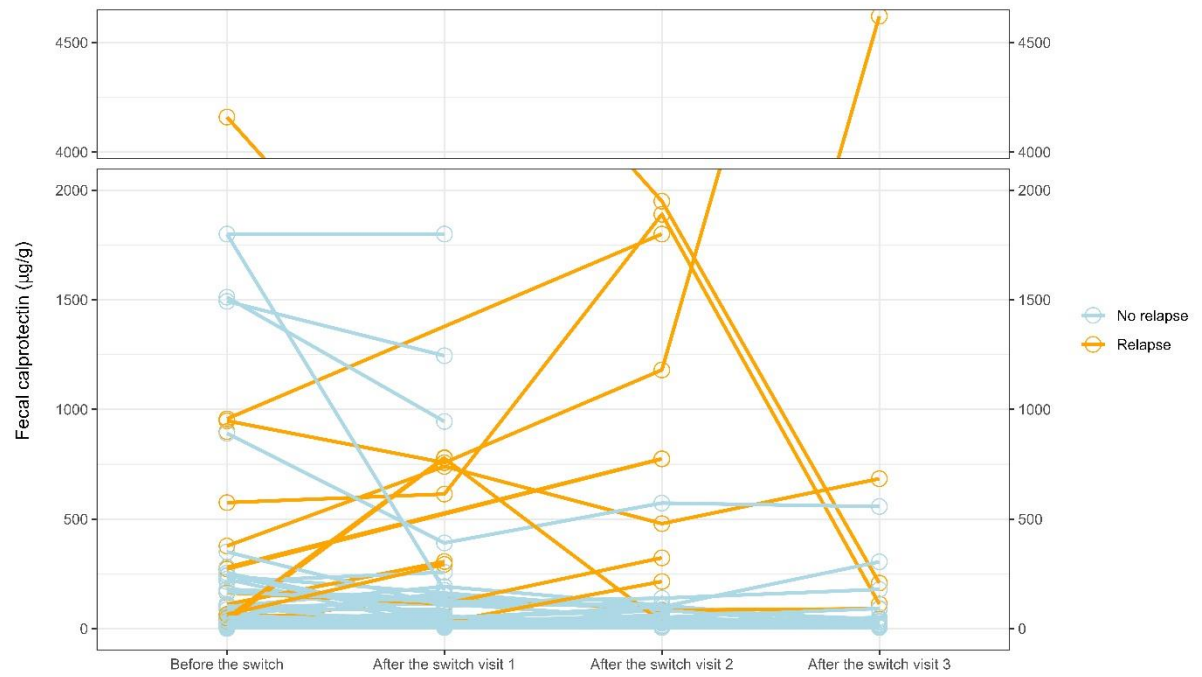


Figure S7. Comparison of fecal calprotectin measurements before and after the switch. Relapse was defined based on clinical recurrence and fecal calprotectin increase of $\geq 150 \mu\text{g/g}$ compared to baseline. Before the switch is considered as baseline, visit 1 is within 4–8 weeks, visit 2 within 8–16 weeks, and visit 3 within 16–24 weeks after the switch. The y-axis breaks at 2000 $\mu\text{g/g}$ and continues at 4000 $\mu\text{g/g}$.

Supplementary Tables

Table S1. Overview of criteria for the relapse cohort used in the prediction outcome model. Biological relapse was defined as an increase of fecal calprotectin of ≥ 150 $\mu\text{g/g}$ compared to baseline and clinical recurrence was defined as partial Mayo score >2 or Harvey–Bradshaw index >4 leading to therapeutic escalation.

ID	Biological relapse	Clinical recurrence	Relapse*	Pre-switch IV regimen	Post-switch SC regimen
1	yes	yes	At visit 2	10 mg/kg q4w	120 mg q2w
2	yes	NA	At visit 1	10 mg/kg q8w	120mg q2w
3	no	yes	At visit 2	10 mg/kg q4w	120 mg q2w
4	yes	no	At visit 2	10 mg/kg q4w	120 mg q2w
5	no	yes	At visit 1	10 mg/kg q6w	120 mg q1w
6	yes	yes	At visit 2	5 mg/kg q8w	120 mg q2w
7	yes	no	At visit 2	10 mg/kg q8w	120 mg q2w
8	yes	yes	At visit 1	10 mg/kg q4w	240 mg q2w
9	yes	no	At visit 1	10 mg/kg q4w	120 mg q2w
10	NA	yes	At visit 1	5 mg/kg q8w	120 mg q2w
11	NA	yes	At visit 1	10 mg/kg q8w	120 mg q2w
12	yes	no	At visit 2	10 mg/kg q4w	120 mg q2w
13	yes	no	At visit 1	5 mg/kg q8w	120 mg q2w
14	yes	no	At visit 2	5 mg/kg q8w	120 mg q8w
15	yes	no	At visit 1	5 mg/kg q8w	120 mg q2w
16	yes	no	At visit 1	10 mg/kg q6w	120 mg q2w

Abbreviations. IV, intravenous; NA, not available; qxw, every x week; SC, subcutaneous. *after the switch.

Table S2. Bland-Altman statistics stratified by relapsed cohort and relapse-free cohort.

	No Relapse (N=57)	Relapse (N=19)
Bias % [95% CI]	0.23 [-0.92 – 1.38]	-2.39 [-5.57 – 0.78]
Lower level of agreement % [95% CI]	-8.46 [-17.23 – 0.30]	-16.21 [-23.40 – -10.52]
Upper level of agreement % [95% CI]	8.91 [0.15 – 17.68]	11.42 [-2.75 – 26.61]

Abbreviations. CI, confidence interval.

Table S3. Parameter estimates for the final logistic regression model.

Parameter	Estimate	Standard error	Pr(> z)
Intercept	-6.9559	1.4459	***
CL effect	8.3953	3.3297	*
fCal effect	0.5422	0.1927	**

* $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$. Abbreviations. CL, clearance; fCal, fecal calprotectin.

Supplementary Information

In the *Dosing* tab, the user can define the administrations of infliximab by providing the required dosing information in each row.

The corresponding simulated concentration–time profile for this particular dosing scenario is shown in the *Exposure Analysis* tab in the main bar panel.

The *Individual* tab can be used to specify patient characteristics such as the categorical covariates; antibodies towards infliximab (present/absent), and co-administration of methotrexate (yes/no), and fecal calprotectin (in $\mu\text{g/g}$).

The *Target* tab allows setting the infliximab trough concentration target via a slider. Additionally, the user can determine a threshold probability of target attainment (PTA), which is then considered in the PTA over time plot. For the actual exposure predictions, the user defines two flexible time points, *Analysis timepoint 1 [day]*, and *Analysis timepoint 2 [day]*. The first time point corresponds to a pre-switch visit (visit 0 [v0]) and the second time point corresponds to a post-switch visit (v1). Infliximab exposure measured at the pre-switch visit can be defined in *Exposure at timepoint 1 [mg/L]*. Blue arrows in the simulated concentration–time profile indicate the two selected time points.

The *Settings* tab provides options to adjust and refine the simulations. It contains input boxes where the number of Monte–Carlo simulations can be adjusted (default = 250), as well as the prediction interval around the simulated median (default = 90%), the total simulation time (default = 200 days), and the sources of variability (default includes inter-individual variability, but no residual variability). The simulated dataset can be downloaded in this tab for further analysis outside the software tool.

The *Comparison* tab in the main bar panel contains diagnostic plots to analyze the predicted infliximab exposure after switch. Based on the scatterplot, the user can evaluate a trend in infliximab concentrations between previously defined timepoints v0 (trough concentration

during intravenous therapy) and v1 (trough concentration during subcutaneous therapy). The boxplot next to it serves as an additional visualization tool to support the finding, showing simulated trough concentrations at v0 and v1.

The *Bayesian forecasting* tab contains a concentration–time profile for the population prediction, as well as the individual concentration–time profile based on individualized pharmacokinetics parameters obtained from the Bayesian forecasting maximum *a posteriori* step. A table at the bottom provides a summary of the output including administration time points, observations, predictions, covariates, and individual PK parameters.

The *Relapse Predictor* tab includes a heatmap (see **Figure 11** in the manuscript) where our logistic regression pharmacodynamics model is included. Based on the Bayesian forecasted infliximab clearance before the switch and the measured fecal calprotectin concentration before the switch, the probability of relapse is calculated, and the result is shown. The grey dashed line represents the identified cutoff of 20%, and the black cross indicates the calculated probability of relapse based on the patient characteristics provided by the user.

Model code integrated into the software tool

```

### start model code #####
$PARAM @annotated
TVKA: 0.21648 : Absorption rate
TVCL: 0.3024 : Clearance
TVV1: 2.89 : Central volume
TVV2 : 2.06 : Peripheral volume of distribution
TVQ : 0.8616 : Intercompartmental clearance

ETA1: 0 : Clearance (L/h)
ETA2: 0 : Central volume (L)

$PARAM @annotated @covariates
BW : 70 : Body weight (kg)
ATI : 0 : Anti body towards infliximab
MTX : 0 : Methotrexate coadministration

$OMEGA 0.1490 0.0562
$SIGMA
0.353 // proportional
0 // additive

$CMT @annotated
DEPOT : Depot compartment () [ADM]
CENT : Central compartment (mg/L) [ADM, OBS]
PERIPH: Peripheral compartment ()

$TABLE
double DV = (CENT/V1) * (1 + EPS(1)) + EPS(2) ;
double IPRED = CENT/V1 ;

$MAIN
F_CENT = 1 ;
double CL = TVCL * exp(ETA1 + ETA(1)) * pow(BW / 70, 0.639) * (1+0.31*ATI) * ;
(1+0.04*MTX) ;
double V1 = TVV1 * exp(ETA2 + ETA(2)) * pow(BW / 70, 0.606) ;
double Q = TVQ * pow(BW / 70, 0.639) ;
double V2 = TVV2 * pow(BW / 70, 0.606) ;
double KA = TVKA ;
F_DEPOT = 0.576 ;

double k20 = CL/V1 ;
double k23 = Q/V1 ;
double k32 = Q/V2 ;

$ODE
dxdt_DEPOT = -KA*DEPOT ;
dxdt_CENT = KA*DEPOT - k23*CENT + k32*PERIPH - k20*CENT ;
dxdt_PERIPH = k23*CENT - k32*PERIPH ;

$CAPTURE DV CL
### end model code
#####

```


Chapter 2

Population pharmacokinetics–pharmacodynamics of infliximab in patients with acute severe ulcerative colitis

This chapter is based on

Emmanuel Niyigena*, **Yannick Hoffert***, Waqqas Afif, Alessandro Pedicelli, Xavier Roblin, Jurij Hanžel, Konstantinos Papamichael, Taku Kobayashi, Laure Elens, Zhigang Wang, Marc Ferrante, Bram Verstockt, Séverine Vermeire, Niels Vande Casteele, Robert Battat, Erwin Dreesen. Personalised Infliximab Rescue Therapy to Maximise Colectomy-Free Survival in Patients with Acute Severe Ulcerative Colitis. *J Crohns Colitis*. 2026 Mar 10;20(3). [*Shared co-first authorship].

Abstract

Background and Aims

Infliximab is an established rescue therapy for patients with steroid-refractory acute severe ulcerative colitis, yet optimal dosing strategies minimizing colectomy risk remain unclear. We aimed to develop a model-informed risk stratification algorithm to identify patients at high risk of colectomy within 90 days of initiating infliximab to support personalized dosing.

Methods

We conducted a multicenter, retrospective population pharmacokinetics and exposure–response study using data from patients with acute severe ulcerative colitis. A parametric time-to-event model was developed to characterize the 90-day colectomy risk. Patient characteristics and pharmacokinetic projections were evaluated as predictors. These modelling results informed the development of an algorithm for risk stratification and personalized infliximab rescue dosing.

Results

Seven medical centers contributed data from 72 patients with acute severe ulcerative colitis, yielding a total of 152 infliximab serum concentrations. Eleven patients underwent colectomy within 90 days. The strongest predictor of colectomy was the clearance-normalized exposure between weeks 2 and 4 (area under the concentration–time curve, AUC_{w2-4} to Bayesian-forecasted infliximab clearance, CL), with an area under the receiver operating characteristic curve of 0.79 (95%CI: 0.52–1.00). The AUC_{w2-4}/CL ratio was calculated by individualizing the population pharmacokinetics model using the patient's body weight, baseline C-reactive protein, and infliximab concentrations. Patients with a log-transformed AUC_{w2-4}/CL ratio <5.79 were classified as high risk for colectomy (sensitivity 83%, specificity 85%). Overall classification accuracy was 85% [95%CI: 74%–92%].

Conclusions

We developed a model-based dose–exposure–response framework to predict colectomy risk in acute severe ulcerative colitis. We integrated the algorithm into an interactive tool to enable individualized infliximab rescue therapy.

1. Introduction

ASUC affects approximately one-quarter of patients with ulcerative colitis, either at initial presentation or later in the disease course.⁹⁴ ASUC represents a medical emergency with substantial morbidity. Despite advances in treatment, up to 36% of patients experience multiple episodes and around 40% ultimately require colectomy.⁹⁴ Corticosteroids are the primary treatment for ASUC, yet one in three patients fail to respond.^{95,96} In such cases, infliximab emerged as the second-line treatment option.

Following the well-established general principles of the dose–exposure–response paradigm, achieving rapid and adequate infliximab exposure is essential for treatment success.^{97,98} However, point-of-care TDM of infliximab is particularly challenging in the ASUC setting. Disease-related factors such as target-mediated infliximab clearance and protein-losing enteropathy (“leaky gut”) significantly alter the PK of infliximab.^{89,99} Increased infliximab clearance results in a shorter half-life and lower trough concentrations compared to patients with less active disease.^{6,7} These disease-driven changes are associated with more severe symptoms and higher colectomy rates.¹⁰⁰ Consequently, treating patients with ASUC with infliximab requires a specific dosing strategy that accounts for disease activity.

About 25 years ago, the first infliximab dose-finding study in steroid-refractory ASUC was conducted by Sands and colleagues.¹⁰¹ The study was terminated prematurely due to slow patient enrolment.¹⁰¹ Despite the pressing medical need, the optimal dosing strategy has remained uncertain, causing the adoption of various clinical practices using intensified dosing regimens (higher doses and/or shorter intervals) to compensate for increased infliximab clearance.^{102,103} Nevertheless, recent evidence from the PREDICT-UC trial suggests that intensified and accelerated infliximab rescue therapy does not improve remission or colectomy-free survival, compared to standard dosing.¹⁰⁴ Yes, results from the PREDICT-UC PK analysis trial show that day 3 infliximab exposure and day 7 infliximab clearance predict 3-month colectomy.¹⁰⁵ However, there is currently a lack of supportive

evidence and a clinically actionable personalized dosing algorithm for risk stratification to address increased clearance at the point of care.

In this multicenter study, we pooled real-world clinical data from patients with steroid-refractory ASUC to develop and validate a model-informed dose-exposure-response framework. We used this model to construct a risk stratification algorithm that informs personalized infliximab rescue therapy. The algorithm has been implemented in an interactive software tool designed to support future clinical validation and maximize the success of infliximab rescue therapy in ASUC.

2. Materials and Methods

2.1 Patients

We performed a multicenter, retrospective study using pooled data from patients 18 years and older hospitalized with corticosteroid-refractory ASUC diagnosed according to Truelove and Witts criteria.¹⁰⁶ Fifteen medical centers participating in the spECTRUM consortium—an international collaboration of TDM specialists in inflammatory bowel diseases, rheumatology, and dermatology spanning 23 countries across five continents—were contacted to query whether they agreed to participate and share individual-level data of patients who received infliximab as a rescue therapy as from 1 January 2013 and prior to 1 March 2023. Patients were eligible if they had not received anti-tumor necrosis factor therapy before and had at least one infliximab serum concentration measured within the first fourteen weeks after the start of the infliximab rescue therapy.

The study was approved by the Ethics Committee Research UZ / KU Leuven (S64627), and written informed consent for the secondary use of data was obtained from participants of the original studies. We performed the study in accordance with the Declaration of Helsinki, Good Clinical Practice, and applicable regulatory requirements including individual data protection. Patients or the public were not involved in the design, conduct, reporting, or dissemination plans of this research.

2.2 Infliximab Pharmacokinetics Model Development and Evaluation

We characterized the infliximab dose–exposure relationship using population PK (popPK) modelling. Infliximab serum concentrations below the lower limit of quantification (LLOQ) and above the upper limit of quantification (ULOQ) were handled using the M3 method.¹⁰⁷ To address data heterogeneity across centers, we tested inter-center variability and various residual error models during model development. SCM was performed to evaluate baseline covariate effects of body weight, body mass index (BMI), C-reactive protein (CRP), serum

albumin, and between-center variability. Calibration of the final model was evaluated using prediction-corrected visual predictive check (pcVPC) plots ($n = 1,000$ simulated replicates of the original dataset). Parameter uncertainty of the final model was evaluated using a nonparametric bootstrap ($n = 2,000$ bootstrap runs). Missing covariates were handled using median imputation.

2.3 Time-To-Colectomy Model Development and Evaluation

We developed a parametric time-to-event model to describe time to colectomy. Exponential, Gompertz, Surge, and Weibull models have been tested as baseline hazard models. Predictors of hazard risk parameters were evaluated using SCM, in which baseline factors (body weight, BMI, CRP, and serum albumin) and infliximab exposure metrics were tested. The infliximab exposure metrics include empirical Bayes estimate (EBE)-based PK parameters—TDM-based individual clearance—and cumulative infliximab exposure across various time intervals (area under the concentration–time curve; AUC). AUC-to-clearance ratios (AUC/CL) were also considered in the covariate search. Calibration and uncertainty of the final model were evaluated using pcVPC plots and a nonparametric bootstrap. ROC curves were used to evaluate the discriminatory ability of the covariates tested in the time-to-event model.

2.4 Risk Stratification and Precision Dosing Algorithm

The maximum Youden J statistic was used to classify patients into low-risk and high-risk groups. The popPK and time-to-event models were used to calculate the optimal infliximab dosing regimen for each individual, required to reach a specified target probability of colectomy corresponding to the Youden J statistic.

2.5 Interactive Risk Stratification and Precision Dosing Tool

An interactive risk stratification and precision dosing tool based on the final popPK and time-to-event model for colectomy was developed. The *Shiny* package (version 1.7.4) in R was

used for developing the web application, which we published as an open-source application. In compliance with Regulation (EU) 2017/745, free access to the web application is provided under a license agreement for non-commercial research purposes.

3. Results

3.1 Patient Characteristics

Data from 72 patients with ASUC were contributed by seven medical centers, providing a total of 152 infliximab concentrations. The step-by-step inclusion process that yielded this final cohort is depicted in **Figure S8**. The median [IQR] infliximab serum concentration was 6.7 [3.9–10.9] mg/L. Eleven patients (15%) underwent colectomy within the first 90 days (median time to colectomy 32 [range 11–84] days). At baseline, disease activity assessed by the Mayo endoscopic subscore was predominantly severe (81%), with the remainder classified as moderate (19%). The median first infliximab dose was 9.9 mg/kg (range 3.8–15.0), and patients received a median of 3 infusions (range 1–7) over a dosing period spanning a median of 7 days (IQR 0–42 days). Anti-infliximab antibodies were detected in 6 of 152 samples (4%). A detailed overview of the baseline characteristics, on-treatment characteristics, and outcomes is provided in **Table 3**.

Table 3. Summary of baseline, treatment, and outcome characteristics of studied patients (N=72)

Characteristic	Value
<i>Baseline demographics and phenotype</i>	
Age, years, median [IQR]	34 [22–48]
Sex, female, n (%)	18 (35)
Body weight [†] , kg, median [IQR]	63 [56–75]
Body mass index [†] , kg/m ² , median [IQR]	22 [19–26]
Active smoker : previous smoker : non-smoker, n (%)	1 (3) : 9 (23) : 29 (74)
<i>Disease phenotype at baseline</i>	
Disease duration, years, median [IQR]	3 [1–7]
Bowel movements, /day, median [IQR]	10 [8–15]
Disease extension, extensive : left sided, n (%)	23 (70) : 10 (30)
Mayo endoscopic subscore, 3 (severe disease) : 2 (moderate disease), n (%)	30 (81) : 7 (19)
<i>Serology at baseline</i>	
C-reactive protein [‡] , mg/L, median [IQR]	30 [7–82]
Serum albumin [§] , g/L, median [IQR]	31 [27–38]
Hemoglobin, g/dL, median [IQR]	12.5 [10.4–13.0]
White blood cell count, G/L, median [IQR]	11 [8–14]
<i>Previous medication</i>	
Previous exposure to aminosaliclates, n (%)	32 (74)
Previous exposure to immunomodulators, n (%)	14 (33)
<i>Rescue therapy</i>	
Number of infliximab infusions per patient [¶] , n, median [range]	3 [1–7]
Follow-up of the dosing period, days, median [IQR]	7 [0–42]
First infliximab dose, mg/kg, median [range]	9.9 [3.8–15.0]
Sampling time since the most recent infusion, days, median [IQR]	26 [11–42]
Number of infliximab concentration measurements per patient [#] , n, median [range]	2 [1–5]
Number of patients with 1: 2: 3: 4: 5 infliximab concentrations, n (%)	33(46):18(25):7(10):8(11):6(8)
Exposure to thiopurine ^{**} , n (%)	14 (47)
Concomitant methotrexate ^{††} , n (%)	0 (0)
Concomitant aminosaliclates ^{‡‡} , n (%)	3 (30)
Concomitant steroids, despite steroid refractoriness ^{§§} , n	13 (87)
Anti-infliximab antibodies, n (%)	6 (4)
<i>Treatment outcome</i>	
Number of patients with colectomy performed within 90 days, n (%)	11 (15)
Time to colectomy, days, median [range]	32 [11–84]

All percentages are based only on non-missing data.

* Body weight was missing for two patients.

† Body mass index was missing for 29 patients.

‡ C-reactive protein was missing for three patients.

§ Serum albumin was missing for four patients.

¶ Schedules did not necessarily follow classical induction (e.g., 0–2–6 weeks); timing was adapted by each center based on local expertise and patient response.

Including 11 below and 56 above the limit of quantification (44% censored).

** One patient received thiopurines solely prior to the start of infliximab therapy, without concurrent administration. †† Methotrexate was missing for 42 patients.

‡‡ Aminosaliclates was missing for 62 patients.

§§ Concomitant steroids was missing for 57 patients.

Abbreviations. IQR, interquartile range.

3.2 Infliximab Pharmacokinetics

The infliximab popPK was best described using a one-compartmental model with first-order elimination (**Table 4, Supplementary Code**). A higher body weight was associated with an increased infliximab clearance and volume of distribution (allometric scaling). A lower CRP at baseline was associated with a decreased clearance. The median [IQR] estimated a posteriori (i.e., TDM-based) infliximab clearance in the population was 0.51 [0.32–0.73] L/day. The median [IQR] estimated infliximab half-life was 18.1 [12.9–28.7] days. Calibration of the final popPK model was adequate (**Figure S9**). Comprehensive goodness-of-fit diagnostics—encompassing observed versus predicted concentrations, residual plots, and individual concentration–time profiles—are provided in **Figure S10–S11**.

Table 4. Population pharmacokinetics and time-to-event model parameter estimates

	Final model estimates (%RSE) [%shrinkage]	Bootstrap median [95% CI]
<i>Population pharmacokinetics model</i>		
Typical values		
CL (L/day)	0.49 (13)	0.49 [0.33; 0.63]
V (L)	13.2 (16.8)	12.9 [9.0; 18.1]
Covariate effects		
Baseline body weight on CL	0.75 (fixed)	0.75 (fixed)
Baseline body weight on V	1 (fixed)	1 (fixed)
Baseline C-reactive protein on CL	0.56 (32)	0.64 [0.21; 1.56]
Interindividual variability		
on CL (%CV)	72.1 (19) [32]	68.6 [28.5; 108.0]
on V (%CV)	87.8 (37) [41]	90.0 [3.9; 207.7]
Residual variability		
Proportional error (%)	44 (10)	43.5 [31.7; 52.8]
<i>Time-to-event model</i>		
Typical values		
Baseline hazard	0.001 (135)*	0.001 [0.000; 0.006]
Shape parameter	1.08 (26)	1.10 [0.70; 2.29]
Covariate effects		
AUC _{w2-4} /CL on baseline hazard	-0.56 (22)	-0.56 [-1.00; -0.13]

The number of successful bootstrap runs is 2000 out of 2000 attempts.

*RSE was back-calculated via the delta method with baseline hazard = 10^{θ} where $\theta = -2.95$ (19.8%).

Abbreviations. CI, confidence interval; CL, clearance; CV, coefficient of variation calculated as $CV = \sqrt{e^{\omega^2} - 1}$ with ω^2 being the interindividual variability; RSE, relative standard error (for random effects and residual variability reported on the approximate standard deviation scale; standard error/variance estimate/2).

3.3 Time-To-Colectomy Model

Information on colectomy was available for 71 patients. The hazard risk for colectomy was best described with a Weibull hazard function. Calibration of the final predictive model was adequate (**Figure S12**). The hazard risk for colectomy decreased as the week 2 to week 4 AUC-to-infliximab clearance ratio (AUC_{w2-4}/CL) increased (**Table 4**).

Because the ratio of AUC_{w2-4}/CL predicted colectomy after week 4, five patients who experienced colectomy before week 4 have been excluded when internally evaluating the predictive performance of AUC_{w2-4}/CL . As compared to AUC_{w2-4} and CL alone, AUC_{w2-4}/CL improved discrimination with an increase in area under the ROC curve (AUROC) from 0.76 (95% CI 0.46–1.00) (AUC_{w2-4}) and 0.71 (95% CI 0.44–0.97) (CL) to 0.79 (95% CI 0.52–1.00).

3.4 Colectomy Risk Stratification Algorithm

The logarithmic ratio of AUC_{w2-4}/CL of 5.79 was used to discriminate between patients at low and high risk for colectomy (sensitivity 83%, specificity 85%) (**Figure 13, Figure S13**). Patients with a logarithmic ratio of AUC_{w2-4}/CL below 5.79 were classified as high-risk, with ratio values ranging from 1.11 to 5.79 (14/66 patients; 21%). Patients with a logarithmic ratio of AUC_{w2-4}/CL of 5.79 and higher were classified as low-risk, with ratio values ranging from 5.79 to 8.84 (52/66 patients; 79%). The colectomy rate was 36% (5/14 patients) and 2% (1/52 patients) in the high- and low-risk group, respectively. Classification accuracy was 85% (95% CI 74%–92%), and the associated Fisher's exact test p-value was 0.001. The effectiveness of the risk stratification algorithm is presented in **Figure 14**. The model-simulated proportion of patients without colectomy (%) aligns well with the observed proportion in the low-risk group (**Figure 14, panel A**) and the high-risk group (**Figure 14, panel B**).

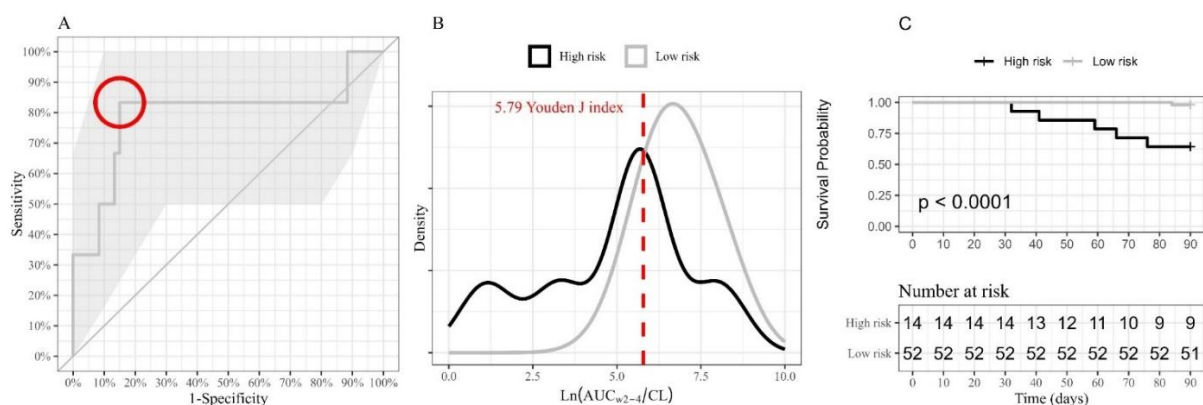


Figure 13. Discriminatory performance of the colectomy risk stratification algorithm. (A) Receiver operating characteristic curve representing the ability of clearance-normalized infliximab exposure to discriminate between patients with a high and a low colectomy risk. The 95% nonparametric confidence interval is represented by the grey shaded area. (B) Density plot representing the distributions of clearance-normalized infliximab exposure for patients with a high colectomy risk (black line) and a low colectomy risk (grey line). The optimal threshold derived from the Youden J index is represented by a red circle and a red dashed line in Panels A and B, respectively. (C) Kaplan–Meier estimates of the time to colectomy for patients with a high colectomy risk (black line) and a low colectomy risk (grey line). The P-value is calculated using the Log-Rank test.

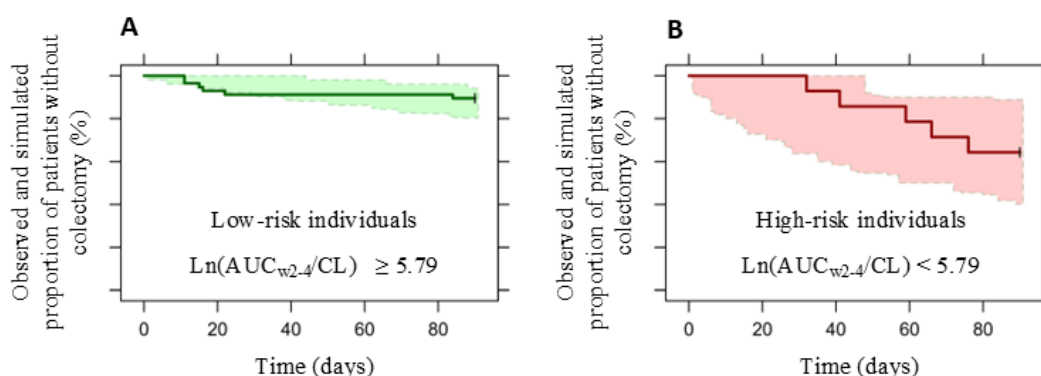


Figure 14. Predictive performance of the colectomy risk stratification algorithm. Kaplan–Meier Visual predictive checks for the final time-to-colectomy model for patients with low-risk (A) and high-risk (B). The solid line represents the observed study data using Kaplan–Meier estimators, and the shaded area represents the simulated 95% prediction interval of the final time-to-colectomy model. Risk stratification is based on the clearance-normalized infliximab exposure ($\text{Ln}(\text{AUC}_{w2.4}/\text{CL})$) threshold of 5.79 (Youden J index).

3.5 Interactive Precision Dosing Tool

We developed an evidence-based, interactive risk stratification and precision dosing tool to guide individualized infliximab rescue therapy for patients with ASUC. Input variables include baseline body weight and CRP, and infliximab serum concentration(s). By combining infliximab TDM and Bayesian forecasting, our tool performs individual patients' dosing simulations, targeting a user-defined 90-day colectomy risk (default 20%, corresponding to

a logarithmic ratio of AUC_{w2-4}/CL of 5.79) (**Figure 15**). Additional details and user instructions are provided in the **Supplementary Results**. A snapshot of the tool is shown in **Figure 16**. Free access to the web application is provided at <https://gitlab.kuleuven.be/pharmacometrics>.

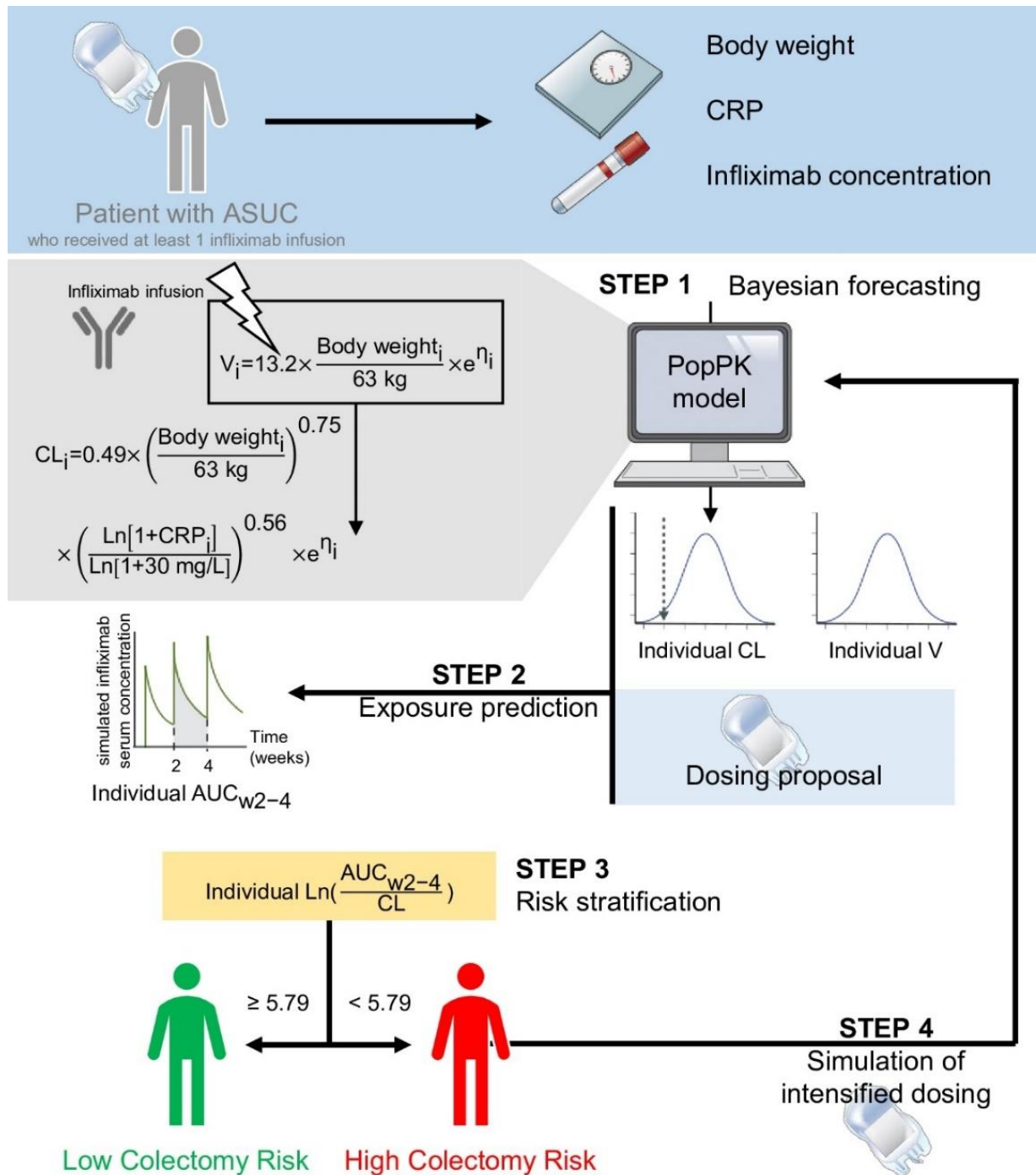


Figure 15. Schematic representation of the colectomy risk stratification algorithm. (Step 1) The infliximab clearance (CL) and volume of distribution (V) are estimated using Bayesian forecasting by leveraging the population pharmacokinetics (popPK) model and three patient-specific characteristics: (1) body weight, (2) C-reactive protein (CRP) at the start of infliximab rescue therapy, and (3) an infliximab concentration obtained in the first two weeks of infliximab rescue therapy. (Step 2) The overall infliximab exposure between weeks 2 and 4 (i.e., area under the infliximab concentration–time curve between weeks 2 and 4, AUC_{w2-4}) is calculated for a proposal dosing regimen by leveraging the popPK model. (Step 3) The clearance-normalized infliximab exposure is calculated as the natural logarithm of the AUC_{w2-4} -to-clearance ratio ($\ln[AUC_{w2-4}/CL]$). A threshold of 5.79 distinguishes between patients at low risk for colectomy (≥ 5.79), who have relatively high infliximab exposure considering their infliximab clearance, and patients at high risk for colectomy (< 5.79), who have relatively low infliximab exposure considering their infliximab clearance. (Step 4) If a patient is at high risk for colectomy in Step 3, intensified infliximab dosing schedules can be simulated in Step 2, thereby enabling personalized infliximab dosing at week 2 and potentially reducing the patient’s colectomy risk.

ASUC : Infliximab dosing tool

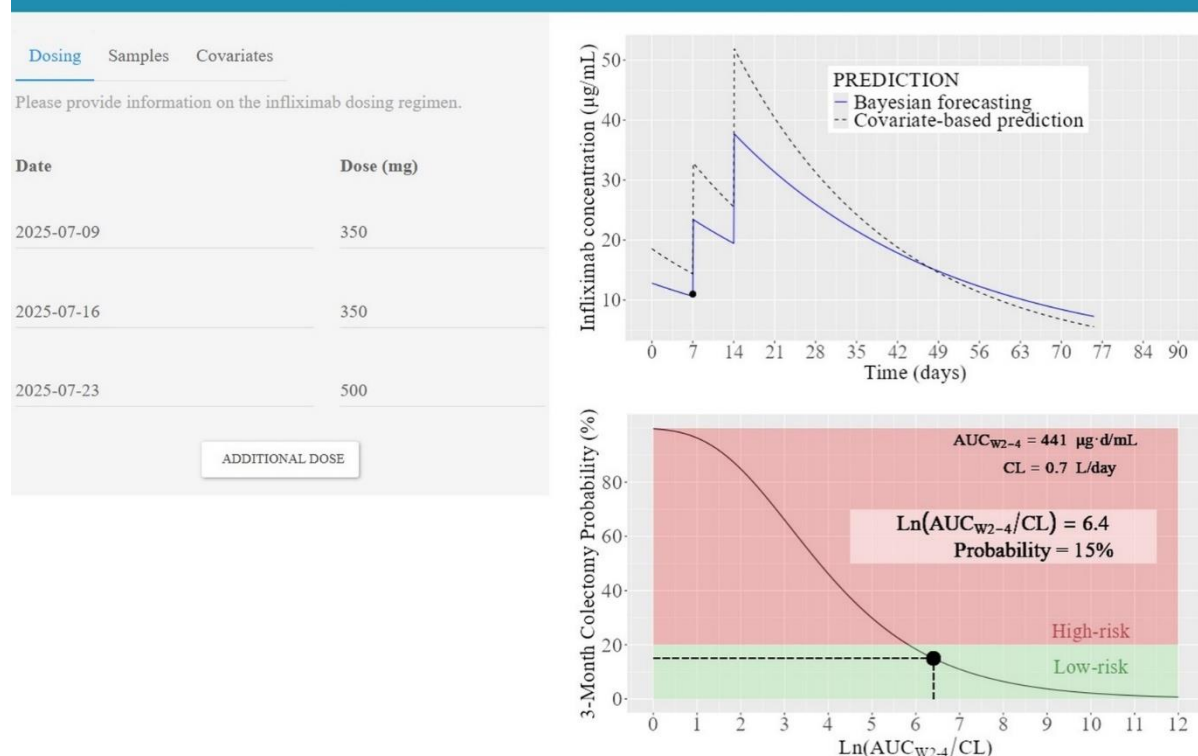


Figure 16. Interactive risk stratification and infliximab dosing tool. The tool consists of three tab panels that allow the user to provide patient-specific information: the infliximab dosing regimen (“Dosing” tab), the infliximab concentration(s) measured in serum samples (“Samples” tab), and the patient’s body weight and C-reactive protein (CRP) at the start of infliximab rescue therapy (“Covariates” tab). This information is used in combination with the infliximab population pharmacokinetics model to predict the patient’s infliximab pharmacokinetic parameters (*a posteriori* Bayesian forecasting, solid line; *a priori* covariate-based prediction; dashed line), visualize the predicted infliximab concentration–time course under the provided dosing regimen, and the probability of colectomy within 90 days after start of infliximab rescue therapy.

4. Discussion

The present work introduces a model-based algorithm to inform infliximab dosing in patients with ASUC. We demonstrated that clearance-normalized infliximab exposure best predicts time-to-colectomy. This is the first dose–exposure–response study of infliximab in patients with ASUC. Our findings pave the way towards personalized infliximab treatment in a highly vulnerable group of patients.

ASUC is a severe medical condition of low incidence, and PK data are scarce. Through an international collaborative effort, we were able to pool PK data of 72 patients with ASUC, allowing us to build a popPK model, describing the relation between infliximab dosing and infliximab exposure. We identified CRP as a statistically significant covariate in the popPK model. Serum albumin is a more commonly withheld covariate in popPK models of monoclonal antibodies, possibly explained by its non-competitive shared FcRn-mediated recycling pathway. However, high correlation between CRP and serum albumin and a lack of standardization of serum albumin quantification across medical centers may explain why CRP outperformed serum albumin in our study. Moreover, we log-transformed CRP data, which is not commonly done, yet strongly advised, to account for its well-described non-normal distribution.¹⁰⁸ This resulted in a better model fit and more precise parameter estimates compared to models based on the untransformed CRP distribution and serum albumin (**Table S4**).

We combined infliximab exposure (AUC_{w2-4}) and infliximab PK (clearance) to predict colectomy within 90 days from the start of infliximab rescue therapy. The AUC_{w2-4}/CL ratio best predicted time-to-colectomy and was the sole covariate retained in our time-to-event model. Two patients may have equally high overall exposure, resulting in the same AUC_{w2-4} , yet one may have a higher clearance – due to higher target-mediated drug disposition and protein-losing enteropathy – and thus an increased risk of colectomy. Similarly, two patients may have the same infliximab clearance, yet one may have a lower infliximab exposure – due

to insufficient dosing – and consequently will be at a higher risk of colectomy. Unlike for small molecules, drug clearance of monoclonal antibodies serves as a biomarker for disease severity in inflammatory bowel disease.^{89,99,100} The AUC_{w2-4}/CL ratio combines both exposure (modifiable risk factor; exposure drives disease activity) and PK (unmodifiable risk factor; disease activity drives exposure), thereby accounting for the bidirectional relation between infliximab PK and pharmacodynamics.³¹ Moreover, the AUC_{w2-4}/CL ratio may also remind us of the pharmacokinetic–pharmacodynamic (PKPD) index used to guide dosing of antimicrobials. Yet, not all patients will benefit from higher exposure, especially those who are inherently non-responsive to anti-TNF therapy. Future studies should focus on identifying patients with early therapy failure and developing alternative treatment strategies.

Unlike Battat et al. (2021), we did not identify baseline infliximab clearance as a predictor for colectomy. Battat *et al.* worked with the *a priori* estimated clearance, solely relying on baseline serum albumin and patient sex. Consequently, their clearance estimates are highly correlated with baseline serum albumin (**Figure S14, panel A**). Li Wai Suen *et al.* identified day 7 infliximab clearance as a predictor for 3-month colectomy, estimating individual patient clearance using a previously published popPK model for ulcerative colitis, while keeping other PK parameters fixed. Incorporating observed data to refine an *a priori* popPK model is a valuable approach to better estimate a patient's true drug clearance. However, based on general PK principles, infliximab concentrations measured as early as days 5 and 7 may not provide sufficient information to reliably re-estimate true clearance, suggesting that further validation of this method is needed.

In contrast, our *a posteriori* (i.e., TDM-based) clearance estimates are mainly informed by infliximab serum concentrations, alongside baseline CRP and body weight using a dedicated in-house developed popPK model in patients with ASUC (**Figure S14, panel C**). In our study, *a posteriori* clearance, as such, did not have sufficient statistical power to predict colectomy

alone. However, the clearance-normalized overall exposure between weeks 2 to 4 (AUC_{w2-4}/CL), was the strongest predictor of colectomy in our study (**Table S5**). By estimating CL *a posteriori*, we achieve a more accurate characterisation than an *a priori* estimate could provide; however, this advantage comes with the limitation that it requires observed data and therefore cannot be obtained at baseline. Consequently, our workflow only applies to patients with a TDM sample available before experiencing colectomy. Our findings reinforce the observations from the PREDICT-UC trial, suggesting a combined exposure–clearance parameter. This further supports the reliability of clearance as a predictive biomarker in patients with ASUC. For a detailed comparison between Battat *et al.* (2021), Li Wai Suen *et al.* (2025), and the present study, we kindly refer to **Table S6**.

Personalized medicine approaches are increasingly being employed throughout clinical drug development. Lately, increasing focus has been given to model-informed dose optimization strategies such as MIPD of infliximab.^{109,110} Identifying and establishing early predictors of treatment outcome is an important asset to support dose optimization and early decision-making. Dedicated MIPD trials in patients with ASUC, such as the TITRATE trial¹¹¹, did not meet statistical significance. However, their pioneering work paves the way for the design and execution of new studies. Historically, infliximab exposure has been used to guide dose selection, and the introduction of rapid assays has tremendously improved the traditional monitoring. In recent years, their availability has increased while costs have decreased, making them more accessible. Measuring drug exposure in combination with our developed popPK model enables Bayesian forecasting and the estimation of an individual patient's clearance and subsequent calculation of an individual patient's AUC. Our developed software tool streamlines this process, providing an easy and efficient approach for risk stratification by relying on available TDM data.

The present work has some limitations. First, the secondary use of multicenter, real-world data from a low-prevalence disease prevented control over the sampling design, resulting in

sparse and unbalanced sampling. Consequently, we were unable to capture the biphasic terminal elimination of infliximab or account for clearance changes over time (due to disease progression or recovery) using IOV and/or time-varying covariates. Additionally, a popPK approach presumes that pharmacokinetic parameters remain constant over time, which may not always hold true in the setting of ASUC. Despite heterogeneity in sampling schedules and assay methods, the gold standard nonlinear mixed effects (NLME) modelling approach showed to be fit-for-purpose. Between-center variability was explicitly tested and found negligible. Model robustness was confirmed through bootstrap validation and pcVPC diagnostics, supporting reproducibility despite heterogeneity. Future studies should aim to collect more samples, along with time-varying covariate measurements, to better characterize disease dynamics. Additionally, no histologic data were available, limiting our ability to correlate mucosal pathology with PK and outcome data. We relied on Mayo endoscopic subscores as a surrogate for disease severity. In this regard, results from dedicated infliximab PK studies in patients with ASUC, such as the IGNITE trial (EU CT 2023-503509-12-00), are awaited to provide further insights and address these limitations. Second, the time-to-colectomy model includes uncertainty regarding the (clinically indicated) time point of colectomy. The day of a colectomy is not purely driven by disease activity but also depends on clinical workflows and preferences of the patient. However, shifts of a few days only have a minor influence, considering the 90-day follow-up period. Additional uncertainty stems from the low number of events, bearing the risk of overfitting and issues with model stability. As additional data accrue, model performance should be reassessed and, if warranted, the model updated and recalibrated to refine parameter estimates. The low number of events was also reflected in the wide confidence intervals around the AUROC using the identified predictor (**Figure 13**). Although the threshold reflects the optimal mathematical solution, its clinical applicability must be weighed carefully before it is used for decision-making. Third, we did not externally validate our risk stratification and

dosing algorithm. We invite the scientific community to make use of our proposed workflow for external validation. Therefore, we provide access to model code and supporting resources. The developed software tool facilitates accessibility and clinical validation.

To conclude, we performed the first model-based dose–exposure–response analysis of infliximab in patients with ASUC. We provide supportive evidence for clearance monitoring of infliximab in patients with ASUC. We demonstrated methodological advances in predictive modelling that enable truly personalized infliximab dosing. This novel approach to precision medicine holds the potential to pave the way for more effective, personalized infliximab dosing.

Chapter 2 Supplementary Material

Supplementary Figures

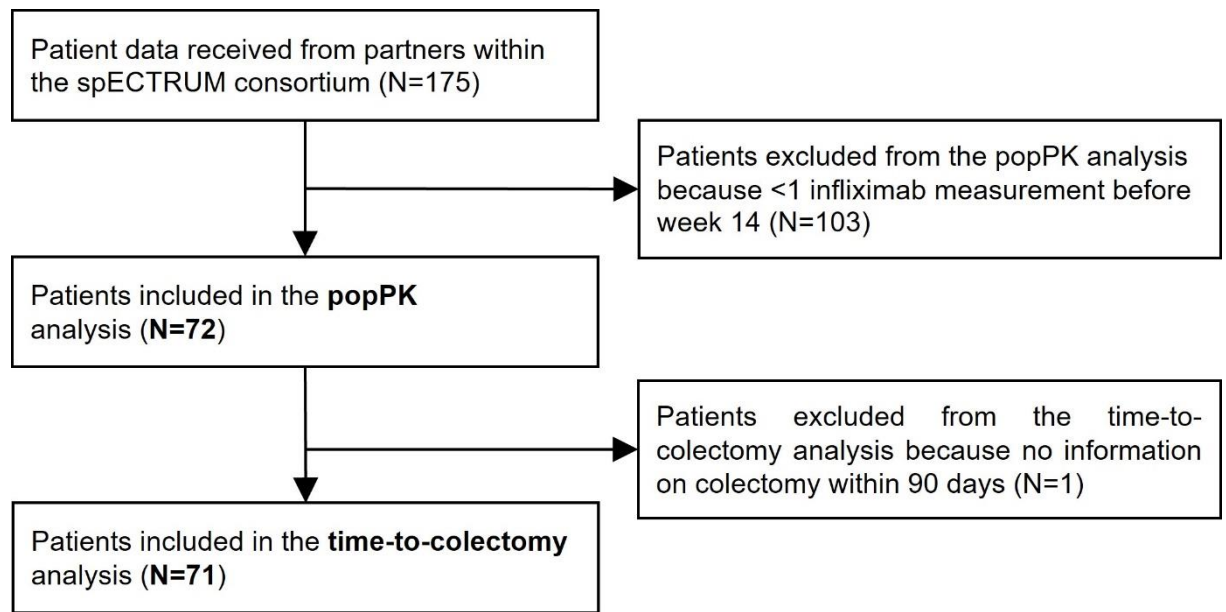


Figure S8. Flowchart depicting the selection of the study cohort for the popPK and time-to-colectomy analysis. Abbreviations. popPK, population pharmacokinetics.

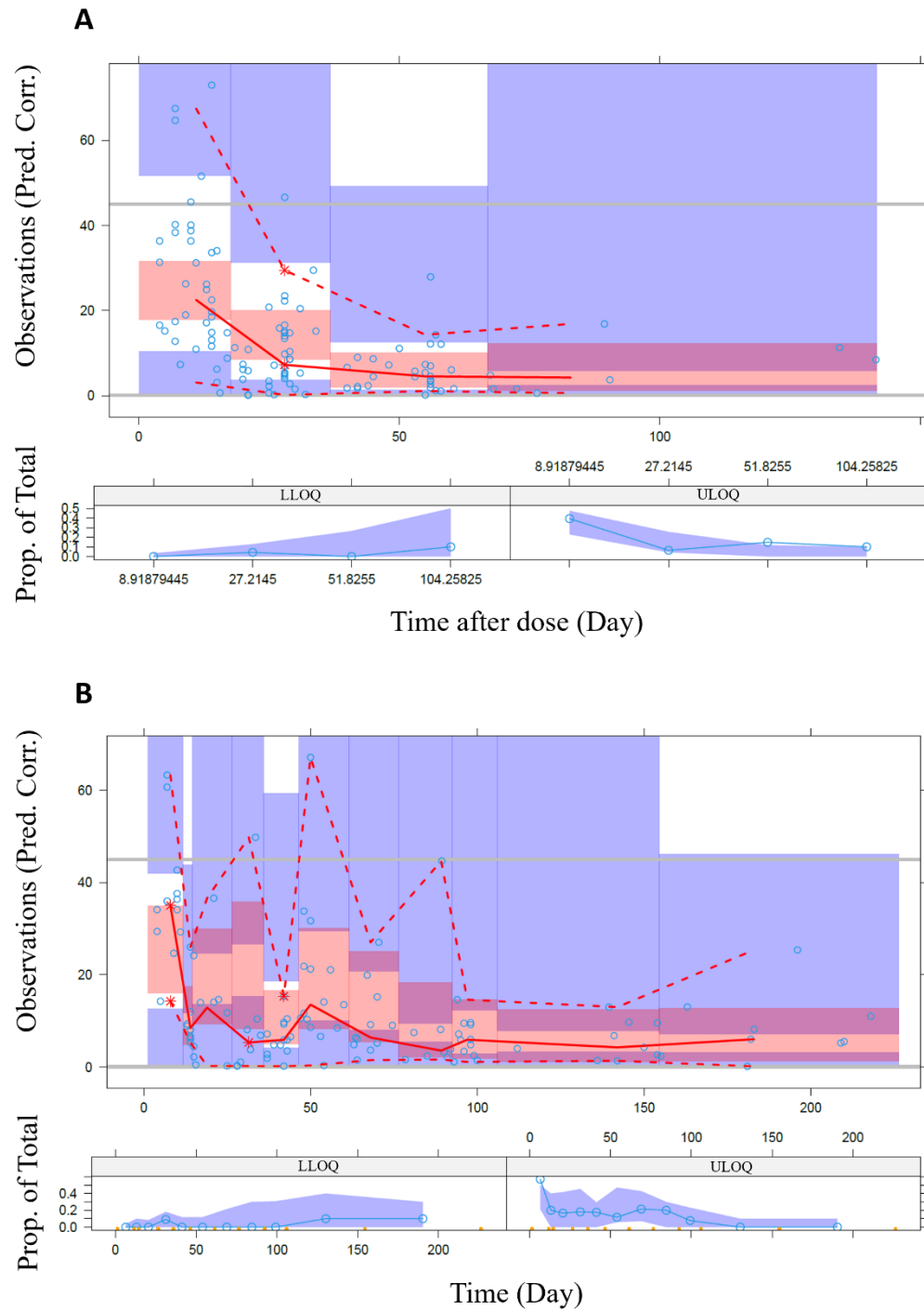


Figure S9. Visual predictive check of the final population pharmacokinetics model of infliximab. The Y-axis shows the infliximab serum concentrations, and the X-axis shows time after dose (A) and time since start of infliximab rescue therapy (B). Open circles represent (prediction-corrected) observed infliximab serum concentrations; The red solid and dashed lines represent the median, 5th, and 95th percentile of the observed data; The red and blue boxes represent the 95% prediction interval around the median, 5th, and 95th percentile of the simulated data (1000 simulations). The time-course of observed and predicted proportions of infliximab serum concentrations below the lower limit of quantification (LLOQ) and above the upper limit of quantification (ULOQ) are presented below the infliximab concentration–time courses.

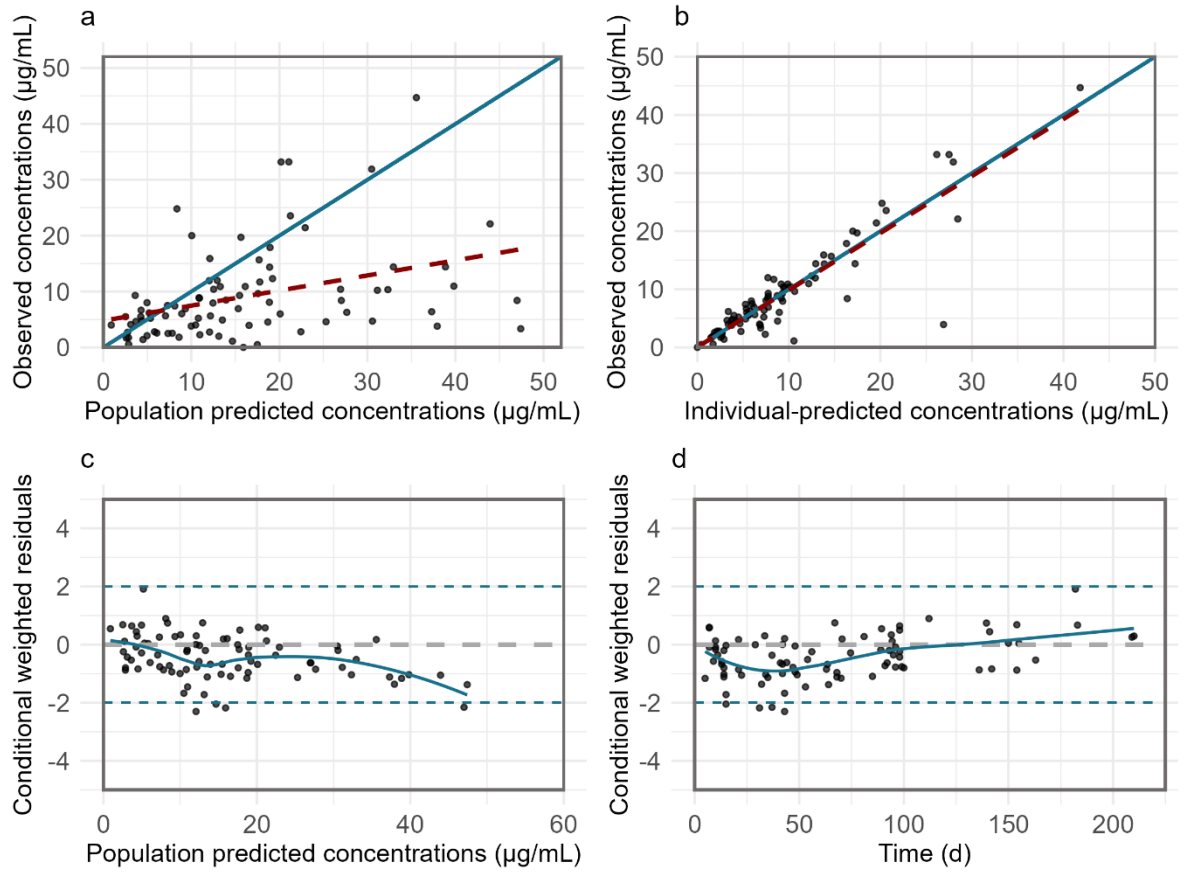
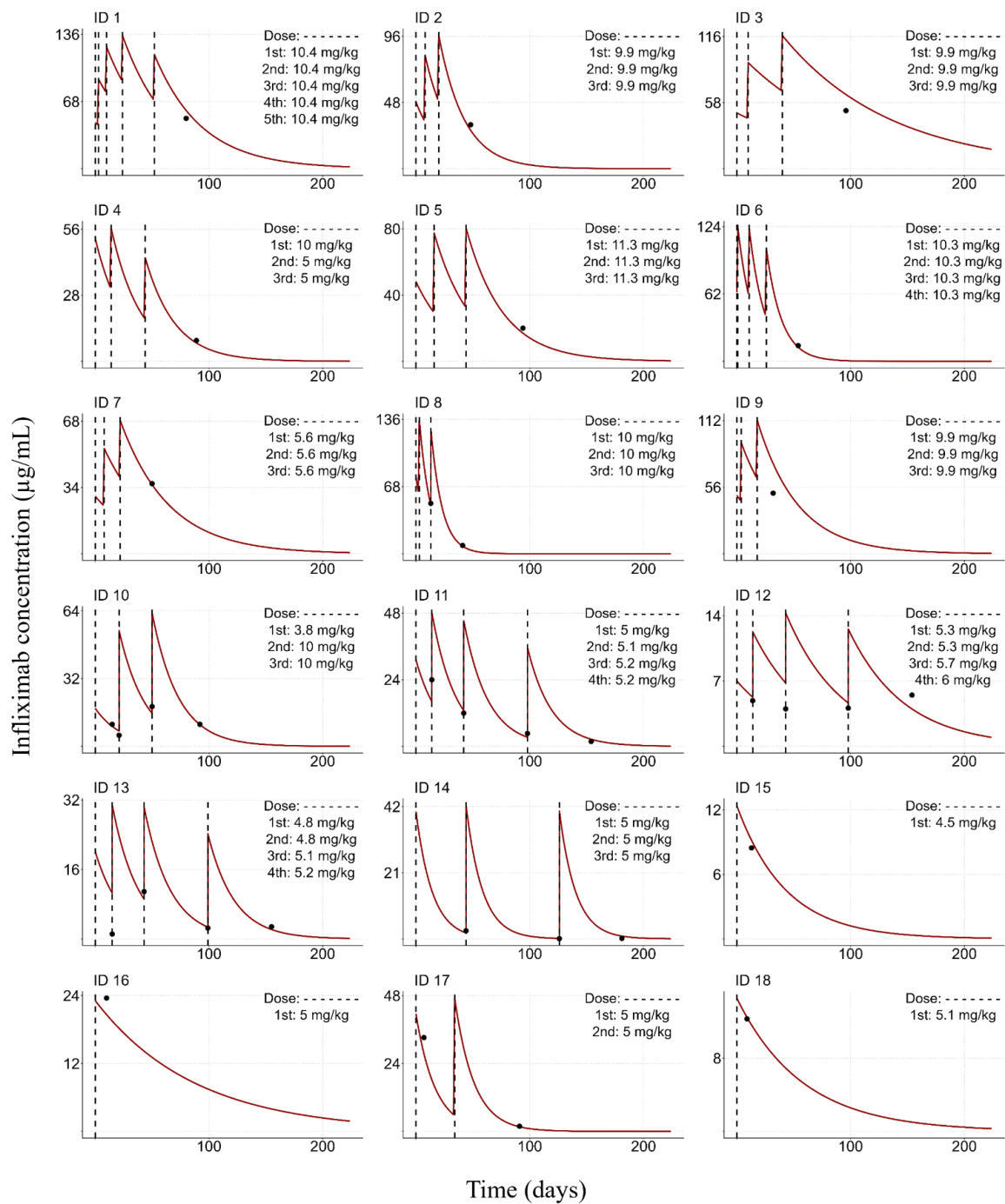
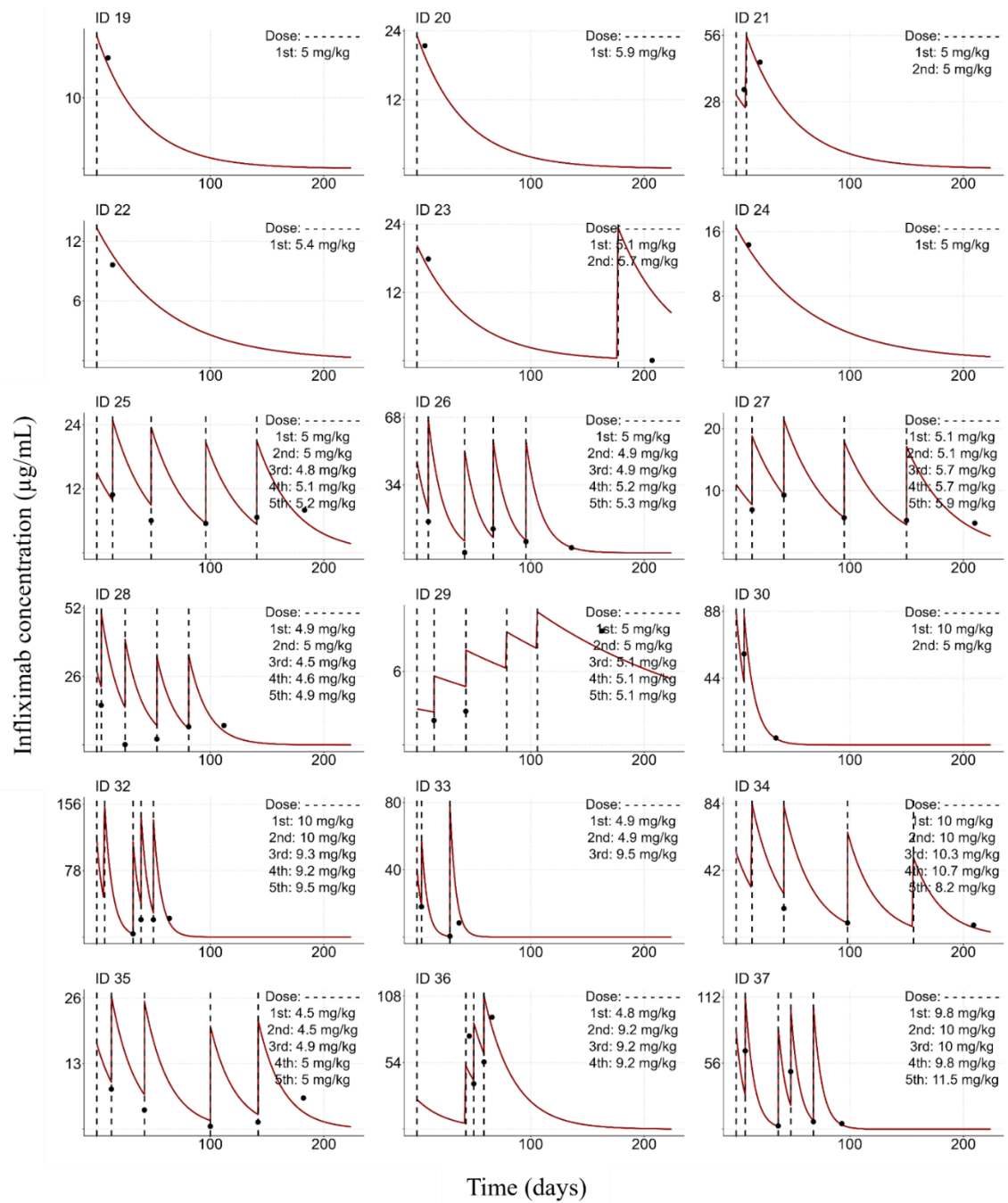
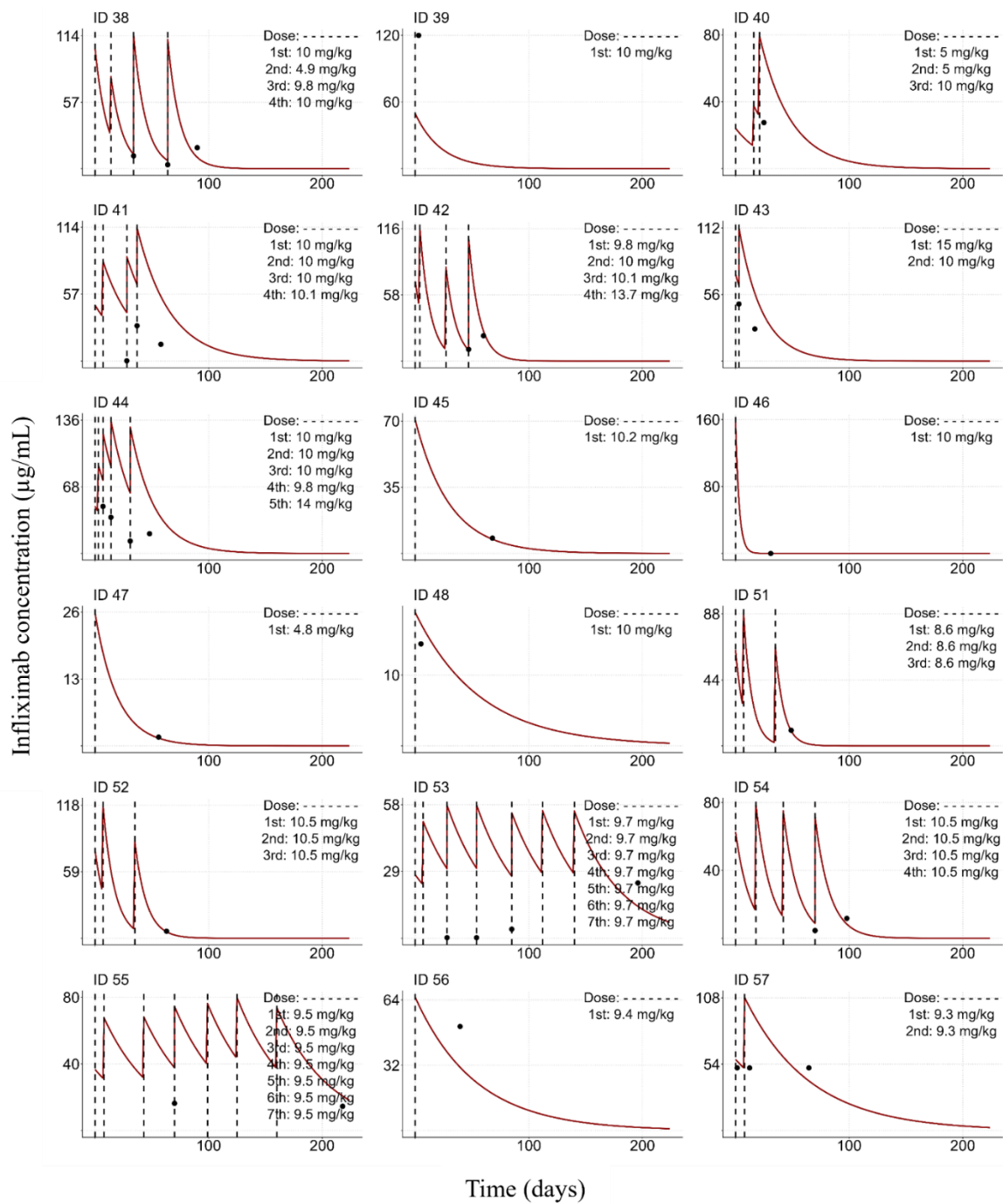


Figure S10. Residual-based diagnostic plots of the final population pharmacokinetics model: Observed infliximab serum concentrations versus population-predicted serum concentrations (a); Observed infliximab serum concentrations versus individual-predicted infliximab serum concentrations (b); Conditional weighted residuals versus population-predicted infliximab serum concentrations (c); Conditional weighted residuals versus time (d). Unity lines are represented by solid blue lines (panels a and b), trend lines are represented by red dashed lines (panels a and b) and solid blue lines (panels c and d).







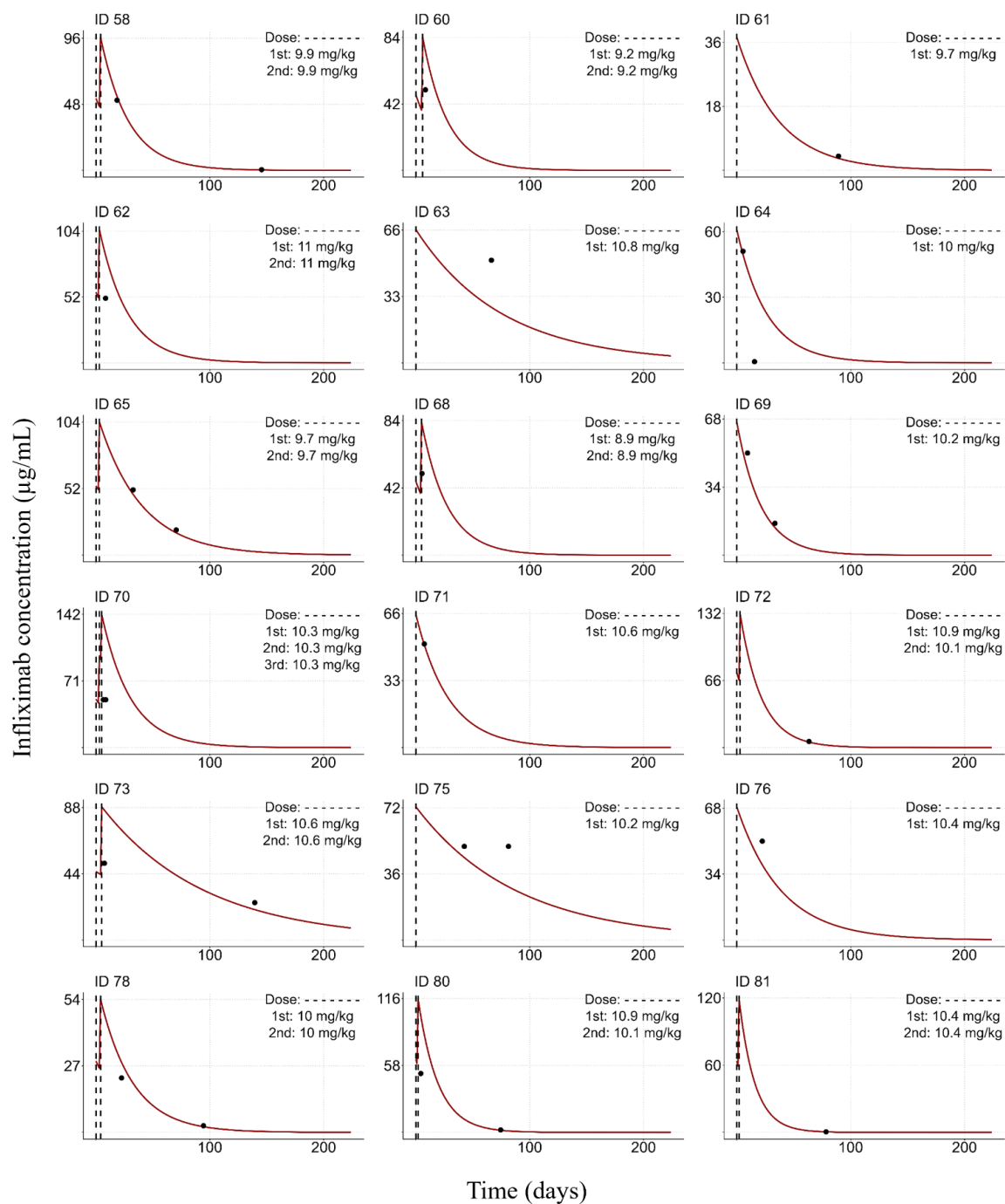


Figure S11. Individual infliximab serum concentration–time profiles: Each panel refers to a single individual. Red lines represent the model-predicted infliximab concentration–time courses; Black dots are the infliximab serum concentrations as observed in the study dataset; Black dotted vertical lines denote the timing of each infusion, while the legend reports the infliximab dose administered at each infusion event.

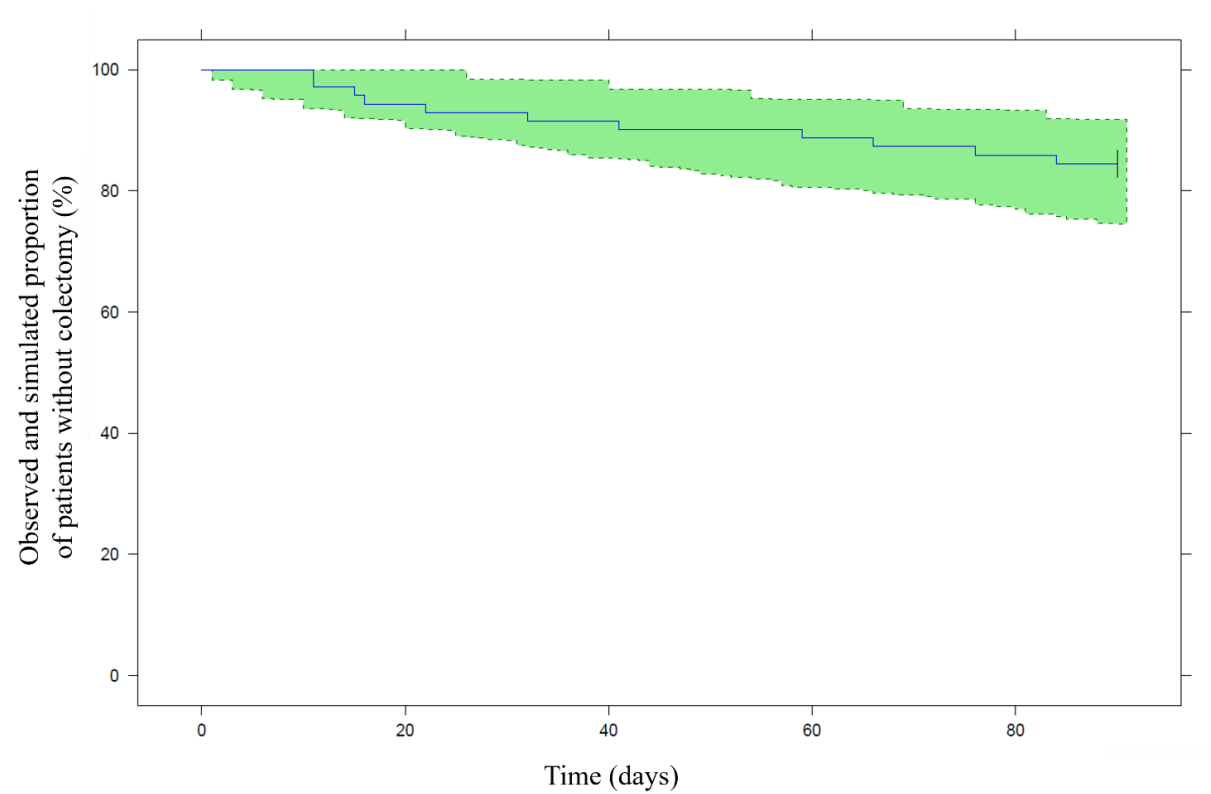


Figure S12. Kaplan–Meier curve and visual predictive check of the final time-to-event model for colectomy-free survival. The solid blue line represents the Kaplan–Meier curve of the observed data, with the green area representing the 95% confidence interval of the simulated data from the time-to-event model (1000 simulations).

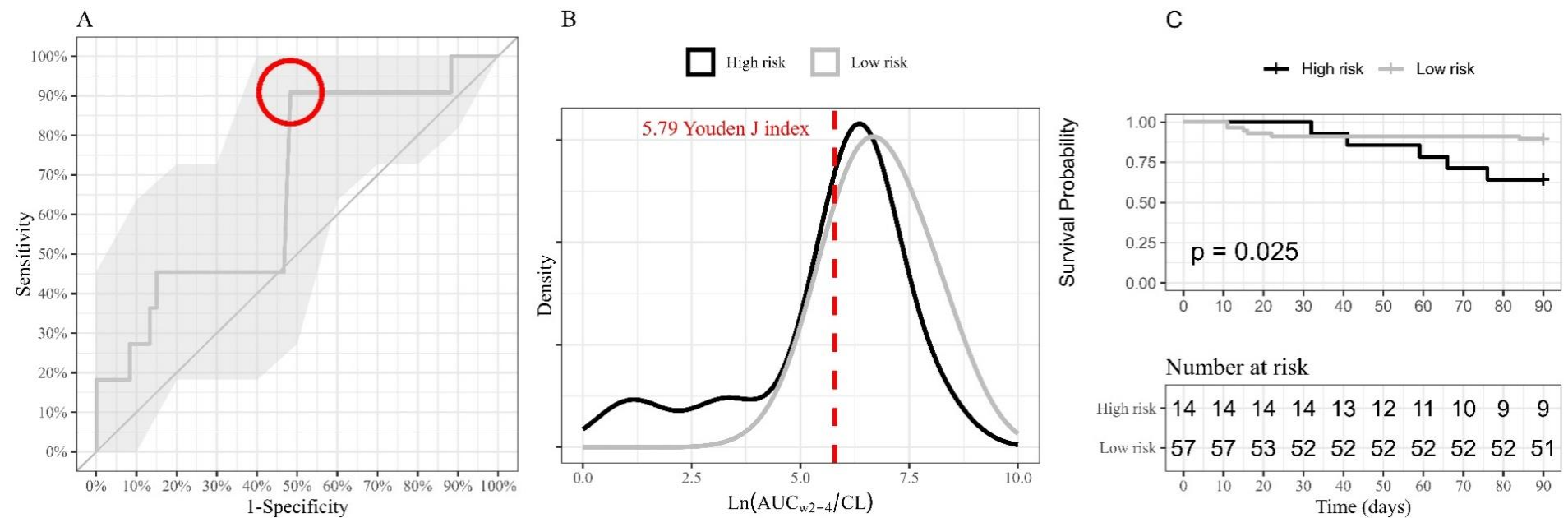


Figure S13. Discriminatory performance of the colectomy risk stratification algorithm based on the data from 71 patients, including the five patients who had colectomy before week 4. (A) Receiver operating characteristic curve representing the ability of clearance-normalized infliximab exposure to discriminate between patients with a high a low colectomy risk. The 95% nonparametric confidence interval is represented by the grey shaded area. (B) Density plot representing the distributions of clearance-normalized infliximab exposure for patients with a high colectomy risk (black line) and a low colectomy risk (grey line). The optimal threshold derived from the Youden *J* index is represented by a red circle and a red dashed line in Panels A and B, respectively. (C) Kaplan–Meier estimates of the time to colectomy for patients with a high colectomy risk (black line) and a low colectomy risk (grey line). The P-value is calculated using the Log-Rank test.

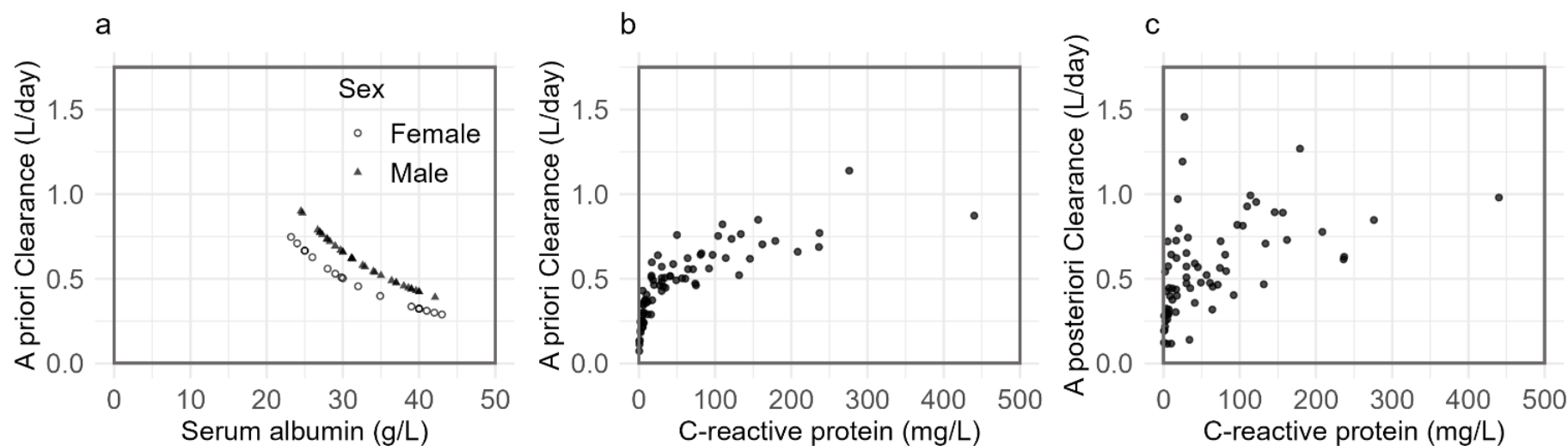


Figure S14. Comparison of clearance estimates between Battat et al. (2021) and our work. The a priori predicted infliximab clearances (Y-axis) versus serum albumin (X-axis) of each patient in our study cohort, computed using the equation from Battat et al. (2021) (a). The a priori predicted infliximab clearances (Y-axis) versus C-reactive protein (X-axis) of each patient in our study cohort, computed using the equation from Battat et al. (2021) (b). The a posteriori predicted infliximab clearances (Y-axis) versus C-reactive protein (X-axis) of each patient in our study cohort, computed using our final population pharmacokinetics model (c).

Supplementary Tables

Table S4. Stepwise covariate selection during population pharmacokinetics model development.

Model	Reference Model	Description	OFV	OFV change relative to Reference Model	Significance Level
1		Base Model	546.15	0	
Forward Inclusion					
2	1	Base Model +CRP on CL	537.36	- 8.79	<0.01
3	1	Base Model +Ln(CRP) on CL	537.01	- 9.14	<0.01
4	1	Base Model +ALB on CL	543.95	- 2.20	N.S.
5	1	Base Model +BMI on CL	548.06	1.91	N.S.
6	1	Base Model +CRP on V	550.13	3.99	N.S.
7	1	Base Model +Ln(CRP) on V	562.61	16.46	N.S.
8	1	Base Model +ALB on V	547.32	1.18	N.S.
9	1	Base Model +BMI on V	549.77	3.62	N.S.
Backward Elimination					
10	3	Base Model –Ln(CRP) on CL	546.17	9.16	<0.01

Each row represents an individual step. The decrease in the objective function value is measured in relation to the reference model.

Abbreviations. ALB, serum albumin at baseline; BMI, body mass index at baseline; CL, clearance; CRP, C-reactive protein at baseline; Ln, Natural logarithm; N.S., not significant; OFV, objective function value; V, volume of distribution.

Table S5. Stepwise covariate selection during time-to-colectomy model development.

Model	Reference Model	Description	OFV	OFV change relative to Reference Model	Significance Level
1		Base model	160.00		
Forward Inclusion					
2	1	Base model +CRP	158.14	- 1.86	N.S.
3	1	Base model +Ln(CRP)	159.95	- 0.05	N.S.
4	1	Base model +ALB	157.46	- 2.54	N.S.
5	1	Base model +BMI	159.96	- 0.04	N.S.
6	1	Base model +BW	159.98	- 0.02	N.S.
7	1	Base model +CL	156.01	- 3.99	<0.05
8	1	Base model +AUC _{w01}	159.52	- 0.48	N.S.
9	1	Base model +AUC _{w02}	159.99	- 0.01	N.S.
10	1	Base model +AUC _{w04}	159.22	- 0.78	N.S.
11	1	Base model +AUC _{w24}	157.64	- 2.36	N.S.
12	1	Base model +Ln(AUC _{w0-1} /CL)	159.83	- 0.17	N.S.
13	1	Base model +Ln(AUC _{w0-2} /CL)	159.39	- 0.61	N.S.
14	1	Base model +Ln(AUC _{w0-4} /CL)	156.80	- 3.20	N.S.
15	1	Base model +Ln(AUC _{w2-4} /CL)	152.65	- 7.35	<0.01
16	15	Base model +Ln(AUC _{w2-4} /CL) + CL	152.27	- 0.37	N.S.
Backward Elimination					
17	15	Base model -Ln(AUC _{w2-4} /CL)	160.00	7.35	<0.01

Each row represents an individual step. The decrease in the objective function value is measured in relation to the reference model.

Abbreviations. ALB, serum albumin at baseline; AUC_{wx-y}, Bayesian forecasted area under the infliximab concentration–time curve between week x and y; BMI, body mass index at baseline; BW, body weight at baseline; CL, clearance; CRP, C-reactive protein at baseline; Ln, Natural logarithm; N.S., not significant; OFV, objective function value; V, volume of distribution.

Table S6. Comparison of Battat et al. (2021), Li Wai Suen et al (2025) and the present work.

	Battat et al. (2021)	Li Wai Suen et al. (2025)	Present study
Number of centers	1 (single-center study)	13 (multicenter study)	7 (multicenter study)
Number of patients	39	135	72
PopPK model population	Moderately-to-severely active UC*	UC	In-house developed model for ASUC [†]
Colectomy prediction model	Logistic regression model	Poisson regression model	Time-to-event model
Event of interest	Colectomy <i>at</i> 6 months	Colectomy <i>within</i> 3 months	Colectomy <i>within</i> 3 months
Clearance estimation	A priori clearance	Re-estimated using Fasanmade model [#]	A posteriori clearance (based on infliximab serum concentrations)
Covariates on clearance	Baseline serum albumin and sex	–	Baseline CRP and body weight
Colectomy predictor	Baseline predictor (not modifiable)	On treatment predictor (modifiable)	On treatment predictor (modifiable)

*Diagnosed using Mayo score

[†]Diagnosed using Truelove and Witts' criteria

[#]Fasanmade AA, Adedokun OJ, Ford J, et al. Population pharmacokinetic analysis of infliximab in patients with ulcerative colitis. Eur J Clin Pharmacol 2009;65:1211-28.

Abbreviations. ASUC, acute severe ulcerative colitis; CRP, C-reactive protein; PopPK, population pharmacokinetics; UC, ulcerative colitis.

Supplementary Results

The app interface is composed of two main areas, a side bar on the left and a main area on the right. The side bar is where the user can enter the baseline phenotype, infliximab doses, and infliximab serum concentrations. The main area shows the simulated output. The user can see the population-based and individual-based simulated infliximab serum concentration–time profile of the respective infliximab dosing. The risk of colectomy is calculated and shown with respect to the corresponding risk class (high or low risk of colectomy).

Supplementary Code

NONMEM Control Stream for the Final Infliximab Population Pharmacokinetics

Model

```
$SIZES PD=-100
```

\$PROBLEM

```
$INPUT ID TIME Dummy DV AMT TAD OCC OCCb2 CMT MDV EVID LOQ UOQ Mthree  
DISDURb DISDURb_imp DISEXTb DISEXTb_imp PRE5ASAb PRE5ASAb_imp PREIMMUb PREIMMUb_imp  
CRP CRP_imp CRPbase CRPmod LOGCRP LOGCRPbase LOGCRPmod ALB ALB_imp ALBbase ALBmod HB  
HB_imp MESb MESb_imp WBC WBC_imp WT WT_imp CRPSTR BMI BMI_imp BOWELMOV COLECTOW12  
COLECTOW12Y1 GENDER GENDER_imp SID
```

```
$DATA pktab8b.txt IGNORE=@ IGNORE=(Dummy=1);
```

\$SUBROUTINES ADVAN1 TRANS2

```
$PK LTCL = LOG(THETA(1))  
MU_1 = LTCL  
CL = EXP(MU_1+ETA(1)) * ((WT/63)**0.75) * (((LOGCRPbase)/3.433987)**THETA(3))  
  
LTVV = LOG(THETA(2))  
MU_2 = LTVV  
V = EXP(MU_2+ETA(2)) * ((WT/63)**1)  
  
S1 = V
```

```
$THETA 0.487 ; CL (L/day)  
13.2 ; V (L)  
0.556 ; CRP effect on CL
```

```
$OMEGA 0.419 ; IIV on CL  
0.571 ; IIV on V
```

```
$SIGMA 0.173 ; Proportional Residual Error
```

```
$ERROR IPRED = F  
IRES = DV-IPRED  
W = (SQRT(IPRED**2*SIGMA(1,1)))  
IWRES = IRES/W  
ESPi = DV/F  
DUM1 = (LOQ-IPRED)/W  
PROB1 = PHI(DUM1)  
DUM2 = (IPRED-UOQ)/W  
PROB2 = PHI(DUM2)
```

```
IF (Mthree.EQ.0) THEN  
F_FLAG = 0  
Control = 0  
Y = IPRED*(1+ERR(1))  
ENDIF
```

```
IF (Mthree.EQ.1) THEN  
F_FLAG = 1 ;F_FLAG=1 indicates Likelihood ;F_FLAG=2 indicates -2log likelihood  
Control = 1  
MDVRES = 1  
Y = PROB1  
ENDIF
```

```
IF (Mthree.EQ.2) THEN  
F_FLAG = 1 ;F_FLAG=1 indicates Likelihood ;F_FLAG=2 indicates -2log likelihood  
Control = 2
```

```

MDVRES  = 1
Y       = PROB2
ENDIF

```

```

$EST METHOD=SAEM INTERACTION LAPLACIAN PRINT=5 NBURN=400 NITER=300
$EST METHOD=IMP EONLY=1 PRINT=1 NITER=15 ISAMPLE=1000

```

```

$COV MATRIX=R UNCOND PRINT=E

```

```

$TABLE ID DV TIME MDV EVID PRED RES IPRED IRES RES WRES CWRES IWRES Control LOQ UOQ
WT W ESPi DISDURb DISEXTb PRE5ASAb PREIMMUb CRP ALB HB MESb WBC BMI BOWELMOV COLECTOW12
COLECTOW12Y1 GENDER SID CL V ETA(1) ETA(2) AMT CMT ONEHEADER NOPRINT NOAPPEND
FILE=mytablr27

```

NONMEM Control Stream for the Final Time-To-Colectomy Model

\$PROBLEM

```
$INPUT ID TIME DV EVID SIM CRPbase LOGCRPbase ALBbase BMI WT CLM AUC02 Yratio02 AUC04  
        Yratio04 AUC24 Yratio24 AUC01 Yratio01
```

```
$DATA ttetab.txt IGNORE=@  
        ;Sim_start  
        IGNORE(SIM.EQ.1)  
        ;Sim_end
```

```
$SUBROUTINE ADVAN=6 TOL=5
```

```
$MODEL COMP=(HAZARD)
```

```
$PK BASE      = (10**THETA(1))*EXP(ETA(1)) ; ETA is a placeholder here  
    SHP        = THETA(2) ; shape  
    BASECOV    = EXP(THETA(3)*(Yratio24-6. 609))  
    LAM        = BASE*BASECOV ; lambda=scale
```

```
$DES DEL=1E-6 ; to keep from taking 0**power  
    DADT(1) = LAM*SHP*((T+DEL)**(SHP-1))
```

\$ERROR

```
    DELX = 1E-6  
    IF(NEWIND.NE.2) OLDCHZ=0 ;reset the cumulative hazard  
    CHZ = A(1)-OLDCHZ ;cumulative hazard from previous time point in data set  
    OLDCHZ = A(1) ;rename old cumulative hazard  
    SUR = EXP(-CHZ) ;survival probability  
    HAZNOW=LAM*SHP*((TIME+DELX)**(SHP-1))  
    IF(DV.EQ.0) Y=SUR ;censored event (prob of survival)  
    IF(DV.EQ.1) Y=SUR*HAZNOW ;prob density function of event
```

```
IF(ICALL.EQ.4) THEN ; for simulation  
    CALL RANDOM (2,R)  
    DV=0  
    RTTE = 0  
    IF(TIME.EQ.90) RTTE = 1 ; for the censored observation  
    IF(R.GT.SUR) THEN  
        DV=1  
        RTTE = 1  
    ENDIF  
ENDIF
```

```
$THETA -2.95 ; (BASE)  
        (0, 1.08) ; (SHAPE)  
        -0.557 ; Yratio24
```

```
$OMEGA 0 FIX ; (0, 0.1)
```

```
;Sim_start : add/remove for simulation  
;$SIMULATION (5988566) (39978 UNIFORM) ONLYSIM NOPREDICTION SUB=  
$ESTIMATION MAXEVAL=9990 METHOD=CONDITIONAL LAPLACE LIKE PRINT=1  
$COVARIANCE PRINT=E  
;Sim_end
```

```
$TABLE ID TIME HAZNOW WT BMI CLM Y SUR NOPRINT ONEHEADER FILE=sdtab49
```


Population pharmacokinetics of apixaban in patients with short bowel syndrome

This chapter is based on

Barbara Deleenheer*, **Yannick Hoffert***, Thomas Vanassche, Lucas Wauters, Lorenz Van der Linden, Erwin Dreesen, Tim Vanuytsel. Apixaban Absorption Is Reduced in Short Bowel Syndrome and Correlates With Citrulline: A Prospective Population Pharmacokinetic Study. [In Draft]

[*Shared co-first authorship].

Abstract

Objectives Patients with short bowel syndrome commonly require anticoagulation therapy for prevention of venous thromboembolism or secondary prevention after acute mesenteric ischemia. Insights into the pharmacokinetics of apixaban, a commonly used direct oral anticoagulant, for patients with short bowel syndrome are scarce.

Methods We performed a prospective cross-over pharmacokinetic study evaluating single-dose and steady-state for 2.5 mg and 5 mg apixaban, followed by a dense sampling scheme for participants with and without short bowel syndrome. A population pharmacokinetics model was built to investigate the dose-exposure relationship, and participants' characteristics were tested to explain the variability in the population.

Results Seventy participants, of whom 34 had short bowel syndrome, were included in the study, providing a total of 1,858 apixaban concentrations. A 1-compartmental model with first-order absorption and elimination best described the data. Participants with short bowel syndrome had a lower rate and extent of apixaban absorption. Citrulline was positively correlated with the rate and extent of apixaban absorption. Female participants had a lower volume of distribution, resulting in a lower apixaban half-life compared to male participants.

Conclusions We performed the largest prospective apixaban population pharmacokinetic study in participants with short bowel syndrome. We provide quantitative evidence that apixaban absorption in patients with short bowel syndrome is reduced, necessitating dose escalation to maintain similar exposure.

1. Introduction

Venous thromboembolism (VTE) is a common complication in patients with SBS, primarily because long-term central venous access is often required for parenteral support (PS).¹¹² Anticoagulation is also routinely indicated for secondary prevention after acute mesenteric ischemia, a common precursor of SBS.¹¹³ As a result, many patients with SBS require long-term anticoagulant therapy.

SBS is a rare condition characterized by a residual small-bowel length of less than 200 cm, often following major intestinal resection.¹⁰ Three anatomical configurations are recognized based on the presence of colon and ileocecal valve: end-jejunostomy without colon in continuity (type 1), jejunocolic anastomosis without the ileocecal valve (type 2), and jejunoileal anastomosis with an intact colon and ileocecal valve in continuity (type 3). Types 2 and 3 are collectively referred to as SBS with colon-in-continuity.¹⁰ The resulting reduction in mucosal surface area and altered gastrointestinal physiology can impair the absorption of nutrients, fluids, and orally administered drugs.^{114–118} Although the intestine undergoes functional and structural adaptation over several years, these changes are often insufficient to restore full absorptive capacity, leading to chronic intestinal failure (IF) requiring PS.^{12,14} In recent years, the GLP-2 analogue teduglutide has become available for the treatment of SBS. GLP-2 analogues accelerate the natural intestinal adaptation process after resection through structural (i.e., villus growth) and functional effects (e.g., reduced motility), potentially reducing or discontinuing PS.¹¹⁹

Oral drug absorption in SBS is highly variable and poorly characterized, with most evidence limited to case reports.^{114–118} This is particularly relevant for antithrombotic therapy, where subtherapeutic exposure poses clear clinical risks. Direct oral anticoagulants (DOACs) such as apixaban are standard of care in the general population because fixed dosing without routine monitoring has proven safe and effective.^{120–124} However, their absorption depends on

small intestinal integrity, and data in SBS are sparse. Case reports describe either markedly reduced apixaban exposure requiring substantial dose escalation or normal peak concentrations without the need for dose adjustment.^{125,126} The only prospective popPK study to date suggested that apixaban absorption may be preserved in some patients with SBS, but it included individuals without PS dependence and did not evaluate key markers of IF severity such as residual bowel length or colon-in-continuity.⁷² Moreover, it is hypothesized and anecdotally observed in clinical practice that teduglutide improves drug absorption, but prospective and quantitative data are lacking.¹²⁷

This leaves a critical evidence gap regarding the reliability of oral apixaban absorption in patients with SBS. Therefore, we conducted a prospective study to compare the PK profile of apixaban between patients with and without SBS, and to identify anatomical or clinical covariates, such as residual small-bowel length, colon-in-continuity, PS dependence, citrulline, and teduglutide therapy, that may explain interindividual variability.

2. Materials and Patient Data

2.1 Study design and population

We conducted a prospective, non-randomized, controlled, cross-over PK study between October 2021 and September 2025 in participants with and without SBS, complemented by a steady-state arm (**Figure 17**). In arm 1 (single-dose), apixaban-naïve participants (with and without SBS) received a single dose of 2.5 mg and 5 mg apixaban in a cross-over manner, separated by a ≥ 1 -week washout (≈ 14 apixaban half-lives). In arm 2 (steady-state), participants (with and without SBS) were on chronic apixaban therapy of 5 mg twice daily. Participants with and without SBS were matched for age and BMI. Participants with SBS who started teduglutide were re-enrolled to repeat the same protocol no earlier than six months after teduglutide initiation. Inclusion and exclusion criteria are provided in the *Supplementary Materials*. The study was approved by the Ethics Committee Research of the University Hospitals (UZ) / KU Leuven (S63950) and conducted in accordance with the Declaration of Helsinki.



Figure 17. Diagram of the cross-over study design.
Abbreviations. SBS, short bowel syndrome.

2.2 Recruitment

Participants with SBS were recruited from the LIFT (Leuven Intestinal Failure and Transplantation) outpatient clinic. Participants without SBS and without apixaban therapy

were recruited online and via flyers, and those without SBS on apixaban were recruited from the cardiology outpatient clinic.

2.3 Study procedures

Participants fasted from three hours before to four hours after apixaban administration. Water was withheld from 1 hour before dosing; the dose was administered with 250 mL of water, after which water intake was permitted ad libitum. Baseline blood tests for kidney and liver function were obtained. Apixaban was administered with 250 mL still water. For the single-dose visits, blood samples were collected at 0.5, 1, 1.5, 2, 2.5, 3, 3.5, 4, 4.5, 5, 6, 8, and 10/24 hours after dosing, as per protocol. Samples were centrifuged (1955 g, 10 min, 20°C) within four hours and plasma was stored immediately at 2–8°C and then at –80°C within 24 hours, until analysis.¹²⁸ Details on the analytical methods are provided in the *Supplementary Materials*.

2.4 Data collection

At the first study visit (inclusion), we recorded for all participant: sex, age, self-reported race, weight, and height. For participants with SBS, we additionally documented time since resection, anatomy type, small-bowel length, colon length according to Cummings classification, fasting citrulline, and mean daily stoma output or stool frequency according to anatomy type.¹²⁹ Concomitant medication histories were obtained for all participants. SBS-specific PS data included days per week, volume per week and kcal per week. Serum creatinine on the day of apixaban administration and liver parameters (serum albumin, total protein, aspartate transferase (AST), alanine transferase (ALT), alkaline phosphatase (ALP), gamma glutamyl transferase (GGT), direct and total bilirubin) measured within the preceding six months were collected for every participant.

2.5 Population pharmacokinetics modeling

Base model. A base popPK model was fitted to apixaban concentration–time data, exploring one- and two- compartment disposition structures with first-order absorption and elimination, and optionally a lag time. IIV and interoccasion variability IOV were modeled using exponential random effects to constrain parameters to positive values (log-normal assumption). An occasion was defined as a dosing and sampling visit: 2.5 mg apixaban with teduglutide coadministration, 2.5 mg apixaban without teduglutide coadministration, 5 mg apixaban with teduglutide coadministration, and 5 mg apixaban without teduglutide coadministration (i.e., four occasions). Residual unexplained variability was explored using additive, proportional, and combined error models. PopPK parameters were parameterized as follows using exponential functions (Eq. 12):

$$CL_{i,j} = TVCL \times \exp(\eta_i + \eta_{ij}) \text{ with } \eta_i \sim N(0, \omega_{IIV}^2) \text{ and } \eta_{ij} \sim N(0, \omega_{IOV}^2) \quad (\text{Eq. 12})$$

where $CL_{i,j}$ denotes the apixaban clearance for participant i in occasion j , TVCL denotes the typical population value of the clearance, $\eta_i + \eta_{ij}$ denotes the individual participant's deviation from the typical value in occasion j , and ω_{IIV}^2 and ω_{IOV}^2 are the variances of IIV and IOV.

Bioavailability parameterization. Consistent with prior knowledge and to keep estimates within biological bounds, the absolute oral bioavailability of SBS participants was fixed at 50%²¹, and covariate and random-effect contributions to bioavailability were applied on the logit scale, with back-transformation for interpretation.

Model selection criteria. The most parsimonious base model was withheld based on the OFV (corresponding to the -2LL), precision and plausibility of parameter estimates, and diagnostic plots.

Covariate testing. For the stepwise covariate search with stage-wise filtering⁵⁴, we tested the impact of SBS on absorption-related parameters (e.g., lag time, absorption rate constant, and bioavailability), along with small-bowel length, presence of colon, fasting citrulline, teduglutide coadministration, stoma output, stool frequency, days on PS, PS volume, PS kcal, loperamide co-administration, time since resection, and apixaban dose (on bioavailability only). Body weight and sex were evaluated on volume of distribution. Age, sex, serum creatinine, and loperamide coadministration were tested on clearance. Covariates were grouped a priori in categories (mechanistic, structural, and exploratory) and tested in order of importance (**Table S7**).⁵⁴ A drop in OFV of >3.84 points in the forward addition and an increase in OFV >6.63 points in the backward elimination correspond to *p* values of <0.05, and <0.01, respectively (1 degree of freedom).

Model evaluation. First-order conditional estimation with interaction (FOCE+I) was used for parameter estimation. A condition number below 100 was considered acceptable to avoid overparameterization.⁵⁰ Goodness-of-fit (GOF) plots were used to evaluate model adequacy and VPC plots were used to evaluate model calibration (2,000 replicate simulations per stratum). Bootstraps (n = 2,000) were performed to obtain non-parametric CIs to assess parameter uncertainty.

Software. Data formatting and exploration were performed using custom scripts in R (v.4.4.1; R Core Team, Vienna, Austria), interfaced with RStudio (v.2024.12.1) using the *tidyverse* collection of packages. PopPK analyses were conducted using NONMEM (v.7.5.1; ICON Development Solutions, Gaithersburg, MD, USA) in conjunction with the GNU Fortran 95 compiler and the Perl-speaks-NONMEM (PsN; v.5.2.6) toolkit interfaced with the Pirana software (v.21.11.1; Certara, Inc., Princeton, NJ, USA). Computations were accelerated using high-performance cluster computing (HPC), provided by the ‘Vlaams Supercomputer Centrum’ (VSC).

3. Results

3.1 Study cohort

Seventy participants were enrolled: 53 participants in arm 1 (single-dose) and 17 in arm 2 (steady-state). Thirty-four participants had SBS and 36 did not have SBS (**Figure S15**). Across dosing conditions, 48 participants received 2.5 mg and 5 mg apixaban without teduglutide, 16 received 5 mg without teduglutide, 5 received 2.5 and 5 mg apixaban with and without teduglutide, and 1 received 5 mg with and without teduglutide (**Figure S16**).

3.2 Participant characteristics

At baseline (first study visit), the median age was 64 years (range 36–80), and participants were predominantly female (40/70, 57.1%) (**Table 5**). Self-reported race was mainly white (69/70, 98.6%). Underlying causes of SBS included mesenteric ischemia (14/34, 41.2%), Crohn's disease (13/34, 38.2%), and surgical complications (3/34, 8.8%). SBS anatomy was type 1 in 14 (41.2%), type 2 in 17 (50.0%), and type 3 in 3 (8.8%) participants. Six participants (8.6%) received teduglutide. Detailed demographics and SBS-specific characteristics are provided in **Tables S8–S9**.

Table 5. Patient characteristics for participants with and without SBS.

	SBS (N = 34)	Non SBS (N = 36)
<i>Demographics and serology</i>		
Sex, female, <i>n</i> (%)	22 (64.7)	18 (50.0)
Age, years, median [IQR]	63.5 [55–71]	64.5 [56–71]
Race, white, <i>n</i> (%)	33 (97.1)	36 (100)
Weight, kg, median [IQR]	65.5 [52.8–70.8]	72.7 [63.8–79.8]
BMI, kg/m ² , median [IQR]	22.25 [21.12–25.2]	24.75 [23–26.85]
Serum creatinine, mg/dL, median [IQR]	0.91 [0.80–1.44]	0.85 [0.78–1.04]
Citrulline, μmol/L, median [IQR]	27 [17–34] (n=29)	–
<i>Disease characteristics</i>		
Crohn's disease, <i>n</i> (%)	13 (38.2)	–
Mesenteric ischemia, <i>n</i> (%)	14 (41.2)	–
Surgical complication, <i>n</i> (%)	3 (8.8)	–
Other, <i>n</i> (%)	4 (11.8)	–
Small bowel length, cm, median [IQR]	100 [42.5–120]	–
<i>Colon length</i>		
0%, <i>n</i> (%)	15 (44.1)	–
>0–50%, <i>n</i> (%)	4 (11.8)	–
>50–100%, <i>n</i> (%)	15 (44.1)	–
<i>Parenteral support</i>		
IV fluids, <i>n</i> (%)	11 (32.3)	–
IV fluids + PN, <i>n</i> (%)	23 (67.6)	–
PS/week, days, median [IQR]	5 [3–7]	–
PS volume, mL/week, median [IQR]	7000 [3625–10375]	–
PS kcal, kcal/week, median [IQR]	3420 [960–6685]	–

Abbreviations. IQR, interquartile range; PN, parenteral nutrition; PS, parenteral support; SBS, short bowel syndrome.

3.3 Apixaban concentration–time profiles

Across the full cohort, 1,858 apixaban concentrations were measured (~27 per participant). The maximum concentration ($C_{\max}$) was reached around 180 min post-intake (**Figure 18A, Figure S17A**). Relative to participants without SBS, those with SBS had lower $C_{\max}$ and longer time to $C_{\max}$ ($T_{\max}$). The differences were most pronounced at steady state (**Figure 18B, Figure S17B**).

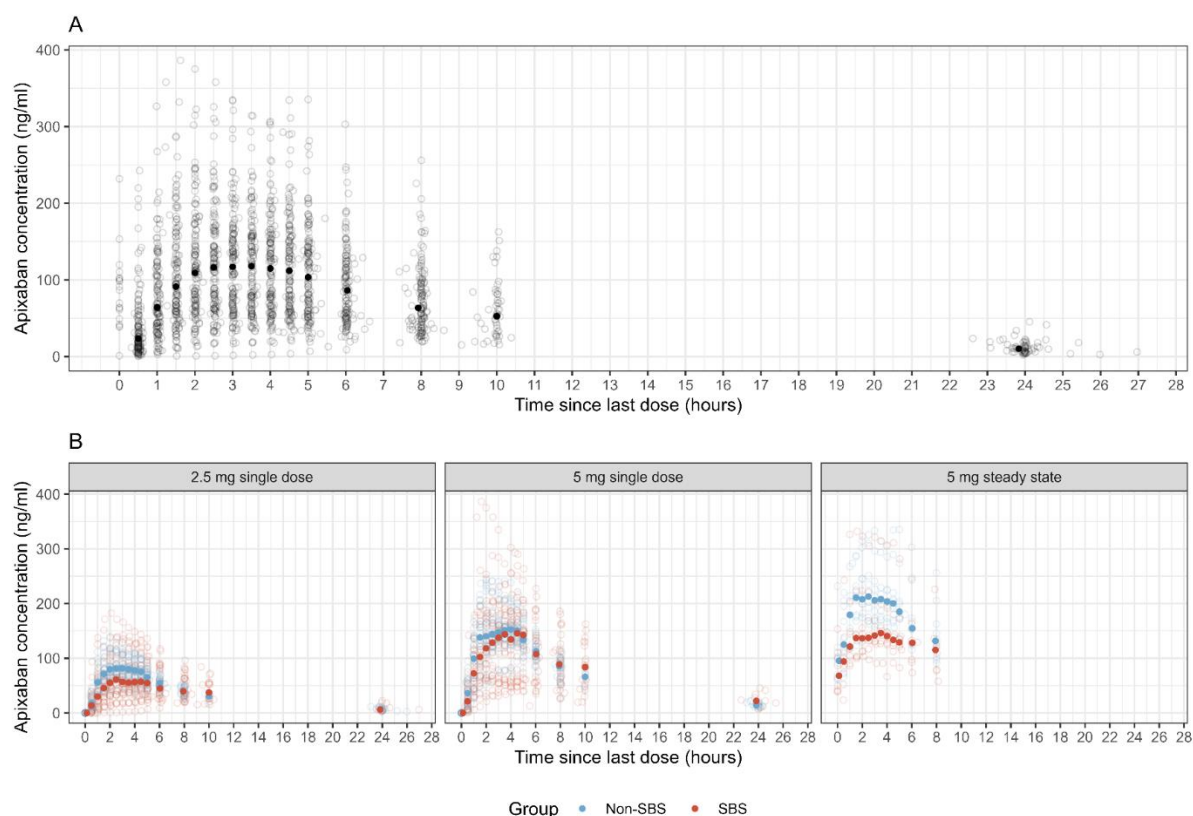


Figure 18. Apixaban plasma concentrations over time since last dose. Open circles depict observed apixaban plasma concentration and filled points depict the median concentration per mid bin with bin breaks at 0, 15, 45, 75, 105, 135, 165, 195, 225, 255, 285, 315, 410, 540, 660, and 2,200 minutes.

Following a single 2.5 mg apixaban dose, the median $C_{\max}$ was 84.6 ng/mL (IQR 74.8–104.8) in non-SBS participants versus 62.9 ng/mL (45.5–101.8) in SBS participants. Following a single 5 mg apixaban dose, the median $C_{\max}$ was 160.3 ng/mL (146.2–192.9) in non-SBS participants versus 157.6 ng/mL (92.7–187.0) in SBS participants. Under twice-daily 5 mg apixaban dosing (steady state), the median $C_{\max}$ was 216.4 ng/mL (200.3–249.4) in non-SBS participants and 143.1 ng/mL (132.6–160.4) in SBS participants. The median overall exposure in the first 720 min after a 2.5 mg single dose of apixaban (AUC_{0-720}) was 29,153.43 min·ng/mL (IQR, 18,494.37–43,313.53) for the SBS cohort, and 36,308.52 min·ng/mL (31,777.76–44,491.45) for the non-SBS cohort. Detailed exposure metrics (by dose and setting) including $T_{\max}$ are provided in the *Supplementary Results* (Figure S18) (Table S10).

3.3 Population pharmacokinetics modeling

A one-compartment popPK model with a lag-time and first-order absorption and elimination best described the data (**Table 6**, **Figure 19**, **Figure S19–S20**). SBS was associated with lower bioavailability and a lower and more variable absorption rate relative to non-SBS. Citrulline was positively associated with both bioavailability and rate of absorption. Sex influenced the volume of distribution (lower in women), resulting in differences in apixaban half-life for women and men of 7.3 h, and 5.7 h, respectively. While the covariate model explained parts of the observed variability, there was a remaining unexplained variability on bioavailability (49.0% coefficient of variation; CV) and absorption rate constant (58.3% CV). Numerical estimates for covariate effects and uncertainty are shown in **Table 6** and **Table S11**, and individual parameter distributions stratified by covariates are shown in **Figure 20**.

Table 6. Parameter estimates for the population pharmacokinetic model

	Base model (RSE%) [%shrinkage]	Final model (RSE%) [%shrinkage]	Bootstrap estimate of the final model* [95%CI]
	OFV 11,660; CN 41.9	OFV 11,571; CN 70.2	
<i>Structural model</i>			
k_a (1/min)	0.0161 (8.4)	–	–
k_a for SBS (1/min)	–	0.0152 (12.6)	0.0164 [0.0124– 0.0237]
k_a for non-SBS (1/min)	–	0.0181 (7.2)	0.0193 [0.0160– 0.0267]
CL (L/min)	0.0256 (4.4)	0.0277 (3.9)	0.0276 [0.0256– 0.0296]
V (L)	14.1 (5.2)	17.6 (5.1)	17.8 [16.2–20.1]
T_{lag} (min)	18.7 (1.3)	18.7 (1.1)	18.7 [18.0–23.5]
F (%)	50 fixed	50 fixed	50 fixed
<i>Covariate model</i>			
Citrulline on k_a	–	0.663 (31.8)	0.650 [0.160–1.084]
Citrulline on F	–	0.818 (18.1)	0.834 [0.461–1.242]
SBS on F	–	-1.52	-1.51 [-1.80– -1.16]
Sex on V	–	-0.224 (20.9)	-0.219 [-0.308– -0.126]
<i>Statistical model</i>			
IIV on k_a (%CV)	50.8% (27.9) [28]		
IIV on k_a for SBS (%CV)	–	58.3 (19.4) [45]	55.7 [26.2–79.3]
IIV on k_a for non-SBS (%CV)	–	0 fixed [†]	0 fixed [†]
IIV on CL (%CV)	11.2% (61.5) [55]	20.8 (17.3) [22]	20.9 [12.6–28.2]
IIV on V_c (%CV)	23.3 (24.0) [21]	16.3 (35.8) [34]	16.8 [4.1–25.1]
IIV on F (%CV)	29.9 (14.1) [17]	49.0 (19.8) [29]	46.2 [24.8–63.1]
[‡] IOV on k_a (%CV)	57.9 (12.5) [27]	54.1 (9.8) [23]	56.6 [44.4–76.5]
[‡] IOV on F (%CV)	21.7 (17.1) [44]	38.1 (10.6) [44]	38.4 [30.2–47.0]
Proportional error (%CV)	18.8 (7.1) [8]	18.8 (7.1) [7]	18.3 [14.1–21.1]
Additive error (ng/mL)	0.277 (12.0) [8]	0.276 (11.8) [7]	0.276 [0.235–2.210]

k_a for non-SBS = 0.0179

k_a for SBS = $0.0152 \cdot (\text{citrulline}/27)^{0.663} \cdot e^{(\text{IIV on } k_a + \text{IOV on } k_a)}$

CL = $0.0277 \cdot e^{(\text{IIV on CL})}$

V = $17.6 \cdot (1 + (-0.224))$ (for female)

F = $e^{((50\% + (-1.52) + (\text{citrulline}/27)^{0.818}) + \text{IIV on F} + \text{IOV on F}) / (1 + e^{((50\% + (-1.52) + (\text{citrulline}/27)^{0.818}) + \text{IIV on F} + \text{IOV on F}))}$

*based on 1,982/2,000 runs (99.1%)

[†]including variability on k_a for the non-SBS cohort did not improve model fit, hence it was fixed to 0.

[‡]up to four occasions.

Abbreviations. CI, confidence interval; CL, clearance; CN, condition number; CV, coefficient of variation

calculated as $\sqrt{\exp(\omega^2)} \times 100\%$; F, absolute oral bioavailability; IIV, interindividual variability; IOV, interoccasion variability; k_a , absorption rate constant; OFV, objective function value; RSE, relative standard error; SBS, short bowel syndrome; T_{lag} , lag time; V, volume of distribution.

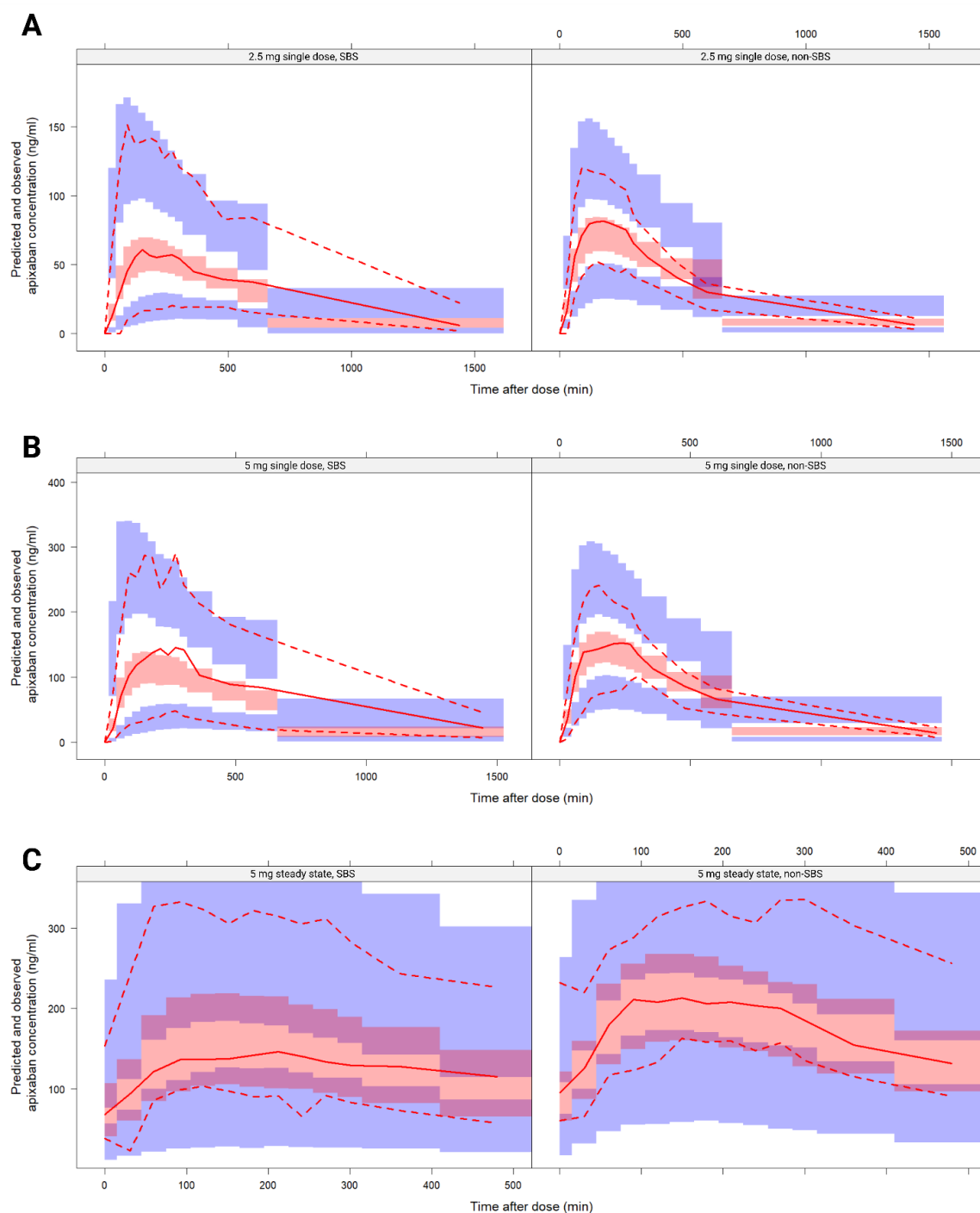


Figure 19. Visual predictive check for (A) 2.5 mg apixaban single dose, (B) 5 mg single dose, and (C) 5 mg steady state, all stratified SBS versus non-SBS. The solid red line is the median of the observed data. The dashed red lines are the 5th and 95th percentiles of the observed data. The red and purple shaded areas indicate the 95% confidence intervals of the median and 5th and 95th percentiles, respectively (based on N = 2000 iterations). Abbreviations. SBS, short bowel syndrome.

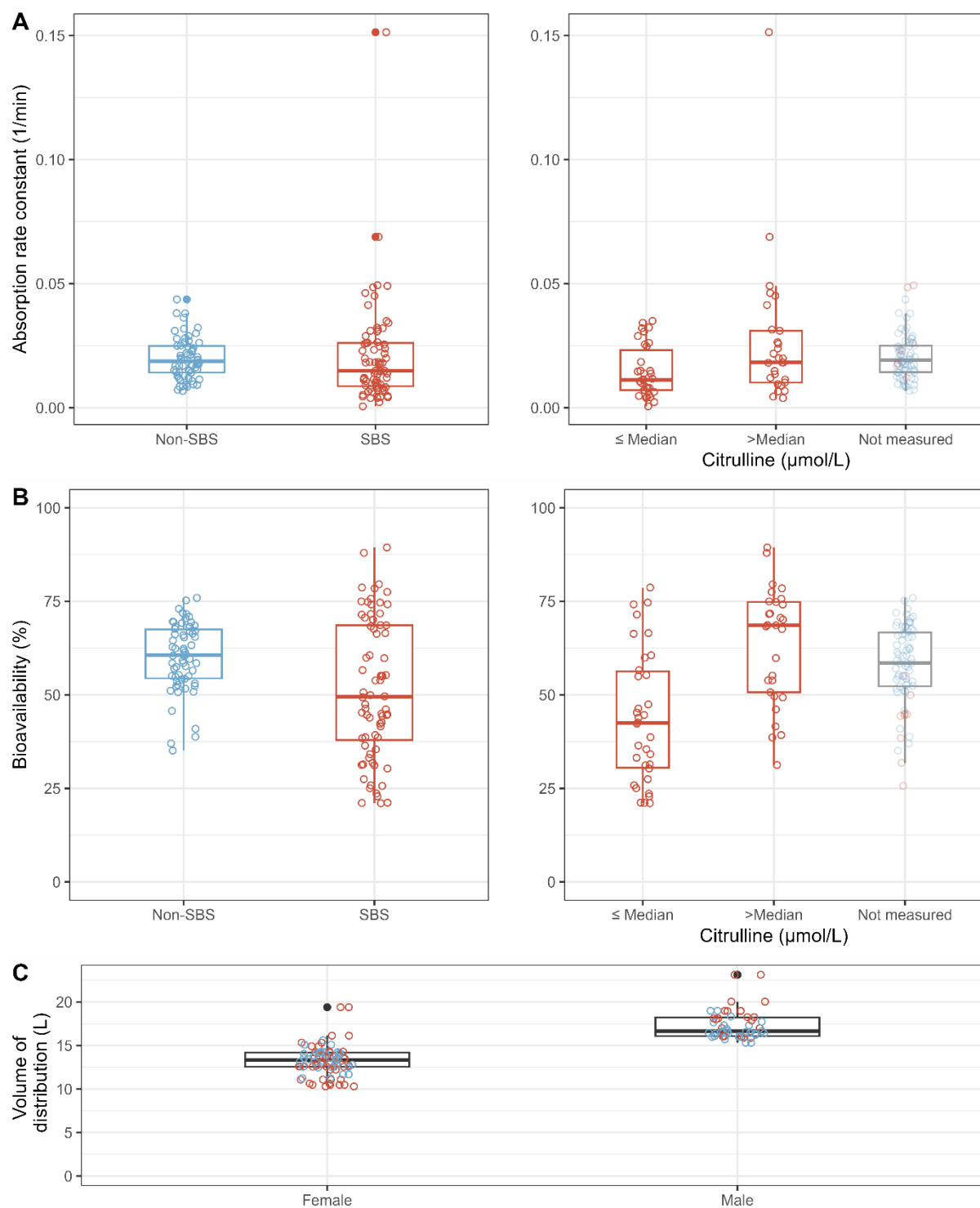


Figure 20. Distribution of estimated pharmacokinetics parameters stratified by the identified covariates. Continuous covariates were categorized into above median, and equal or below median of the population. Blue and orange color code corresponds to patients without and with SBS, respectively. Abbreviations. SBS, short bowel syndrome.

4. Discussion

In this prospective study with dense sampling and a matched control cohort, we found that both the rate and extent of apixaban absorption are reduced in SBS. In the single-dose arms, $C_{\max}$ was lower and $T_{\max}$ was longer in SBS than in non-SBS participants, and at steady-state, the difference in $C_{\max}$ was more pronounced. Population modeling confirmed a lower absorption rate in SBS versus non-SBS and identified a substantial reduction in bioavailability associated with SBS, alongside high IOV on absorption-related parameters. In addition, citrulline, a biomarker of functional enterocyte mass, was positively associated with both the rate and extent of bioavailability, while sex influenced the volume of distribution (higher V in males).¹³⁰ Together, these findings quantitatively substantiate the long-held clinical observation that apixaban absorption is heterogeneous in SBS and partly predictable from patient-level physiology.

Our results extend the limited prospective literature on DOACs in SBS. Case reports have described both markedly attenuated apixaban exposure requiring dose escalation and normal peaks without the need for dose adjustment.^{125,126} These seemingly contradictory patterns are in fact compatible with the pharmacokinetic heterogeneity observed in our study. One prior prospective popPK analysis re-estimated parameters from a non-SBS model and concluded preserved absorption in some SBS patients.⁷² Yet, this study did not evaluate key SBS-specific determinants (residual length, colon-in-continuity, parenteral support dependence, citrulline), nor the potential impact of teduglutide. By enrolling a larger SBS cohort with age/BMI-matched controls, adopting a crossover single-dose design plus a steady-state arm, and prospectively collecting SBS-relevant covariates, we were able to locate part of the between-patient variability in absorption parameters rather than clearance and to quantify physiologically plausible covariate effects. When compared to reference apixaban models in other populations, our lag time, absorption rate, clearance, and volume

of distribution estimates are broadly consistent with prior report, while structural choices (e.g., modeling bioavailability on a logit scale and prioritizing SBS physiology in the covariate structure) reflect our population and design.^{71,70} For a detailed comparison of apixaban popPK studies, we kindly refer to **Table S12**.

The direction and magnitude of the SBS effect on the extent of absorption (bioavailability; F) and rate of absorption (absorption rate constant; k_a) are mechanistically coherent: (i) reduced mucosal surface after resection directly limits the absorptive interface, (ii) reduced enterocyte mass, captured by lower citrulline levels (a biomarker of functional enterocyte mass), likely lowers the transcellular absorption capacity, and (iii) accelerated intestinal transit, particularly in participants without colon-in-continuity, shortens luminal contact time. Conversely, teduglutide, a GLP-2 analogue that boosts structural (villus growth) and functional adaptation, is biologically expected to improve absorption.^{131,132} However, in our study the data did not allow us to determine such an effect on the recovery of apixaban absorption. With only six participants, the sample size was likely too small to reliably determine this effect.

From a clinical standpoint, our data caution against assuming “one-size-fits-all” exposure with standard apixaban dosing in SBS. We predefined “sufficient absorption” as a $C_{max} \geq 30$ ng/mL (5th percentile of phase I–III data^{133,134}) and based on the differences in bioavailability and absorption rate, we anticipate a meaningful fraction of SBS patients may fall below this threshold. These observations support a selective monitoring strategy, particularly in SBS patients with very short bowel-length, absent colon-in-continuity, low citrulline, or high PS dependence, and in those early after teduglutide initiation if absorption is being reassessed. As informed by our popPK model, citrulline monitoring may provide valuable guidance in the management of SBS patients. Given the high variability in T_{max} , fixed-time peak sampling must be interpreted cautiously. Where feasible, short-window

peak sampling (e.g., 2–4 hours post-dose) or model-informed interpretation can reduce timing sensitivity. If exposure is sub-target, options include dose optimization (guided by monitoring and modeling) or alternative anticoagulation in selected cases. However, outcome trials will be needed before recommending population-level dose changes.

Strengths of our study include the prospective design, age/BMI matching, crossover single-dose evaluations, a parallel steady-state arm, sample size, dense sampling, and popPK analyses with pre-specified, physiology-led covariates (bowel and colon length, PS, citrulline, teduglutide). Internal validation by stratified VPC and non-parametric bootstrap supports model stability. Limitations merit discussion. First, steady-state and before/after teduglutide subgroups were modest in size, reducing precision for those contrasts. Second, cytochrome P450 (CYP) 3A4 and P-glycoprotein (P-gp) genotype data were not collected, so we cannot partition variability in bioavailability between absorption versus enteric metabolism/efflux. Third, we fixed F to 50% at the structural level and modeled group and covariate effects on the logit scale, which is an approach that stabilizes estimation but may mask unmodeled dose-dependent bioavailability reported in other settings. Fourth, while IOV captured part of the within-subject fluctuations, the high IOV on k_a and F underscores day-to-day variability in this population. Finally, clinical outcomes such as bleeding and thrombosis were not assessed. Larger, multicenter cohorts would be required to link exposure to outcome in SBS.

Two avenues follow directly from this work. First, model-based dosing simulations will be performed to characterize the probability of target attainment across SBS phenotypes and doses (e.g., 2.5 mg, 5 mg, 7.5 mg, and 10 mg twice daily at steady state). Such simulations may support a “traffic-light” system (low, intermediate, and high risk of underexposure) to help guide clinical decision-making. For instance, standard dosing is likely sufficient in patients at low risk, whereas therapeutic drug monitoring (TDM) and/or dose optimization

may be considered in those at intermediate risk. For patients at high risk, a switch to an alternative DOAC could be considered. This framework is intended solely as a decision-support tool; final treatment decisions remain at the discretion of the treating clinician. Second, pragmatic implementation studies should evaluate sampling strategies (optimized peak windows, partial AUCs) and covariate-informed pathways (e.g., using citrulline, residual length, colon-in-continuity) to triage who needs monitoring or alternative therapy. External validation in multicenter cohorts, inclusion of genetic/phenotypic data for CYP3A4 and P-gp, and head-to-head comparisons across DOACs in SBS would strengthen generalizability and inform practice.

In summary, apixaban absorption in SBS is reduced, variable, and partly predictable from anatomy and biomarkers, with citrulline emerging as an informative correlate of both rate and extent of absorption. These prospective, model-based findings support a move from “uniform dosing” towards risk-stratified management in SBS, anchoring in physiology, sparse monitoring where indicated, and simulation-informed dosing, while underscoring the need for outcome-focused, multicenter studies.

Chapter 3 Supplementary Material

Supplementary Methods

Analytical method, data analysis and software

Standard operating procedure apixaban analysis

Standard curve:

A standard curve from 1000nM to 0.97nM is prepared in plasma with serial dilution starting from a 10mM apixaban stock solution in dimethylsulfoxide (DMSO) and from a 2mM O-desmethyl apixaban sulfate stock solution in DMSO.

Quality control:

Four different quality control (QC) samples (500 – 50 – 5 – 1 nM) are prepared in plasma starting from the previously mentioned stock solutions in DMSO. QC's were aliquoted and stored at -20°C prior to analysis.

Sample preparation

50µl sample (unknown, standard or QC) is added to 1ml acetonitrile containing 2nM ¹³C,²H₈-apixaban (internal standard) for protein precipitation.

Samples are centrifuged for 5 min at 4°C at 20817g. After centrifugation the supernatant is transferred to a clean glass tube and dried under a gentle stream of air.

The residues are redissolved in 200µl mobile phase.

5µl is injected into the LCMS-MS.

UPLC-MS/MS

UPLC Waters ACQUITY UPLC H-Class System

MS/MS Waters Xevo TQ-S micro (Triple Quad)

Software Mass Lynx 4.2

Mobile Phase A: acetonitrile

B: 5 mM Ammonium formate buffer pH 3.0

Gradient elution

Time	Mobile phase A (%)	Mobile phase B (%)
0.0 – 0.5'	23	77
0.5 → 2.0'	23 → 35	77 → 65
2.0 → 2.5'	35 → 95	65 → 5
2.5 – 3.5'	95	5
3.5 → 4.0'	95 → 23	5 → 77
4.0 -5.0'	23	77

Flow 500 µL/min

Injection Volume 5 µL

Column Temperature 40 °C

Sample Temperature 15 °C

Column Kinetex 2,6u XB-C18 100 A (50x2,1 mm)

Detection (*pos mode*) apixaban (m/z 460.4 → 443.4, CE 22 V)

O-D-apixaban-S (m/z 526.4 → 429.4, CE 28 V)

IS (m/z 469.4 → 452.4, CE 22 V)

Interface (ESI)Capillary : 3,5 kV

Cone: 80 V

Desolvation Temp: 600 °C

Desolvation Gas: 1000 L/h

Cone Gas: 50 L/h

Source Temp: 150 °C

During the run, from 0.0 min to 0.35 min and from 3.0 min to 4.5 min the valve is set to waste.

Apixaban was quantified by UPLC-MSMS (ultra performance liquid chromatography with double mass spectrometer) from WatersTM. Apixaban was purchased from Alsachim[®] (labelled), France and Tebubio[®] (unlabeled), Belgium. The standard operating procedure for this analysis can be found in the Supplementary Materials. Individual exposure (C_{max} , T_{max} , C_{720} or C_{min} , AUC, $t_{1/2}$) and PK parameters (V_d , K_{el} and CL) were calculated with PKanalix[®] 2024R1. Population PK parameters k_a and F were obtained via popPK modelling. Descriptive and analytic statistics and graphical representations were obtained with R studio[®] v4.5.2.²⁷ Characteristics and PK parameters are represented as numbers and percentages, median and interquartile ranges (IQR), mean and standard deviations (SD) or geometric mean (GM) and coefficient of variance (CV), as indicated in the tables. Sufficient absorption was defined as a C_{max} above the lowest 5% percentile found in the literature for phase I-III data, i.e. 30 ng/mL for single dose and multiple dose conditions.^{21,28} The significance level was defined as $p < 0.05$.

Participant recruitment

Participants with SBS were recruited from the LIFT (Leuven Intestinal Failure and Transplantation Centre) outpatient clinic of the University Hospital of Leuven, Belgium. Participants without SBS and without apixaban therapy were recruited online and by flyers. Participants without SBS, but on therapy with apixaban were recruited from the cardiology outpatient clinic of the University Hospital of Leuven.

Inclusion- and exclusion criteria per arm

Inclusion criteria arm 1 (single dose)

- SBS: small bowel length of <200 cm after Treitz ligament on long term (> three months) PS; teduglutide-, apixaban- and vitamin K antagonist naive

- controls: healthy individuals without history of gastrointestinal resections or other conditions associated with impaired absorption; apixaban- and vitamin K antagonist naïve

Inclusion criteria arm 2 (steady-state/multiple dose)

- SBS: small bowel length of <200 cm after Treitz ligament on long term (> three months) PS; teduglutide naïve; chronic intake of apixaban 5 mg twice daily for $\geq$ four days
- controls: individuals without history of gastrointestinal resections or other conditions associated with impaired absorption; chronic intake of apixaban 5 mg twice daily for $\geq$ four days

Exclusion criteria

SBS and controls:

- <18 years
- non-Dutch speaking
- recent (< six months) major bleeding according to the International Society on Thrombosis and Haemostasis definition of major bleeding in non-surgical patients⁴⁰
- creatinine clearance of < 15 mL/min or dialysis dependent
- liver failure classified as Child Pugh C
- total bilirubin $\geq$ 1.77 mg/dL (= 1,5 x upper limit of normal)
- presence of coagulopathy and a clinically relevant bleeding risk
- pregnancy or lactation
- concomitant intake of strong combined inhibitors of cytochrome-P 3A4 (CYP3A4) and P-glycoprotein (P-gp)

- participation in a recent (< one month or longer depending on the half-life of the investigational product) trial with an investigational product

SBS:

- recent (< six months) gastrointestinal surgery
- gastrointestinal mucosal disease interfering with absorption (e.g., radio-enteritis, inflammatory bowel disease, coeliac disease,...)
- intestinal fistulae
- SBS with IF resulting from gastric bypass surgery

Data collection and outcomes

The primary endpoint was the maximum or peak plasma apixaban concentration ($C_{\max}$) after a single dose in participants with SBS compared to non-SBS (arm 1). At least 26 participants per cohort were required (effect size 0.8 on $C_{\max}$, power 80%, $\alpha=0.05$, 1/1 allocation, two-tailed unpaired t-test). A convenience sample of five patients with SBS and 10 controls was targeted for arm 2. Secondary endpoints included differences in other exposure parameters such as concentration 12 hours after administration (C_{720}), time to $C_{\max}$ ($T_{\max}$), area under the curve or exposure until 12 hours after administration (AUC_{0-720}) and half-life ($t_{1/2}$), but also differences in the underlying PK parameters such as absorption rate constant (k_a), bioavailability (F), volume of distribution (Vd) and clearance (CL) between participants with and without SBS after a single and multiple doses (i.e., steady-state condition) and before and after teduglutide initiation. Exploratory analyses assessed correlations of SBS-specific covariates with $C_{\max}$, $T_{\max}$ and AUC. Covariates of interest were small bowel length, presence

and length of colon, time since resection, PS volume and citrulline, a non-essential amino acid that is synthesised by the enterocytes and that can be seen as a marker of absorptive function in SBS.

Supplementary Tables

Table S7. Categories of candidate covariates for stepwise covariate search using stage-wise filtering.

Mechanistic covariate-parameter combination	Structural covariate-parameter combination	Exploratory covariate-parameter combination
–	SBS on k_a	Small bowel length on k_a , F
–	SBS on F	Presence of colon on k_a , F
–	–	Colon length on k_a , F
–	–	Citrulline on k_a , F
–	–	Teduglutide coadministration on k_a , F
–	–	Stoma output on k_a , F
–	–	Stool frequency on k_a , F
–	–	Days on PS on k_a , F
–	–	PS volume on k_a , F
–	–	PS kcal on k_a , F
–	–	Loperamide coadministration on k_a , F
–	–	Time since resection on k_a , F
–	–	Body weight on V
–	–	Sex on V
–	–	Age on CL
–	–	Sex on CL
–	–	Serum creatinine on CL
–	–	Loperamide coadministration on CL

Abbreviations. CL, clearance; F, bioavailability; k_a , absorption rate constant; SBS, short bowel syndrome; V, volume of distribution.

Table S8. Patient and parenteral support characteristics for participants with and without SBS, for the single dose and steady-state conditions.

	SBS SD (n = 27)	Non-SBS SD (n = 26)	SBS SS (n = 7)	Non-SBS SS (n = 10)
<i>Patient</i>				
Sex - female	18 (66.7)	15 (57.8)	4 (57.1)	3 (30)
Age - years	64 [55-71] (36-80)	59.5 [55.2-66.8] (49-76)	63 [59-69] (55-80)	71 [67.2-72.8] (57-78)
Race - white	26 (96.3)	26 (100)	7 (100)	10 (100)
Weight – kg	62.2 [53.4-70.4] (45.1-133.5)	72.7 [63.8-77.9] (51-85)	67 [55.2-71.3] (49-82.3)	78.5 [66.4-89.5] (58-96.4)
BMI – kg/m ²	22.1 [20.6-25.2] (18.5-39.9)	24.8 [23.3-26] (19.1-29.8)	22.4 [21.6-25.6] (18.2-29.1)	25.2 [22.4-27.7] (18.4-34.2)
Time since resection ¹ – days	882 [454-2865] (164-9161)		1317 [685.5-2145] (192-3944)	
<i>Underlying pathology¹</i>				
Crohn's disease	12 (44.4)	12 (44.4)	1 (14.3)	1 (14.3)
Mesenteric ischaemia	8 (29.6)	8 (29.6)	6 (85.7)	6 (85.7)
Surgical complications	3 (11.1)	3 (11.1)	0 (0)	0 (0)
Other	4 (14.8)	4 (14.8)	0 (0)	0 (0)
<i>Anatomy¹</i>				
Type 1	11 (40.7)		3 (42.9)	
Type 2	14 (51.9); 3 with colostomy		3 (42.9); 1 with colostomy	
Type 3	2 (7.4); 1 with colostomy		1 (14.3)	
Small bowel length ¹ – cm	100 [50-120] (0-175)		60 [27.5-107.5] (0-150)	
Colon length ¹ - %	43 [0-63.5] (0-100)		57 [0-78.5] (0-100)	
0%	12 (44.4)		3 (42.9)	
>0-50%	4 (14.8)		0 (0)	
>50-100%	11 (40.7)		4 (57.1)	
Stoma output ^{1,2} - mL/day	1908 [878.1-2593.8] (655-4650) n = 14		2097.6 [1348.8-3101.2] (1665-4550) n = 4	
Stool frequency ^{1,2} - #/day	6.5 [3.4-9.6] (1.5-13) n = 12		3.5 [2.2-3.5] (1-3.5) n = 3	
Citrulline ¹ - µmol/L	24 [17.5-37.5] (6-97) n = 23		26.5 [14.8-27.8] (6-45) n = 6	
<i>Parenteral support¹</i>				
Type of PS				
Only IV fluids	9 (33.3)		2 (28.6)	
IV fluids + PN	18 (66.7)		5 (71.4)	
PS/week – days	6 [3-7] (2-7)		5 [3.5-7] (3-7)	
PS volume – mL/week	7000 [3250-11000] (2000-28000)		7000 [4500-8500] (3000-19500)	
PS kcal – kcal/week	3420 [0-5675] (0-14080)		3420 [1620-8365] (0-10120)	

Data are shown in numbers (%) or median [interquartile range] (range) as appropriate. Data only assessed for patients with short bowel syndrome. Stoma output or stool frequency for participants with or without stoma respectively.

Abbreviations. BMI, body mass index; IV, intravenous; PN, parenteral nutrition; PS, parenteral support; SBS, short bowel syndrome; SD, single dose; SS, steady-state after multiple dose.

Table S9. Patient and parenteral support characteristics for participants with SBS before and after teduglutide initiation, for the single dose and steady state conditions.

	SBS SD without teduglutide (n = 5)	SBS SD with teduglutide (n = 5)	SBS without teduglutide (n = 1)	SS SBS SS with teduglutide (n = 1)
<i>Patient</i>				
Sex – female	5 (100)	5 (100)	1	1
Age - years	71 [56-74] (54-74)	72 [57-75] (56-75)	62	63
Race – white	5 (100)	5 (100)	1	1
Weight – kg	53.2 [49-56.8] (45.1-69.8)	54.2 [54-59.6] (49.2-63)	49	53
BMI – kg/m ²	20.5 [19.9-22.8] (18.5-28)	21.6 [20.8-24.2] (20.2-25.2)	22.1	23.9
Time since resection – days	3137 [2861-4336] (882-9138)	3445 [3317-4589] (1253-9565)	1317	1765
<i>Underlying pathology[†]</i>				
Crohn's disease	3 (60)		0	0
Mesenteric ischaemia	0 (0)		1	1
Other	2 (40)		0	0
<i>Anatomy</i>				
Type 1	0 (0)	0 (0)	0	0
Type 2	4 (80); 1 with colostomy	4 (80); 1 with colostomy	0	0
Type 3	1 (20) (with colostomy)	1 (20) (with colostomy)	1	1
Small bowel length– cm	120 [105-125] (80-150)	120 [105-125] (80-150)	35	35
Colon length - %	57 [57-70] (0-71)	57 [57-70] (0-71)	100	100
0%	1 (20)	1 (20)	0	0
>0-50%	0 (0)	0 (0)	0	0
>50-100%	4 (80)	4 (80)	1	1
Stoma output ¹ - mL/day	875 n = 1	967.5 [763.8-1171.2] (560-1375) = 2	/	/
Stool frequency ¹ - #/day	8.5 [5.2-10] (2-11.5) n = 3	1.5 [2.5-5.2] (2-7.5) n = 3	1	2
Citrulline – µmol/L	23 [18.5-28.8] (17-34) n = 4	42 [38-60] (19-73)	27	33
Time since teduglutide initiation - months		8.7 [6.2-10.8] (5.7-15.5)		12.8
<i>Parenteral support</i>				
Type of PS				
Only IV fluids	2 (40)	0 (0)	0	0
IV fluids + PN	3 (60)	2 (40)	1	0
PS/week – days	3.5 [3-4] (3-6)	0 [0-3] (0-3)	3	0
PS volume – mL/week	5250 [3000-7500] (300-7500)	0 [0-2500] (0-4500)	3000	0
PS kcal – kcal/week	4725 [0-4960] (0-6220)	0 [0-1140] (0-4800)	1440	0

Data are shown in numbers (%) or median [interquartile range] (range) as appropriate stoma output or stool frequency for participants with or without stoma respectively. Abbreviations. BMI, body mass index; IV, intravenous; PN, parenteral nutrition; PS, parenteral support; SBS, short bowel syndrome; SD, single dose; SS, steady-state after multiple dose.

Table S10. Comparison of time to reach maximum concentration between dose levels and with and without colon.

Tmax	With colon			Without colon		
	2.5 mg SD	5 mg SD	5 mg SS	2.5 mg SD	5 mg SD	5 mg SS
Mean (min)	268.2	270.1	216.3	138.9	129.9	212.3
SD (min)	132.4	99.4	171.8	64.2	55.5	55.1

Abbreviations. SD, standard deviation; Tmax, time to reach maximum concentration.

Table S11. Model estimates for the base and refined base model, including the structural covariate parameter relationship.

	Base model (RSE%) [%shrinkage] OFV = 11,660.214 CN 41.9	Refined base model (RSE%) [%shrinkage] OFV = 11,631.211 CN 33.2
<i>Structural model</i>		
Absorption rate constant (1/min)	0.0161 (8.4)	–
Absorption rate constant for SBS (1/min)	–	0.0143 (14.9)
Absorption rate constant for non-SBS (1/min)	–	0.0179 (7.3)
Clearance (L/min)	0.0256 (4.4)	0.0277 (3.5)
Volume of distribution (L)	14.1 (5.2)	15.1 (4.4)
Lag time (min)	18.7 (1.3)	18.6 (1.5)
Bioavailability (%)	50 fixed	50 fixed
SBS effect on F	–	-0.429
<i>Covariate model</i>		
Citrulline on k_a	–	–
Citrulline on F	–	–
Sex on V	–	–
<i>Statistical model</i>		
IIV on k_a (%CV)	50.8% (27.9) [28]	–
IIV on k_a for SBS (%CV)	–	73.3 (26.7) [42]
IIV on k_a for non-SBS (%CV)	–	0 fixed
IIV on CL (%CV)	11.2% (61.5) [55]	19.1 (19) [29]
IIV on V_c (%CV)	23.3 (24.0) [21]	21.1 (20) [25]
IIV on F (%CV)	29.9 (14.1) [17]	58.4 (15.7) [28]
[†] IOV on k_a (%CV)	57.9 (12.5) [27]	55.7 (9.7) [29]
[†] IOV on F (%CV)	21.7 (17.1) [44]	45.2 (15.8) [48]
Proportional error (%CV)	18.8 (7.1) [8]	18.8 (7.1) [7]
Additive error (ng/mL)	0.277 (12.0) [8]	0.277 (11.8) [7]

Abbreviations. CL, clearance; CN, condition number; CV, coefficient of variation; F, bioavailability; IIV, interindividual variability; IOV, interoccasion variability; k_a , absorption rate constant; SBS, short bowel syndrome.

Table S12. Comparison of model parameters across published apixaban models and the present study.

	Gaspar <i>et al</i>¹	Cirincione <i>et al</i>²	Hronová <i>et al</i>³	Present study
<i>Patient population</i>	Hospitalized patients	Nonvalvular atrial fibrillation	Short bowel syndrome	Short bowel syndrome
<i>Patients included</i>	100	4,385	13	70
<i>Model parameters</i>				
Bioavailability (%)	–	–	–	50 fixed
Lag time (h)	0.16	–	–	0.32
Absorption rate (1/h)	0.82	0.471	0.471	0.91 (for SBS) and 1.09 (for non-SBS)
Clearance/F (L/h)	3.2	1.6 (renal) and 2.02 (non-renal)	1.6 (renal) and 2.02 (non-renal)	1.66
Central volume of distribution/F	27	30	30	17.8
Intercompartmental clearance/F (L/h)	16	1.92	1.92	–
Peripheral volume of distribution/F (L)	51	27.3	27.3	–
<i>Covariate model on</i>				
Bioavailability	–	Dose	Dose	SBS, citrulline
Absorption rate	–	Dosing time	Dosing time	SBS, citrulline
Clearance	Serum creatinine, P-gp	Age, sex, race, NVAf subject status, ACS subject status	Age, sex, race, NVAf subject status, ACS subject status, presence of CYP3A4/P-gp inhibitor	–
Volume(s) of distribution	–	Body weight, NVAf subject status, ACS subject status	Body weight, NVAf subject status, ACS subject status	Sex

Abbreviations. ACS, acute coronary syndrome, CYP3A4, cytochrome P450 3A4; NVAf, nonvalvular atrial fibrillation; P-gp, p-glycoprotein; SBS, short bowel syndrome.

¹Population pharmacokinetics of apixaban in a real-life hospitalized population from the OptimAT study. 2023. *CPT:PSP*.

²Population Pharmacokinetics of Apixaban in Subjects With Nonvalvular Atrial Fibrillation. 2018. *CPT:PSP*.

³Pharmacokinetics of apixaban in patients with short bowel syndrome: a Bayesian re-estimation approach. 2025. *Expert Opin Drug Metab Toxicol*.

Supplementary Figures

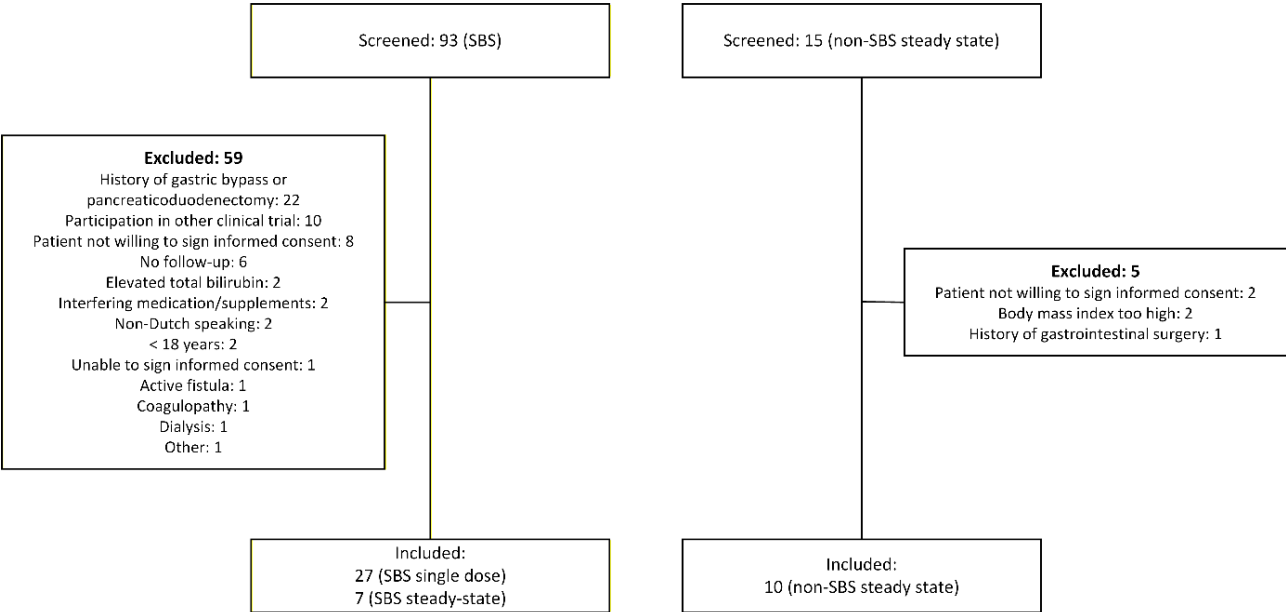


Figure S15. Flowchart of patient inclusion.

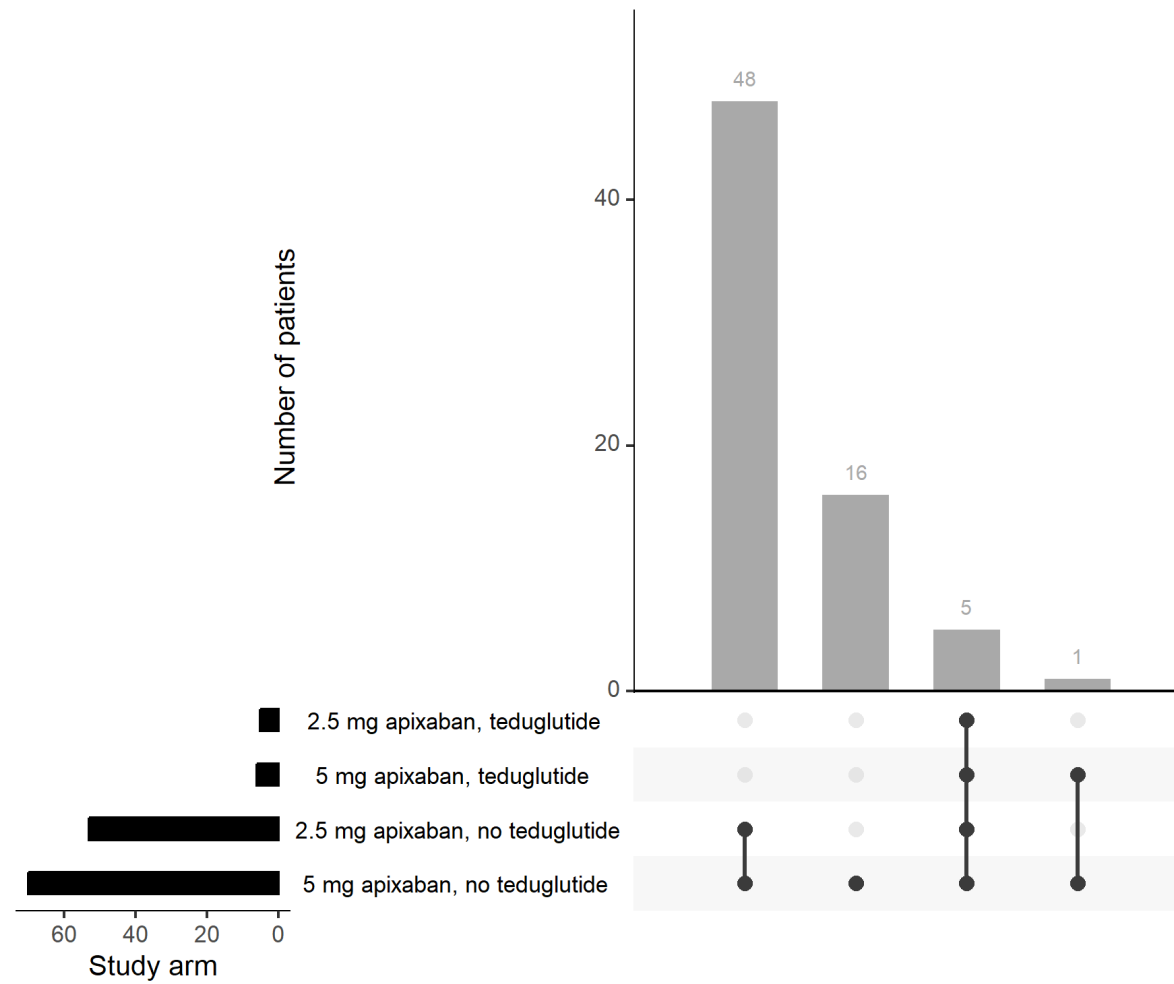


Figure S16. Overview of included patients per study arm.

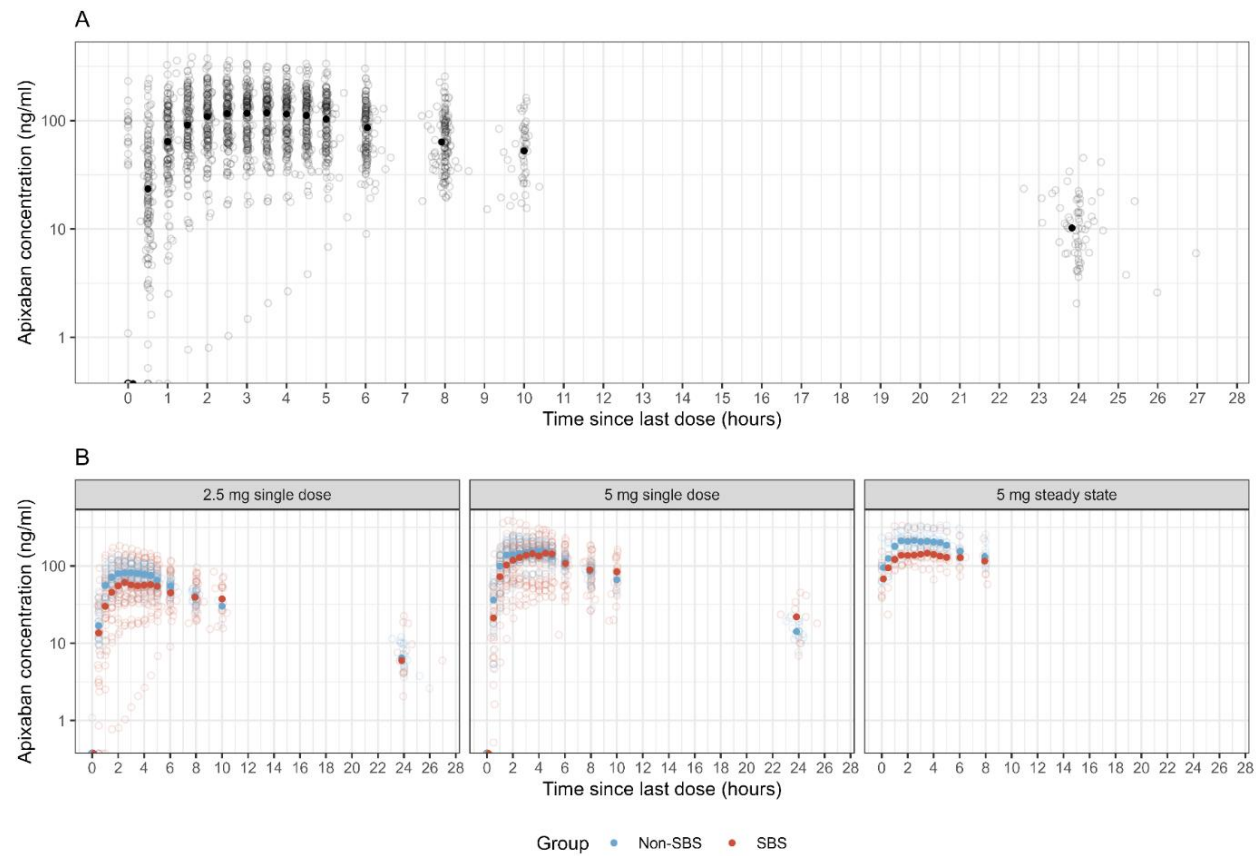


Figure S17. Apixaban concentration (log-scale) over time since last dose. Open circles depict observed apixaban concentration and filled points depict the median concentration per mid bin with bin breaks at 0, 15, 45, 75, 105, 135, 165, 195, 225, 255, 285, 315, 410, 540, 660, and 2,200 minutes.

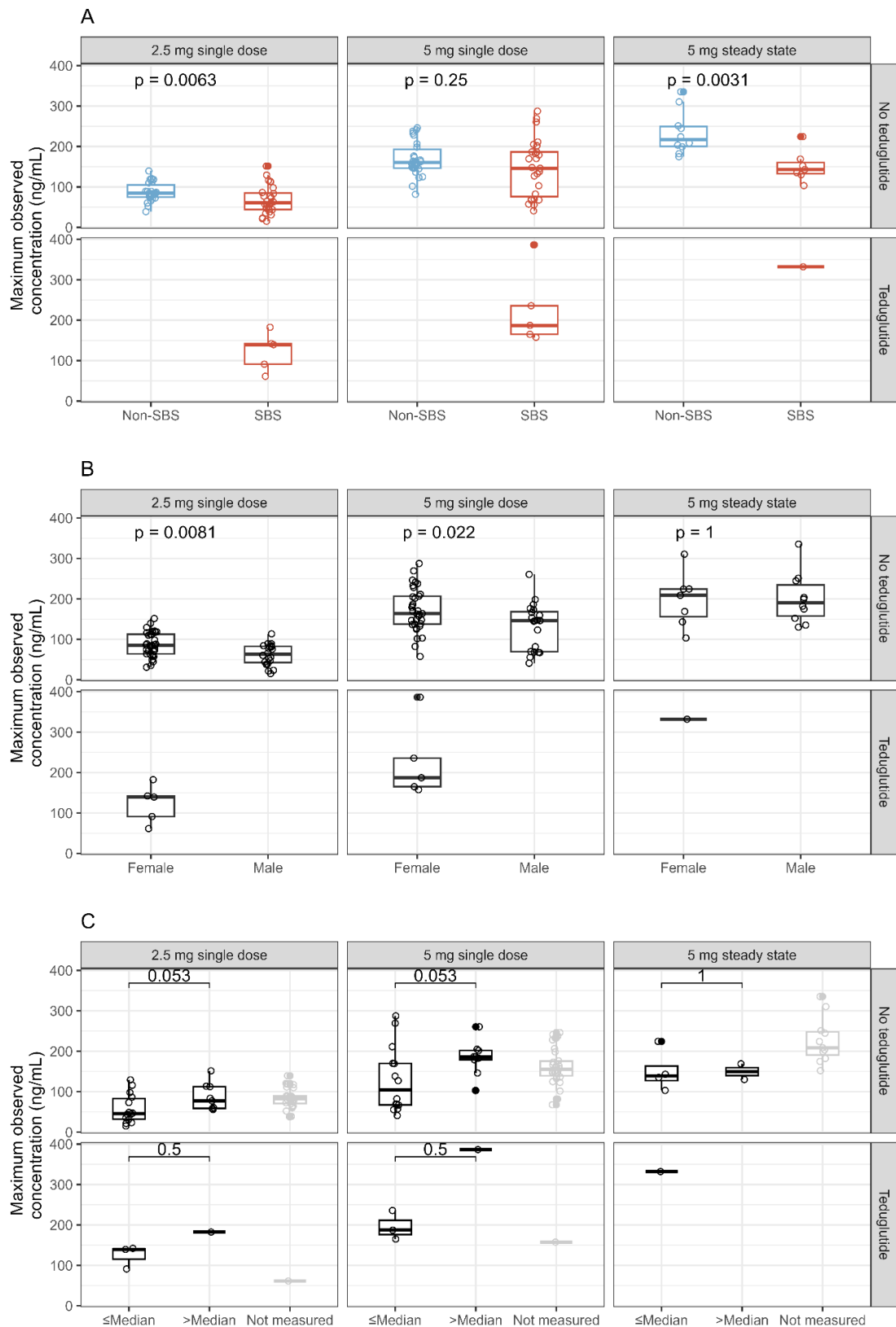


Figure S18. Maximum observed concentration stratified by dose level and the presence of teduglutide coadministration for SBS and non-SBS (A), female and male (B), and citrulline levels above or less than or equal to the median (C). P value based on Wilcoxon signed-rank test.

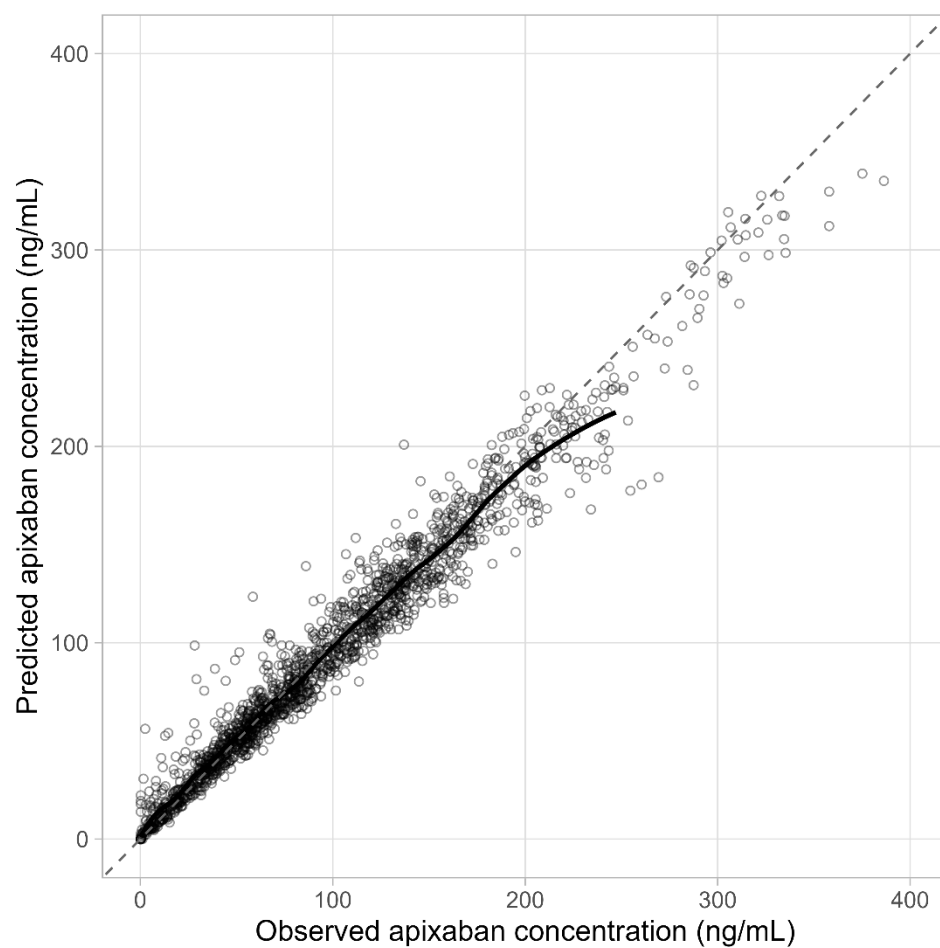
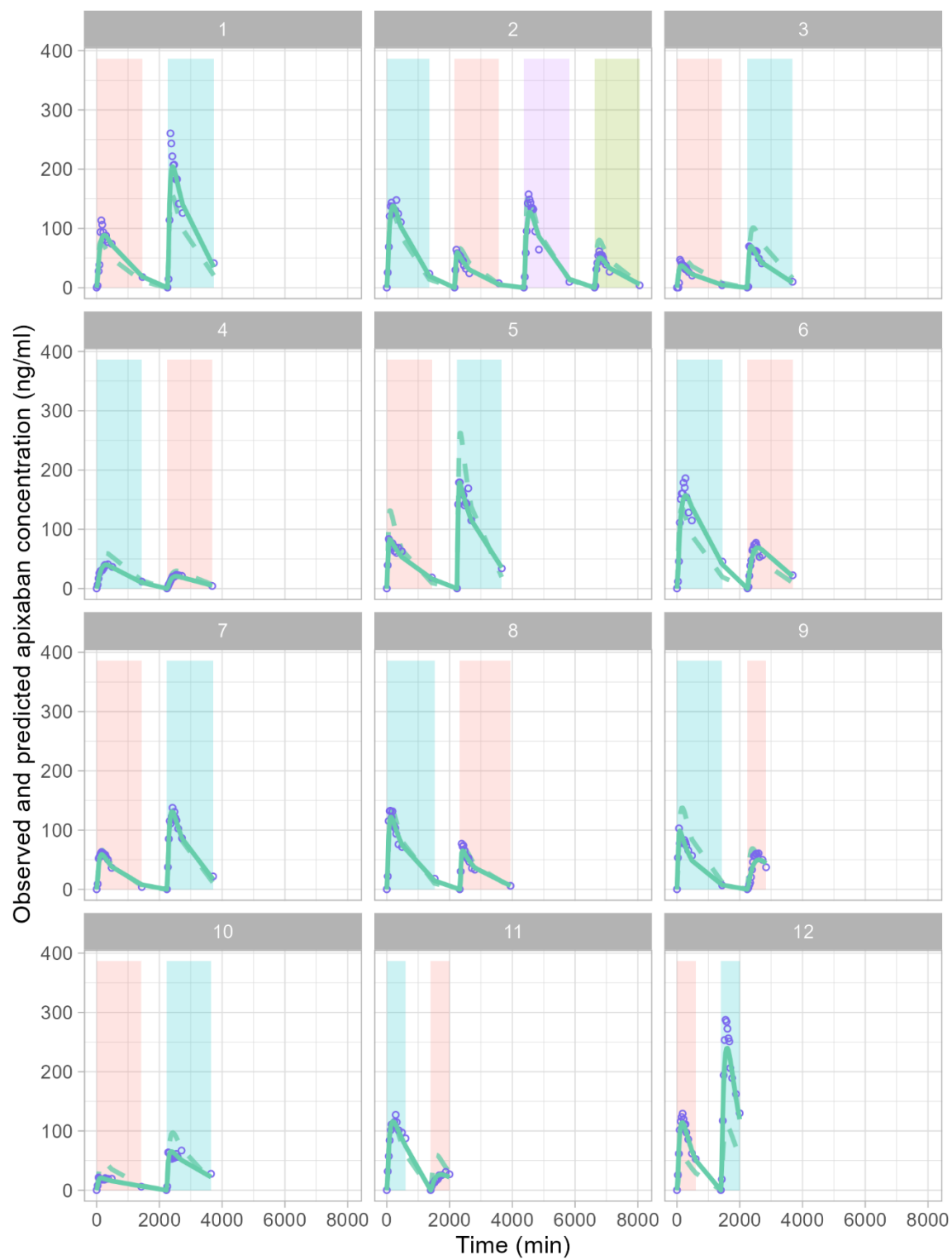
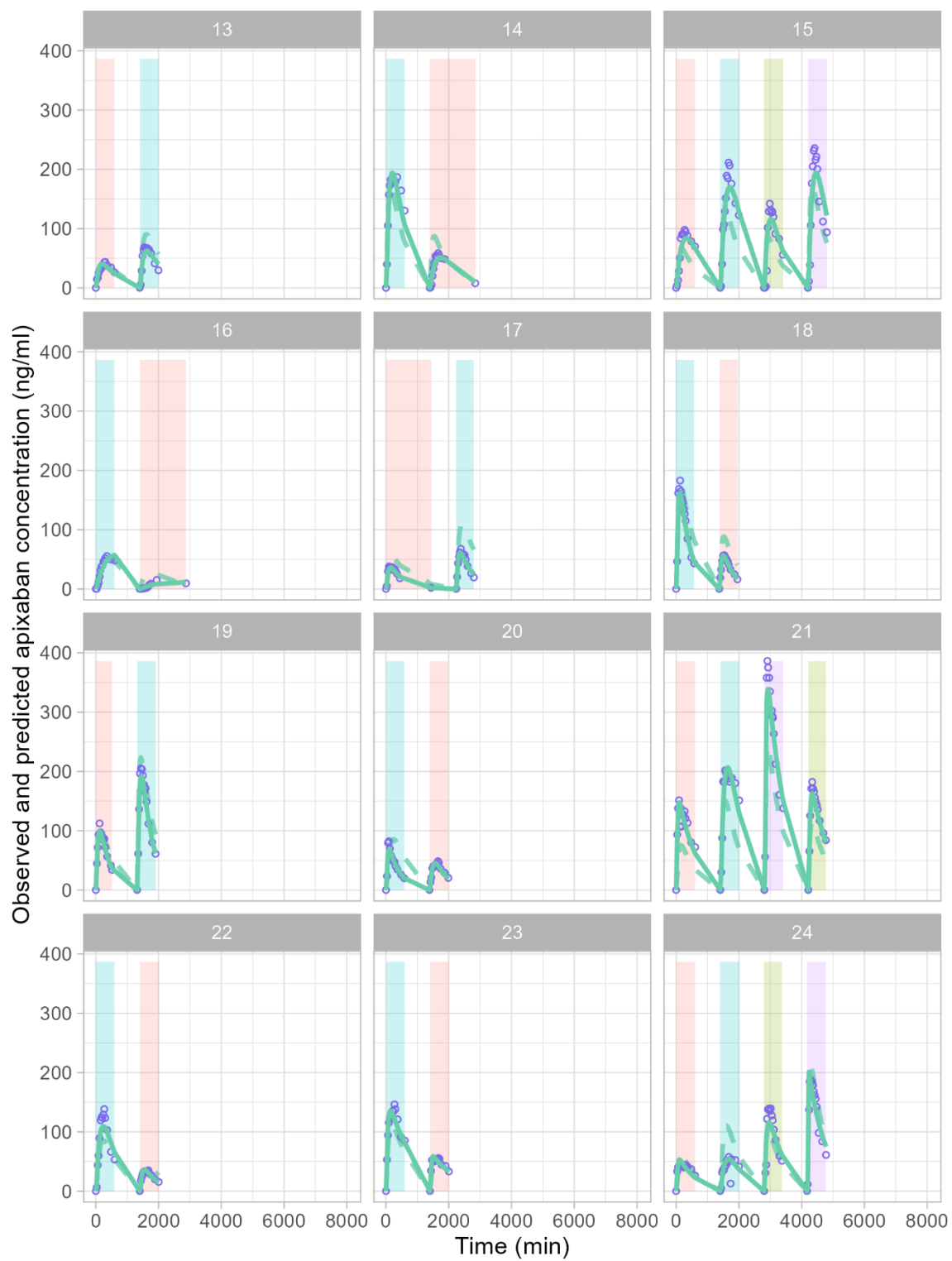
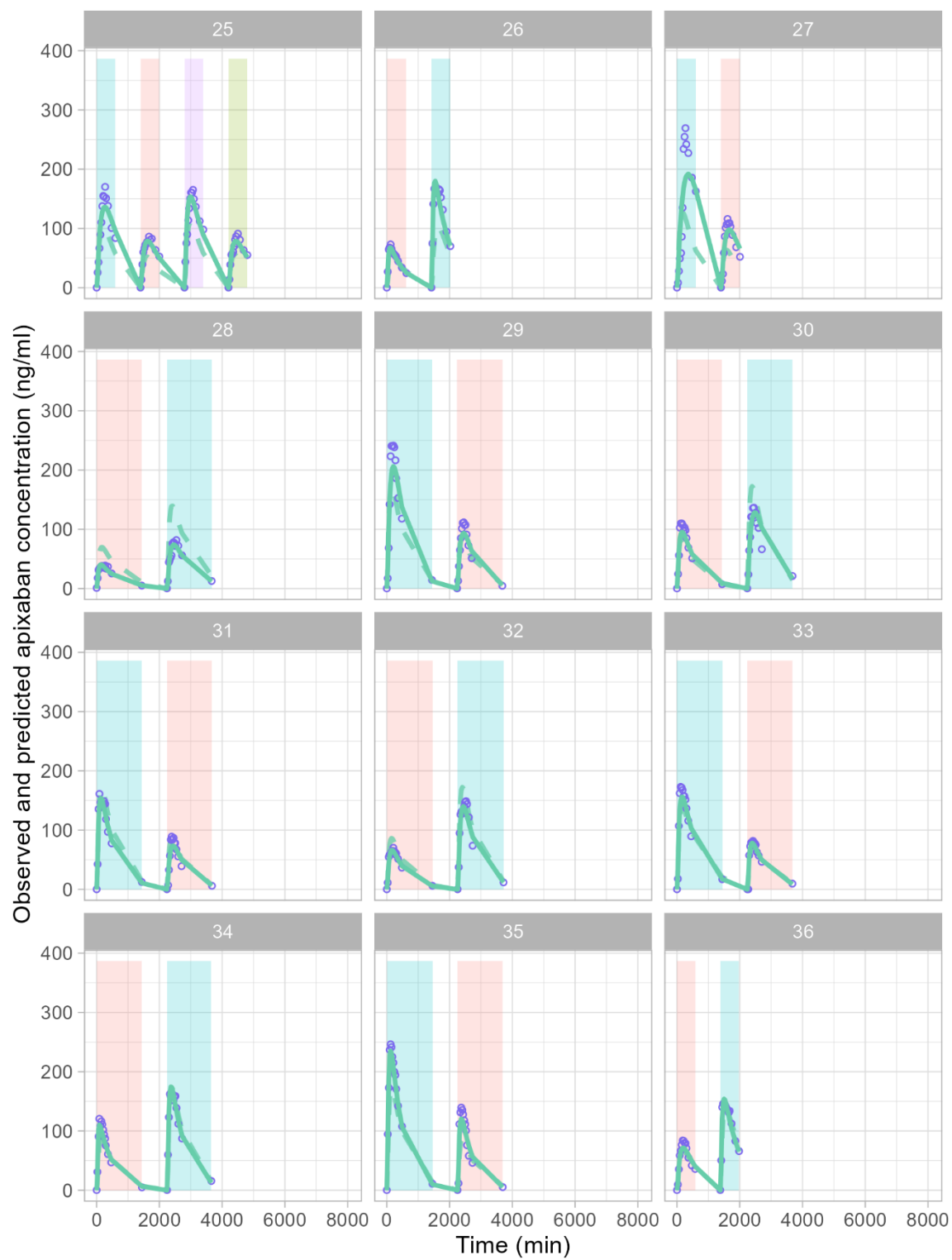
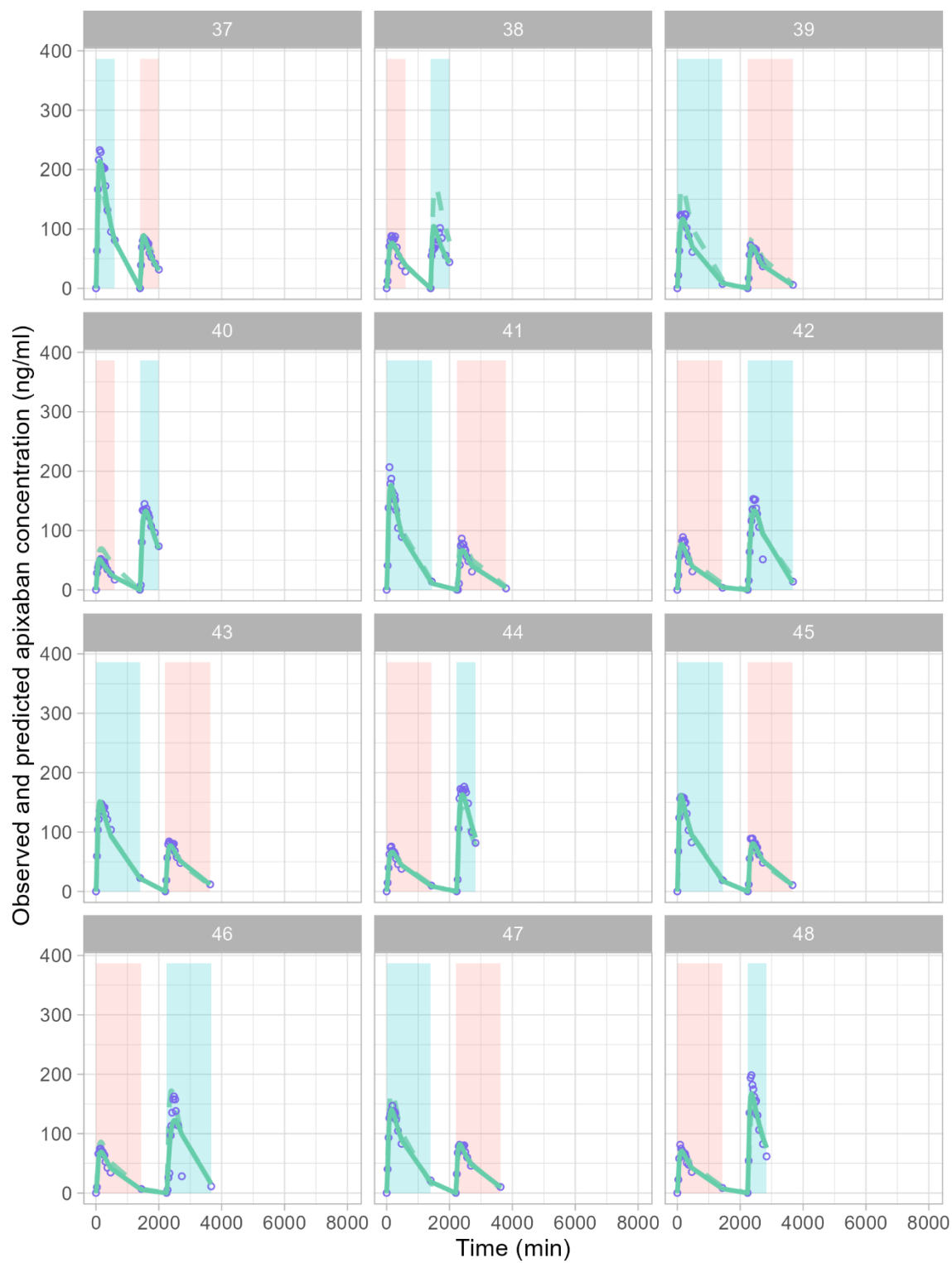


Figure S19. Predicted vs observed apixaban concentration. The black line depicts a local smoother, and the grey dashed line depicts the line of unity.









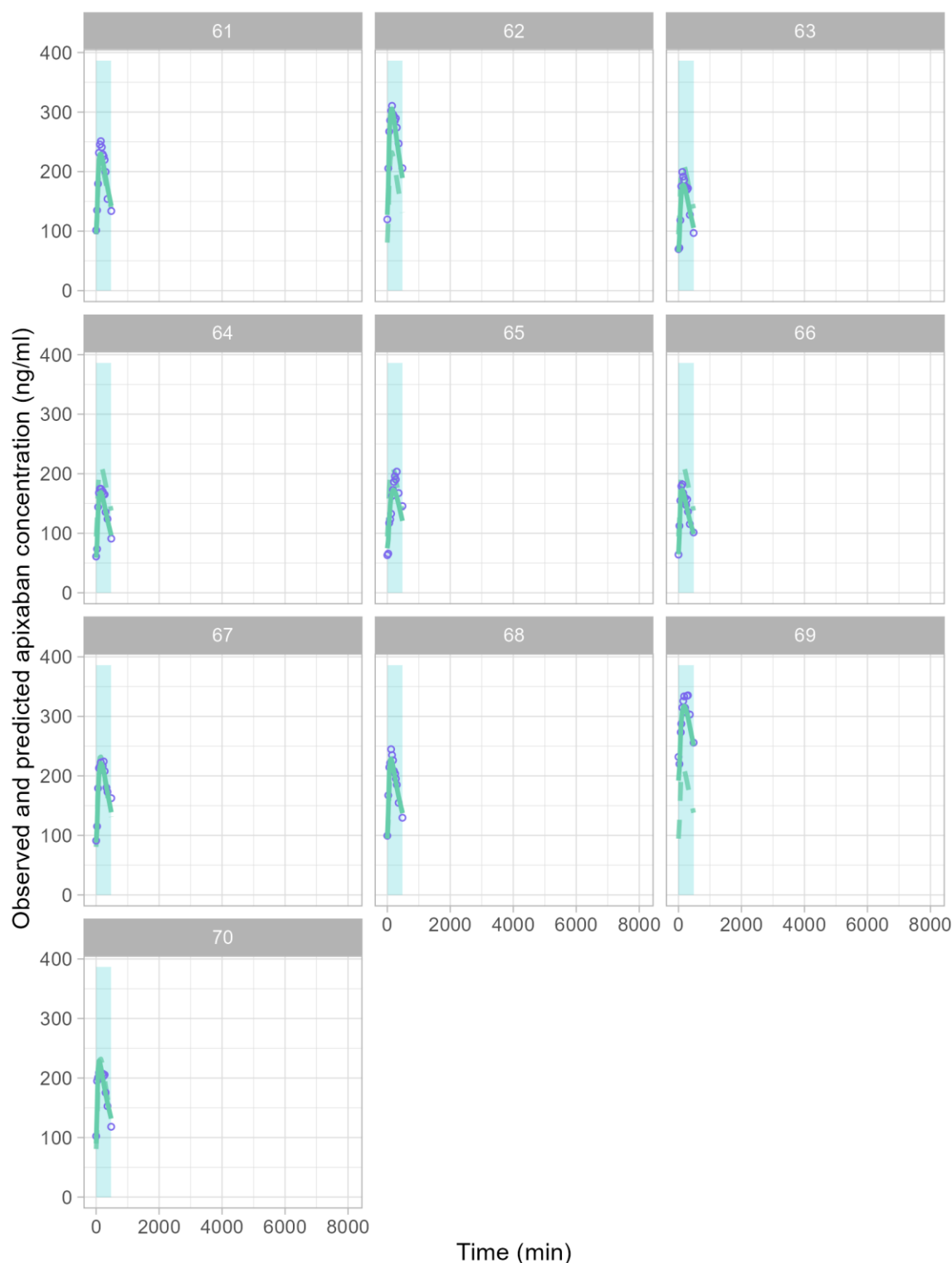


Figure S20. Individual model fits. The open circles represent observed apixaban concentrations. The green solid line represents the individual predicted profile. The green dashed line represents the population prediction. The colour bars in the background represent the four different arms of the cross-over study design: the green panel corresponds to 2.5 mg with teduglutide, the purple panel to 5 mg with teduglutide, the blue panel to 5 mg without teduglutide, and the red panel to 2.5 mg without teduglutide. The time periods between study visits were shrunk together for visualisation purposes

Supplemental Code

\$PROBLEM

```
;; 1. Based on:  
;; 2. Description: Final popPK model run303.mod  
;; x1. Author: YH
```

\$INPUT

```
ID  
record_id = DROP  
TIME  
TAD  
DV = DROP  
AMT  
MDV  
EVID  
CMT  
PARMET = DROP  
TYPE = DROP  
ARM  
ARM_ARM  
ARM_ARM_ARM  
CENS  
TEDU  
AGE  
SEX  
RACE  
WT  
BMI  
ANA  
LSB  
LCOL  
STOT  
DV_MOL  
AMT_MOL  
CREAT  
ALB  
CITRUL  
STOMAOOUTPUT  
STOOLS  
PNDAYS  
PNVOLUME  
PNKCAL  
TSR  
DV  
DOSE_LEVEL
```

```
$DATA          ABSORB_V3_M5.csv IGNORE=@ IGNORE(CMT.EQ.3)
```

```
$SUBROUTINE ADVAN6 TRANS1 TOL=9
```

```
$MODEL NCOMP = 2
```

```
COMP=(DEPOT, DEFDOS)
```

```
COMP=(CENTRAL, DEFOBS)
```

\$PK

```
;;;;;;;;;;;;;  
; IOV block
```

```
FLAG1 = 0
```

```
FLAG2 = 0
```

```
FLAG3 = 0
```

```
FLAG4 = 0
```

```
IF(ARM_ARM_ARM.EQ.1) FLAG1 = 1
```

```
IF(ARM_ARM_ARM.EQ.2) FLAG2 = 1
```

```
IF(ARM_ARM_ARM.EQ.3) FLAG3 = 1
```

```
IF(ARM_ARM_ARM.EQ.4) FLAG4 = 1
```

```

IOV = FLAG1*ETA(7)+FLAG2*ETA(8)+FLAG3*ETA(9)+FLAG4*ETA(10)
IOV_KA = FLAG1*ETA(11)+FLAG2*ETA(12)+FLAG3*ETA(13)+FLAG4*ETA(14)
;;;;;;;;;;;;;
;;; SBS on F
SBS_EFF = 0
IF(ARM.EQ.1) SBS_EFF = THETA(7)
IF(ARM.EQ.3) SBS_EFF = THETA(7)
SBS_F = (SBS_EFF)
;;; SBS on F

;;; CITRUL on F (only for SBS patients available)
CITRUL_IMP = CITRUL
IF(CITRUL.EQ.-99) CITRUL_IMP = 27 ; median
IF(CITRUL.EQ.0) CITRUL_IMP = 0.01
CITRUL_F = 0
IF(ARM.EQ.1) CITRUL_F = (CITRUL_IMP/27)**THETA(8)
IF(ARM.EQ.3) CITRUL_F = (CITRUL_IMP/27)**THETA(8)
;;; CITRUL on F (only for SBS patients available)

;;; SEX on V
SEX_EFF = 0
IF(SEX.EQ.1) SEX_EFF = THETA(9)

SEX_V = (1 + SEX_EFF)
;;; SEX on V

;;; CITRUL on KA (only for SBS patients available)
CITRUL_IMP = CITRUL
IF(CITRUL.EQ.-99) CITRUL_IMP = 27 ; median imp
IF(CITRUL.EQ.0) CITRUL_IMP = 0.01 ; to avoid 0 computation
CITRUL_KA = (CITRUL_IMP/27)**THETA(10)
;;; CITRUL on KA (only for SBS patients available)

F_COV = SBS_F + CITRUL_F
V_COV = SEX_V
KA_COV = CITRUL_KA

;;; split up KA for SBS and no SBS
SBS = 0 ; SBS no
IF(ARM.EQ.1) SBS = 1 ; SBS yes
IF(ARM.EQ.3) SBS = 1 ; SBS yes

IF(SBS.EQ.0) TVKA = THETA(1) ; estimate for KA HV
IF(SBS.EQ.1) TVKA = THETA(2) * KA_COV ; estimate for KA SBS

IF(SBS.EQ.0) ETA_KA = ETA(1) ; ETA for KA HV
IF(SBS.EQ.1) ETA_KA = ETA(2) ; ETA for KA SBS
;;; split up KA for SBS and no SBS

TVCL = THETA(3)
TVV1 = THETA(4)

TVF = THETA(5) + F_COV
TVTLAG = THETA(6)

KA = TVKA * EXP(ETA_KA + IOV_KA)
CL = TVCL * EXP(ETA(3))
V1 = TVV1 * V_COV * EXP(ETA(4))

F1PRE = TVF + ETA(5) + IOV
F1 = EXP(F1PRE)/(1 + EXP(F1PRE))
ALAG1 = TVTLAG * EXP(ETA(6))
K20 = CL/V1
S2 = V1/1000 ; dose in mg and concentration in ng/ml

```

\$DES

DADT(1) = -KA * A(1)
DADT(2) = KA * A(1) - K20 * A(2)

\$ERROR

IPRED = F ; A(2)/V1
IRES = DV-IPRED
Y = IPRED*(1+ERR(1))+ERR(2)
IWRES = IRES/(SQRT(IPRED**2*SIGMA(1,1)+SIGMA(2,2)))

\$THETA

(0, 0.01) ; TVKA HV
(0, 0.01) ; TVKA SBS
(0, 0.02) ; TVCL
(0, 14) ; TVV1
(0.5 FIX) ; TVF
(0, 16) ; TVTLAG
(-0.1) ; SBS on F
(0.1) ; CIT F
(-0.2) ; SEX on V
(0.6) ; CIT KA

\$OMEGA

(0 FIX) ; TVKA HV
(0, 0.1) ; TVKA SBS
(0, 0.05) ; TVCL
(0, 0.1) ; TVV1
(0, 0.8) ; TVF
(0 FIX) ; TVTLAG

\$OMEGA BLOCK(1) (0, 0.26) ; IOV block F

\$OMEGA BLOCK(1) SAME

\$OMEGA BLOCK(1) SAME

\$OMEGA BLOCK(1) SAME

\$OMEGA BLOCK(1) (0, 0.35) ; IOV block KA

\$OMEGA BLOCK(1) SAME

\$OMEGA BLOCK(1) SAME

\$OMEGA BLOCK(1) SAME

\$SIGMA

(0, 0.04) ; PROP
(0, 0.01) ; ADD

\$ESTIMATION MAXEVALS=9999 METHOD=1 INTER NOABORT POSTHOC PRINT=1

\$COVARIANCE PRINT=E

\$TABLE ID DV TIME MDV PRED IPRED IWRES KA CL V1 ETA(1) ETA(2) ETA(3) ARM SBS
ARM_ARM_ARM F1 ONEHEADER NOPRINT NOAPPEND FILE=sdtab303

Establishing a multistate modeling framework for competing risks in pharmacometrics analysis

This chapter is based on

Yannick Hoffert^{*}, Jan Vandewalle^{*}, Maarten Couturier, Lucas Wauters, Erwin Dreesen, Tim Vanuytsel. Long-term outcomes of pull and push percutaneous endoscopic gastrostomy techniques. [Submitted].

[^{*}Shared co-first authorship].

Abstract

Background Enteral feeding is the first choice to treat or avoid malnutrition in patients with impaired oral intake. Percutaneous endoscopic gastrostomy tubes can be placed using a push or pull technique, the former of which is the preference in case pull method is contraindicated. We compared outcomes of push versus pull percutaneous endoscopic gastrostomy.

Methods We performed a retrospective single-center study combining data from patients undergoing pull placement and push placement. The primary outcome was 1-year accidental tube dislodgement, and secondary outcomes were planned removal, peristomal infection requiring antibiotics, and all-cause mortality. A parametric multistate non-linear time-to-event model was developed using stepwise covariate modeling to identify baseline predictors of outcomes.

Results We included 305 patients in our analysis of which 205 underwent the push placement. In multistate analyses, push placement had a higher risk of accidental tube dislodgement than pull placement, whereas peristomal infection, planned tube removal for adequate oral intake, and all-cause mortality did not differ between techniques.

Conclusion Push placement was associated with a higher risk of accidental tube dislodgement. Our multistate approach allows appropriate handling of hazard estimation in competing risk scenarios. These findings support the use of push placement when anatomically indicated.

1. Introduction

Enteral feeding is the first choice to treat or avoid malnutrition in patients with impaired oral intake, such as in patients with neurological disorders, obstructive causes including oropharyngeal, head and neck cancer, esophageal cancer, benign esophageal strictures, and patients in catabolic state secondary to acute or chronic disease.^{135,136} Percutaneous endoscopic gastrostomy tubes are indicated in case of chronic need for enteral nutrition or for decompression (venting) in patients with recurrent bowel obstruction or chronic intestinal pseudo-obstruction.¹³⁷ The use of feeding tubes is increasing, with home enteral nutrition growing in the United States.^{135,136} Two endoscopic placement techniques are used in practice: pull and push.^{138,139} Yet, the technique-specific long-term risks remain uncertain. Prior comparative reports reach conflicting conclusions and often analyze single endpoints without accounting for competing and sequential events.^{140–147} When death precludes later complications, or when an earlier event (e.g., infection) alters the risk of a subsequent one, non-informative censoring can cause bias.

In pull placement, a wire is introduced into the stomach via abdominal puncture, grasped endoscopically, and withdrawn through the mouth. After attaching a feeding tube with a bumper to the wire, this tube is pulled transorally into the stomach and through the abdominal wall until the bumper is fixed to the gastric wall. In push placement, a trocar with an overlying peel-away sheet is placed percutaneously into the stomach under direct endoscopic visualization after fixating the stomach to the abdominal wall with the aid of a gastropexy device. After removing the trocar, a balloon-type feeding tube is introduced into

the stomach through the introducer sheet. Where trans-oral passage is feasible and in the absence of head and neck or esophageal cancer, pull placement is guideline-recommended and is characterized by high technical success (effectiveness) and a well-described safety profile in randomized trials and large cohorts.^{137,142,143,148} Push placement is recommended when trans-oral passage is not feasible or is contraindicated (e.g., head and neck or esophageal cancer, or in severe esophageal strictures) because pull placement may carry risks (e.g., tumor-tract seeding) or technical failure in these settings. This allocation is consistent with international guidance: the American Society for Gastrointestinal Endoscopy and the European Society of Gastrointestinal Endoscopy recommend the introducer (“push”) technique when trans-oral pull-through is contraindicated or not feasible.^{137,148} Consequently, the choice of technique is determined by anatomic feasibility and underlying disease rather than operator preference, which can introduce systematic differences in baseline characteristics between groups. A substantial body of randomized and observational studies has compared pull and push placement, but findings remain inconsistent.^{140–147}

Our aim was to compare push versus pull percutaneous endoscopic gastrostomy with a primary focus on accidental tube dislodgement within 365 days, and to assess differences in three secondary outcomes (planned tube removal for adequate oral intake, peristomal infection requiring antibiotics, and all-cause mortality within 365 days) using a competing-risk/multistate framework. We hypothesize that push placement is associated with a higher 1-year risk of accidental tube dislodgement than pull placement in adults undergoing

gastrostomy. We addressed this gap using a retrospective cohort approach at a tertiary hospital and analyzed technique-specific risks within a competing-risk/multistate framework.

2. Methods

2.1 Patients and data

We performed a retrospective single-center cohort study in 305 adults (18 years and older) who received percutaneous endoscopic gastrostomy at University Hospitals Leuven in Belgium. All adults receiving their first push placement from January 2017 through December 2021 were included (n=205). To provide a historical comparator from an era when push placement was unavailable, we selected 100 consecutive adults who underwent pull placement between January 2014 and December 2015. This sample affords 80% power ($\alpha=0.05$) to detect a 15% absolute difference when the event rate is approximately 10% in one group.¹⁴⁹

Our exposure of interest was push percutaneous endoscopic gastrostomy defined as direct percutaneous introducer placement using a trocar and peel-away sheath with a balloon-retained tube. Our comparator was pull percutaneous endoscopic gastrostomy defined as trans-oral pull-through placement with a bumper-retained tube. We hypothesize that push placement confers a higher 1-year risk of accidental tube dislodgement than pull placement.

Technique selection followed institutional protocol criteria. During the first study period (2014–2015), only the pull technique was available in our center, also for patients with obstructive upper aerodigestive pathology (e.g., head and neck or esophageal cancer, severe esophageal strictures). During the second study period (2017–2021), all patients with these conditions received a gastrostomy tube by the push technique. Patients who had previously undergone a tube placement were not included in the study. All gastrostomy tubes analyzed were placed to provide enteral nutrition, either as full feeding or as supplemental feeding.

Patients were followed until tube removal, death, or 365 days, whichever occurred first. No patients were lost to follow-up due to centralized care and annual enteral-nutrition reimbursement coordinated by the hospital nutrition team. Outcomes were ascertained from multiple sources within the electronic health record (endoscopy and interventional notes, emergency/ward encounters, discharge summaries), device/procedure logs, and medication orders. Accidental tube dislodgement was counted when it prompted unscheduled care with endoscopic or radiologic confirmation or tube replacement.

We collected baseline patient characteristics, including sex, age, body weight, length, primary indication for tube placement, presence of comorbidities, C-reactive protein, and serum albumin. Single median imputation was used in case of missing baseline information. Time-varying measures of nutrition therapy (e.g., prescribed or delivered energy and protein, oral intake during follow-up), feeding-route utilization, or dietitian encounters were not available.

Our study was approved by the Ethics Committee Research University Hospitals / KU Leuven (MP019554), adhering to the principles of the Declaration of Helsinki. Written informed consent was waived because the research posed no more than minimal risk to participants.

2.2 Outcomes

Consistent with the study aim, we compared push versus pull placement with a primary focus on 1-year accidental tube dislodgement and three secondary outcomes within a

competing-risk/multistate framework. The primary outcome was accidental tube dislodgement within 365 days after placement, defined as any unintentional displacement of the gastrostomy tube from the intragastric position (partial or complete) resulting in unscheduled care, endoscopic/radiologic confirmation, or tube replacement; planned removals are excluded. Secondary outcomes were (1) planned tube removal for adequate oral intake within 365 days (documented clinical decision), (2) peristomal infection requiring antibiotics (systemic or topical) within 365 days with chart-documented local signs and an antibiotic prescription, and (3) all-cause mortality within 365 days of placement. Statistical significance for the primary outcome was assessed at two-sided $\alpha=0.05$. Family-wise type-I error across the three secondary outcomes was controlled using the Holm–Bonferroni procedure.

Major complications include free perforation of the stomach or intestine, clinically significant peritonitis, pneumonia, buried bumper syndrome, metastatic seeding at the gastrostomy site, wound infection requiring systemic antibiotics, and major bleeding requiring transfusion and/or endoscopic hemostasis or hospitalization. Minor complications include local bleeding, accidental tube dislodgement managed without peritonitis or surgery (events with severe sequelae are captured under the relevant major category, e.g., peritonitis), granuloma (stomal granulation tissue treated locally, e.g., with silver nitrate/cautery or dressings), excessive pain or leakage managed conservatively, tube fracture, tube blockage, and wound infection managed with local/topical therapy only.

2.3 Setting

University Hospitals Leuven is a tertiary referral center with centralized oncologic care and a hospital nutrition team. Gastrostomy placement is performed within the Department of Gastroenterology and Hepatology by consultant gastroenterologists and by gastroenterologists in training under direct supervision of an experienced staff member. An institutional protocol standardizes pre-procedure assessment, sedation (intravenous midazolam and pethidine, or general anesthesia with propofol in the presence of an anesthesiologist), antibiotic prophylaxis (single-dose cefazolin or clindamycin in case of penicillin allergy), device models, and immediate post-procedural care (overnight observation, local wound care, flushing, and mobilization of the tube). Multidisciplinary follow-up involves gastroenterology, clinical nutrition/dietetics, oncology, and ear-nose-and-throat teams as appropriate. During 2014–2015 the push technique was not available at our center. From 2017 onward it was implemented. Aside from the availability of the push technique and guideline-driven allocation (push when trans-oral passage was not feasible or contraindicated, otherwise pull when feasible), procedural protocols and operator oversight were unchanged across study periods.

2.4 Percutaneous endoscopic gastrostomy placement

All procedures were performed by two gastroenterologists (in training) (ranging from fellows in training to senior doctors) under intravenous sedation (midazolam and pethidine) or general anesthesia (propofol in presence of an anesthesiologist). All patients received

prophylactic antibiotics (a single dose of cefazolin or clindamycin in case of penicillin allergy). The procedures were carried out in sterile conditions at the endoscopy department. The Cook percutaneous endoscopic gastrostomy-24 tube (24 French, Cook Medical) was used for the pull technique. The push technique was performed with the Freka Pexact gastrostomy device (15 French, Fresenius Kabi). Patients routinely stayed in the hospital overnight after placement for observation and initiation of enteral feeding if indicated. Postprocedural instructions regarding local wound care, flushing and mobilization of the tube were provided. These procedural protocols, device models, and post-procedural care instructions were standardized across operators and applied consistently during both study periods.

Management of gastrocutaneous fistula after tube removal followed a standardized pathway. When continued enteral feeding was required and a mature tract was present, the tract was immediately reused with tube replacement. When enteral access was no longer required, the tract was allowed to close spontaneously with short fasting and local care. For persistent leakage, endoscopic closure was performed after local mucosal abrasion (e.g., cautery) using clip-based closure (hemoclips or an over-the-scope clip). Surgical closure was reserved for rare refractory cases. The same pathway applied irrespective of initial placement technique.

2.5 Exploratory data analysis

Data management, post-processing, and exploratory data analysis were performed using the tidyverse collection in R (version 4.3.2, R Foundation for Statistical Computing, R Core Team), interfaced with RStudio (version 2023.12.1; RStudio, Inc.).¹⁵⁰ Kaplan–Meier curves were generated using the survminer and survival package.^{151,152} A P value of <0.05 was considered statistically significant for inferential statistics of the exploratory data analysis.

2.6 Model development

To account for differences in tube dwell time between techniques, all outcome comparisons were analyzed within a multistate time-to-event framework that models time since gastrostomy insertion. This framework estimates transition-specific hazards and cumulative risks under competing events and avoids treating competing events as non-informative censoring, thereby aligning the analysis with real-world clinical pathways. In the multistate model, “ongoing enteral feeding” functions as the initial state rather than an outcome. Prespecified clinical outcomes were accidental dislodgement (primary) and planned tube removal, peristomal infection requiring antibiotics, and all-cause mortality (secondary). Consistent with the study aim, we compared push versus pull placement with a primary focus on 1-year accidental tube dislodgement and three secondary outcomes within a competing-risk/multistate framework.

We fitted time-to-event models to the data using the NONMEM software (version 7.5.0; Icon plc, Gaithersburg, Maryland, USA) with the Laplace method for parameter estimation. Events of interest were (i) ongoing enteral feeding, (ii) on-purpose removal of the tube

because of adequate oral nutrition, (iii) accidental tube dislodgement, (iv) infection requiring systemic or local antibiotic treatment, and (v) death. These five long-term outcomes were defined as states in which a patient can occur at any time point. We developed hazard functions to describe transitions between the states, including a Weibull function (assuming that the hazard rate can change over time; Equation 12) and an exponential function (assuming that the hazard rate is constant; Equation 13). Transition hazards were parameterized as functions of time since gastrostomy insertion (Weibull/exponential forms), thereby inherently adjusting for differences in time-at-risk, including duration of tube presence, between the push and pull cohorts.

$$\text{Transition rate} = \frac{1}{\lambda} \times \gamma \times \left(\frac{1}{\lambda} \times \text{time}\right)^{\gamma-1} \quad (\text{Eq. 13})$$

$$\text{Transition rate} = \frac{1}{\lambda} \quad (\text{Eq. 14})$$

with λ depicting the baseline hazard and γ the shape of the hazard distribution. Among the converged models, the most parsimonious multistate time-to-event model was withheld based on the likelihood ratio test on a 95% significance level.

Sex, age, body mass index, primary indication for tube placement, C-reactive protein, serum albumin, and the placement technique (*pull* vs. *push*) were considered as baseline predictors of the hazard for developing an event. Continuous and categorical covariates were tested with a proportional and linear hazard model, respectively. We performed stepwise covariate modeling with forward inclusion ($p \leq 0.05$) followed by backward elimination ($p \leq 0.01$). A

drop in the objective function value of ≥ 3.84 points corresponds to $p \leq 0.05$, and a drop of ≥ 6.63 points corresponds to $p \leq 0.01$ (1 degree of freedom).

2.7 Model evaluation

The final multistate time-to-event model was evaluated based on the precision and plausibility of parameter estimates. Internal model validation was performed using a bootstrap ($n=1,000$) to obtain non-parametric parameter estimates, and visual predictive checks to assess model calibration.

3. Results

3.1 Patients characteristics

We included 305 patients in our analysis, of which 100 underwent pull placement in the first study period and 205 underwent push placement during the second study period with a follow-up period of 365 days (**Figure S21**). During the first study period, the push technique was not available in our center. To allow comparison with the push cohort we included 100 consecutive adults with head and neck or esophageal cancer who underwent push placement in 2014–2015.

Considering the overall population, the majority of the patients received enteral feeding because of head and neck or esophageal cancer (93.4%). In 141 patients (46.2%), the indication for the gastrostomy tube was symptoms related to the tumor (e.g. dysphagia or inadequate oral intake), while in 144 patients (47.2%) the tube was placed prophylactically in case of anticipated compromised oral intake due to scheduled radio-chemotherapy or surgery, for instance. Less frequent indications in the push placement cohort were benign esophageal stenosis, motility disorders, and intellectual disability with insufficient oral intake, which accounted for 20 patients (6.6%). A detailed and technique-stratified description of the cohort is provided in **Table 7**. No patients were lost to follow-up. Data were complete to tube removal, death, or day 365.

Table 7. Baseline and outcome characteristics of patients undergoing *pull* or *push* placement

Characteristic	<i>Pull</i> placement cohort (n=100)	<i>Push</i> placement cohort (n=205)
<i>Baseline demographics and phenotype</i>		
Age, years, median (IQR)	62 (56, 65)	65 (59, 71)
Male sex, n (%)	64 (64.0)	145 (70.7)
Body mass index, kg/m ² , median (IQR)*	22.3 (19.4, 25.2)	21.6 (18.2, 25.0)
<i>Indication</i>		
Cancer, prophylactic, n (%)	77 (77.0)	67 (32.7)
Cancer, obstructive symptoms, n (%)	23 (23.0)	118 (57.6)
Tube placement for other indication, n (%)	0 (0)	20 (9.8)
<i>Baseline serology</i>		
Serum albumin, g/L, median (IQR) [†]	39.5 (36.5, 43.9)	38.0 (33.3, 39.0)
C-reactive protein, mg/L, median (IQR) [‡]	6.2 (2.2, 20.8)	13 (8.4, 41.5)
<i>Long-term outcomes</i>		
Time to death, days, median (IQR)	203 (118, 251)	109 (41, 211)
Time to infection, days, median (IQR)	25 (20, 123)	14 (9, 30)
Time to tube dislodgment, days, median (IQR)	53 (40, 85)	95 (32, 118)
Time to tube removal, days, median (IQR)	95 (68, 174)	80 (49, 117)
One-year mortality, n (%)	31 (31.0)	86 (41.9)

Abbreviations. IQR, interquartile range.

Cohort denominators: pull n=100; push n=205. Percentages rounded to one decimal.

*BMI was missing for pull n = 4/100 (4.0%), push n = 2/205 (1.0%), total n = 6/305 (2.0%).

[†]Serum albumin was missing for pull n = 16/100 (16.0%), push n = 54/205 (26.3%), total n = 70/305 (23.0%).

[‡]C-reactive protein was missing for pull n = 2/100 (2.0%), push n = 24/205 (11.7%), total n = 26/305 (8.5%).

3.2 Outcomes

In the pull placement cohort within one year, a total of 17 (17.0%) patients had an infection of the insertion point requiring local or systemic use of antibiotics, 62 (62.0%) had the tube removed because of adequate oral intake, eight (8.0%) had an accidental tube dislodgment, and 31 (31.0%) patients died. In the push cohort within one year, a total of 18 (8.8%) patients had an infection requiring local or systemic use of antibiotics, 81 (39.5%) had the tube removed, 76 (37.1%) had a tube dislodgment, and 86 patients died (42.0%) (**Table 7, Table 8**). To facilitate early vs. late comparisons, Supplementary **Table S13** summarizes counts and proportions by technique at day 7, day 30, day 90, and day 365 for tube dislodgement, peristomal infection requiring antibiotics, planned tube removal, and death.

The univariate Kaplan–Meier analysis reveals a statistically significant difference between the tubes for tube dislodgment ($P < 0.001$), infection ($P = 0.04$), and death ($P = 0.04$) (Figure 21).

Table 8. Major and minor complications for patients undergoing pull or push placement

	Pull cohort (n=100)	push placement cohort (n=205)
<i>Major complications</i>		
Perforation, <i>n</i> (%)	0 (0.0)	2 (1.0)
Peritonitis, <i>n</i> (%)	2 (2.0)	2 (1.0)
Pneumonia, <i>n</i> (%)	6 (6.0)	17 (8.3)
Buried bumper syndrome, <i>n</i> (%)	2 (2.0)	1 (0.5)
Metastasis placement site, <i>n</i> (%)	0 (0.0)	0 (0.0)
Wound infection requiring systemic antibiotic, <i>n</i> (%)	7 (7.0)	11 (5.4)
Major bleeding, <i>n</i> (%)	0 (0.0)	0 (0.0)
<i>Minor complications</i>		
Local bleeding, <i>n</i> (%)	0 (0.0)	4 (2.0)
Tube dislodgment, <i>n</i> (%)	8 (8.0)	76 (37.1)
Granuloma, <i>n</i> (%)	1 (1.0)	2 (1.0)
Excessive pain, <i>n</i> (%)	5 (5.0)	12 (5.9)
Excessive leakage, <i>n</i> (%)	3 (3.0)	14 (6.8)
Tube fracture, <i>n</i> (%)	1 (1.0)	7 (3.4)
Tube blockage, <i>n</i> (%)	0 (0.0)	7 (3.4)
Wound infection requiring local antibiotic, <i>n</i> (%)	10 (10.0)	7 (3.4)
Other, <i>n</i> (%)	1 (1.0)	5 (2.4)

Cohort denominators: pull $n=100$; push $n=205$. Percentages rounded to one decimal.

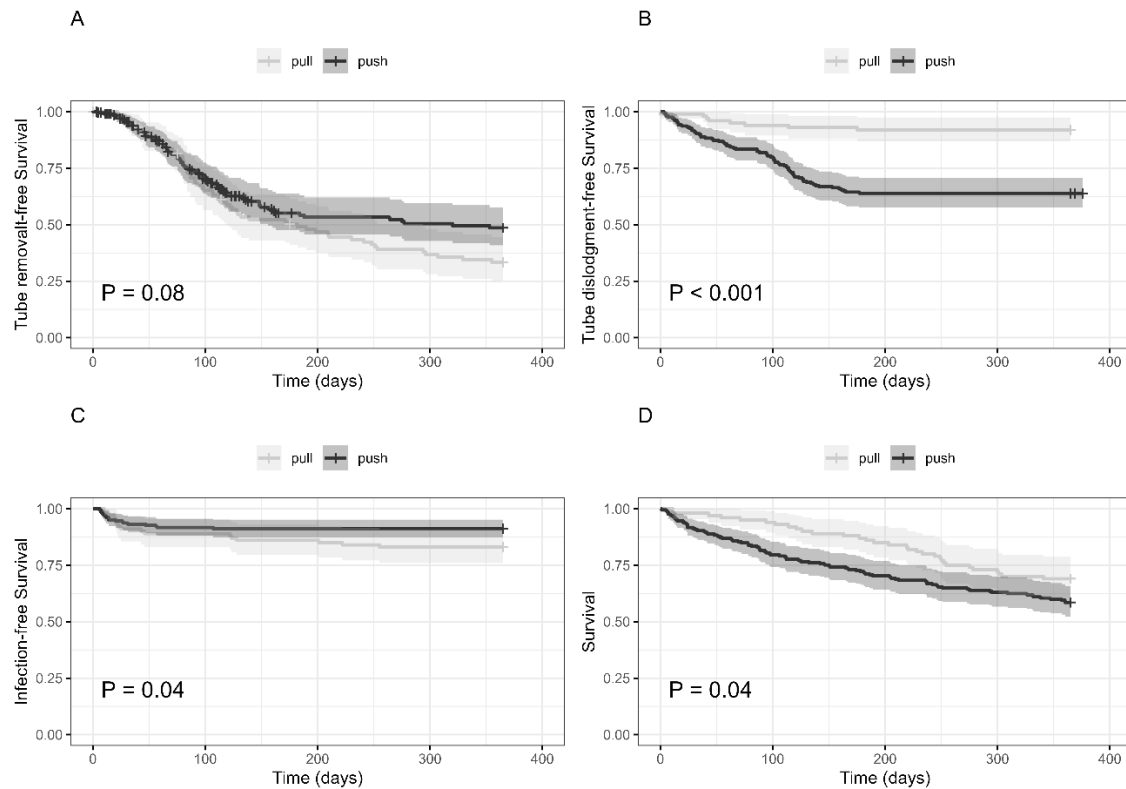


Figure 21. Kaplan–Meier analysis for the four long-term outcomes; tube removal (A), tube dislodgment (B), infection (C), and death (D). All P values are based on a log-rank test; pull n=100, push n=205.

3.3 Multistate time-to-event model

The previous univariate analysis revealed a statistically significant relation between the type of tube and tube dislodgment, infection, and death. The following model-based analysis aimed to identify and quantify predictors for each event while accounting for competing risks to allow a biased-free comparison of the two tubes with respect to tube dislodgment, tube removal, infection, and death. The developed multistate model combines all outcomes in a holistic system, where single outcomes are connected in a unidirectional way (**Figure 22**). The model specifies clinically plausible transition paths between states. For example, transitions were permitted from infection to dislodgement to death, but not from death to any subsequent state. Transitions without empirical support in our data (e.g., dislodgement to

subsequent infection) were not modeled. Allowed transitions are depicted in **Figure 22**. After identifying the overall model structure, covariates were tested one by one on each transition to identify and quantify predictors for developing the corresponding event (**Table S14**).

The push-technique had a significantly higher risk of tube dislodgment compared to the pull technique. The model revealed no significant difference between the tubes in terms of removal of the tube, infection, and death. Patients receiving the tube because of other indication than cancer had a higher risk for infection compared to patients receiving it because of cancer. Patients with a low C-reactive protein at baseline had a higher risk of infection. With an increase in baseline C-reactive protein, the risk of death increased. Furthermore, patients with indication for tube placement other than cancer had the highest risk for mortality, followed by patients with obstructive symptoms from cancer. Patients with cancer receiving the tube prophylactically (i.e., in anticipation of potentially reduced intake during radio-chemotherapy) had the lowest risk for mortality among the investigated population. Patients with a higher baseline C-reactive protein had a lower chance of tube removal. **Figure 22** shows a diagrammatic representation of the final multistate time-to-event model including the identified predictors on each transition. The model is described in more detail in the Supplementary Results (**Table S15**).

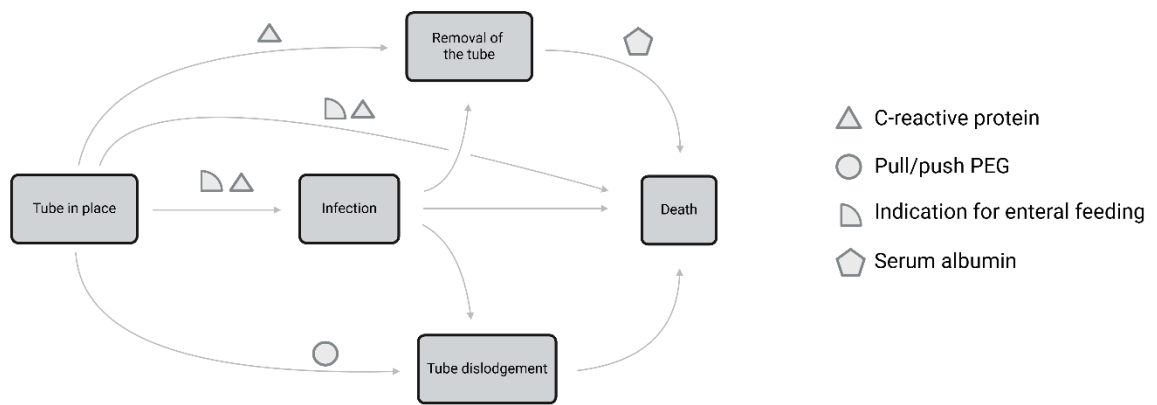


Figure 22. Diagrammatic representation of the multistate time-to-event model. The predictors for transitioning between states are represented by different shapes. Only tube removal and death were considered competing risk events, as these transitions preclude the occurrence of tube dislodgement.

To evaluate how well the model captures the data, we used model simulations and overlaid them with observed data. The model follows the trend of the data well with the data falling within the 95% confidence interval of the model simulation (**Figure S22**). The same analysis stratified by the type of tube (pull or push) shows the same level of agreement (**Figure S23**). Model robustness was evaluated, using non-parametric bootstrap estimates, which shows good agreement. The model estimates align well with the bootstrap estimates (**Table S15**). A detailed overview of the covariate modeling approach, featuring the stepwise inclusion and elimination with the corresponding significance level was included in the Supplementary Results (**Table S14**). Kaplan–Meier analysis depicting the observed trend of the data, and stratified by identified predictors on the corresponding event was appended in the Supplementary Results (**Figure S24–S26**). For reader-facing subgroup views, technique-stratified displays are provided in Supplementary **Figure S23**, and predictor-stratified trajectories (by primary indication, albumin, and C-reactive protein) are shown in

Supplementary **Figures S4–S6**. Covariate selection steps, including the evaluation of body mass index, appear in Supplementary **Table S14**.

4. Discussion

We evaluated whether push versus pull gastrostomy differed in 1-year accidental dislodgement (primary) and in three secondary outcomes within a competing-risk/multistate framework. We demonstrated that the pull technique has a significantly lower risk of tube dislodgment compared to the push technique. Several other studies already demonstrated the increased rate of tube dislodgment when using the push technique^{140,141,149} which is most likely related to the use of a balloon-type tube vs. fixed internal bumper in the push vs. pull technique respectively. Our model-based multivariate analysis revealed no other associated risk factors for tube dislodgment besides the placement technique.

Next to the increased rate of tube dislodgment in the push placement cohort, the multivariate analysis withheld no other differences for minor adverse events between the push- and pull cohorts. The higher rate of wound infection requiring local antibiotics in the pull placement cohort could not be confirmed in the multistate model analysis.^{142,145} Furthermore, multivariate analysis confirmed similar major adverse events rates for both cohorts. Although patients receiving push placement in our cohort had a higher overall mortality rate, multistate modelling analysis demonstrated that the type of tube (push vs. pull technique) is not associated with increased mortality.

Our estimate of a higher risk of accidental tube dislodgement after push placement is consistent with prior comparative reports.^{140,144,149} For peristomal infection, findings in trials and large cohorts have been inconsistent.^{142,143,145} After explicitly accounting for competing

and sequential events in our multistate framework, we did not observe a difference between techniques. Finally, higher crude mortality in push cohorts described elsewhere likely reflects confounding by indication and baseline severity. In our analysis, technique was not an independent predictor of mortality, which is compatible with multicenter comparisons once indication is considered.^{141,153} Between-group differences in crude mortality reflected clinical indication rather than technique: non-cancer indications (only in the push placement group) had the highest mortality, cancer with obstructive symptoms next, and prophylactic cancer the lowest. Besides the indication, we identified C-reactive protein and serum albumin as independent risk factors for death.

Lower baseline albumin (poor nutritional status and inflammation) was associated with increased mortality, in line with prior evidence linking albumin to post-gastrostomy survival.^{154,155} Higher baseline C-reactive protein (reflecting systemic inflammation) was associated with increased mortality and a lower likelihood of planned tube removal. These patterns are clinically plausible because inflammatory burden tracks with disease severity and frailty. The negative association between C-reactive protein and peristomal infection likely reflects patient mix and event competition: individuals with high inflammatory burden are more likely to die earlier or remain too unwell for elective removal, which reduces their opportunity time to develop minor infections despite the competing-risk modeling framework. These covariate findings should therefore be interpreted as prognostic signals that warrant prospective confirmation.

Our study has several strengths. Multistate modeling was performed to account for competing risks to correct for potential confounders. We highlight the importance of multistate modeling in clinical research and demonstrate its use in the context of gastrostomy-related outcome analysis. Already introduced in 1978¹⁵⁶, the Aalen–Johansen estimator, the multistate counterpart of the Kaplan–Meier estimator, remains underused in medical research likely due to limited computational power and integration challenges. Available software packages^{152,157} aim to facilitate the integration in medical research, however, their use remains limited. Analyzing the long-term outcomes in a traditional way using Kaplan–Meier analysis reveals an apparent significant relation of the type of tube and tube dislodgment, infection, and death. However, the initially observed significant association was likely confounded by the sequence of competing events (e.g., infection → dislodgement → death). Another strength of our study is the timing of recruitment. Patients were recruited in two time periods, where patients in the earlier period solely received the pull placement because the push was not available. The later period recruited a similar cohort receiving the push placement since it became available at the time of recruitment. Since the push and pull technique is used for different indications, it is difficult to set up a prospective, randomized controlled study nowadays.

Our study has several limitations. First, the monocentric study design in a tertiary care referral center can potentially limit the generalizability of our results. However, the procedures were performed by a broad spectrum of operators with various levels of experience in endoscopy, including fellows in training, under direct supervision.

Additionally, external validation is typically required for prediction models intended to estimate individual risks in new populations. In contrast, our multistate model was designed as a descriptive and analytical tool to quantify differences in hazards between the two placements. As such, the focus was on estimating associations rather than generating transportable predictions, making external validation less applicable in this context. Second, our data were collected, and our outcome model was built for patients having gastrostomy in place. It may happen that patients had an infection after the removal of the tube, but these data were not collected. It was out of the scope and clinically less relevant to compare these outcomes in a setting where no tube is in place. Third, we combined the outcome of infection requiring local antibiotic (typically defined as minor adverse events), and infection requiring systemic antibiotic (typically defined as major adverse events) in one outcome to increase the sample size. The isolated outcomes were too rare to identify predictors and draw relevant conclusions. Furthermore, the follow-up period of 365 days may have been too short to observe additional major complications such as metastasis. While baseline nutritional (body mass index, albumin) and inflammatory status (C-reactive protein, albumin) were considered in the model (and albumin and C-reactive protein were retained as predictors of mortality) time-varying aspects of nutrition therapy (energy/protein prescriptions, delivered intake, and interruptions due to device issues) were not captured. Because device-related complications can transiently reduce delivered nutrition, these unmeasured factors may mediate associations between placement technique and clinical events. Prospective studies should record standardized intake and therapy metrics

alongside device outcomes and clinical endpoints. We did not quantify nutritional adequacy after placement (particularly in patients with partial oral intake) and tube-related complications may influence delivered nutrition. This warrants prospective evaluation.

5. Conclusion

This is the first long-term outcome analysis in a cohort undergoing percutaneous endoscopic gastrostomy tubes using a state-of-the-art multistate modeling approach. We demonstrate that the indication, low albumin, and high C-reactive protein, but not the use of push vs. pull technique, were associated with increased mortality. This emphasizes the importance of assessing the patient's inflammatory and nutritional status before the procedure. Furthermore, push and pull placement showed similar major complication rates. Finally, our analysis confirms the higher rate of tube dislodgment in the push placement.

Chapter 4 Supplementary Material

Supplementary Figures

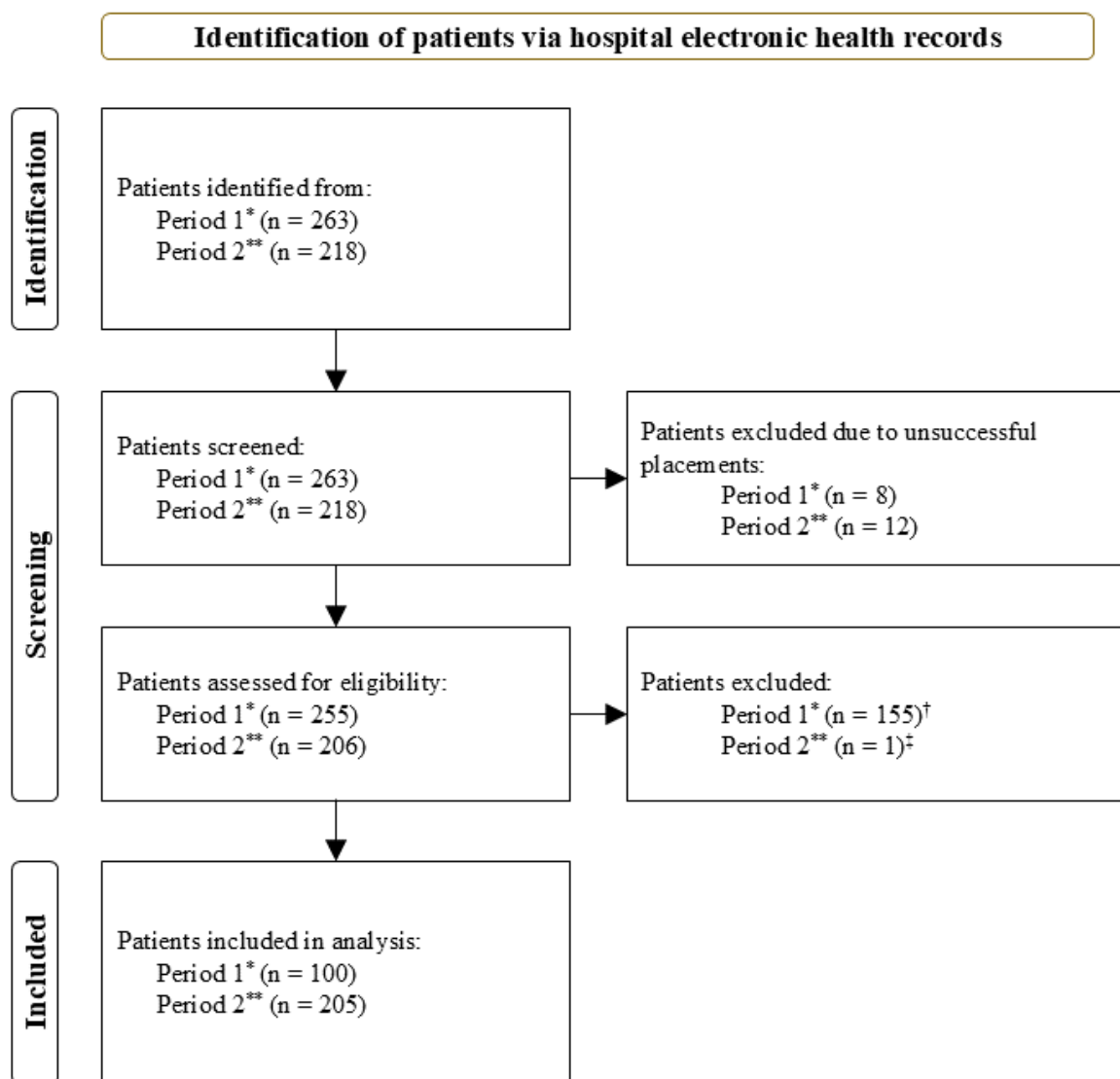


Figure S21. Flowchart depicting the inclusion and exclusion of the cohort.

* January 2014 to December 2015: pull technique

** January 2017 to December 2021: push technique

[†]No head and neck cancer

[‡]Was removed due to model fitting issues

Source: Page MJ, et al. BMJ 2021;372:n71. doi: 10.1136/bmj.n71.

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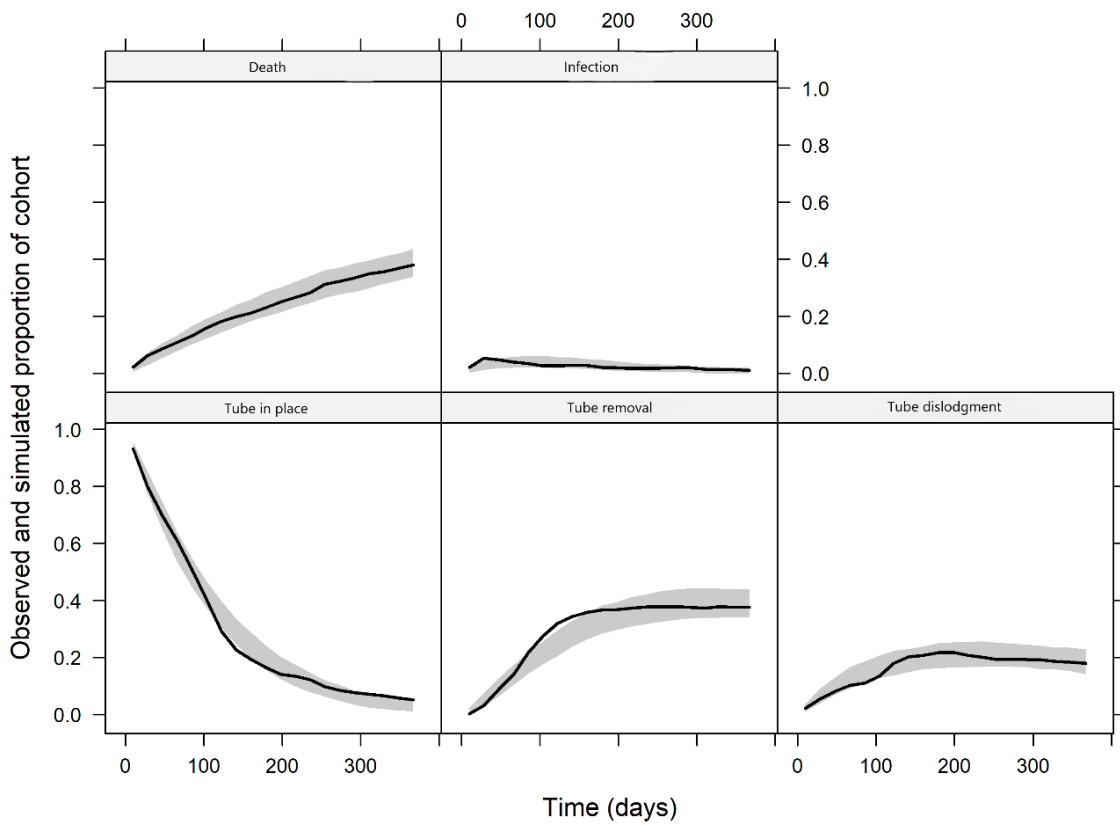
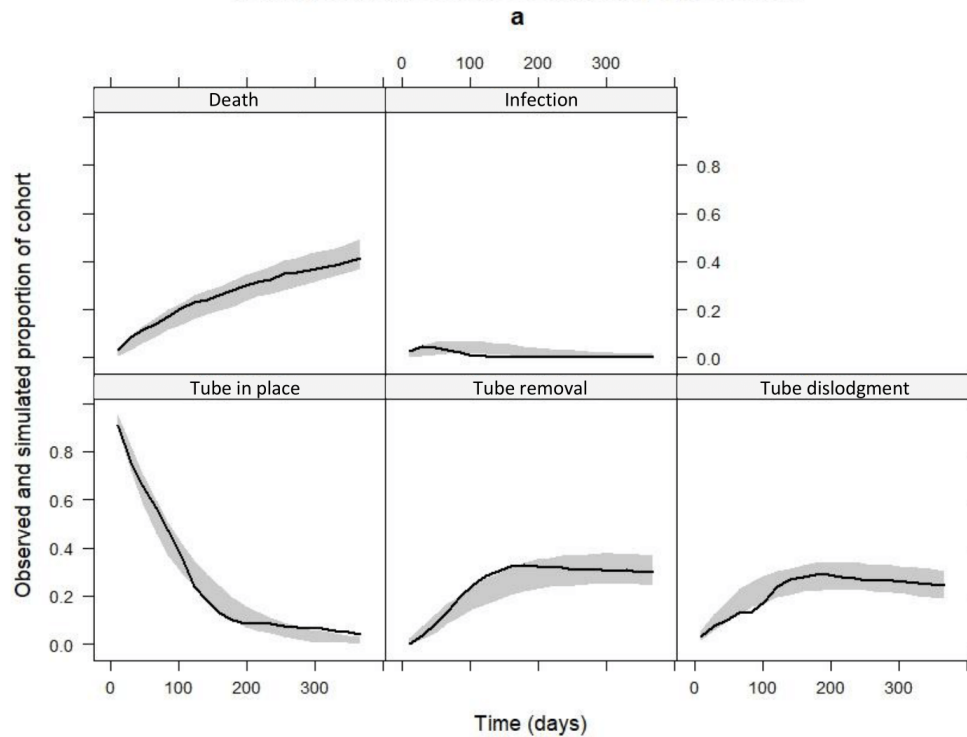


Figure S22. Visual predictive checks of the multistate time-to-event model. Each panel corresponds to one of the five distinct states/outcomes of interest. The black solid line shows the observed proportion of the cohort and the grey shaded area shows the 95% confidence interval of the model prediction.

Visual predictive check stratified by type of tube

page 1 of 2



Visual predictive check stratified by type of tube

page 2 of 2

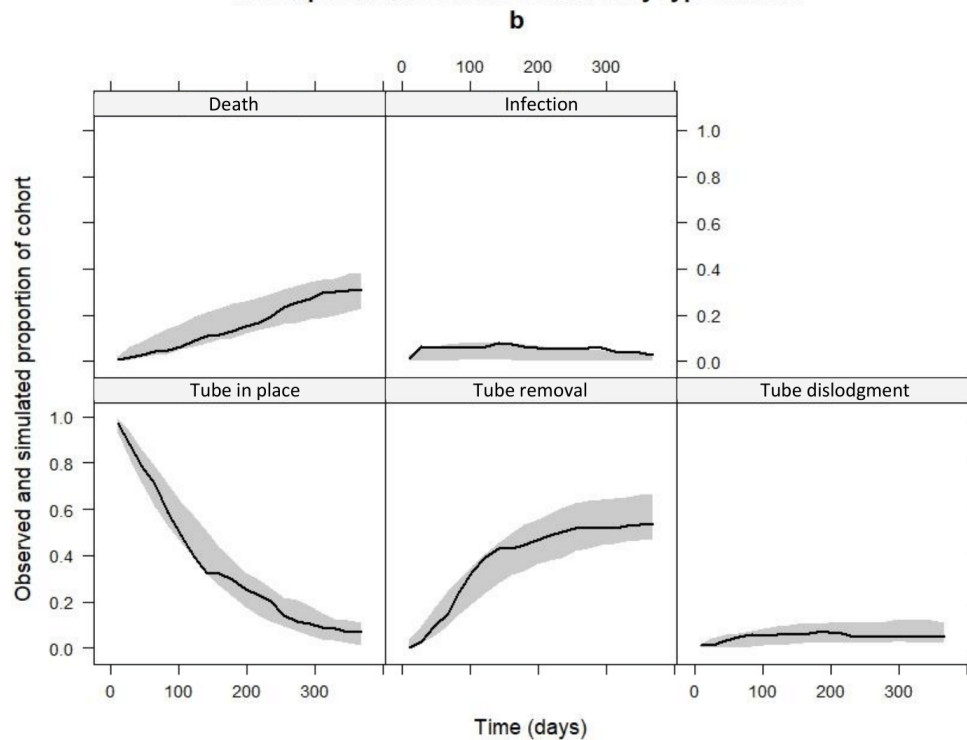


Figure S23. Visual predictive check of the final multistate model. The five panels correspond to the five distinct states. The black solid line shows the observed proportion of the cohort and the grey shaded area shows the 95% confidence interval of the model prediction. The plot is stratified by type of tube where panel a corresponds to the *push*-placement and panel b corresponds to the *pull*-placement.

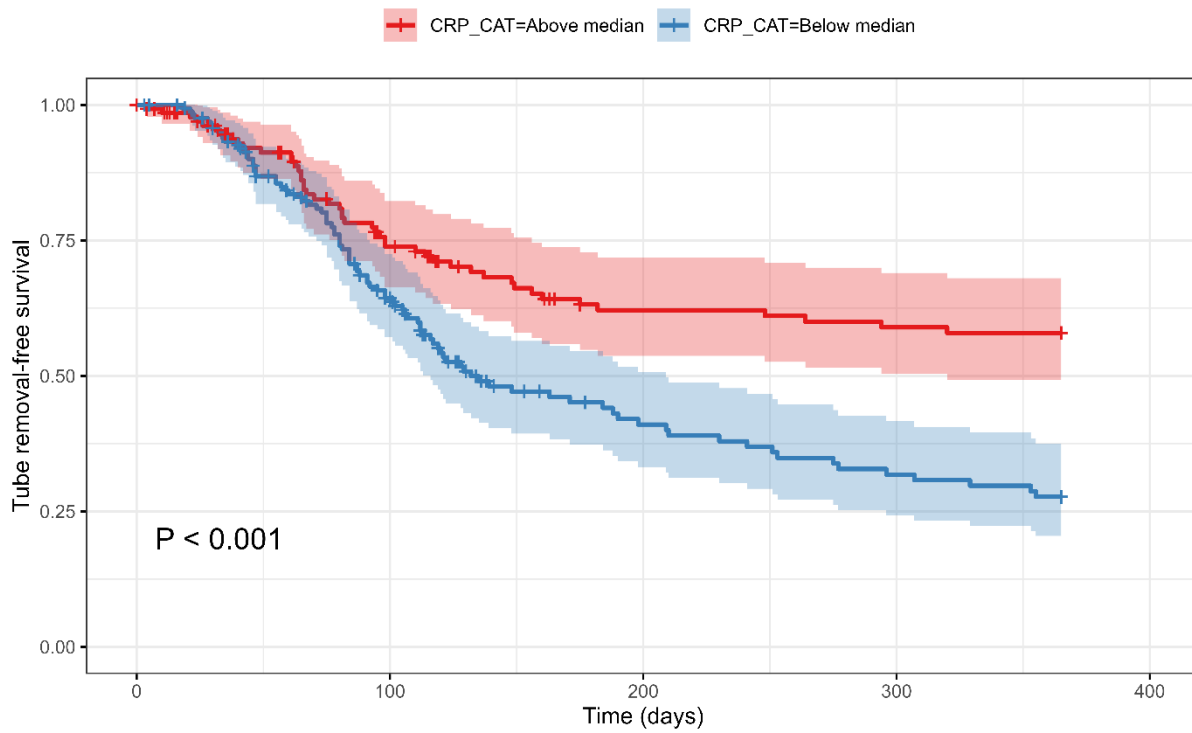


Figure S24. Kaplan Meier curve for tube removal-free survival. The cohort was stratified by c-reactive protein above and below the population median. P values were based on a log-rank test. Abbreviations. CRP_CAT, C-reactive protein category.

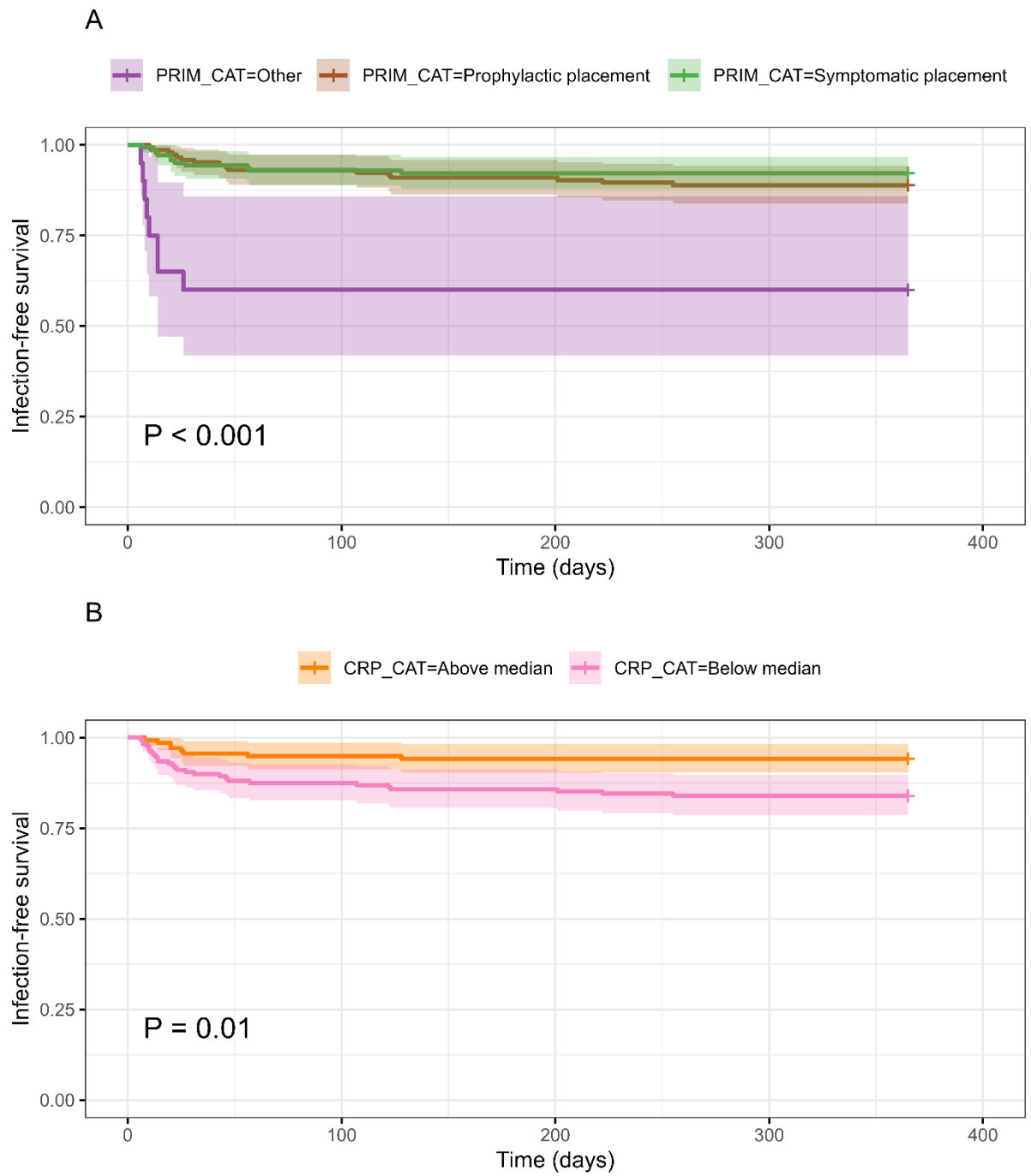


Figure S25. Kaplan Meier curve for tube infection-free survival. The cohort was stratified by primary indication (A) and by c-reactive protein above and below the population median (B). All P values were based on a log-rank test.

Abbreviations. CRP_CAT, c-reactive protein category; PRIM_CAT, primary indication category.

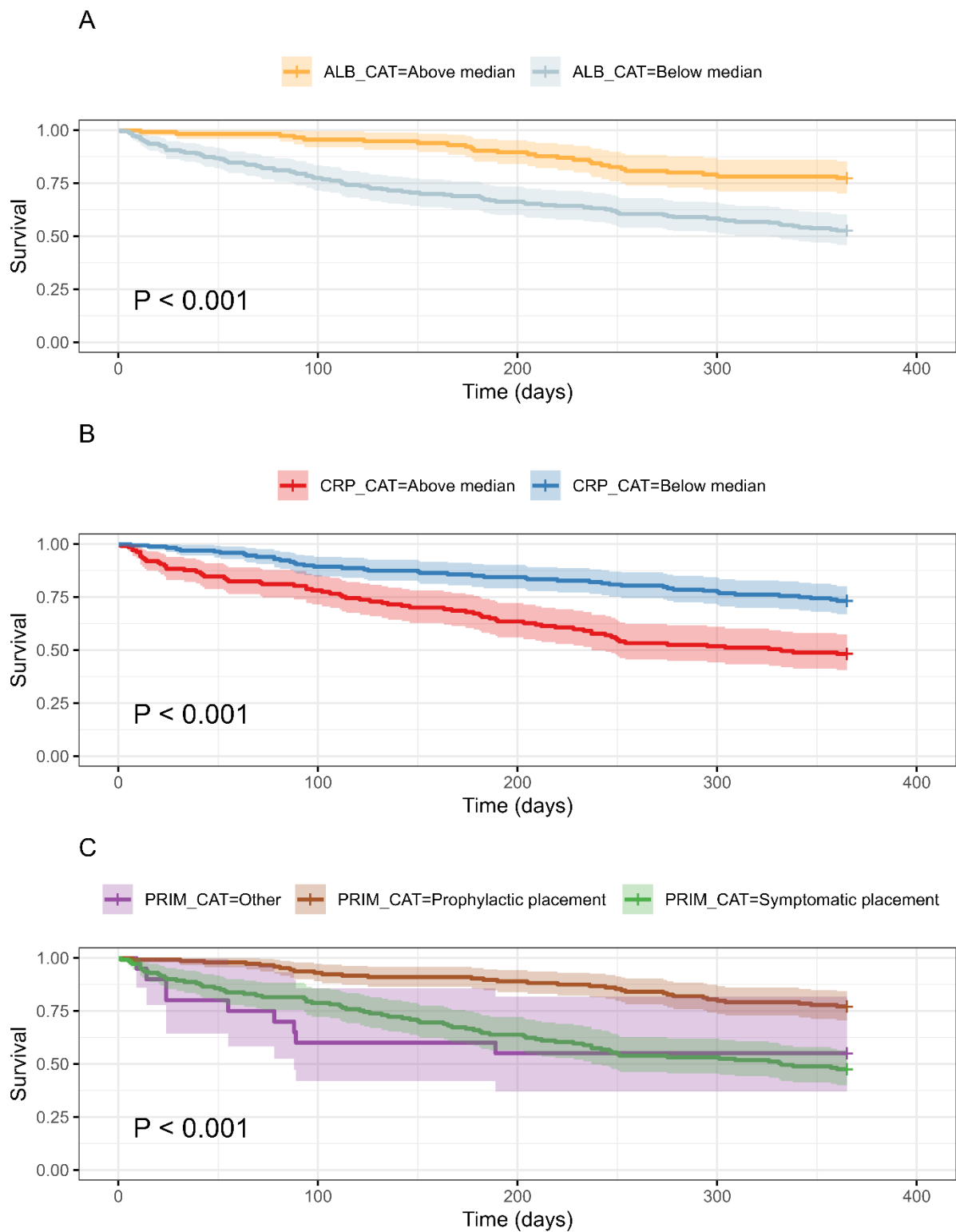


Figure S26. Kaplan Meier curve for survival. The cohort was stratified by albumin above and below the population median (A), by c-reactive protein above and below the population median (B) and by primary indication (C). All P values were based on a log-rank test. Abbreviation. ALB_CAT, albumin category; CRP_CAT, c-reactive protein category; PRIM_CAT, primary indication category.

Table S13. Early outcome rates for pull PEG and push PEG cohort

Outcome, n (%)	Pull placement cohort (n=100)	Push placement cohort (n=205)
Peristomal infection requiring antibiotics		
Day 7	0 (0)	3 (1.5)
Day 30	9 (9.0)	13 (6.3)
Day 60	11 (11.0)	17 (8.3)
Day 365	17 (17.0)	18 (8.8)
Planned tube removal for adequate oral intake		
Day 7	0 (0)	1 (0.5)
Day 30	3 (3.0)	9 (4.4)
Day 60	12 (12.0)	25 (12.2)
Day 365	62 (62.0)	81 (39.5)
Tube dislodgment		
Day 7	1 (1.0)	5 (2.4)
Day 30	1 (1.0)	18 (8.8)
Day 60	4 (4.0)	31 (15.1)
Day 365	8 (8.0)	76 (37.1)
All-cause mortality		
Day 7	0 (0.0)	5 (2.4)
Day 30	2 (2.0)	18 (8.8)
Day 60	4 (4.0)	27 (13.2)
Day 365	31 (31.0)	86 (42.0)

Counts (n) and proportions (%) at day 7, day 30, day 90, and day 365 for peristomal infection requiring antibiotics, planned tube removal for adequate oral intake, tube dislodgement, and all-cause mortality, stratified by pull versus push placement.

Table S14: Overview of the model development. Every row shows a single step of the stepwise covariate modeling approach. The drop in the objective function value (dOFV) is relative to the reference model from the previous modeling step.

Model run	Reference	Description	OFV	dOFV	sign. level
5	–	Base model	25362.278	–	–
forward inclusion					
6	5	Base model + K12(TYPE)	25362.276	-0.002	NS
7	5	Base model + K13(TYPE)	25319.015	-43.263	<0.001
8	5	Base model + K14(TYPE)	25360.243	-2.035	NS
9	5	Base model + K15(TYPE)	25361.056	-1.222	NS
10	5	Base model + K24(TYPE)	25359.515	-2.763	NS
11	5	Base model + K34(TYPE)	25362.261	-0.017	NS
12	5	Base model + K52(TYPE)	25357.161	-5.117	<0.05
13	5	Base model + K53(TYPE)	25355.988	-6.29	<0.05
14	5	Base model + K54(TYPE)	25360.735	-1.543	NS
15	5	Base model + K12(SEX)	25361.202	-1.076	NS
16	5	Base model + K13(SEX)	25357.914	-4.364	<0.05
17	5	Base model + K14(SEX)	25360.853	-1.425	NS
18	5	Base model + K15(SEX)	25353.398	-8.88	<0.01
19	5	Base model + K24(SEX)	25361.973	-0.305	NS
20	5	Base model + K34(SEX)	25362.075	-0.203	NS
21	5	Base model + K52(SEX)	25360.93	-1.348	NS
22	5	Base model + K53(SEX)	25361.948	-0.33	NS
23	5	Base model + K54(SEX)	25359.369	-2.909	NS
24	5	Base model + K12(IND12)	25362.001	-0.277	NS
25	5	Base model + K13(IND12)	25351.917	-10.361	<0.01
26	5	Base model + K14(IND12)	25340.399	-21.879	<0.001
27	5	Base model + K15(IND12)	25343.556	-18.722	<0.001
28	5	Base model + K24(IND12)	25360.028	-2.25	NS
29	5	Base model + K34(IND12)	25353.099	-9.179	<0.01
30	5	Base model + K52(IND12)	25359.364	-2.914	NS
31	5	Base model + K53(IND12)	25360.469	-1.809	NS

32	5	Base model + K54(IND12)	25353.695	-8.583	<0.01
33	5	Base model + K12(CRP)	25352.145	-10.133	<0.01
34	5	Base model + K13(CRP)	25360.014	-2,264	<0.001
35	5	Base model + K14(CRP)	25343.782	-18.496	<0.001
36	5	Base model + K15(CRP)	25353.017	-9.261	<0.01
37	5	Base model + K24(CRP)	25356.905	-5.373	<0.05
38	5	Base model + K34(CRP)	25358.801	-3.477	NS
39	5	Base model + K52(CRP)	25359.819	-2.459	NS
40	5	Base model + K53(CRP)	25361.118	-1.16	NS
41	5	Base model + K54(CRP)	25358.871	-3.407	NS
42	5	Base model + K12(ALB)	25359.664	-2.614	NS
43	5	Base model + K13(ALB)	25357.055	-5.223	<0.05
44	5	Base model + K14(ALB)	25347.093	-15.185	<0.001
45	5	Base model + K15(ALB)	25362.278	0	NS
46	5	Base model + K24(ALB)	25353.66	-8.618	<0.01
47	5	Base model + K34(ALB)	25357.633	-4.645	<0.05
48	5	Base model + K52(ALB)	25355.677	-6.601	<0.05
49	5	Base model + K53(ALB)	25362.214	-0.064	NS
50	5	Base model + K54(ALB)	25356.999	-5.279	<0.05
51	5	Base model + K12(AGE)	25356.365	-5.913	<0.05
52	5	Base model + K13(AGE)	25362.263	-0.015	NS
53	5	Base model + K14(AGE)	25359.246	-3.032	NS
54	5	Base model + K15(AGE)	25356.003	-6.275	<0.05
55	5	Base model + K24(AGE)	25356.924	-5.354	<0.05
56	5	Base model + K34(AGE)	25350.985	-11.293	<0.001
57	5	Base model + K52(AGE)	25361.566	-0.712	NS
58	5	Base model + K53(AGE)	25359.377	-2.901	NS
59	5	Base model + K54(AGE)	25361.001	-1.277	NS
60	5	Base model + K12(BMI)	25360.758	-1.52	NS
61	5	Base model + K13(BMI)	25358.383	-3.895	<0.05

62	5	Base model + K14(BMI)	25360.014	-2.264	NS
63	5	Base model + K15(BMI)	25362.278	0	NS
64	5	Base model + K24(BMI)	25358.429	-3.849	<0.05
65	5	Base model + K34(BMI)	25360.779	-1.499	NS
66	5	Base model + K52(BMI)	25360.017	-2.261	NS
67	5	Base model + K53(BMI)	25362.264	-0.014	NS
68	5	Base model + K54(BMI)	25361.342	-0.936	NS
69	5	Base model + K12(HOSP)	25361.814	-0.464	NS
70	5	Base model + K13(HOSP)	25361.866	-0.412	NS
71	5	Base model + K14(HOSP)	25361.725	-0.553	NS
72	5	Base model + K15(HOSP)	25362.264	-0.014	NS
73	5	Base model + K24(HOSP)	25362.216	-0.062	NS
74	5	Base model + K34(HOSP)	25356.378	-5.9	<0.05
75	5	Base model + K52(HOSP)	25361.943	-0.335	NS
76	5	Base model + K53(HOSP)	25361.792	-0.486	NS
77	5	Base model + K54(HOSP)	25358.953	-3.325	NS
78	7	Base model + K13(TYPE) + K14(IND12)	25297.144	-21.871	<0.001
79	7	Base model + K13(TYPE) + K15(IND12)	25300.307	-18.708	<0.001
80	7	Base model + K13(TYPE) + K14(CRP)	25300.522	-18.493	<0.001
81	7	Base model + K13(TYPE) + K14(ALB)	25303.829	-15.186	<0.001
82	7	Base model + K13(TYPE) + K34(AGE)	25315.985	-3.03	NS
83	7	Base model + K13(TYPE) + K13(IND12)	25317.638	-1.377	NS
84	7	Base model + K13(TYPE) + K12(CRP)	25308.881	-10.134	<0.01
85	7	Base model + K13(TYPE) + K15(CRP)	25309.752	-9.263	<0.01
86	7	Base model + K13(TYPE) + K34(IND12)	25309.83	-9.185	<0.01
87	7	Base model + K13(TYPE) + K15(SEX)	25310.143	-8.872	<0.01
88	7	Base model + K13(TYPE) + K24(ALB)	25310.396	-8.619	<0.01
89	7	Base model + K13(TYPE) + K54(IND12)	25310.432	-8.583	<0.01
90	7	Base model + K13(TYPE) + K52(ALB)	NC		
91	7	Base model + K13(TYPE) + K53(TYPE)	25312.792	-6.223	<0.05

92	7	Base model + K13(TYPE) + K15(AGE)	25312.727	-6.288	<0.05
93	7	Base model + K13(TYPE) + K12(AGE)	25313.102	-5.913	<0.05
94	7	Base model + K13(TYPE) + K34(HOSP)	25313.112	-5.903	<0.05
95	7	Base model + K13(TYPE) + K24(CRP)	25313.642	-5.373	<0.05
95	7	Base model + K13(TYPE) + K24(CRP)	25313.642	-5.373	<0.05
96	7	Base model + K13(TYPE) + K24(AGE)	25313.661	-5.354	<0.05
97	7	Base model + K13(TYPE) + K54(ALB)	25313.736	-5.279	<0.05
98	7	Base model + K13(TYPE) + K13(ALB)	25317.878	-1.137	NS
99	7	Base model + K13(TYPE) + K52(TYPE)	25313.898	-5.117	<0.05
100	7	Base model + K13(TYPE) + K13(SEX)	25316.903	-2.112	NS
101	7	Base model + K13(TYPE) + K13(BMI)	25316.881	-2.134	NS
102	7	Base model + K13(TYPE) + K24(BMI)	25315.166	-3.849	<0.05
103	78	Base model + K13(TYPE) + K14(IND12) + K15(IND12)	25278.455	-18.689	<0.001
104	78	Base model + K13(TYPE) + K14(IND12, CRP)	25288.058	-9.086	<0.01
105	78	Base model + K13(TYPE) + K14(IND12, ALB)	25292.608	-4.536	<0.05
106	78	Base model + K13(TYPE) + K14(IND12) + K12(CRP)	25287.003	-10.141	<0.01
107	78	Base model + K13(TYPE) + K14(IND12) + K15(CRP)	25287.872	-9.272	<0.01
108	78	Base model + K13(TYPE) + K14(IND12) + K34(IND_merge)	25296.644	-0.5	NS
109	78	Base model + K13(TYPE) + K14(IND12) + K15(SEX)	25288.275	-8.869	<0.01
110	78	Base model + K13(TYPE) + K14(IND12) + K24(ALB)	25288.541	-8.603	<0.01
111	78	Base model + K13(TYPE) + K14(IND12) + K54(IND_merge)	25294.884	-2.26	NS
112	78	Base model + K13(TYPE) + K14(IND12) + K53(TYPE)	25290.921	-6.223	<0.05
113	78	Base model + K13(TYPE) + K14(IND12) + K15(AGE)	25290.851	-6.293	<0.05
114	78	Base model + K13(TYPE) + K14(IND12) + K12(AGE)	25291.226	-5.918	<0.05
115	78	Base model + K13(TYPE) + K14(IND12) + K34(HOSP)	25291.257	-5.887	<0.05
116	78	Base model + K13(TYPE) + K14(IND12) + K24(CRP)	25291.785	-5.359	<0.05
117	78	Base model + K13(TYPE) + K14(IND12) + K24(AGE)	25291.797	-5.347	<0.05
118	78	Base model + K13(TYPE) + K14(IND12) + K54(ALB)	25291.881	-5.263	<0.05
119	78	Base model + K13(TYPE) + K14(IND12) + K52(TYPE)	25292.027	-5.117	<0.05
120	78	Base model + K13(TYPE) + K14(IND12) + K24(BMI)	25293.302	-3.842	<0.05

121	78	Base model + K13(TYPE) + K14(IND12) + K15(IND_merge)	25278.508	-18.636	<0.001
122	121	Base model + K13(TYPE) + K14(IND12, CRP) + K15(IND_merge)	25269.427	-9.081	<0.01
123	121	Base model + K13(TYPE) + K14(IND12, ALB) + K15(IND_merge)	25273.977	-4.531	<0.05
124	121	Base model + K13(TYPE) + K14(IND12) + K15(IND_merge) + K12(CRP)	25268.373	-10.135	<0.01
125	121	Base model + K13(TYPE) + K14(IND12) + K15(IND_merge, CRP)	25269.174	-9.334	<0.01
126	121	Base model + K13(TYPE) + K14(IND12) + K15(IND_merge, SEX)	25276.12	-2.388	NS
127	121	Base model + K13(TYPE) + K14(IND12) + K15(IND_merge) + K24(ALB)	25269.906	-8.602	<0.01
128	121	Base model + K13(TYPE) + K14(IND12) + K15(IND_merge) + K53(TYPE)	25272.277	-6.231	<0.05
129	121	Base model + K13(TYPE) + K14(IND12) + K15(IND_merge, AGE)	25274.957	-3.551	NS
130	121	Base model + K13(TYPE) + K14(IND12) + K15(IND_merge) + K12(AGE)	25272.605	-5.903	<0.05
131	124	Base model + K13(TYPE) + K14(IND12) + K15(IND_merge, CRP) + K12(CRP)	25259.077	-9.296	<0.01
132	124	Base model + K13(TYPE) + K14(IND12, CRP) + K15(IND_merge) + K12(CRP)	25259.275	-9.098	<0.01
133	124	Base model + K13(TYPE) + K14(IND12, ALB) + K15(IND_merge) + K12(CRP)	25263.84	-4.533	<0.05
134	124	Base model + K13(TYPE) + K14(IND12) + K15(IND_merge) + K12(CRP) + K24(ALB)	25259.774	-8.599	<0.01
135	124	Base model + K13(TYPE) + K14(IND12) + K15(IND_merge) + K12(CRP) + K53(TYPE)	25262.141	-6.232	<0.05
136	124	Base model + K13(TYPE) + K14(IND12) + K15(IND_merge) + K12(CRP, AGE)	25264.925	-3.448	NS
137	131	Base model + K13(TYPE) + K14(IND12, CRP) + K15(IND_merge, CRP) + K12(CRP)	25249.963	-9.114	<0.01
138	131	Base model + K13(TYPE) + K14(IND12, ALB) + K15(IND_merge, CRP) + K12(CRP)	25254.542	-4.535	<0.05
139	131	Base model + K13(TYPE) + K14(IND12) + K15(IND_merge, CRP) + K12(CRP) + K24(ALB)	25250.478	-8.599	<0.01
140	131	Base model + K13(TYPE) + K14(IND12) + K15(IND_merge, CRP) + K12(CRP) + K53(TYPE)	25252.852	-6.225	<0.05
141	137	Base model + K13(TYPE) + K14(IND12, CRP) + K15(IND_merge, CRP) + K12(CRP) + K24(ALB)	25241.36	-8.603	<0.01
142	137	Base model + K13(TYPE) + K14(IND12, CRP, ALB) + K15(IND_merge, CRP) + K12(CRP)	25248.081	-1.882	NS
143	137	Base model + K13(TYPE) + K14(IND12, CRP) + K15(IND_merge, CRP) + K12(CRP) + K53(TYPE)	25243.739	-6.224	<0.05
144	141	Base model + K13(TYPE) + K14(IND12, CRP) + K15(IND_merge, CRP) + K12(CRP) + K24(ALB) + K53(TYPE)	25235.136	-6.224	<0.05
backward elimination					
146	141	Base model + K14(IND12, CRP) + K15(IND_merge, CRP) + K12(CRP) + K24(ALB)	25284.608	43.248	<0.001
147	141	Base model + K13(TYPE) + K14(CRP) + K15(IND_merge, CRP) + K12(CRP) + K24(CRP)	25253.814	12.454	<0.001
148	141	Base model + K13(TYPE) + K14(IND12) + K15(IND(CRP) + K12(CRP) + K24(ALB)	25250.478	9.118	<0.01
149	141	Base model + K13(TYPE) + K14(IND12, CRP) + K15(IND_merge, CRP) + K12(CRP) + K24(ALB)	25260.047	18.687	<0.001

150	141	Base model + K13(TYPE) + K14(IND12, CRP, ALB) + K15(IND_merge, CRP) + K12(CRP) + K24(ALB)	25250.672	9.312	<0.01
151	141	Base model + K13(TYPE) + K14(IND12, CRP) + K15(IND_merge, CRP) + K12(CRP) + K24(ALB)	25251.474	10.114	<0.01
152	141	Base model + K13(TYPE) + K14(IND12, CRP) + K15(IND_merge, CRP) + K12(CRP) + K24(ALB)	25249.963	8.603	<0.01

Abbreviations. AGE, age; ALB, albumin; CRP, c-reactive protein; dOFV, drop objective function value; HOSP, days of hospitalization after placement; IND12, indication; IND_merge, merged indication; K_{xy} , transition rate between two outcomes (1=tube in place, 2=tube removal, 3=tube dislodgment, 4=death, 5=infection); OFV, objective function value; NC, not converged; NS, not significant; SEX, sex; TYPE, type of tube.

Table S15. Overview of parameter estimates of the final multi-state time-to-event model.

Parameter	Base estimate (RSE%) OFV 25362.278	model Final model estimate (RSE%) OFV 25241.36	*Bootstrap estimate [95% CI]
<i>Structural model parameter</i>			
λ_{12}	236 (7)	242 (7)	240 [208–279]
λ_{13}	439 (15)	269 (13)	264 [206–352]
λ_{14}	563 (19)	216 (43)	216 [84–847]
λ_{15}	961 (17)	165 (36)	160 [51–501]
λ_{24}	1300 (20)	1320 (20)	1315 [906–2024]
λ_{34}	667 (19)	666 (19)	663 [467–1002]
λ_{52}	139 (22)	139 (22)	137 [85–211]
λ_{53}	396 (38)	396 (38)	396 [190–1089]
λ_{54}	463 (41)	464 (41)	467 [237–1411]
γ_{13}	0.961 (9)	1.01 (9)	1.03 [0.85–1.22]
γ_{12}	1.36 (7)	1.39 (7)	1.39 [1.24–1.56]
γ_{14}	1.02 (11)	1.05 (10)	1.06 [0.84–1.34]
<i>Covariate model parameter</i>			
c-reactive protein effect on K12	–	-0.194 (31)	-0.192 [-0.31– -0.06]
type of tube effect on K13	–	-2.11 (20)	-2.16 [-20.2– -1.39]
prophylactic tube placement effect on K14	–	-1.64 (32)	-1.68 [-2.95– -0.21]
symptomatic tube placement effect on K14	–	-0.776 (63)	-0.808 [-1.926–0.735]
c-reactive protein effect on K14	–	0.356 (33)	0.361 [0.150–0.642]
prophylactic and symptomatic tube placement effect on K15	–	-2.16 (19)	-2.19 [-3.39– -1.05]
c-reactive protein effect on K15	–	-0.357 (32)	-0.356 [-0.590– -0.130]
albumin effect on K24	–	-0.101 (34)	-0.100 [-0.160– -0.049]

*Based on 992/1,000 successful bootstrap runs.

Abbreviations. CI, confidence interval; OFV, objective function value; RSE, relative standard error.

Model code of the final multistate model

```
### start model code ###
$PROBLEM
;; 1. Based on: run141
;; 2. Description: Base model + K13(TYPE) + K14(IND12, CRP) + K15(IND_merge, CRP) +
K12(CRP) + K24(ALB)
;; x1. Author: Yannick Hoffert
$INPUT ID TIME          EVID CMT AMT DV SIM DUMM SIM_ONE DUMM_ONE DUMM_TWO CRP_BL ALB_BL
TYPE PRIM_IND_NUM SEX AGE HOSP BMI
$DATA msm_inf.csv IGNORE=@
$SUBROUTINES ADVAN13 TOL=6
$MODEL NCOMP=5
$PK
BSV = ETA(1) ; place holder

IF(NEWIND.NE.2) DV1 = 1
IF(NEWIND.NE.2) XDV = 1
IF(NEWIND.NE.2) PDV = 1

;Sim_start: add for simulation
IF(DV.GE.1) DV1 = DV
IF(DV.GE.1) XDV = DV
;Sim_end

OTYPE=TYPE

OIND1=0
OIND2=0
IF(PRIM_IND_NUM.EQ.1) OIND1 = 1
IF(PRIM_IND_NUM.EQ.2) OIND2 = 1

OIND=0
IF(PRIM_IND_NUM.LE.2) OIND = 1 ; ENT-SYMP, ENT-PROF for K15
; IF(PRIM_IND_NUM.GE.3) OIND = 1 ; anything else for K15

OCRP=CRP_BL

OALB=ALB_BL

;;-----add covariate prospectively-----
;;-----mean transit times in days-----
MTT12=THETA(1)
MTT13=THETA(2)
MTT14=THETA(3)
MTT15=THETA(4)

MTT24=THETA(5)
MTT34=THETA(6)
MTT52=THETA(7)
MTT53=THETA(8)
MTT54=THETA(9)

;;-----Weibull parameters-----
HZL13=1/MTT13
HZA13=THETA(10)

HZL12=1/MTT12
HZA12=THETA(11)

HZL14=1/MTT14
HZA14=THETA(12)

; HZL15=1/MTT15
; HZA15=THETA(13)
;;-----Initialize location probabilities-----
IF(NEWIND.NE.2) THEN
BIO1 = 1
```

```

BIO2 = 0
BIO3 = 0
BIO4 = 0
BIO5 = 0

ENDIF
;-----
;----- "Bioavailability" factors -----
F1 = BIO1
F2 = BIO2
F3 = BIO3
F4 = BIO4
F5 = BIO5

$DES
DEL = 1E-16
WEIBULL12 = HZL12*HZA12*(HZL12*(T+DEL))**(HZA12-1)
WEIBULL13 = HZL13*HZA13*(HZL13*(T+DEL))**(HZA13-1)
WEIBULL14 = HZL14*HZA14*(HZL14*(T+DEL))**(HZA14-1)
; WEIBULL15 = HZL15*HZA15*(HZL15*(T+DEL))**(HZA15-1)
;-----probability transfer constants-----
K12=WEIBULL12*EXP(THETA(13)*(OCRP-2.565))
K13=WEIBULL13*EXP(THETA(14)*OTYPE)
K14=WEIBULL14*EXP(THETA(15)*OIND1 + THETA(16)*OIND2)*EXP(THETA(17)*(OCRP-2.565))
; K15=WEIBULL15
K15=1/MTT15*EXP(THETA(18)*OIND)*EXP(THETA(19)*(OCRP-2.565))

K24=1/MTT24*EXP(THETA(20)*(OALB-38))
K34=1/MTT34
K52=1/MTT52
K53=1/MTT53
K54=1/MTT54

;--Differential equations for the probability of each score
DADT(1) = - A(1) * (K12 + K13 + K14 + K15) ; ongoing tube treatment
DADT(2) = A(1) * K12 - A(2) * K24 + A(5) * K52 ; removal of tube
DADT(3) = A(1) * K13 - A(3) * K34 + A(5) * K53 ; tube
dislodgement
DADT(4) = A(1) * K14 + A(2) * K24 + A(3) * K34 + A(5) * K54 ; death
DADT(5) = A(1) * K15 - A(5) * K52 - A(5) * K54 - A(5) * K53 ; infection

$ERROR

P_1 = A(1) ; Probability of observing ongoing tube treatment
P_2 = A(2) ; Probability of observing removal of tube
P_3 = A(3) ; Probability of observing tube dislodgement
P_4 = A(4) ; Probability of observing death
P_5 = A(5) ; Probability of observing infection

; ; -----Cumulative probabilities -----
CUP1 = A(1) ; 1
CUP2 = A(1) + A(2) ; 1 + 2
CUP3 = A(1) + A(2) + A(3) ; 1 + 2 + 3
CUP4 = A(1) + A(2) + A(3) + A(4) ; 1 + 2 + 3 + 4
CUP5 = A(1) + A(2) + A(3) + A(4) + A(5) ; 1 + 2 + 3 + 4 + 5

;----- Model prediction -----

IF(DV.EQ.1.AND.EVID.EQ.0) Y = P_1
IF(DV.EQ.2.AND.EVID.EQ.0) Y = P_2
IF(DV.EQ.3.AND.EVID.EQ.0) Y = P_3
IF(DV.EQ.4.AND.EVID.EQ.0) Y = P_4
IF(DV.EQ.5.AND.EVID.EQ.0) Y = P_5

TOT = P_1 + P_2 + P_3 + P_4 + P_5
;-----

```

```

; Sim_start
; IF(ICALL.EQ.4) THEN
;
; IF (EVID.EQ.0) THEN
; CALL RANDOM (2,R)
; USUR=R
; ENDIF
;
; DV1=PDV
; IF(USUR.LE.CUP1.AND.EVID.EQ.0) DV1=1
; IF(USUR.GT.CUP1.AND.USUR.LE.CUP2.AND.EVID.EQ.0) DV1=2
; IF(USUR.GT.CUP2.AND.USUR.LE.CUP3.AND.EVID.EQ.0) DV1=3
; IF(USUR.GT.CUP3.AND.USUR.LE.CUP4.AND.EVID.EQ.0) DV1=4
; IF(USUR.GT.CUP4.AND.USUR.LE.CUP5.AND.EVID.EQ.0) DV1=5
;
; XDV = DV1
; IF(EVID.EQ.0) PDV=XDV
; ENDIF
;Sim_end
;----- Update Bioavailability factors -----
BIO1=0
BIO2=0
BIO3=0
BIO4=0
BIO5=0

IF(XDV.EQ.1) BIO1 = 1
IF(XDV.EQ.2) BIO2 = 1
IF(XDV.EQ.3) BIO3 = 1
IF(XDV.EQ.4) BIO4 = 1
IF(XDV.EQ.5) BIO5 = 1

$THETA
;;-----multi-state model parameters-----
(0,200) ;MTT12      1
(0,400) ;MTT13      2
(0,500) ;MTT14      3
(0,400) ;MTT15      4

(0,1000) ;MTT24      5
(0,600)           ;MTT34 6
(0,600)           ;MTT52 7
(0,600)           ;MTT53      8
(0,500)           ;MTT54      9

(0,2) ;Shape13      10
(0,2) ;Shape12      11
(0,2) ;Shape14      12

(-0.01)           ;CRP on K12  13
(-2) ;TYPE on K13   14
(-2) ;IND1 on K14   15
(-2) ;IND2 on K14   16
(-0.01)           ;CRP on K14  17
(-3) ;IND on K15    18
(-0.01)           ;CRP on K15  19
(-0.01)           ;ALB on K24  20

$OMEGA 0 FIX ; eta as a placeholder
; Sim_start
;$SIGMA 0 FIX
;$SIMULATION (12345) (123 UNIFORM) (123 UNIFORM) NSUB=1 ONLYSIM
$ESTIMATION METHOD=1 LIKE LAPLACE NUMERICAL SLOW PRINT=1 SIGL=9
MAXEVALS=9999 CTYPE=4 NSIG=2
$COVARIANCE MATRIX=R
;Sim_end

```

```

$TABLE ID DV1 XDV TIME AMT EVID CMT DV BIO1 BIO2 BIO3 BIO4 BIO5 F1 F2 F3 F4 F5
P_1 P_2 P_3 P_4 P_5 K12 K13 K14 K15 K24 K34 K52 K53 K54 Y TOT
CRP_BL ALB_BL TYPE PRIM_IND_NUM SEX AGE HOSP BMI NOPRINT FILE=sdtab00141
### end model code ###

# code to create a visual predictive check for a multistate time-to-event model
-samples=200 -flip_comments -dv=XDV -levels=1,2,3,4 -seed=1234 -rplots=2 -
auto_bin=auto -dir=vpc_run1 run141.mod

```


Systematic review of tacrolimus population pharmacokinetics models and a benchmark study of software tools

This chapter is based on

Yannick Hoffert, Nada Dia, Tim Vanuytsel, Robin Vos, Dirk Kuypers, Johan Van Cleemput, Jef Verbeek, Erwin Dreesen. Model-Informed Precision Dosing of Tacrolimus: A Systematic Review of Population Pharmacokinetic Models and a Benchmark Study of Software Tools. *Clin Pharmacokinet*. 2024 Oct. 63(10):1407-1421.

Abstract

Background and Objective Tacrolimus is an immunosuppressant commonly administered after solid organ transplantation. It is characterized by a narrow therapeutic window and high variability in exposure, demanding personalized dosing. In recent years, population pharmacokinetics models have been suggested to guide model-informed precision dosing of tacrolimus. We aimed to provide a comprehensive overview of population pharmacokinetics models and model-informed precision dosing software modules of tacrolimus in all solid organ transplant settings, including a simulation-based investigation of the impact of covariates on exposure and target attainment.

Methods We performed a systematic literature search to identify population pharmacokinetics models of tacrolimus in solid organ transplant recipients. We integrated selected population pharmacokinetics models into an interactive software tool that allows dosing simulations, Bayesian forecasting, and investigation of the impact of covariates on exposure and target attainment. We conducted a web survey amongst model-informed precision dosing software tool providers and benchmarked publicly available tools in terms of models, target populations, and clinical integration.

Results We identified eighty population pharmacokinetics models, including 44 one-compartment and 36 two-compartment models. The most frequently retained covariates on clearance and distribution parameters were cytochrome P450 3A5 polymorphisms and body weight, respectively. Our simulation tool, hosted at <https://lpmx.shinyapps.io/tacrolimus/>, allows thorough investigation of the impact of covariates on exposure and target attainment. We identified 15 model-informed precision dosing software tool providers, of which ten offer a tacrolimus solution and nine completed the survey.

Conclusion Our work provides a comprehensive overview of the landscape of available tacrolimus population pharmacokinetics models and model-informed precision dosing

software modules. Our simulation tool allows interactive, thorough exploration of covariates on exposure and target attainment.

1. Introduction

Tacrolimus is the indisputable cornerstone immunosuppressant in solid organ transplantation to avoid allograft rejection.¹⁵⁸ TDM of tacrolimus is ubiquitously used in routine clinical practice to account for inter- and intra-individual PK variability. Tacrolimus is characterized by a narrow therapeutic window.¹⁵⁹ Subtherapeutic exposure encompasses the risk of graft rejection, which is generally the main concern early after transplantation. Long-term supratherapeutic exposure increases the risk of adverse events, including nephrotoxicity, neurotoxicity, diabetogenesis, and malignant complications.^{160,161} To some extent, PK variability is explained by intrinsic patient characteristics such as cytochrome P450 (CYP) genotype, hematocrit, and serum protein levels, that influence clearance, and distribution of tacrolimus.¹⁶²

Finding the right dose that balances between subtherapeutic and supratherapeutic exposure resembles a tightrope walk. Precision dosing is an absolute requirement for drugs with a narrow therapeutic window and high PK variability.¹⁶³ Therefore, EMA recommended concentration-guided dose titration of tacrolimus in the label.¹⁵⁹ While recognizing the need for TDM of tacrolimus, the label does not recommend exposure targets nor provides a dosing algorithm to reach the exposure target. In recent years, popPK models have garnered interest. Besides quantitatively describing the variability in the dose–exposure relationships of drugs, these data-driven models are also the basis for MIPD. After taking blood samples and determining the drug concentration, Bayesian forecasting is used to deduce the individual patient's PK parameters and personalized, model-informed dosing recommendations can be derived. Hence, popPK models are a powerful tool to potentially improve TDM of tacrolimus.

Despite the lack of formal cost-effectiveness data, routine TDM of tacrolimus has improved therapeutic efficacy and safety.¹⁵⁹ Further there is increasing evidence, that MIPD can

significantly improve trough concentration target attainment.^{164–167} Since then, MIPD modules have been increasingly employed. Healthcare technology companies focusing on MIPD provide Bayesian forecasting software tools for PK parameter estimation and dose individualization. Tacrolimus modules are typically built upon a selected popPK model, with the possibility of applying it across a wide range of solid organ transplant settings. However, employing the right model, in the right way, for the right patient, at the right time, is crucial for a successful implementation of MIPD.¹⁶⁸ Therefore, the present work aims to provide a systematic overview of published popPK models of tacrolimus in all solid organ transplant settings, including a quantitative assessment of covariates and a simulation-based investigation of the impact of covariates on exposure and target attainment. Furthermore, software tool providers were contacted to benchmark publicly available tools in terms of the integrated popPK model, foreseen target population, and clinical integration of the tacrolimus MIPD module.

2. Methods

2.1 Population Pharmacokinetics Models Search and Data Extraction

A systematic literature search of PubMed and Web of Science was performed to identify research articles reporting on popPK models of tacrolimus in solid organ transplant recipients, published from inception until 15 January 2024. The query was {'tacrolimus[Title]' AND 'transplant*' AND 'pharmacokinetic*' AND 'population'} for PubMed and {tacrolimus pharmacokinetic model} for Web of Science. Models were considered for inclusion if PK parameters were estimated with a parametric approach. Models were excluded if they were built on (i) datasets exclusively including pediatric data, (ii) non-solid organ transplantation data, (iii) animal or *in vitro* data, or (iv) if no popPK approach was used.

Information on (i) the characteristics of publication (e.g., name of the author, year of publication), (ii) clinical trial/setting (e.g., number of patients included, amount of samples), and (iii) popPK model (e.g., model structure, covariates that influence PK parameters, typical values of PK parameters) was summarized in a database (**Supplementary Methods**). The search strategy and database were reviewed independently by two investigators (YH and ND). Data collected in the standardized database were further processed for analysis. Data cleaning and data exploration were conducted in R (version 4.3.0) and interfaced with RStudio (version 2023.03.1) using the *tidyverse* (version 2.0.0) collection of packages.¹⁵⁰

2.2 Software Provider Questionnaire

A 6-question web survey was designed and developed in Google Forms (**Supplementary Methods**). Providers of publicly available software tools for MIPD tools were contacted in February 2024 via contact details mentioned on the provider websites. A disclaimer in the accompanying email stated that by filling in the survey, the responder consented to the collection and temporary storage of responses. Responses were manually curated.

2.3 Software Development and Simulation Study

Based on the highest sample-to-patient ratio per graft type, models were selected to be included in a simulation software tool. Model code was reconstructed in R (version 4.3.0; R Foundation for Statistical Computing, R Core Team) with the RStudio integrated development environment (version 2023.09.1; RStudio, Inc.), and independently reviewed by two authors (YH and ND). To enable a responsive and flexible analysis, a web application was developed using the *Shiny* package (version 1.7.4) in R.

Concentration–time profiles and probability of target attainment (PTA) over time profiles were used to illustrate the simulated model output. The default simulation conditions in the software tool were set to 250 Monte Carlo simulations with a stepsize (resolution) of 0.25 hours, which generated a simulated data point every 15 minutes. Stochastic simulations are, by default, performed with random effects (interindividual variability) and without residual variability. The normal distribution of the interindividual variability parameters (η -distribution) for each PK parameter was used to illustrate the random effects distribution. Individual PK parameters were obtained by sampling 250 times (default) from the η -distribution. Simulations were performed with the open-source package *mrgsolve* (version 1.1.1)¹⁶⁹, and Bayesian forecasting was performed with the open-source package *mapbayr* (version 0.10.0).⁸⁵

A simulation study was performed to demonstrate how the software tool offers a thorough investigation of tacrolimus PK, including the impact of covariates on exposure and target attainment. The Liver model was used for demonstration purposes using the target range of 6–10 ng/ml.¹⁷⁰ Default exposure target ranges per graft type are summarized in **Table S16**. Adjustable sliders were integrated into the software tool to account for center-specific and protocol-specific exposure target ranges.

3. Results

3.1 Literature Review and Model Selection Process

In total, 873 papers were screened for eligibility. The results of the search strategy are summarized in the PRISMA flow diagram (**Figure 23**). Screening yielded 80 articles with one popPK model per article.^{164,166,171–248}

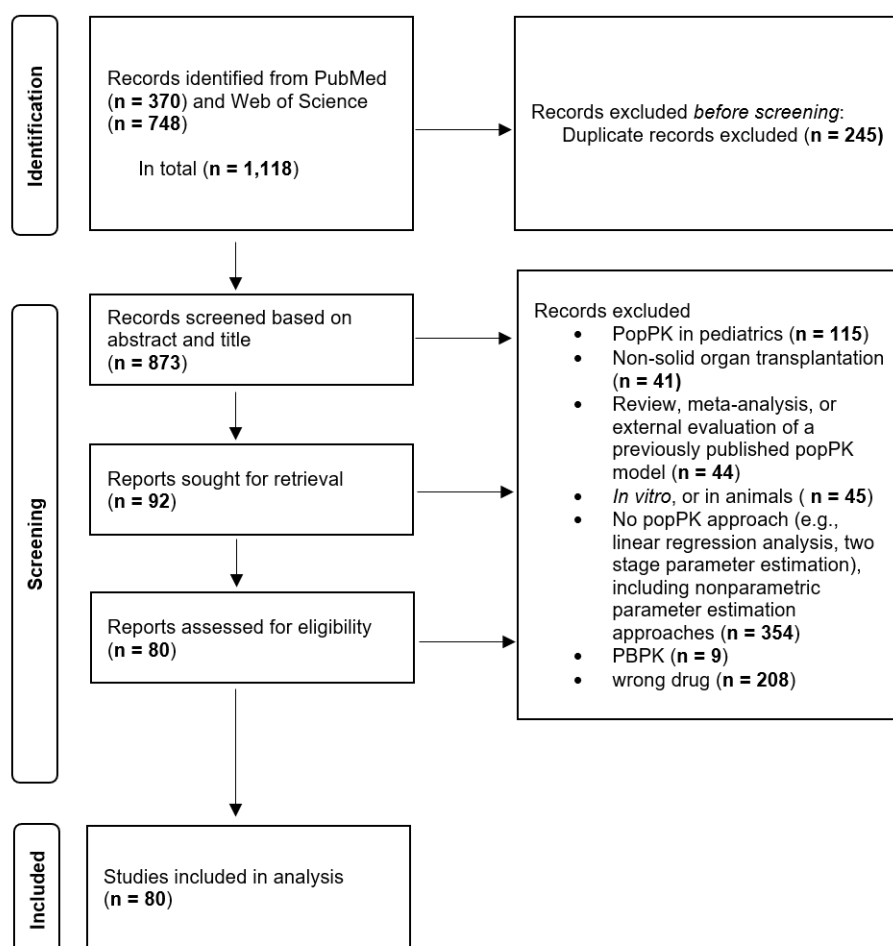


Figure 23. PRISMA flow diagram of the literature search and model selection process. PBPK, physiologically-based pharmacokinetics; PopPK, population pharmacokinetics.

The majority of the included models were built on studies from kidney transplant recipients (51.3%), followed by liver (32.5%), heart (6.3%), and lung (3.8%) transplant recipients. Two models were built on a mixed heart–lung transplant cohort (2.5%), one model was built on a mixed liver–kidney (1.3%) transplant cohort, one model was built on a mixed kidney–lung transplant cohort (1.3%), and one model was built on a mixed liver–kidney–heart transplant cohort (1.3%) (**Figure 24a**). No tacrolimus popPK model was built on data from bowel

transplant recipients. The number of yearly popPK model publications exponentially increased with time, with 12 out of 80 models (15.0%) published in 2022 (**Figure 24b**).

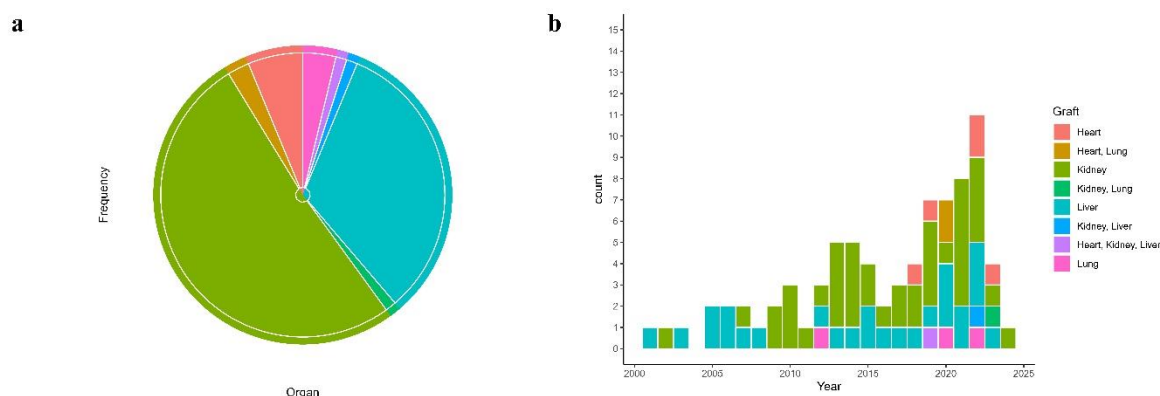


Figure 24. Overview of (a) the distribution of underlying transplanted graft type in the identified population pharmacokinetics (popPK) models, and (b) the number of tacrolimus popPK models published per year.

The clinical study designs underlying the identified popPK models varied greatly. The median number of participants in the studies was 77 [interquartile range, 49–143] (**Figure 25a**). The tacrolimus popPK model of Koomen *et al.*²⁴³ was built on the largest cohort, including 1,060 subjects undergoing kidney or lung transplantation. However, Andreu *et al.* demonstrated a successful popPK model development with only 16 patients.¹⁸⁰ The median number of samples that was used for model building was 824 [455–1742] (**Figure 25b**). With 23,176 tacrolimus concentration records, the popPK model of Lu *et al.* was built on the largest number of tacrolimus samples, yet only 408 participants contributed to this data.²¹⁷ Just over half of the models were one-compartment models (44/80; 55.0%) and the remaining models had a two-compartmental structure (**Figure 3c**). The median samples-to-patient ratio in the model-building datasets for one-compartment models was 11.5 and 8.9 for two-compartment models.

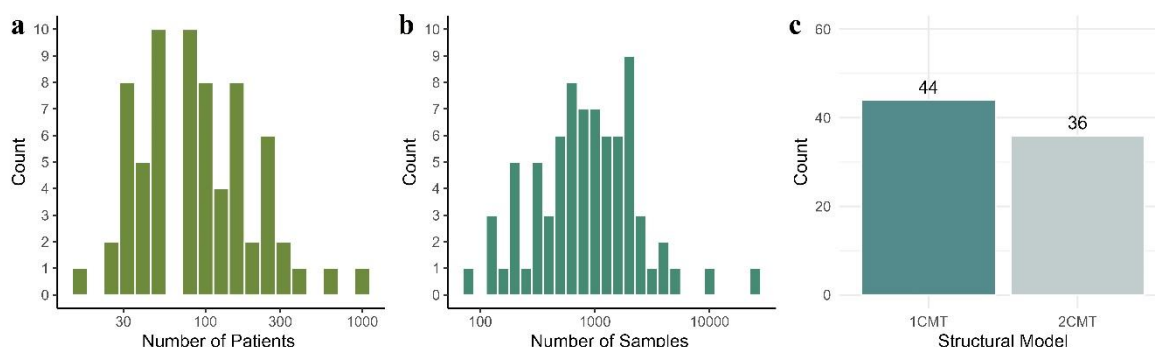


Figure 25. Distribution of included patients in the clinical trials used to develop the population pharmacokinetics (popPK) model (a), and number of samples taken (b). The bar chart represents the number of one-compartment and two-compartment models identified during the systematic literature review (c). Abbreviations. 1CMT, one-compartment; 2CMT, two-compartment.

In total, 50 covariates were identified to have a statistically significant impact on tacrolimus clearance, 21 covariates impacted the (central and/or peripheral) volume(s) of distribution, and one covariate impacted intercompartmental clearance. The most frequently *tested* covariates on clearance were age (69/80; 86.3%), body weight (68/80; 85.0%), and hematocrit (63/80; 78.8%). The most frequently *tested* covariates on volumes of distribution were body weight (68/80; 85.0%), hematocrit (63/80, 78.8%), and sex (61/80; 76.3%). The most frequently *retained* covariates on clearance were CYP3A5 genotype of the transplant recipient (39/80; 48.8%), postoperative day (26/80; 32.5%), hematocrit (26/80; 32.5%), and body weight (11/80; 13.8%). The most frequent *retained* covariates on volumes of distribution were body weight (11/80; 13.8%), postoperative day (5/80; 6.3%), lean body weight (including fat-free mass) (5/80; 6.3%), and hematocrit (3/80; 3.8%).²⁴⁹ **Figure 26a** and **Figure 26b** provide a visual representation of the retained and tested covariates per PK parameter.

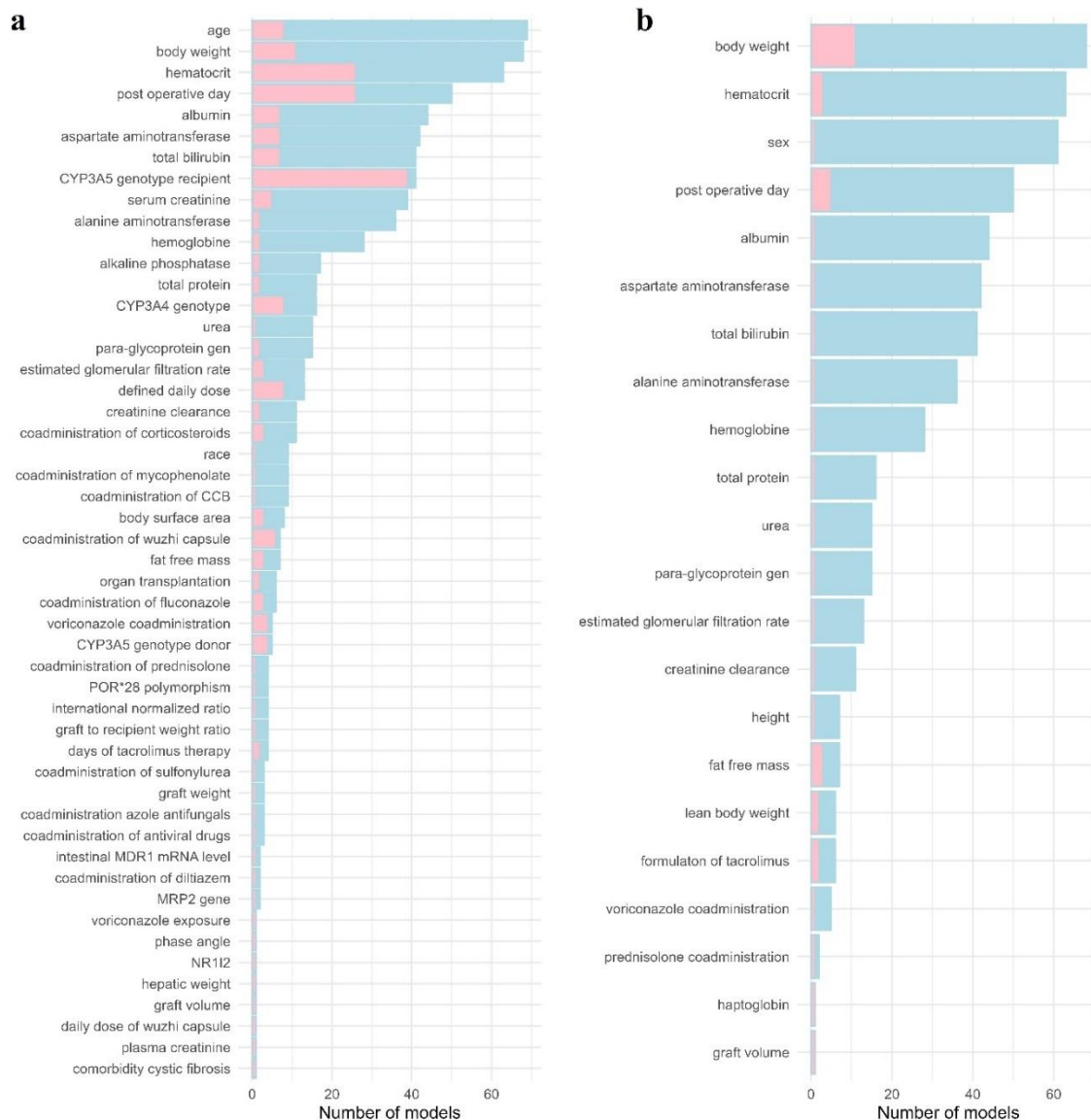


Figure 26. The bar charts show covariates that were tested and retained in the final model with influence on clearance (a) and volume of distribution (central and/or peripheral) (b), respectively. The blue bars represent the total number of models in which the covariates were tested. The pink bars indicate the number of times the covariates were retained in the final models. Abbreviations. CCB, calcium channel blocker; CL, clearance; CYP3A4, cytochrome P450 3A4; CYP3A5, cytochrome P450 3A5; MDR1, multidrug resistance protein 1; mRNA, messenger ribonucleic acid; MRP2, multidrug resistance associated protein 2; POR28, P450 oxidoreductase 28 variant; NR112, nuclear receptor subfamily 1 group I member 2; V, volume of distribution.

3.2 Characteristics of Publicly Available Software Modules For MIPD of

Tacrolimus

We identified 15 software tool providers, of which ten offer a tacrolimus module (Rx Studio, PrecisePK, InsightRx Nova, MWPharm, Tuxuci, DosemeRx, JPKD, NextDose, BestDose, ISBA) (**Figure 27**). Nine provided information on their tacrolimus model, target population, clinical use, and regulatory aspects (**Table 9**). The remaining software tools focus on other disease areas or are currently in the process of tacrolimus module integration (TDMx). RxStudio, InsightRx Nova, MWPharm, Tucuxi, and ISBA include more than one tacrolimus popPK model, accounting for adult and pediatric populations or different types of underlying transplantation/disease. The InsightRx Nova software tool covers a broad range of transplantation types in the tacrolimus landscape and includes kidney, hematopoietic stem cell transplantation, and liver transplantation cohorts. ISBA from CHU Limoges integrated various transplantation types including kidney, liver, and lung recipients, and several unpublished models. Similarly, MWPharm covers kidney, lung, liver, heart, and patients with systemic lupus erythematosus, all treated with tacrolimus.

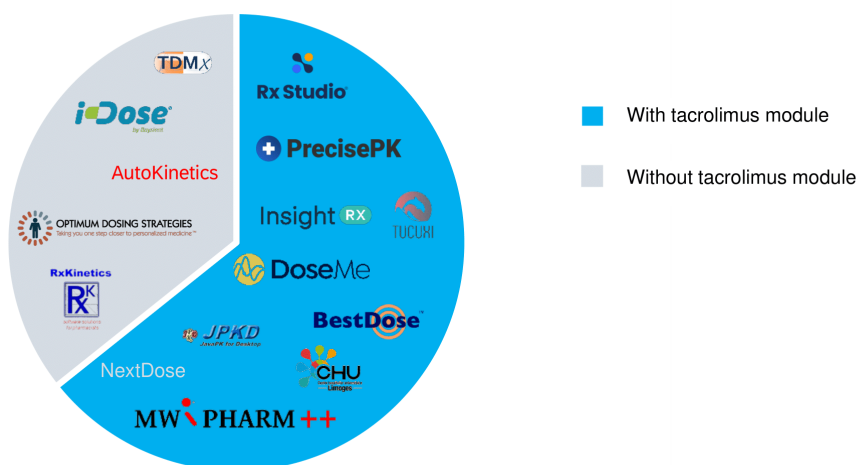


Figure 27. Overview of the fifteen software tool providers that were contacted. Ten of them have a tacrolimus module integrated.

Nine software tools are currently in clinical use (RxStudio, PrecisePK, InsightRx Nova, MWPharm, DosemeRX, JPKD, Bestdose, and ISBA) to guide tacrolimus dosing. The

estimated number of patients that are dose individualized varied, between less than 100 patients per year up to more than 1,000 patients per year (MWPharm, and ISBA) as reported by the software tool providers. PrecisePK, MWPharm, DosemeRX, and BestDose reported to have performed a prospective validation of the tacrolimus module prior to integration in clinical routine. A CE-marking, displaying conformity with EU regulations, was obtained by PrecisePK, InsightRx Nova, MWPharm, and DoseMeRX.

Table 9. Characteristics of publicly available software modules for model-informed precision dosing of tacrolimus

Software tool	Population	Transplantation type/disease	Model(s)
Rx Studio	Adult	Liver	Oteo <i>et al</i> ²²⁹
	Adult	Liver	Riff <i>et al</i> ²⁵⁰
	Adult	Kidney, Liver	Woillard <i>et al</i> ²⁵¹
PrecisePK	Adult	Kidney	–
InsightRx Nova	Pediatric	Kidney	Andrews <i>et al</i> ²⁵²
	Adult	Kidney	Andrews <i>et al</i> ¹⁷³
	Adult, Pediatric	HSCT	Brooks <i>et al</i> ²⁵³
	Adult	HSCT	Jacobson <i>et al</i> ²⁵⁴
	Adult	Liver	Martial <i>et al</i> ²¹⁵
	Adult	Liver	Moes <i>et al</i> ²¹⁶
	Adult	Kidney	Musuamba <i>et al</i> ²⁵⁵
	Pediatric	Kidney	Schijvens <i>et al</i> ²⁵⁶
	Pediatric	HSCT	Wallin <i>et al</i> ²⁵⁷
	Adult	Kidney	Woillard <i>et al</i> ²⁵⁰
MwPharm	Adult, Pediatric	Liver, Kidney, HSCT, Healthy volunteers	Nanga <i>et al</i> ²⁵⁸
	Adult, Pediatric	Heart, Liver, Kidney	Lu <i>et al</i> ²¹⁷
	Adult	Systemic lupus erythematosus	Wang <i>et al</i> ²⁵⁹
	Adult	Kidney	Franken <i>et al</i> ¹⁷²
	Adult	Lung	Monchaud <i>et al</i> ²⁰⁸
	Adult	Liver	Cai <i>et al</i> ²⁶⁰
	Adult	Lung	Monchaud <i>et al</i> ²⁰⁸
Tuxuci	Adult	Lung	Cai <i>et al</i> ²¹⁴
	Adult	Lung	Monchaud <i>et al</i> ²⁰⁸
DoseMeRX	Adult	Kidney	Woillard <i>et al</i> ²⁰⁶
JPKD	Adult	Liver	Zahir <i>et al</i> ²⁰⁵
BestDose	Adult	Kidney	Åsberg <i>et al</i> ²⁶¹
ISBA	Adult	Liver	Riff <i>et al</i> ²⁵⁰
	Adult	Kidney, Liver	Woillard <i>et al</i> ²⁵¹
	Adult	Lung	Monchaud <i>et al</i> ²⁰⁸
	Adult	Lung	Saint-Marcoux <i>et al</i> ²⁶²
	Adult	Kidney	Woillard <i>et al</i> ²⁰⁶
	Adult	Kidney	Saint-Marcoux <i>et al</i> ²⁶³
	Adult	Kidney	Saint-Marcoux <i>et al</i> ²⁶⁴
	Adult	Kidney, Liver	Ponthier <i>et al</i> ²⁶⁵
	Adult	Kidney	Benkali <i>et al</i> ²¹¹

The table only contains software tools with an integrated tacrolimus module.
Abbreviations. HSCT, hematopoietic stem cell transplantation; –, not disclosed.

3.3 Software Development and Simulation Study

We developed an interactive software tool to perform popPK model simulations and Bayesian forecasting. The four models with the highest sample-to-patient ratio for each graft type have been integrated into the software tool. These were the models by Lu *et al.* (2019; kidney; sample-to-patient ratio [SPR]=56.8), Kirubakaran *et al.* (2022; heart; SPR=12.6), Cai *et al.* (2020; lung; SPR=15.5), and Lee *et al.* (2006; liver; SPR=50.7).^{175,188,214,217} The two-compartmental Kidney model included covariates aspartate aminotransferase, serum

albumin, sex, and race. The two-compartmental Heart model included covariates fat-free mass, hematocrit, and coadministration of azithromycin. The one-compartmental Lung model included covariates body weight, defined daily dose, postoperative days, hematocrit, CYP3A5 genotype, and coadministration of voriconazole. The one-compartmental Liver model included as covariates total bilirubin, international normalized ratio, graft-to-weight-ratio, and body weight.

Detailed instructions for the use of the software tool are available in the web application hosted at <https://lpmx.shinyapps.io/tacrolimus/>. In short, the sidebar panel on the left allows the user to select the model of interest and provide dosing details, covariate characteristics, exposure targets, and simulation settings. Server-sided, the user input is the basis for popPK simulations and calculations, and the output is presented in the main panel right of the sidebar (Figure S27). The main panel contains the simulated time course of the tacrolimus concentrations, the PTA over time profile, and the Bayesian forecasting results (Figure 28).

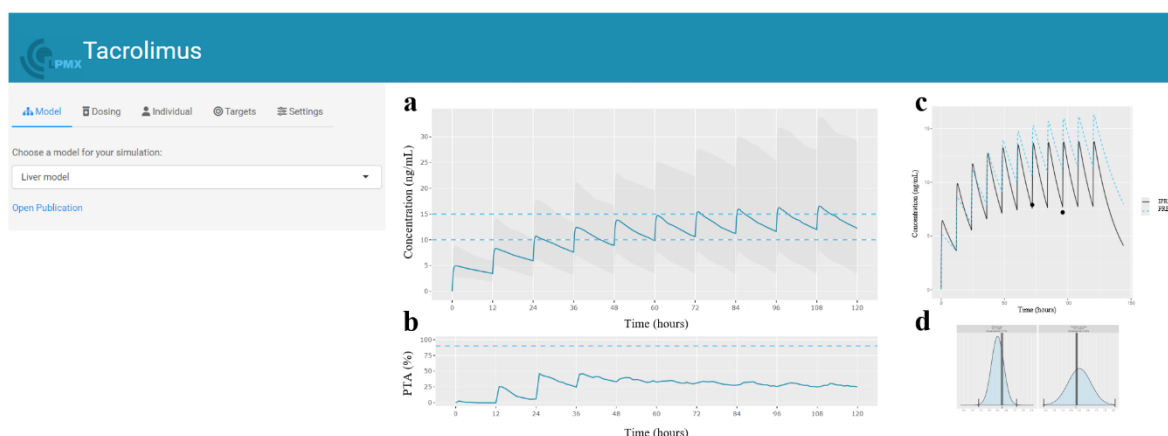


Figure 28. Snapshot of the user interface of the simulation software tool. On the left, the user can select a model, provide dosing details, individual characteristics, targets, and change the default simulation and visualization settings. User-inputted information is transferred for the server-sided calculations. The result of the server-sided calculations is outputted on the right side of the software tool. Visualizations include (a) a concentration-time profile and (b) a probability of target attainment–time plot. The blue solid line indicates the median trend, and the grey shaded area is the 90% prediction interval. The blue dashed lines indicate user-specified boundaries of the exposure target range and the desired minimal probability of target attainment. The Bayesian forecasting results are shown as a concentration-time profile (c). Black dots represent historical tacrolimus concentrations. The black solid line represents the individually predicted profile (IPRED). The blue dashed line represents the typical individual (population prediction, PRED). In panel (d), grey bars indicate the location of the posthoc pharmacokinetics parameters in the η distributions. Abbreviations: IPRED, individually predicted concentration; LPMX, KU Leuven Pharmacometrics research group; PRED, population predicted concentration; PTA, Probability of target attainment.

We illustrate the use of the software tool using the default simulation conditions with the Liver model (**Table 10**).

Table 10. Default simulation conditions for the Liver model in the simulation software tool

Transplanted organ	Liver
Tacrolimus dose	3 mg
Dosing interval	12 hours
Total bilirubin	1.3 mg/dL
Body weight	57 kg
Graft:recipient weight ratio	1.10%
International normalized ratio	1.3

Providing measured tacrolimus trough concentrations of 7.9 ng/mL and 7.2 ng/mL at 72 hours and 96 hours post-transplantation, respectively, individual PK parameters were obtained by maximum *a posteriori* Bayesian estimation. Clearance was estimated to be 22.7 L/h ($\Leftrightarrow$ typical value 20.5 L/h) and the central volume of distribution was estimated to be 440 L ($\Leftrightarrow$ typical value 568 L). The corresponding empirical Bayes estimates of inter-individual variability parameter η of clearance and volume of distribution were 0.31 and -0.25, respectively. The predicted tacrolimus concentration at postoperative day 5 was 8.0 ng/mL and thus outside the exposure target range for liver transplant recipients.

The software tool can be used to explore “what-if” scenarios and their impact on the predictions. For instance, if body weight was 75 kg instead of 57 kg, the individually predicted tacrolimus trough concentration on postoperative day 5 would have been 7.8 ng/mL (i.e., -0.2 ng/mL), with no impact of weight gain in this hypothetical individual (PTA on postoperative day 5 remained 25.2%) With total bilirubin decreasing from 1.3 mg/dL to 1.1 mg/dL, the predicted tacrolimus trough concentration on postoperative day 5 remained unchanged.

4. Discussion

The present work systematically reviewed tacrolimus popPK models and compared structural characteristics across the different types of solid organ transplantation. We identified tacrolimus popPK models that may be used at the patient's bedside for MIPD to support clinical decision-making. We reached out to software tool providers and benchmarked tacrolimus MIPD modules, focusing on the integrated popPK models and the current clinical integration. Additionally, we developed a freely accessible tacrolimus simulation tool, which allows to explore clinically relevant scenarios for four types of solid organ transplantation.

In recent years, more popPK models in heart and lung transplant patients have been added to the tacrolimus popPK repository in solid organ transplant recipients. While kidney and liver transplantation are more frequently performed and hence form the basis of most models, lung and heart transplantation are less frequent and therefore less represented in the models.²⁶⁶ In addition, to this date, no tacrolimus popPK model has been developed for bowel transplant recipients. Bowel transplantation is a high-risk surgery and is therefore performed very rarely and only as a last resort in patients in whom parenteral nutrition cannot be continued.

We identified large multicenter studies including a variety of solid organ transplantation recipients, as well as small monocenter trials focussing on a single graft type. There was no significant association between the sample-to-patient ratio and the popPK model structure, which seems counterintuitive at first glimpse. However, the ratio does not reveal any information on the balancedness of the dataset, i.e., the distribution of when, how often, and for which patient samples were taken. In an extreme case scenario, only a few patients were closely monitored with numerous tacrolimus samples (intermediate, peak, trough), resulting

in a higher sample-to-patient ratio, whereas more stable patients are less closely monitored, resulting in a lower sample-to-patient ratio.

The terms TDM and MIPD are commonly used in the field of model-guided dose selection. TDM has historically focused on the quantification of the exposure when measuring the drug concentration was the primary challenge. In addition to these established terms, concentration control (CC) was introduced to stress the need for an appropriate exposure target focusing, and target concentration intervention (TCI) has been introduced to draw attention to the process of dose calculation/selection.²⁶⁷ Today, the term TDM is frequently used to describe the holistic process including bioanalysis, interpretation of the assay result, and calculation and administration of the optimal next dose. Therefore, model-guided TDM and MIPD have become inseparable terms.^{268,269}

External evaluation of a popPK model prior to its implementation in a software tool is crucial. However, for tacrolimus, it is not comprehensively performed.²⁷⁰ We contacted developers of precision dosing software tools to query their perspectives and insights on web-based precision dosing approaches. Of the nine providers that offer a tacrolimus module, four declared that they performed a prospective validation of their tacrolimus module.

Four providers obtained CE marking in the European Union (EU) for their tacrolimus module. Software tools for clinical decision-making can obtain CE marking, displaying conformity with EU regulations. CE marking is not necessarily required for tools that support clinical decision-making for tacrolimus dosing. The devices serve as a supporting tool and propose dosing regimens to the clinician who makes the final decision, making CE marking nonobligatory.

Our in-house-developed software tool includes four popPK models, one for each solid organ transplantation type (kidney, heart, lung, and liver). The tool allows flexible model selection to investigate the impact of patient characteristics on tacrolimus exposure and target

attainment under user-defined dosing regimens. The integrated software environment allows popPK simulations and Bayesian forecasting. For the latter, our software tool includes the default setting where two trough concentrations are used to obtain individual PK parameter estimates, but the user can adjust these sampling time points. However, we would like to point out the importance of informative sampling when performing Bayesian forecasting. The full concentration–time curves may be misleading when suboptimal sampling schedules are used. It was outside the scope of our work to externally evaluate the predictive performance of the models in our simulation tool. Therefore, our research tool is not intended for dose individualization during clinical trials or in routine clinical care.

Worldwide clinical implementation of MIPD of tacrolimus is currently lacking and prospective clinical trials are warranted. An integrated MIPD service in clinical routine requires automated workflows, suitable infrastructure, and trained healthcare professionals. Specifically for tacrolimus, a few prospective clinical trials have demonstrated their significant benefit in facilitating target attainment,^{164–166} or in minimizing large deviations from predefined exposure targets when using computer-guided personalized dosing.¹⁶⁷ Especially the latter was recently introduced and discussed as a novel marker for clinical outcomes.^{271,272} Additionally, there is increasing evidence that the overall exposure, expressed as an area under the concentration–time curve (AUC_{0-12h}) or AUC-to-trough ratio, is better associated with treatment outcomes than trough concentrations.²⁷³ In the 2019 Consensus Report of the *Immunosuppressive Drugs* Scientific Committee of the International Association of Therapeutic Drug Monitoring and Clinical Toxicity (IATDMCT), Brunet *et al.* proposed monitoring of tacrolimus AUC early after transplantation, at the time of immunosuppression minimization, for special populations, and specific clinical situations.¹⁵⁹ However, prospective confirmation is awaited. Moreover, to determine AUCs with sufficient accuracy, the combination of three sampling times (predose, 1 and 3 hours post-dose) was recommended.²⁷⁴ Therefore, in clinical settings, the

determination of an AUC_{0-12h} is time and resource intensive and thus not commonly performed. Machine learning approaches have been introduced to estimate tacrolimus AUC, showing excellent performance based on two or three blood concentrations.²⁷⁵ The ISBA software is dedicated solely to AUC estimation based on at least three samples. Other software tools like DoseMe allow estimation of trough concentrations or AUCs even from one sample. Additionally, with the rise of dried blood spot sampling, AUC monitoring will become more feasible.¹⁵⁹ Although evidence is accumulating it remains crucial to validate the performance. Overall the model-guided dosing trials mainly focused on kidney or liver transplant recipients since these cohorts display the largest populations within solid organ transplant recipients. For heart and lung transplant recipients, we could not identify results of prospective clinical trials of this kind.

The accuracy and, therefore, safety and effectiveness of MIPD is highly dependent on accurate information on (i) the time of sample collection, (ii) the drug administration time and (iii) the chosen popPK model. However, numerous popPK models that we identified indicate a “fit for purpose” modeling task without such thing as “one model fits all”. The generalizability of models across graft types has not been investigated and is currently lacking.²⁷⁶ Besides assessing the additional benefit of MIPD in terms of clinical efficacy, it is crucial to evaluate safety, feasibility, and cost-effectiveness.²⁷⁷

The present work has some limitations. First, due to incomplete reporting, we had to make some assumptions when pooling and analyzing the data from publications. For example, the *tested* covariates during model building that were not retained in the final model are often reported in a less transparent way compared to the *retained* covariates. Co-medication is a commonly reported covariate that is tested on PK parameters, but it is not always very clear which co-medication it includes. Another example is the CYP3A5 genotype, for which it was not always clearly described if it was about the genotype of the recipient, the genotype of the

donor, or both. Hence, we made the assumption (if not explicitly mentioned) that the tested covariate of CYP3A5 is only the genotype of the recipient. Additionally, the genotype can be reported as a binary variable (expressor or non-expressor), but also as a categorical variable with more categories (poor, intermediate, or extensive metabolizer). We had to reconstruct model codes based on information available in the publications. Even though this step was backed up by independently reviewing the model code by two authors, it bears the risk of a potentially incomplete model. To allow correct understanding and wide applicability of published popPK models, it is highly recommended to report model code, for instance, in a supplementary file. Second, the survey was sent to fourteen software tool providers that we identified based on a previous publication²⁷⁸ and a literature and internet search. There is a risk that we might have missed MIPD software tools. Also, it was outside the scope of our survey to inquire about PK parameter estimation processes and dose selection algorithms. Third, due to the computational limitations of the server, it was not feasible to include all 80 popPK models in a software tool. Hence we decided to include one model per graft type based on an arbitrary selection criterion. However, we will explore the possibility of including more models in the future, making our simulation software tool a repository of all popPK models of tacrolimus. Fourth, we did not externally validate our software tool since it is not intended for clinical use. It is a research tool that can be used to illustrate the behavior of previously published popPK models.

In conclusion, our work provides a comprehensive overview of the landscape of tacrolimus popPK models. It features characteristics of identified models with a dedicated part on potential applications for MIPD. The survey shed light on available tacrolimus MIPD software tools and benchmarks the characteristics of each provider. Furthermore, we developed a simulation tool to explore tacrolimus dose–exposure relationship. We provide free access to our software tool and intend to operate it as a platform for transdisciplinary research collaborations to advance scientific knowledge on tacrolimus dosing. We anticipate

continued maintenance and more features to be included in our software platform. We invite the scientific community to explore our software tool for research purposes. Furthermore, we hail opportunities for external evaluation and prospective validation of our tool. We welcome any suggestions to improve the quality and user experience of the tool, and we advocate for open-source research to increase reliability and transparency in pharmacometrics research.

Supplementary Methods

6-question web survey that was sent out to providers of software tools

Name of the software tool and the company: _____

Does the software tool include a module to support precision dosing of tacrolimus? In case you answer with 'No', can you elaborate why?

- ☐ Yes
- ☐ No
- ☐ Additional information _____
- ☐

Which tacrolimus population pharmacokinetics model is implemented in your software tool? If applicable please provide a traceable publication identifier (PMID, DOI, ...). If the model is not published, please provide a short description including the following information:

1. Foreseen target population (e.g., HSCT, kidney/liver/heart transplant recipients, or all patients receiving tacrolimus)?

2. Structural model?

3. Covariates?

Did you perform a prospective clinical validation of the tacrolimus module? If applicable, please provide a clinical trial identifier.

- ☐ Yes
- ☐ No
- ☐ Additional information _____

Is your software tool with the tacrolimus module currently in clinical use to guide tacrolimus dosing? If yes, can you provide a rough estimate of the number of patients per year?

- ☐ Yes
- ☐ No
- ☐ Additional information _____

Is the tacrolimus module in your software tool registered as a medical device in the European Union? Feel free to provide details.

- ☐ Yes
- ☐ No
- ☐ Additional information _____

Supplementary Results

Table S16. Default tacrolimus exposure target ranges for the software tool in adult solid organ transplant recipients

Transplanted organ	Exposure target [ng/ml]	Reference
Kidney	4–12	Brunet <i>et al.</i> 2019 ¹⁵⁹
Heart	5–15	Brunet <i>et al.</i> 2019 ¹⁵⁹
Lung	10–25	Brunet <i>et al.</i> 2019 ¹⁵⁹
Liver	10–15	Brunet <i>et al.</i> 2019 ¹⁵⁹
Bowel	5–25	Jain <i>et al.</i> 1994 ²⁷⁹
		Andres <i>et al.</i> 2010 ²⁸⁰
		Jain <i>et al.</i> 1992 ²⁸¹
		Reyes <i>et al.</i> 2005 ²⁸²
		Trevizol <i>et al.</i> 2012 ²⁸³

Table S17. Summary characteristics of the identified tacrolimus population pharmacokinetics models

Publication	PMID	Model structure	N patients	N samples	Ratio	Covariate effect on CL	Covariate effect on V	Covariates tested	Graft type
Abderahmene 2024	38018879	2 CMT	196	1,901	9.7	CYPR AGE BSA	—	AGE BSA CYPR CYP3A4 ABCB1 POR SEX HAT WT BMI LBW SCR AST HCT TBIL CRP ALB HLAM	kidney
Al Kofahi 2021	33512723	1 CMT	608	10,992	18.1	AGE CYPR CYP3A4 COR CCB POD AVD	—	WT AGE POD SMO DDST SEX RACE eGFR COR CCB AVD ACE AZO DM CYPR CYP3A4	kidney
Ali 2023	38140040	2 CMT	98	655	6.7	CYPR CYP3A4	—	AGE SEX WT BMI HCT CYP3A4 CYPR ABCB1	kidney
Allard 2019	31003735	2 CMT	24	221	9.2	—	—	AGE SEX WT LBW HCT eGFR AST ALT ALB TBIL	liver
Alqahtani 2021	34145119	1 CMT	139	149	1.1	AST	—	AGE SEX WT SCR CrCL DDS AST ALT ALB TBIL	kidney
Andreu 2015	25254416	2 CMT	16	588	36.8	—	—	WT BMI AGE SEX CrCL ALB AST ALT TBIL HGB HCT RBC	kidney
Andreu 2017	28050888	2 CMT	304	1,891	6.2	CYPR CYP3A4 AGE	—	WT BMI AGE SEX HGB HCT RBC AST ALT CYPR CYP3A4	kidney
Andrews 2019	30552703	2 CMT	337	4,527	13.4	CYPR CYP3A4 AGE ALB BSA SCR HCT	LBW	WT HT POD SEX AGE RACE HCT SCR eGFR AST ALB CRP TP TBIL CYP3A4 CYPR ABCB1 CCB COR PKD RRT NKX DGF HLAM PRA BMI LBW IBW FFM BSA	kidney
Antignac 2005	15991041	1 CMT	37	728	19.7	ALB AST	—	ALB AST	liver
Antignac 2007	17425625	1 CMT	83	1,589	19.1	POD COR	—	POD COR	kidney
Benkali 2009	19902988	2 CMT	32	1,504	47.0	HCT	—	SEX WT COR HCT PXR	kidney
Benkali 2010	20818834	2 CMT	41	492	12.0	CYPR	—	WT AGE SEX COR HCT HGB SCR CYPR	kidney
Bergmann 2013	24089074	2 CMT	173	1,554	9.0	CYPR HCT WT POD	PREDN	AGE RACE WT HAT DDST ALB SCR UREA TBIL APH CrCL HCT PPI PREDN CYPR POD	kidney
Cai 2020	32621968	1 CMT	52	807	15.5	WT DDS HCT POD CYPR VRCZ	WT	AGE SEX HAT WT HCT ALB AST TBIL CrCL DDS POD CYPR VRCZ	lung
Campagne 2018	29775201	2 CMT	67	594	8.9	CYPR	WT	RACE SEX AGE WT BMI ALB GLU TG TCH HDL LDL eGFR SCR HGB HCT PLA LEU ABCB1 POD DM ACE ARB BB CCB PRO STA CYPR	kidney
Chen 2017	28833329	2 CMT	125	1,234	9.9	POD CrCL ABCB1	ABCB1	SEX AGE WT POD TBIL ALT AST APH GLT URA UREA SCR ALB HGB RBC HCT PLA LEU CrCL ABCB1	liver
Chen 2020	30995884	1 CMT	80	NA	—	—	POD	AGE BMI FFM TP POD BCM	liver
Chen 2021	34511980	1 CMT	142	1,378	9.7	CrCL HCT WZC CYPR DDS	—	WT HAT AST TBIL HCT CrCL DDS POD CYPR WZC	kidney
Chen 2022	35485761	1 CMT	52	351	6.8	WT VOZ DDS HCT	WT	AGE SEX HAT WT HCT ALB AST TBIL CrCL DDS POD CYPR VOZ	lung
Cheng 2022	36482138	1 CMT	53	547	10.3	ALB POD CYP3A4	CrCL	AGE SEX WT BSA POD HCT HGB TP ALB AST ALT GLT INR SCR UREA URA BNP	heart

Choi 2022	35830880	1 CMT	269	1,829	6.8	POD AGE TRANSP	WT	LVEF DM HTE HUR HLI PRO MEP DEX AZO AMIO SU CCB SEX AGE WT TRANSP POD FORM	liver kidney
Du 2022	35502774	1 CMT	116	1,327	11.4	DDS eGFR FLU TBIL VRCZ DWZC POD ALT HGB	TBIL VRCZ ALT UREA	AGE WT DDS DWZC POD AST ALT ALB APH GLT LADH TBIL DBIL GLOB GLOR TBA ADD GLU UREA SCR URA TCO2 TG TCH HDL LDL eGFR HGB HCT PTT TTI INR SEX FLU VRCZ PPI CCB FLU	liver
Faelens 2022	35020971	1 CMT	279	3,551	12.7	HCT WT	WT	AGE WT HCT FORM	kidney
Francke 2022	35567629	2 CMT	46	284	6.2	CYPR CYP3A4 AGE ALB CREA HCT PA	—	ATM FTI LTM LTI OVH PA BMI BSA WT CYPR CYP3A4 AGE ALB CREA HCT	kidney
Franken 2022	35442010	2 CMT	184	590	3.2	CYPR CYP3A4 HCT SCR ALB AGE BSA	LBW	WT SEX AGE HCT SCR ALB CYP3A4 CYPR ABCB1 BMI LBW IBW BSA	kidney
Fukatsu 2001	11699612	1 CMT	35	824	23.5	POD TBIL SCR HW	WT	POD TBIL SCR SEX AGE HW	liver
Fukudo 2008	18408564	1 CMT	60	1,827	30.5	POD SCR TBIL ALB MDR1	—	GRWR SCR TBIL ALB POD MDR1 AGE SEX	liver
Golubovic 2014	24184751	1 CMT	105	1,999	19.0	POD WT TP AST HCT	—	POD WT AGE DDST SEX RRT SCR HCT TP APH AST MPA COR	kidney
Gong 2019	32296499	1 CMT	146	897	6.1	POD WZC	WT	POD AGE WT HAT SCR HGB HCT ALB TBIL UBIL APH ALT AST GLT SEX WZC	heart
Grover 2011	21849516	2 CMT	23	184	8.0	—	—	DDS AGE SEX WT BMI DOT	kidney
Han 2013	22660440	1 CMT	80	2,788	34.9	CYPR CYPD HCT POD	WT	SEX AGE WT DDST GWT POD COR CrCL HGB HCT ALB AST ALT CYPR ABCB1 CYPD	kidney
Han 2014	24238261	1 CMT	102	1,626	15.9	CYPR POD	—	AGE SEX WT KDI DDST GWT HGB HCT SCR AST ALT TBIL ALB TP CYPR ABCB1 POD	kidney
Han 2018	30730287	1 CMT	107	707	6.6	CYPR VRCZ FLU WZC	—	SEX AGE WT HDI DDS POD MPA PREDN FLU VRCZ WZC CYPR RBC HGB HCT PLA ALT AST TBIL TP ALB UREA SCR	heart
Henin 2021	34515977	1 CMT	69	1,951	28.3	WT CYPR	WT	AGE SEX CYPR DM WT BMI SCR eGFR HCT HGB ALB	kidney
Ji 2018	30498355	1 CMT	58	605	10.4	CYPD CYPR POD	POD	AGE SEX GRWR WTR HCT TBIL ALT ALB WT POD CYPR CYPD	liver
Jing 2021	33768546	1 CMT	165	824	5.0	HCT POD CYPR WZC	—	CYPR ABCB1 SEX PPI WZC NIF WT POD AGE AST HCT UREA TP TBIL ALB GLT URA HGB ALT RBC SCR	kidney
Kim 2019	31409869	2 CMT	32	128	4.0	CYPR MPA	—	WT AGE SCR eGFR HCT CYPR MPA	kidney
Kirubakaran 2022	36239542	2 CMT	87	1,099	12.6	FFM HCT AZO	FFM	FFM HCT AZO	heart
Koomen 2023	37306899	1 CMT	1,060	1,973	1.9	eGFR HCT TRANSP	HPG	POD DDS FORM WT FFM HCT AGE ALB APH ALT AST TBIL BSA BMI FFM eGFR RACE HPG PRO SEX TP TRANSP SU	kidney lung
Lee 2006	16863483	1 CMT	35	1,775	50.7	POD TBIL INR GRWR GWT	—	SEX AGE TBIL ALT APH INR ALB HCT DDST GWT GRWR POD COR FLU	liver
Li 2007	17875118	1 CMT	72	1,106	15.4	TBIL CYPD CYPR	—	AGE WT POD HCT INR ALT AST TP ALB TBIL SCR BDR SEX FLU DIL CYPR MDR1 CYPD	liver

Li 2023	36097320	1 CMT	145	1,287	8.9	DDS HCT POD VRCZ AST	POD HCT	WT SEX GRWR ALB AST GLT TBIL SCR HCT DDS POD VRCZ CYP3A4 CYPR	liver
Ling 2020	32453653	1 CMT	234	936	4.0	HCT POD CYPR	—	CPYR ABCB1 POR SEX PREDD CCB AGE WT LEU RBC HGB HCT ALT AST ALB	kidney
Lu 2015	25500866	2 CMT	112	1,100	9.8	ALT	—	TBIL UREA SCR POD URA SEX AGE WT POD LEU HGB HCT PLA	liver
Lu 2019	30950096	2 CMT	408	23,176	56.8	RACE AST	SEX ALB AST	ALT AST TBIL WT BMI LBW AGE RACE SEX TRANSP	liver kidney heart
Lu 2021	34133842	1 CMT	41	418	10.2	POD NR1I2	—	ALT AST ALB TBIL POD SEX AGE WT VRCZ WZC MEP ALT	liver
Martial 2021	33786892	2 CMT	53	748	14.1	—	—	AST TBIL DBIL GLT APH CEST SCR HCT HGB PLA LEU NR1I2	liver
Moes 2016	26521259	2 CMT	49	282	5.8	CYPR	—	WT HCT CYPR CYP3A4 CYPD AGE WT SEX HCT HGB ALB HAT SCR	liver
Monchaud 2012	22339449	2 CMT	78	182	2.3	CYPR CF	—	IBW BSA BMI LBW RACE CYP3A4 CYPR PREDD MPA NIF AZA	lung
Musuamba 2012	22207766	2 CMT	65	455	7.0	HCT CYPR ABCB1	—	CF SEX AGE WT POD SCR HCT CYPR AGE SEX RACE WT HCT CYPR ABCB1	kidney
Nakagawa 2020	32514937	1 CMT	57	540	9.5	GV WT	GV WT	ABCC2 ALT AST TBIL GLT APJ SCR UREA eGFR TP	liver
Ogasawara 2013	23633119	2 CMT	102	500	4.9	AGE CYPR ABCC2	—	GV WT SEX RACE DM AGE WT POD HCT HGB	kidney
Oteo 2013	22660442	1 CMT	75	335	4.5	—	—	GLU ALT APH ALB TBIL UREA SCR CrCL PREDD MPA CYPR CYP3A4 ABCB1	liver
Paschier 2023	37477064	2 CMT	38	433	11.4	CYPR	—	ABCC2 POD AST ALB HCT	heart
Press 2009	19258929	2 CMT	31	NA	—	CYPR	REGI	AGE WT HCT CYPR SEX POD HCT ALB LDL HDL PREDD REGI ABCB1	kidney
Quintairos 2021	33481955	2 CMT	58	1,102	19.0	—	—	CYPR CYP3A4 PXR WT AGE SEKS AGE SEX DDST eGFR AGE CIT TOD DM	kidney
Resendiz 2019	30260084	1 CMT	52	600	11.5	HCT CYPR	—	LYM AGE WT HT BMI SCR eGFR GLU UREA	kidney
Rong 2019	30377942	2 CMT	49	320	6.5	eGFR	eGFR	HCT POD SEX AVD PPI CCB DDST FORM CYPR CYP3A4	kidney
Sam 2006	16430311	1 CMT	31	213	6.9	APH WT SCR	HT HCT	AGE WT HT ALB SCR eGFR POD MPA SEX	liver
Shao 2020	31888346	2 CMT	43	721	16.8	CYPR CYPD	—	APH WT CBCP CECF HCT HT SCR AGE WT POD RBC HGB HCT TBIL ALT	liver
Shi 2022	36444212	1 CMT	150	NA	—	CYPR WT TBIL	—	AST GLT TP ALB UREA SCR CYPR CYPD AGE SEX HAT WT TBIL DBIL ALT AST	liver
Sikma 2020	31745812	2 CMT	30	119	4.0	—	—	HGB SEX AGE RTR TRANSP HT WT SOFA	heart lung
Sikma 2020a	31840222	2 CMT	30	119	4.0	—	—	ARBC HCT ALB HDL AGP PH SEX AGE RTR TRANSP HT WT SOFA	heart lung
Staatz 2002	12496747	1 CMT	70	1,060	15.1	DOT AST	—	ARBC HCT ALB HDL AGP PH WT AGE SEX POD DOT TBIL APH AST	kidney
								GLT ALT CrCL HCT COR	

Staatz 2003	12548506	1 CMT	68	1,742	25.6	AST	WT	WT AGE SEX TRANSP BRP POD DOT TBIL APH AST GLT ALT CrCL HCT COR	liver
Storset 2014	25279405	2 CMT	242	3,100	12.8	CYPR FFM	FFM	CYPR WT FFM AGE SEX PREDD POD HCT SCR AST ALT TBIL ALB APH	kidney
Storset 2014a	24071959	2 CMT	69	514	7.4	FFM	FFM	SEX CYPR AGE HT WT FFM HCT SCR CR{ ALB TBIL AST ALT APH	kidney
Teng 2022	35220592	1 CMT	151	2,397	15.9	CYPR WZC POD COR	—	CYPR AGE SEX POD HGB HCT TBIL UREA URA ABCB1 WZC COR MPA AVD	liver
Vadcharavivad 2016	27191538	1 CMT	96	1,183	12.3	HGB DOT	—	WT DOT ALB HGB eGFR PREDD	kidney
Velickovic Radovanovic 2010	20497746	1 CMT	46	77	1.7	PRO DDS	—	WT SEX AGE POD HGB CRP ALT AST DDS MPA PPI PRO CTM	kidney
Woillard 2010	21284698	2 CMT	73	186	2.5	HCT CYPR	FORM	SEX WT AGE POD PREDD FORM HCT HGB SCR CYPR	kidney
Zahir 2005	16044097	1 CMT	67	694	10.4	ALB HCT DIL FLU	—	AGE WT ALT AST GLT TBIL HCT ALB TP HDL LDL POD DIL FLU	liver
Zhang 2012	22377746	1 CMT	262	3,763	14.4	HCT TP SU DDS	HCT TP	HGB RBC HCT LEU NGR LYM PLA GLU UREA SCR TP ALB ALT AST APH GLT TBIL DBIL TG HDL LDL TBA SEX HBV MPA SU DDS	liver
Zhang 2015	26227094	1 CMT	83	246	3.0	CYPR POR	—	SEX AGE WT POD ALT AST POR CYP3A4 CYPR ABCB1	kidney
Zhang 2017	28401703	1 CMT	83	2,109	25.4	POD HCT CYPR	POD	AGE WT POD HCT TBIL SEX CYPR	kidney
Zhang 2022	35395919	2 CMT	240	1,950	8.1	WT WZC CYPR POD HCT	—	CYPR SEX AGE POD HCT WZC ECD WT	kidney
Zhu 2014	26556199	2 CMT	47	435	9.3	DDS POD	—	DDS AGE POD APH ALT AST TBIL SCR HCT RBC LEU	liver
Zhu 2015	25869250	1 CMT	95	2,285	24.1	POD TBIL HCT CYPR APH UREA	POD HGB	AGE WT POD HCT HGB TP ALT AST GLT ALB APH TBIL SCR UREA CYPR	liver
Zhu 2018	30040022	2 CMT	141	1,232	8.7	WT HCT POD CYPR	—	AGE SEX WT BSA SCR HGB HCT ALT AST TP ALB POD PREDD CCB DDS CYPR	kidney
Zuo 2013	23459029	1 CMT	161	873	5.4	HCT CYPR CYP3A4	—	CYP3A4 POR AGE SEX WT POD RR CYP3A4 CYPR ALT AST TP TBIL DBIL UREA SCR URA LEU RBC HGB HCT PREDD CCB PPI	Kidney

Abbreviations. ABCB1, ATP binding cassette subfamily B member 1; ABCC2, ATP binding cassette subfamily C member 2; ACE, coadministration of angiotensin converting enzyme inhibitor; ADD, adenosine deaminase; ADM, administration of tacrolimus; AGE, age of patient; AGP, alpha glycoprotein; ALB, albumin; ALT, alanine transaminase; APH, alkaline phosphatase; ARB, angiotensin receptor blocker; ARBC, administration of red blood cells; AST, aspartate aminotransferase; ATM, adipose tissue mass; AVD, coadministration of an antiviral drug; AZA, coadministration of azathioprine; AZO, coadministration of azole antimycotic; BB, coadministration of beta blocker; BCM, body cell mass; BDR, bile drainage; BMI, body mass index; BNP, NT-pro BNP N-terminal pro-brain natriuretic peptide; BRP, biliary reconstructive procedure (duct-to-duct anastomosis or Roux loop surgery); BSA, body surface area; CAS, coadministration of caspofungin; CBCP, blood to plasma ratio of tacrolimus concentration; CCB, coadministration of calcium channel blocker; CECP, erythrocyte to plasma ratio of concentration; CEST, choline esterase; CF, cystic fibrosis; CIT, cold ischemia time; CL, clearance; COR, coadministration of corticosteroids; CrCL, creatinine clearance; CREA, plasma creatinine; CRP, c-reactive protein; CTM, coadministration of cotrimoxazole; CYP3A4, cytochrome P450 3A4; CYPD, cytochrome P450 3A5 genotype of donor; CYPR, cytochrome P450 3A5 genotype of recipient; CYPRPOR, combined genotype for CYP3A5, and POR*28; DBIL, direct bilirubine; DDS, defined daily dose; DDST, donor deceased status; DEX, coadministration of dexamethasone; DIL, coadministration of diltiazem; DGF, deployed graft function; DM, diabetes mellitus; DOT, days of tacrolimus therapy; DWZC, daily dose of wuzhi capsule; ECMO, extracorporeal membrane oxygenation; ECD, coadministration of echinocandines; eGFR, estimated glomerular filtration rate; FFM, fat free mass; FLU, coadministration of fluconazole; FORM, formulation of tacrolimus;

GLOB, globulin; GLOR, albumin globuline ratio; GLT, glutamyl transpeptidase; GLU, glucose level; GRWR, graft:recipient weight ratio; GV, graft volume; GWT, graft weight; HBV, hepatitis B virus surface antibody; HCT, hematocrit; HDI, cause of heart disease; HGB, haemoglobin; HLAM, human leucocyte antigen mismatch; HLI, comorbidity hyperlipidemia; HPG, haptoglobin; HT, height; HTE, comorbidity hypertension; HUR, comorbidity hyperuricemia; HW, hepatic weight; IBW, ideal body weight; ICYP, coadministration of CYP3A54 inhibitors; INC, intake condition (fasted, fed); HDL, high-density lipoprotein; INR, international normalized ratio; KDI, cause of kidney disease; LADH, lactate dehydrogenase; LBW, lean body weight; LDL, low-density lipoprotein; LEU, leucocytes; LVEF, left ventricular ejection fraction; LYM, lymphocyte count; MDR1, intestinal MDR1 mRNA level; MEP, coadministration of methylprednisolone; MPA, coadministration of mycophenolic acid; NGR, neutrophile granulocytes; NIF, coadministration of nifedipine; NKX, number of previous kidney transplantation; NR1I2, nuclear receptor subfamily 1 group I member 2; OCC, occasion; PA, phase angle; PH, pH blood; PKD, primary kidney disease; PLA, platelet count; POD, postoperative day; PMID, PubMed identifier; POR, POR*28 polymorphism; PPI, coadministration of proton pump inhibitors; PRA, panel reactive antibodies; PREDD, prednisolone dose; PREDN, maximum concentration value for free prednisolone; PRO, coadministration of prednisolone; PTT, prothrombin time; PXR, human pregnane X receptor; RACE, race of recipient; RBC, red blood count; RR, blood pressure; RRT, renal replacement therapy prior to transplantation; RTR, reason for graft transplantation; SCR, serum creatinine, SEX, sex of the recipient; SOFA, sequential organ failure assessment score; STA, coadministration of statins; SU, coadministration of sulfonylurea; TBA, total bile acid; TBIL, total bilirubin; TCH, total cholesterol; TCO2, total CO2; TG, triglyceride; TOD, time of dialysis; TP, total protein; TRANSP, type of transplantation; TTI, thrombin time; UBIL, unconjugated bilirubin; URA, uric acid; UREA, urea; V, volumes of distribution; VOZ, voriconazole concentration; VRCZ, coadministration of voriconazole; WT, body weight; WTR, body weight ratio donor to recipient; WZC, coadministration of wuzhi capsule.

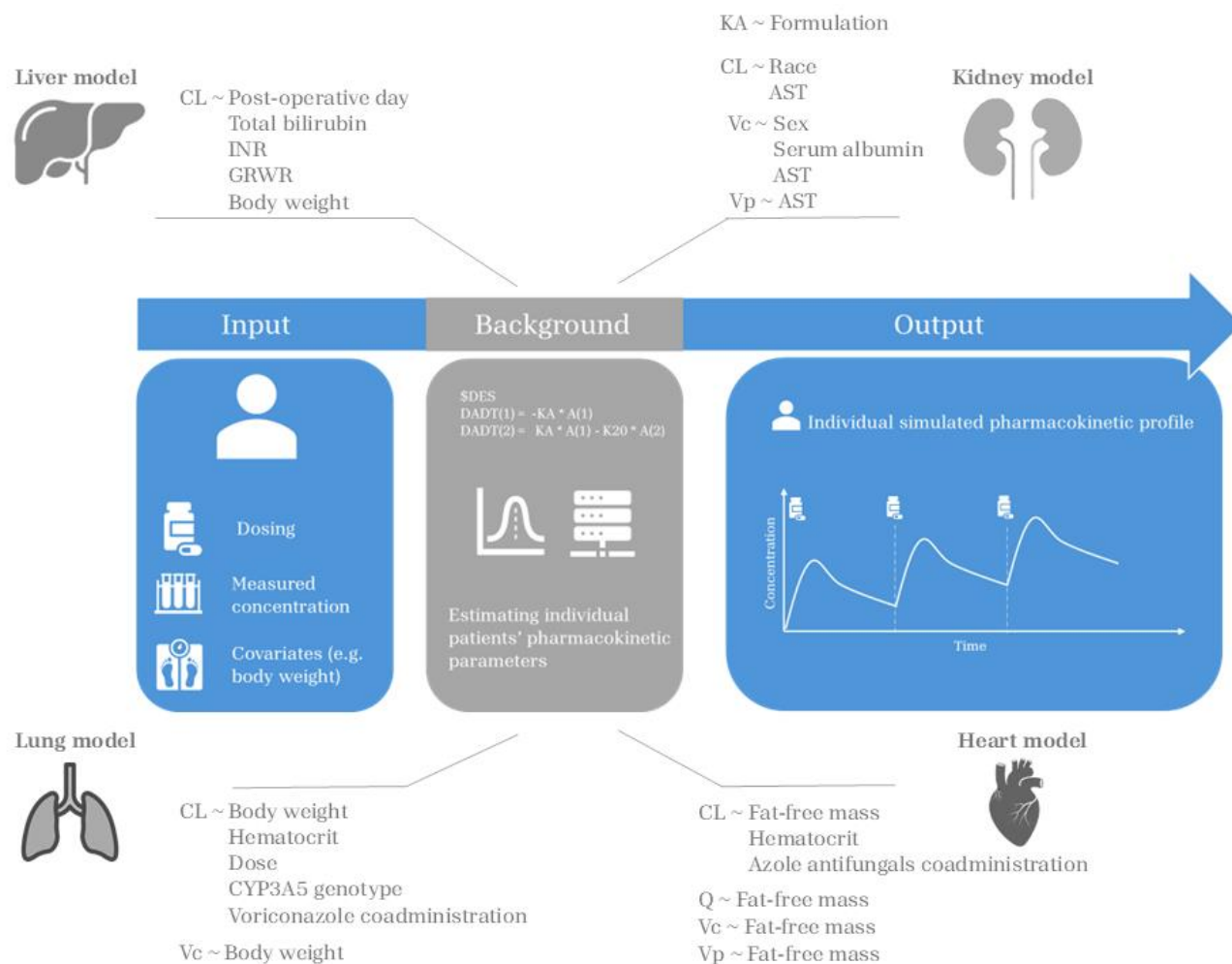


Figure S27. Model scheme of the developed software tool to illustrate the user-required input, server-sided calculation, and app-based simulated output. Abbreviations. AST, aspartate aminotransferase; CL, clearance; CYP3A5, cytochrome P450 3A5; GRWR, graft-to-recipient-weight ratio; INR, international normalized ratio; KA, absorption rate constant; Q, intercompartmental clearance; Vc, central volume of distribution; Vp, peripheral volume of distribution.

Tacrolimus population pharmacokinetics and exposure–response modeling in intestinal transplant recipients

This chapter is based on

Yannick Hoffert^{*}, Barbara Deleenheer^{*}, Caroline Boelhouwer, Nele Van den Eede, Emmanuel Niyigena, Laurens Ceulemans, Laure Elens, Diethard Monbaliu, Jacques Pirenne, Lucas Wauters, Tim Vanuytsel, Erwin Dreesen. Early Tacrolimus Exposure Predicts Rejection Risk in Intestinal Transplant Recipients. [In Draft]
[*Shared co-first authorship].

Abstract

Background and Objective Intestinal transplantation remains a rare yet life-saving intervention for patients with intestinal failure. Tacrolimus, the cornerstone immunosuppressant in this setting, poses major dosing challenges due to its narrow therapeutic index and highly variable pharmacokinetics.

Methods We conducted a 15-year single-center retrospective study (2007–2022) of routine care data from 20 intestinal transplant recipients (median age 41 years; 11 males) to develop a population pharmacokinetic and multistate time-to-event model framework for tacrolimus.

Results A one-compartment pharmacokinetic model with time-dependent clearance and allometrically scaled volume of distribution best described the data. Clearance increased after transplantation to a plateau by day 84 (8.62 L/h), and oral bioavailability was 26.8%. Early tacrolimus exposure (cumulative area under the concentration-time curve during the first 13 days) and the tacrolimus inpatient variability of the daily area under the concentration-time curve were predictors of biopsy-proven rejection within the first year after transplantation. An exposure threshold of 4,888.7 ng·h/mL classified patients into high and low rejection risk groups (area under the receiver operating characteristic curve of 0.77, sensitivity 0.79, and specificity 0.83), leading to a 1-year probability of rejection of 30.8%.

Conclusion Our findings highlight the critical importance of early and stable tacrolimus exposure for rejection prevention and provide a basis for model-informed dose initiation of tacrolimus in this rare setting. External validation is warranted.

1. Introduction

Intestinal transplantation is considered for patients with chronic intestinal failure who face life-threatening complications on long-term parenteral nutrition, such as intestinal failure-associated liver disease, recurrent catheter-related bloodstream infections, or progressive loss of central venous access. In 2019, the International Intestinal Rehabilitation and Transplant Association (IIRTA) updated consensus indications, reflecting advances in intestinal rehabilitation and peri-transplant care.²⁸⁴ Over the past decades, global intestinal transplantation activity has declined, and intestinal transplantation remains the least common solid organ transplant.²⁸⁵ Nevertheless, it is life-saving for selected patients and improves quality of life after successful engraftment.

Immunosuppressive management to prevent graft rejection after intestinal transplantation is particularly challenging due to complex intestinal graft immunobiology and high risk of rejection and infection.²⁸⁶ Tacrolimus is the cornerstone calcineurin inhibitor in this setting.^{39–42} Its clinical use is complicated by a narrow therapeutic window: underexposure increases the risk of graft rejection due to inadequate immunosuppression, whereas overexposure contributes to nephrotoxicity, diabetogenesis, and malignancy.^{44,161} Consequently, safe and effective tacrolimus dosing resembles a tightrope walk as tacrolimus PK varies widely. A central challenge is the pronounced variability in tacrolimus exposure, both between patients and within individual patients over time. Pronounced inpatient fluctuations in tacrolimus exposure have been associated with adverse outcomes across several solid organ transplant cohorts, including increased rejection, donor-specific antibody formation, graft loss, and mortality.^{271,287,288} To manage this variability, concentration-guided dose titration, also known as TDM, is routinely performed.²⁸⁹

In recent years, MIPD has emerged as a promising strategy to individualize tacrolimus dosing, which integrates PK–PD models and patient data to optimize dosing.⁶² Although

increasingly applied in other solid organ transplantation settings, its potential remains unstudied in intestinal transplantation due to the lack of dedicated tacrolimus PK-PD models.²⁹⁰

In this exploratory study, we retrospectively analyze 15 years of routine clinical data from a single center. We explore the dose-exposure-response relationship of tacrolimus using state-of-the-art modeling approaches and simulate optimal dosing regimens for improved rejection-free survival (RFS). Our goal is to provide a PK-PD framework that informs personalized immunosuppressive therapy for a highly vulnerable patient population.

2. Materials and methods

2.1 Patients and data

We conducted a single-center retrospective study at University Hospitals Leuven (UZ Leuven, Belgium), covering the period from electronic registry initiation in March 2007 through May 2022. The study was approved by the Ethics Committee Research UZ / KU Leuven (S65900). Patients were informed about the secondary use of their data, and no dedicated informed consent was required for this retrospective analysis.

The EHR systems ('Klinisch werkstation', KWS and 'Patiënt Data Management Systeem', PDMS) were screened to identify recipients of isolated intestinal transplantation, combined liver-intestinal transplantation, and (modified) multivisceral transplantation. For the first year post-transplantation, we extracted tacrolimus dosing, whole-blood tacrolimus concentrations, dosing times, hematocrit, serum albumin, hemoglobin, CRP, direct bilirubin, white blood cell count, AST, ALT, GGT, serum creatinine, total protein, and lactate dehydrogenase (LDH).

Therapy outcomes included time-to-first biopsy-proven rejection and time-to-death. Rejection was defined by significant epithelial apoptosis on surveillance biopsies according to center practice. Data formatting, analyses, and visualization were performed in R (version 4.4.1) in RStudio (version 2024.12.1+563) using the *tidyverse* suite.²⁹¹

2.2 Immunomodulatory protocol

Patients received triple immunosuppression comprising tacrolimus, methylprednisolone, and either azathioprine or everolimus according to the Leuven Immunomodulatory Protocol. Tacrolimus was initiated by continuous intravenous (IV) infusion at 0.04 mg/kg/day for adults and 0.06 mg/kg/day for pediatric patients. Oral administration (Prograf®, Modigraf®, compounded suspension by the hospital pharmacy) was considered once enteral feeding

had stabilized after clinical assessment. The adult oral starting dose was calculated as four times the IV dose divided into two daily doses (8 am and 8 pm). Pediatric oral dosing was age-adjusted: 0.3 mg/kg/day in two doses (<14 years) or 0.2 mg/kg/day in two doses (≥ 14 years). Whole-blood tacrolimus concentrations were monitored twice daily following first IV and first oral dose, with dose-titration aiming for protocol-defined trough targets (**Table S18**).

2.3 Bioanalytical assay

From 2007 to 2010, whole-blood tacrolimus concentrations were measured by enzyme-conjugated immunoassay (Dade Behring Siemens Dimension RxL; lower limit of quantification, LLOQ 2 ng/mL). In December 2010, the method was changed to ARCHITECT chemiluminescent microparticle immunoassay (Abbott and Fujirebio Diagnostics Inc., Malvern PA, USA; LLOQ 1 ng/mL). From April 2017 onwards, automated extraction and liquid chromatography–tandem mass spectrometry (LC–MS/MS; LLOQ 0.5 ng/mL) was used. Internal comparability procedures indicated consistency across assay transitions. Methodology and calibration details are provided in *Supplementary Methods* (**Table S19**).

2.4 Population pharmacokinetic modeling

We analyzed the tacrolimus concentration-time data using parametric NLME popPK modeling. Different structural and residual error models were evaluated. Interindividual variability was tested on all PK parameters, assuming log-normal (non-linear) distributions to avoid negative estimates. Stepwise covariate modeling with stage-wise filtering (SCM+) was performed to explain PK variability (**Table S20**).⁵⁴ Candidate covariates included demographic and clinical laboratory variables extracted above. OFV criteria were $\Delta\text{OFV} \geq 3.84$ ($p \leq 0.05$) for forward inclusion and backward elimination (1 degree of freedom). Parameter precision and plausibility were assessed. Details on model building, covariate testing, and model validation are provided in *Supplementary Methods*. PopPK analysis was carried out using NONMEM (version 7.5.1; ICON plc, Gaithersburg, MD, USA) using a GNU

Fortran 95 compiler and the Perl-speaks-NONMEM (PsN; version 5.3.0; Uppsala University, Uppsala, Sweden) toolkit. Pirana (version 21.11.1; Certara, Radnor PA, USA) was used for program administration and interface generation. Computations were accelerated using HPC, provided by the VSC.

2.5 Exposure-response: Rejection-death model

We developed a parametric multistate time-to-event model to describe and predict rejection within the first year post-transplant. The secondary event of interest was death, which represents a competing risk of the primary outcome, i.e., patients who die early on have less time to experience rejection. Consequently, we developed a multistate framework to ensure a bias-free analysis in the presence of competing risks. PK projections derived from the popPK model included cumulative exposure over prespecified intervals (area under the concentration–time curve from start of tacrolimus therapy to day t , AUC_{0-t}) and coefficient of variation of daily cumulative exposure in the first t days ($CV\%_t$). Age, body weight, hematocrit, serum creatinine and CRP were also evaluated as median values from start of therapy to day t . The $CV\%_t$ for an individual i was calculated as:

$$CV\%_{t,i} = \frac{\sqrt{\frac{1}{t-1} \sum_{d=1}^t (AUC_{d,i} - \overline{AUC}_i)^2}}{\frac{1}{t} \sum_{d=1}^t AUC_{d,i}} \times 100\% \quad (\text{Eq. 15})$$

with t depicting the total number of days, d the day, $AUC_{d,i}$ the daily AUC for an individual patient i , $\overline{AUC}_i$ the average AUC across all days for an individual i .²⁹² ROC analysis and Youden's J index were used to identify optimal thresholds (*pROC*, version 1.19.0.1).

2.6. Dosing simulations

We performed simulations using the final popPK model to illustrate tacrolimus dosing strategies and their impact on exposure. Various IV dosing strategies for therapy initiation were evaluated. Furthermore, we illustrate the impact of a hypothetical measured

tacrolimus concentration on an individual patient's simulated exposure using a Bayesian approach. Details on dosing simulations are provided in *Supplementary Methods*. A freely available, interactive software tool was developed to allow customized tacrolimus dosing simulations and calculated risk of rejection (*Shiny*, version 1.7.4).

3. Results

3.1 Patients and data

Between March 2007 and May 2022, 23 patients underwent intestinal transplantation at UZ Leuven. Three were excluded due to severe complications following allograft enteropathy and transplantectomy (**Figure S28**), yielding a final cohort of 20 recipients (11 were male). The median age at transplant was 41 years [interquartile range, IQR, 30.5–47.3] and median body weight was 55.4 kg [IQR, 47.7–63.9] (**Table 11**). Three patients (15%) were under the age of 18 years. Transplant types included isolated intestinal (n = 10), combined liver–intestine (n = 3), and multivisceral (n = 7). Within the first year post-transplantation, fourteen patients had biopsy-proven rejection, and four patients died (**Figure 29**).

Table 11. Baseline characteristics of the patient cohort (N = 20)

Parameter	Value
Sex, female, <i>n</i> (%)	9 (45)
Age, <i>years</i> , median [IQR]	41 [30.5–47.3]
Pediatric patients (<18 years old), <i>n</i> (%)	3 (15)
Weight, <i>kg</i> , median [IQR]	55.4 [47.7–63.9]
Height, <i>cm</i> , median [IQR]	170 [163–174]
BMI, <i>kg/m²</i> , median [IQR]	19.3 [16.7–22.6]
Short bowel syndrome, <i>n</i> (%)	10 (50)
*Days on PN pre Tx, <i>days</i> , median [IQR]	1372 [951–4427]
Transplant type, ITx:cLi-ITx:MVTx	10:3:7
Diagnosed diabetes pre-Tx, <i>n</i> (%)	3 (15)
Native pancreas in place, <i>n</i> (%)	13 (65)

Abbreviations. cLi-ITx, combined liver-intestine transplantation; ITx, isolated intestinal transplantation; IQR, interquartile range; MVTx, multivisceral transplantation; PN, parenteral nutrition; Tx, transplantation.

*not applicable for six patients

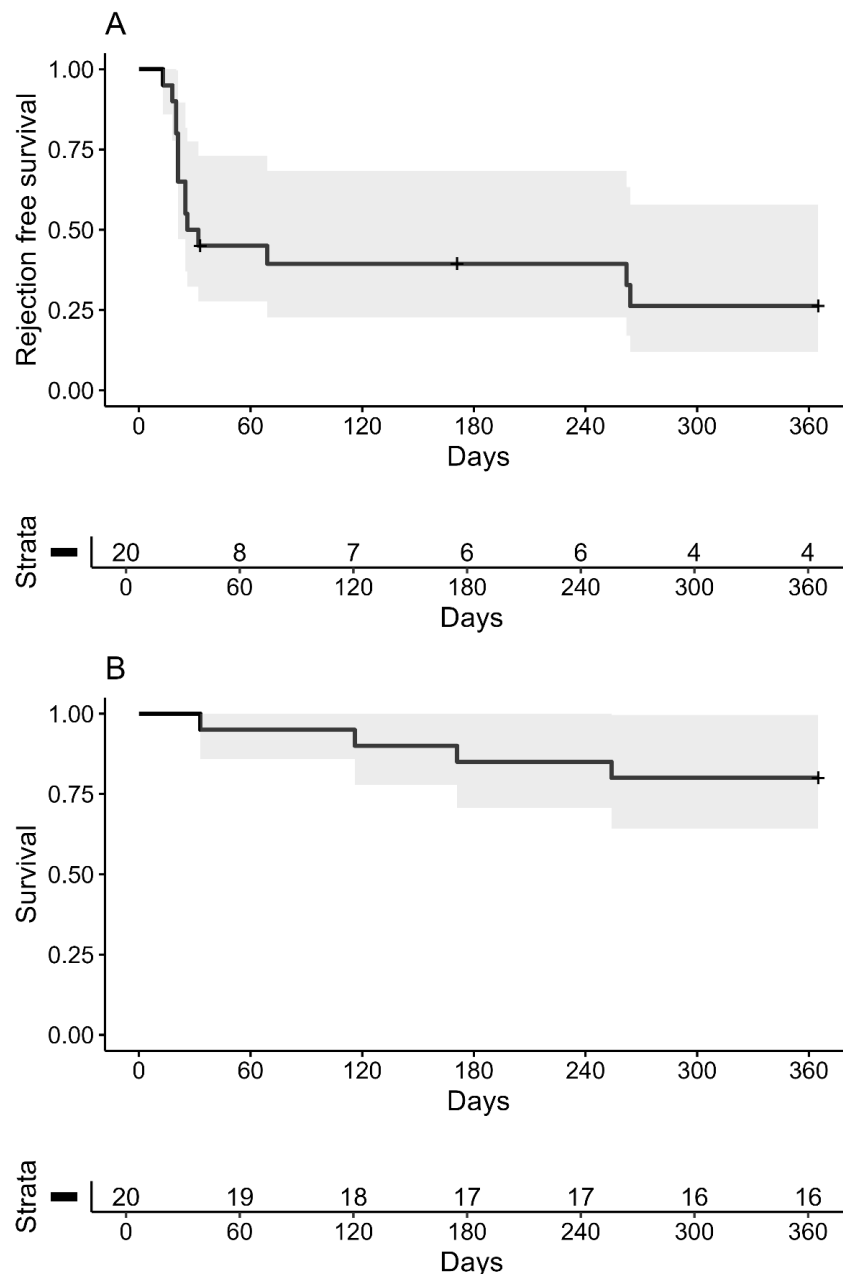


Figure 29. Kaplan–Meier curve for overall survival (A) and rejection-free survival (B). Tick marks indicate right-censoring due to dropout and therefore loss of follow-up. Grey shaded areas indicate the confidence interval of the Kaplan–Meier estimator.

Patients contributed 2,189 tacrolimus whole-blood concentrations (median, 109 [IQR, 83–149] per patient). Tacrolimus concentrations were highest in the first 14 days post-transplantation (median, 18.0 [IQR, 14.8–22.8] ng/mL), decreased over time, and were lowest in months 6–12 (median, 8.3 [6.2–12.0] ng/mL) (**Table 12**). Biomarkers reflecting systemic illness, anemia, and liver function varied substantially, mirroring the unstable postoperative physiology and dynamic clinical course of these patients (**Figure S29–S32**).

Table 12. Tacrolimus on-treatment characteristics

Parameter	Value
<i>Tacrolimus dosing</i>	
IV rate, mg/h, median [IQR]	0.050 [0.030–0.074]
[‡] Oral dose 0-1 month, mg, median [IQR]	2 [1–4]
[‡] Oral dose 1-3 months, mg, median [IQR]	4 [2.5–5]
[‡] Oral dose 3-12 months, mg, median [IQR]	2.5 [2–4]
Co-administration of everolimus, n (%)	4 (20)
Co-administration of sirolimus, n (%)	2 (10)
Co-administration of mycophenolic acid, n (%)	3 (15)
Co-administration of azathioprine, n (%)	19 (95)
Co-administration of corticosteroids, n (%)	20 (100)
Co-administration of cyclosporin, n (%)	3 (15)
<i>On tacrolimus treatment</i>	
Total tacrolimus samples, n	2,189
Tacrolimus exposure 0–14 days, ng/mL, median [IQR]	18.0 [14.8–22.8]
Tacrolimus exposure 0–2 months, ng/mL, median [IQR]	14.8 [11.6–18.4]
Tacrolimus exposure 0–3 months, ng/mL, median [IQR]	14.0 [10.9–17.9]
Tacrolimus exposure 3–6 months, ng/mL, median [IQR]	9.6 [7.0–13.4]
Tacrolimus exposure 6–12 months, ng/mL, median [IQR]	8.3 [6.2–12.0]
One-year survival, n (%)	16 (80%)
One-year rejection-free survival	6 (30%)

[‡]including prograf, modigraf, and oral suspension.

3.2 Population pharmacokinetics

Tacrolimus concentration–time data were best described by a one-compartment popPK model with first-order absorption and linear elimination. Tacrolimus clearance increased over time, stabilizing at a plateau by day 84 post-transplantation (**Table 13**). Typical CL rose from 1.59 L/h on Day 0 to a plateau of 8.62 L/h (relative standard error, RSE, 37.7%), corresponding to a shortening of half-life from 16.2 h to 3.0 h. The typical V was 37.2 L (RSE 34.4%) using allometric scaling with fixed exponent (0.75), the k_a was 0.0132 h⁻¹ (RSE 15.2%, estimated using a power function), and F was 26.8% (RSE 31%) but varied considerably between patients (range of F, 7.8% to 59.5%). Also CL, V, and k_a varied widely between patients, yet no covariates could explain this interindividual variability, consistent with the limited cohort size. The model captured the central tendency and dispersion of the data well (**Figure 30, Figure S33**).

Table 13. Parameter estimates for the final tacrolimus population pharmacokinetics and exposure-response model.

	Estimate	RSE (%) [shrinkage %]
Population pharmacokinetics model		
<i>Structural model</i>		
CL (L/h)	1.59	37.7
V (L)	37.2	34.4
Absorption estimate θ	-1.88	15.2
k_a (1/h), 10^6	0.0132	
F (%)	26.8	31
<i>Covariate model</i>		
ε , Inflection point for time on CL (days)	84.3	2.4
β , Increase for time on CL	0.0838	31.1
<i>Statistical model</i>		
IIV on CL (%CV)	41.8	34 [11]
IIV on V (%CV)	176.6	22 [19]
IIV on k_a (%CV)	176.6	20 [5]
IIV on F (%CV)	51	20 [11]
Proportional error (%CV)	32.9	13 [1]
Additive error (ng/mL)*	3.09	16 [1]
Rejection-death model		
<i>Structural model</i>		
Baseline hazard $1/\lambda$	138	33
Shape γ	0.889	22
MTT_{13} ($=1/k_{13}$)	1250	71
MTT_{23} ($=1/k_{23}$)	1960	70
<i>Covariate model</i>		
AUC ₀₋₁₃ effect	-0.000547	33
CV% ₁₃ effect	0.0428	41

Coefficient of variation (%CV) was calculated as: $\sqrt{\omega^2} \cdot 100\%$

*reported as transformed η value

Individual tacrolimus clearance was parameterized as: $CL_{ind} = TVCL_{pop} + \beta \cdot \text{postoperative day} \cdot (1-L) + \beta \cdot \varepsilon \cdot L$
(with $L = 0$ if postoperative day < 84 , and $L = 1$ if postoperative day ≥ 84)

Abbreviations. CL, clearance; CV, coefficient of variation; F, bioavailability; IIV, interindividual variability; k_a , absorption rate constant; MTT, mean transit time; SIR, Sampling importance resampling; V, volume of distribution.

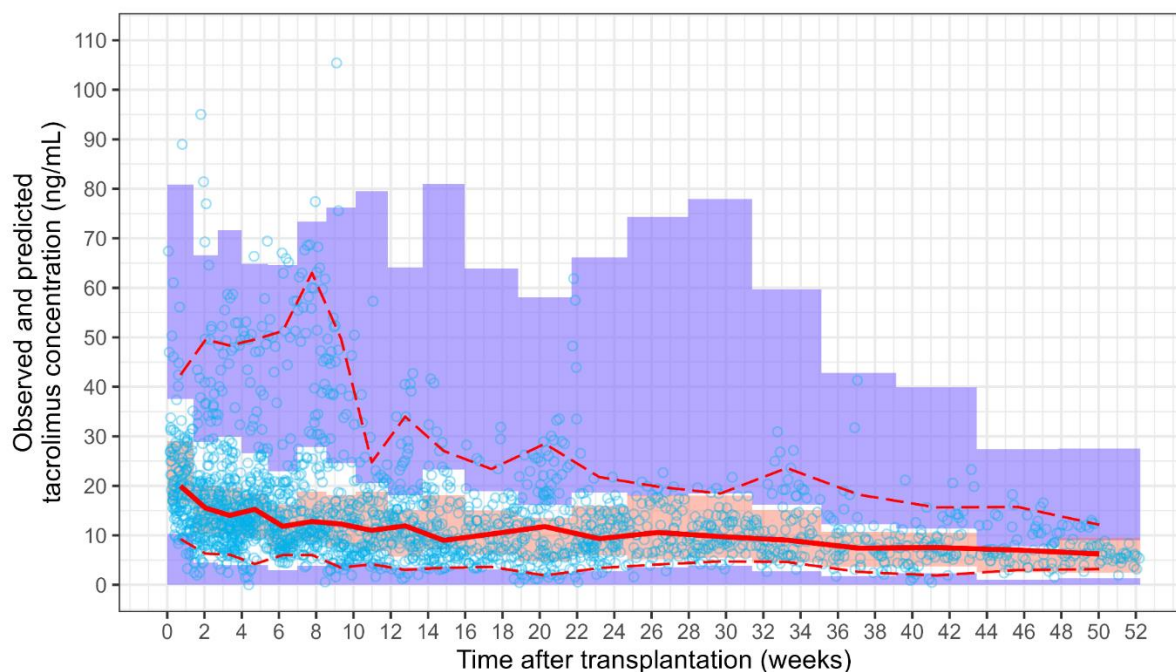


Figure 30. Visual predictive check plots of the final population pharmacokinetic model. Open circles represent prediction- and variance-corrected observed data. The solid red line is the median of the observed data. The

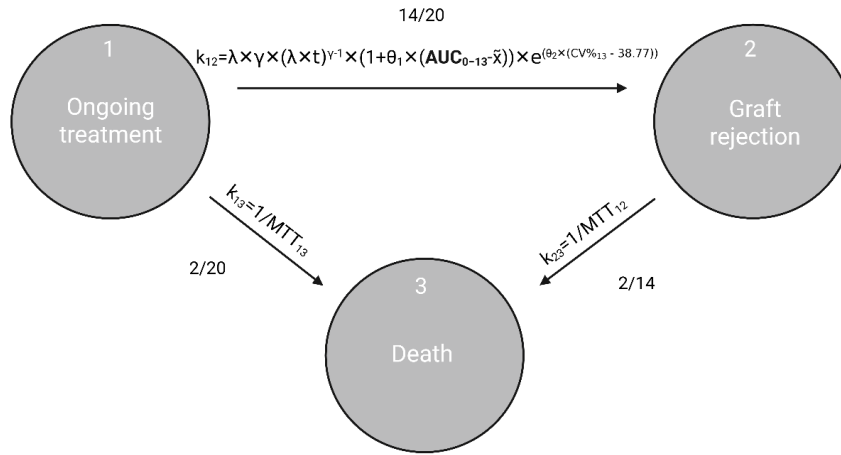
dashed red lines are the 5th and 95th percentiles of the observed data. The red and purple shaded areas indicate the 95% confidence intervals of the median and 5th and 95th percentiles, respectively (N = 2000 simulations). Negative predictions due to the error model were set to zero.

Individual model fits demonstrate acceptable agreement between observations and predictions (**Figure S34**). Individual CL trajectories illustrated the time-dependent increase and stabilization by day 84 (RSE 2.4%) (**Figure S35**).

3.3 Exposure–response: rejection prediction

The developed multistate model includes three distinct states (vertex): i) ongoing tacrolimus treatment (state 1), ii) rejection (state 2) and iii) death (state 3). The vertices were connected using one-way edges, based on clinical plausibility (ongoing tacrolimus treatment → rejection, rejection → death, ongoing tacrolimus treatment → death) (**Figure 31, panel A**). Fourteen out of 20 patients (70.0%) experienced rejection, two patients (10.0%) died after a rejection episode, and two patients (10.0%) died without a rejection episode. A Weibull function described the transition from state 1 to state 2, whereas simple transition rates described the transition from state 1 to state 3, and state 2 to state 3. We identified the cumulative tacrolimus exposure within the first 13 days (AUC_{0-13}) and the coefficient of variation of the daily AUC in the first 13 days as statistically significant predictors in our model. The higher the AUC_{0-13} , the lower the hazard risk of rejection, and the lower the CV of the daily AUC in the first 13 days ($CV\%_{13}$), the lower the risk of rejection. The model captures the trend of the data well for all three states (**Figure 31, panel B**). Details on the stepwise covariate search were included in the *Supplementary Results* (**Table S21**). The following threshold identification was performed by the actionable predictor AUC_{0-13} , while fixing the $CV\%_{13}$ to the population median.

A



B

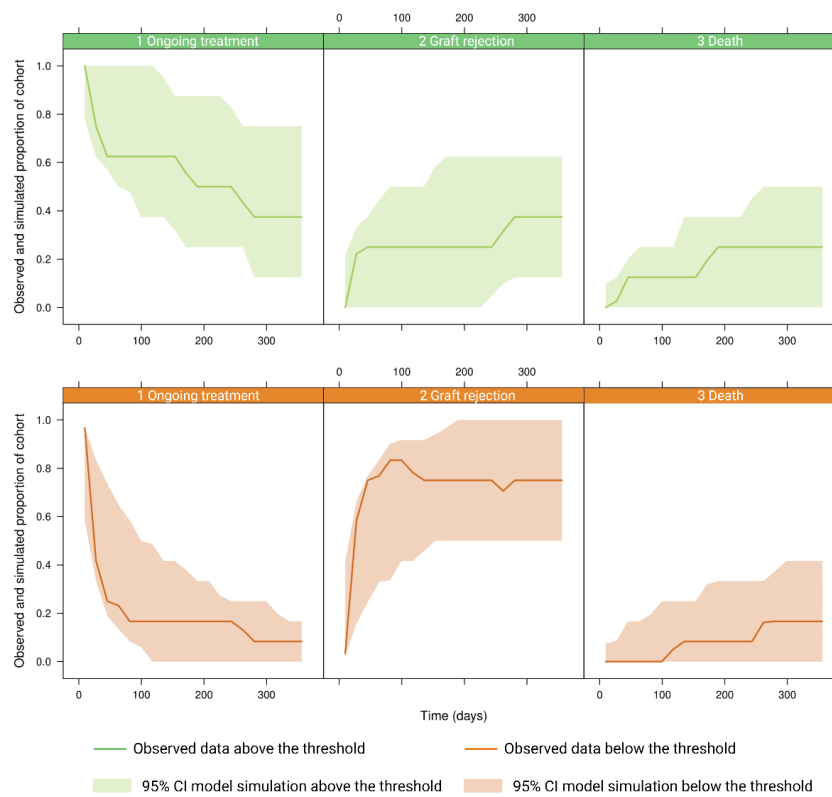


Figure 31. Multistate model structure depicting the pathway of the included patients (e.g., Ongoing treatment → Graft rejection → Death) (A) and the corresponding Kaplan–Meier Visual predictive check plots of the final model, including the observed data and the 95% confidence interval of the model simulation (N = 2000 simulations) (B). Fourteen patients had a rejection, two patients died, and two patients died after having a rejection. The green colored panels represent the strata above the identified AUC_{0-13} threshold, and the orange panel the strata below the identified threshold. AUC, area under the curve; CI, confidence interval; CV, coefficient of variation; MTT, mean transit time for transition rates; t, time; γ , shape parameter of the Weibull function; λ , hazard parameter of the Weibull function, $\tilde{x}$, population median of the predictor.

The ROC analysis yielded an area under the ROC curve (AUROC) of 0.77 [95%CI 0.48–1.00]

(Figure 32). The Youden-optimized threshold of 4888.7 ng·h/mL classified patients into high-

and low-risk groups, and translates to a continuous tacrolimus level of 15.7 ng/mL. A formulation stratified analysis (IV vs oral initiation) yields a similar trend (**Figure S36**).

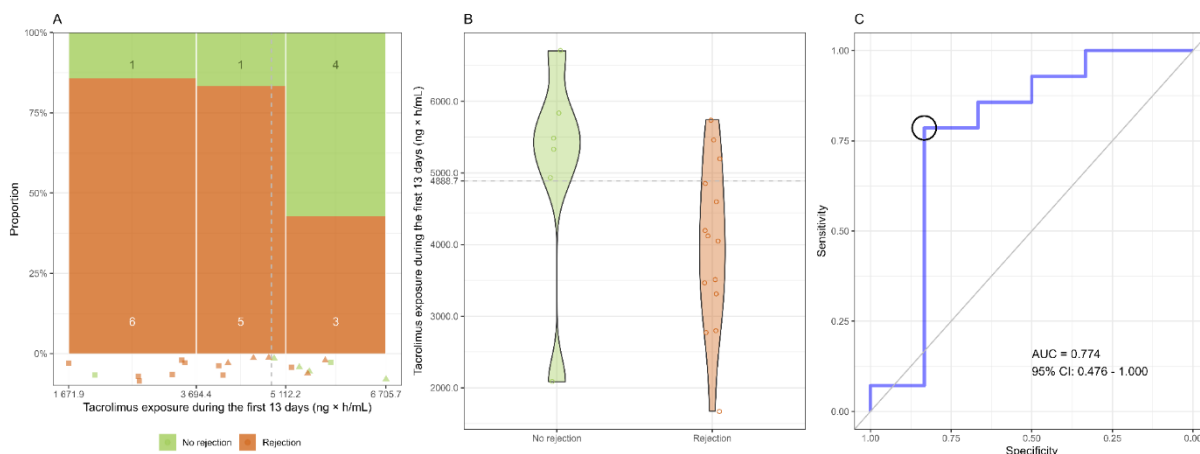


Figure 32. Model-predicted cumulative tacrolimus exposure (A) and its distribution (B) stratified by patients with and without rejection during the first year post intestinal transplantation and the corresponding receiver operating characteristic curve (C). The dashed line in panel (A) and panel (B) marks the identified optimal threshold ($AUC_{0-13} = 4888.7$ ng·h/mL) and is depicted as a black circle in panel (C). The triangles and squares in panel (A) indicate intravenous and oral tacrolimus initiation, respectively. AUC, area under the curve; CI, confidence interval.

The hazard risk of rejection for an individual patient was calculated as:

$$\text{Hazard risk(biopsy-proven rejection)}_i = \lambda \times \gamma \times (\lambda \times \text{time})^{\gamma-1} \times (1 + (-0.000573 \times (AUC_{0-13,i} - 4401.2))) \times e^{(0.0428 \times (CV\%_{13} - 38.77))} \quad (\text{Eq. 16})$$

With λ depicting the baseline hazard of the Weibull function, γ depicting the shape parameter of the Weibull function, AUC_{0-13} the cumulative tacrolimus exposure for the first 13 days post-transplantation, and $CV\%_{13}$ the variation coefficient of the daily AUC variations during the first 13 days.

Below the threshold (<4888.7 ng·h/mL), 12/20 patients (60%) were classified as high-risk, of whom 11 patients had rejection. At or above the threshold (≥ 4888.7 ng·h/mL), 8/20 (40%) were classified as low risk, of whom five patients had no rejection (**Figure 32**). The corresponding analysis using the $CV\%_{13}$ was included in the Supplementary Results (**Figure S37**). The resulting confusion matrix indicated sensitivity 0.79 [95% CI 0.49–0.95, Clopper-Pearson interval], specificity 0.83 [95% CI 0.36–1.00], positive predictive value

(PPV) 0.92 [95% CI 0.62–1.00], and negative predictive value 0.63 [95% CI 0.25–0.92]. The model performance stratified by the identified predictor (above and below the AUC_{0-13} threshold) yields good model calibration (**Figure 31**).

The threshold AUC_{0-13} of 4888.7 ng·h/mL corresponds to a probability of rejection at 1 year of 30.8% (**Figure 33**).

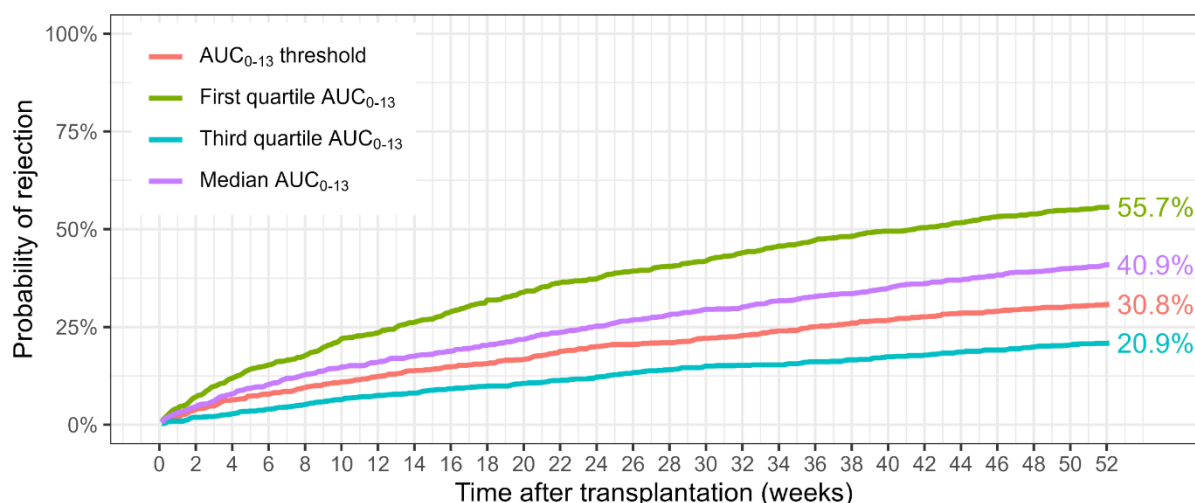


Figure 33. Probability of rejection over time for four different AUC_{0-13} scenarios. The AUC_{0-13} threshold refers to the Youden J index-derived threshold, and the median, first and third quartile refers to the observed ranges in our cohort. Proportions over time were based on 1,000 stochastic simulations.

3.4 Dosing simulations and risk stratification

We developed an interactive tool based on the integrated popPK and rejection prediction model. A hypothetical scenario was constructed and simulated with the following parameters:

i) 0.96 mg tacrolimus IV for 24 h for the first 11 days, followed by 3 mg oral tacrolimus BID for two days, ii) measured tacrolimus concentration of 12.5 ng/mL (day 1 post ITx), 18.0 ng/mL (day 2), 19.7 ng/mL (day 4), 17 ng/mL (day 6), 22.0 ng/mL (day 8), and 13 ng/mL (day 12) and iii) body weight of 55.4 kg (**Figure 34**). The investigated scenario yields an individual AUC_{0-13} of 5008.1 ng·h/mL, and would therefore be assigned to the low risk cohort. The

software tool can be accessed through the following link:

<https://gitlab.kuleuven.be/pharmacometrics/>.

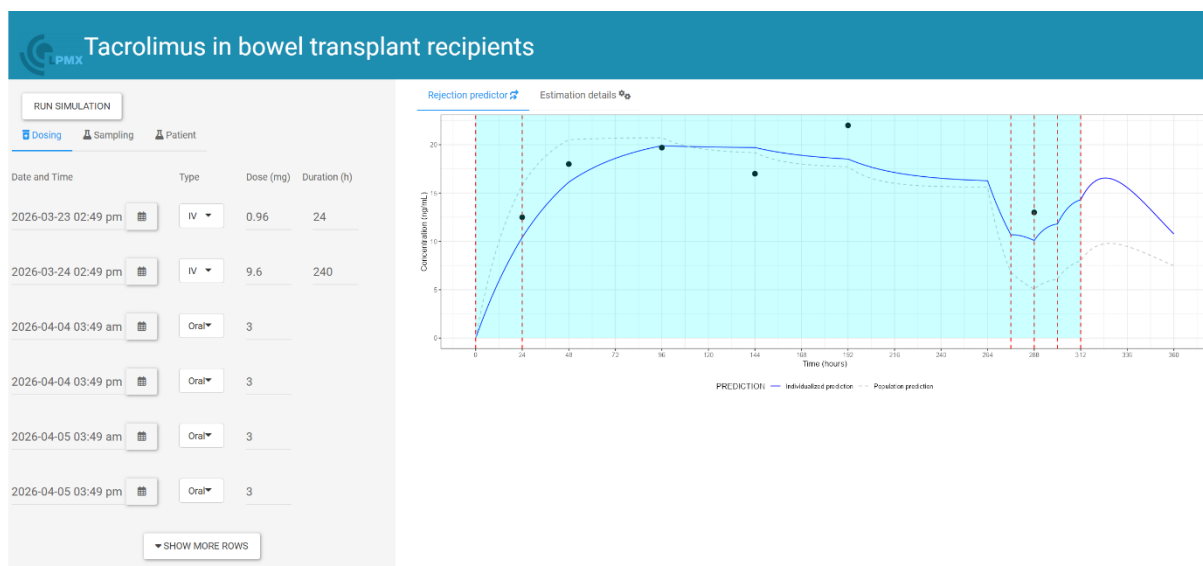


Figure 34. Snapshot of the software tool interface (v.1.0). On the left-hand side, the user inputs tacrolimus dosing and sampling history for a hypothetical patient. The right-hand side outputs the simulated output. The left plot shows the simulated concentration time profile for the population (grey dashed line), and the individual patient (blue solid line) based on the observed concentration (black dots), red dashed lines indicate dosing occasions.

4. Discussion

We present the first popPK and the first exposure–response framework for tacrolimus in intestinal transplant recipients. By using a multistate model to account for competing risks, our approach minimizes bias, offering a more accurate and reliable exposure-response analysis. In this single-center retrospective study spanning 15 years of routine care, we offer unique insight into tacrolimus disposition with time-dependent clearance that stabilized by day 84, and early cumulative exposure (AUC_{0-13}) as a predictor of rejection within the first year. Simulations illustrated that protocol-based IV initiation increased the likelihood of reaching the early exposure target. Our findings emphasize the clinical importance of early tacrolimus exposure in intestinal transplantation and offer a quantitative foundation for model-informed dose initiation.

Tacrolimus disposition in intestinal transplantation was well described by a one-compartment model with first-order absorption and time-dependent clearance that rose after surgery and stabilized by day 84, a trajectory that is biologically plausible given postoperative inflammation (e.g., interleukin-6) transiently suppressing CYP3A4/5 activity before recovery.²⁹³ The oral bioavailability of 26.8% is consistent with prior population studies^{213,254,294} and supportive of the 4× IV-to-oral conversion used in the Leuven protocol. The absorption rate constant was 0.0132 h^{-1} , which is notably lower than values in other tacrolimus models, likely reflecting the early postoperative context.^{189,295,296} Finally, although pediatric recipients often exhibit higher clearance (necessitating higher doses) due to greater CYP3A5 activity, age was not retained as a covariate in our cohort, which underscores the role of center-specific physiology and emphasizes the need for exposure-guided dose adjustment in all intestinal transplant recipients.^{297–300}

We found that cumulative tacrolimus exposure over the first 13 days (AUC_{0-13}) was a predictor of rejection within the first year after intestinal transplantation. We explored

several candidate AUC windows up until day 13, which marked the latest time point before the earliest rejection in our cohort. Evaluating exposure periods extending beyond this point would have required incorporating information that occurs after the event we aim to predict. A simple threshold of 4888.7 ng·h/mL separated high and low risk patients. Furthermore, we demonstrate that $CV\%_{13}$ is predictive of rejection. Because overall exposure (e.g., AUC) targets can be achieved through many different dosing and exposure patterns, this finding highlights that not only attaining the target is important, but also how it is achieved. Ideally, a daily AUC target is reached rapidly and then maintained. In contrast, highly fluctuating exposure, such as periods of excessive AUC followed by minimal or absent exposure, is less desirable. The relevance of tacrolimus inpatient variability (IPV) in solid organ transplantation has been described before, and there is increasing recognition of IPV as a biomarker for therapy outcomes.^{271,292,301,302} Although there were only 14 events for the primary outcome and 4 events for the secondary outcome, considered sparse, our contemporary multistate approach, which incorporates a competing risks model, is robust, allows for relatively precise parameter estimation, and provides excellent fit to the observed data. Multistate models allow the simultaneous analysis of recurrent, competing, and right-censored events, making them indispensable for long-term outcomes and clinical data analysis. In recent years, PK/PD model-guided dosing support has been recognized, including MIPD. Several prospective clinical studies, as those summarized by Minichmayr *et al.*, have demonstrated the value of model-informed strategies in improving tacrolimus target attainment and individualizing immunosuppressive therapy.⁶² While the majority of the trials evaluate exposure-based metrics, increasing attention is paid to response-based dosing decisions. To facilitate later confirmation of our work, we integrated our quantitative dose-exposure-response modeling framework into an actionable software tool. Under the current MDR, we wish to emphasize that the software tool is not intended to guide individual patients' dosing. Moreover, we provide a full software tool and wish to facilitate clinical

validation with an external patient cohort. Given the absence of MDR approval, our tool is intended for research purposes only.

The present work comes with some limitations. First, the data were collected retrospectively from routine clinical care, and therefore are subject to limitations such as unbalanced opportunistic sampling schemes, and measurement noise including timing and documentation errors. These factors may limit the generalizability of the findings to cohorts from prospective studies. Second, information on CYP3A5 and CYP3A4 genotypes is not routinely available at our center. Such data would be valuable, as these enzymes depict the main metabolic pathway for tacrolimus and are often retained as significant covariates on tacrolimus clearance.²⁹⁰ Third, our covariate analysis only allowed the identification of time as a significant predictor of clearance and allometric scaling on volume of distribution. Our mixed-effects PK modeling approach is a powerful tool for analyzing small n studies, and data from 20 patients with over 2,000 observations and long follow-up were sufficient to reliably estimate the fixed-effects model structure. However, we encountered limitations in identifying random effects, including inter-individual variability. Consequently, we chose not to include additional covariates to explain variability within this small cohort. Furthermore, the data did not allow to identify a significant threshold for our AUC_{0-13} rejection predictor, and the resulting AUROC included 0.5 on a 95% CI, likely driven by the small sample size. However, our parametric multistate approach still picked it up as a significant predictor, indicating reliability of our finding. In addition to nonlinear mixed-effects modeling, which enables efficient use of sparse and unbalanced data, several complementary strategies are recognized to support robust PK/PD analyses in small cohorts. These include the use of Bayesian approaches or literature-informed priors to stabilize parameter estimation, integration of external or historical data where appropriate, and application of optimal design principles to maximize information content per subject. Finally, we wish to emphasize the lack of external validation of our developed models. Given

the small sample size, we decided to use the entire cohort for model development rather than dividing it into separate training and evaluation datasets. An ideal external validation data requires dosing information, measured tacrolimus exposure, and information on rejection and death within the first year post transplantation. Global collaboration and data sharing will be essential for further advancing model-informed precision dosing in this particular setting.

In conclusion, our study demonstrates opportunities for model-guided dosing of tacrolimus and a robust quantitative framework together with an actionable software tool to pave the way towards personalized therapy for intestinal transplant recipients.

Supplementary Methods

2.2. Immunomodulatory protocol

Table S18. Tacrolimus trough exposure targets over time for isolated and combined intestinal transplantation based on the Leuven Immunomodulatory Protocol.

Time period	Isolated intestinal transplantation (ng/ml)	Combined and multivisceral intestinal transplantation (ng/ml)
Day 0-14	18-20 (PO), 23-25 (IV)*	13-15 (PO), 18-20 (IV)*
Day 15-28	14-16	10-12
(Month 2.) Day 29-56	12-14	8-10
Month 3	10-12	6-8
Month 6	8-10	4-6
Maintenance without everolimus	6-8	4-5
Maintenance with everolimus	4-5	2.5-3.5

*refers to a steady-state concentration

2.3. Bioanalytical assay

Table S19. Assays for tacrolimus quantification.

	Enzyme conjugated immunoassay (Dade Behring Siemens Dimension RxL)	ARCHITECT chemiluminescent microparticle immunoassay (Abbott and Fujirebio Diagnostics Inc)	LC-MS/MS
Time period (year)	2007 to 2010	2010 to 2017	From 2017
LLOQ	2 ng/mL	1 ng/mL	0.5 ng/mL
Comparability to previous assay	Yes	Yes	Yes

Abbreviations. LC-MS/MS, Liquid chromatography tandem mass spectrometry; LLOQ, Lower limit of quantification.

2.4. Population pharmacokinetics modeling

One and two compartmental models were tested to describe the tacrolimus concentration–time profile. Pharmacokinetics parameters were assumed to follow a log–normal (non-linear) distribution, achieved using an exponential function following the subsequent relationship;

$$PK = TVPK \times e^{\eta_i} \text{ with } \eta_i \sim N(0, \omega^2)$$

Where TVPK denotes the typical value of the population for a given pharmacokinetic (PK) parameter (e.g. CL or V) and PK denotes the individual patient's PK parameter taking into account the individual subjects deviation from the population value denoted by η_i and the variance of the inter-subject variability ω^2 . Stepwise covariate modeling with stage-wise filtering (SCM+) was performed to identify statistically significant covariates in a multivariate analysis. SCM+ followed a number of objective function value (OFV) comparisons. Covariates were retained following a forward inclusion and backward elimination procedure on a $p = 0.05$ threshold, respectively, and precision and plausibility of parameter estimates guided by a condition number $\leq 1,500$ to avoid overparameterization.

Table S20. Covariates tested in SCM+ with stage-wise filtering.

Mechanistic	Structural	Exploratory
Post-operative day on CL	–	Hematocrit on CL, V, k_a , F
Allometric scaling on V		Hemoglobin on CL, V, k_a , F
		White blood cell count on CL, V, k_a , F
		Age on CL, V, k_a , F
		Body weight on CL, V, k_a , F
		AST on CL, V, k_a , F
		ALT on CL, V, k_a , F
		Albumin on CL, V, k_a , F
		C-reactive protein on CL, V, k_a , F
		Serum creatinine on CL, V, k_a , F

Abbreviations. ALT, alanine aminotransferase; AST, aspartate aminotransferase; CL, clearance; F, bioavailability; k_a , absorption rate constant.

2.6. Dosing simulations

Stochastic dosing simulations were performed using Monte Carlo simulations ($N = 1,000$) while sampling from the interindividual variability distribution. Individual patients' pharmacokinetic parameters were estimated using Bayesian forecasting using observed tacrolimus concentration and the developed population pharmacokinetics model. The pharmacokinetics parameter estimation relies on the Markov chain Monte Carlo (MCMC) algorithm integrated in NONMEM.

Supplementary Tables

3.1 Exposure–response: rejection prediction

Table S21. Covariate selection for the population pharmacokinetics model.

Model run	dOFV	Significance level
<i>First round of SCM</i>		
Base model	–	–
Base model + AUC _{0–3}	-0.901	ns
Base model + AUC _{0–7}	-3.311	ns
Base model + AUC _{0–13}	-6.45	<0.05
Base model + AUC _{3–7}	-4.029	ns
Base model + AUC _{7–13}	-1.63	ns
Base model + CV% daily AUC	-6.943	<0.01
Base model + C-reactive protein	-0.006	ns
Base model + age	-2.649	ns
Base model + weight	-0.011	ns
Base model + serum creatinine	-2.292	ns
Base model + HCT	-5.609	<0.05
<i>Second round of SCM with Base model + CV% daily AUC</i>		
New model + AUC_{0–13}	-4.393	<0.05
New model + HCT	-3.902	<0.05
New model + AUC _{3–7}	-1.076	ns
<i>Third round of SCM with Base model + CV% daily AUC + AUC_{0–13}</i>		
New model + HCT	Convergence issue	

Abbreviations. AUC, area under the curve; CV, coefficient of variation; dOFV, difference objective function value; ns. not significant.

Supplementary Figures

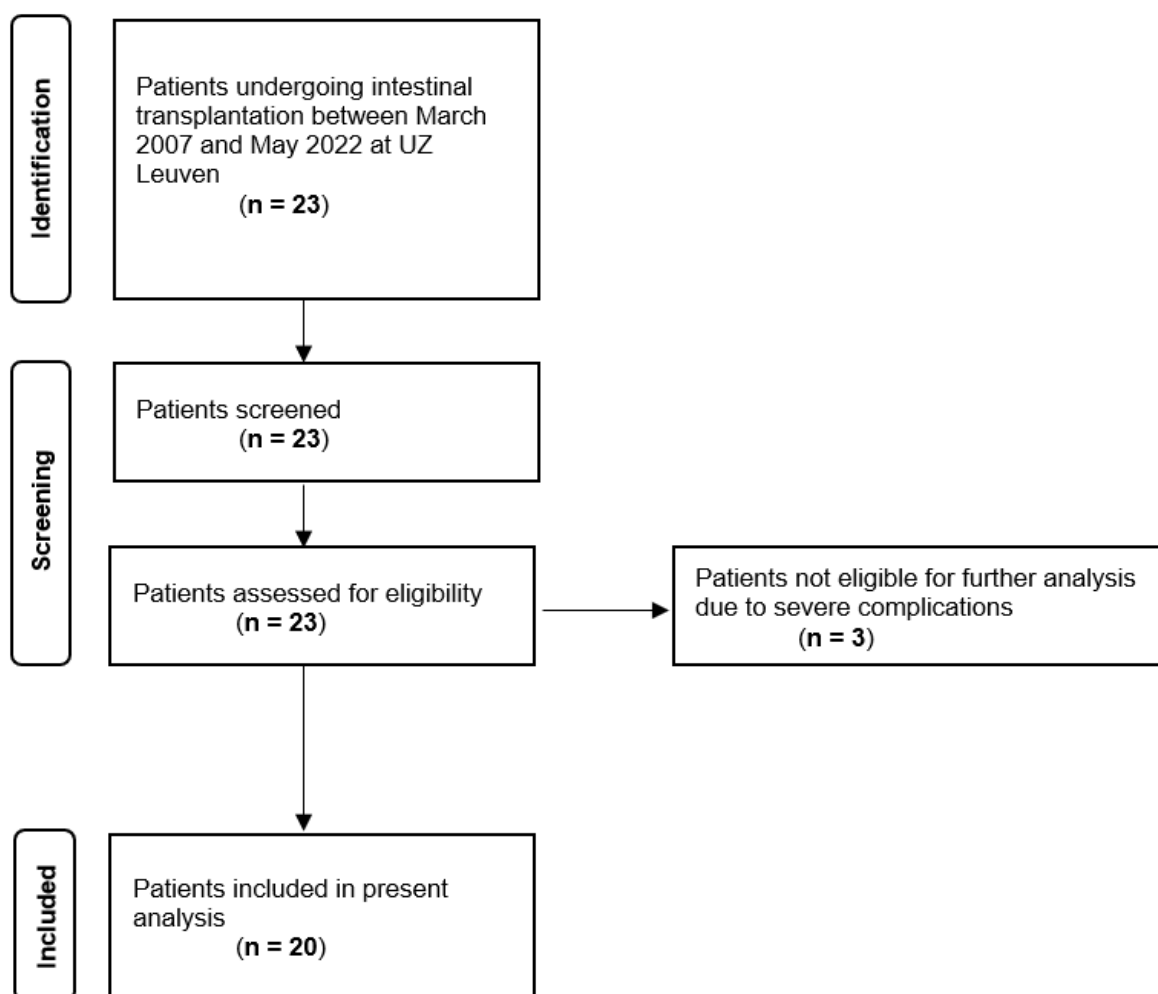


Figure S28. Flowchart depicting the selection of the final cohort, including the identification, screening and inclusion step.

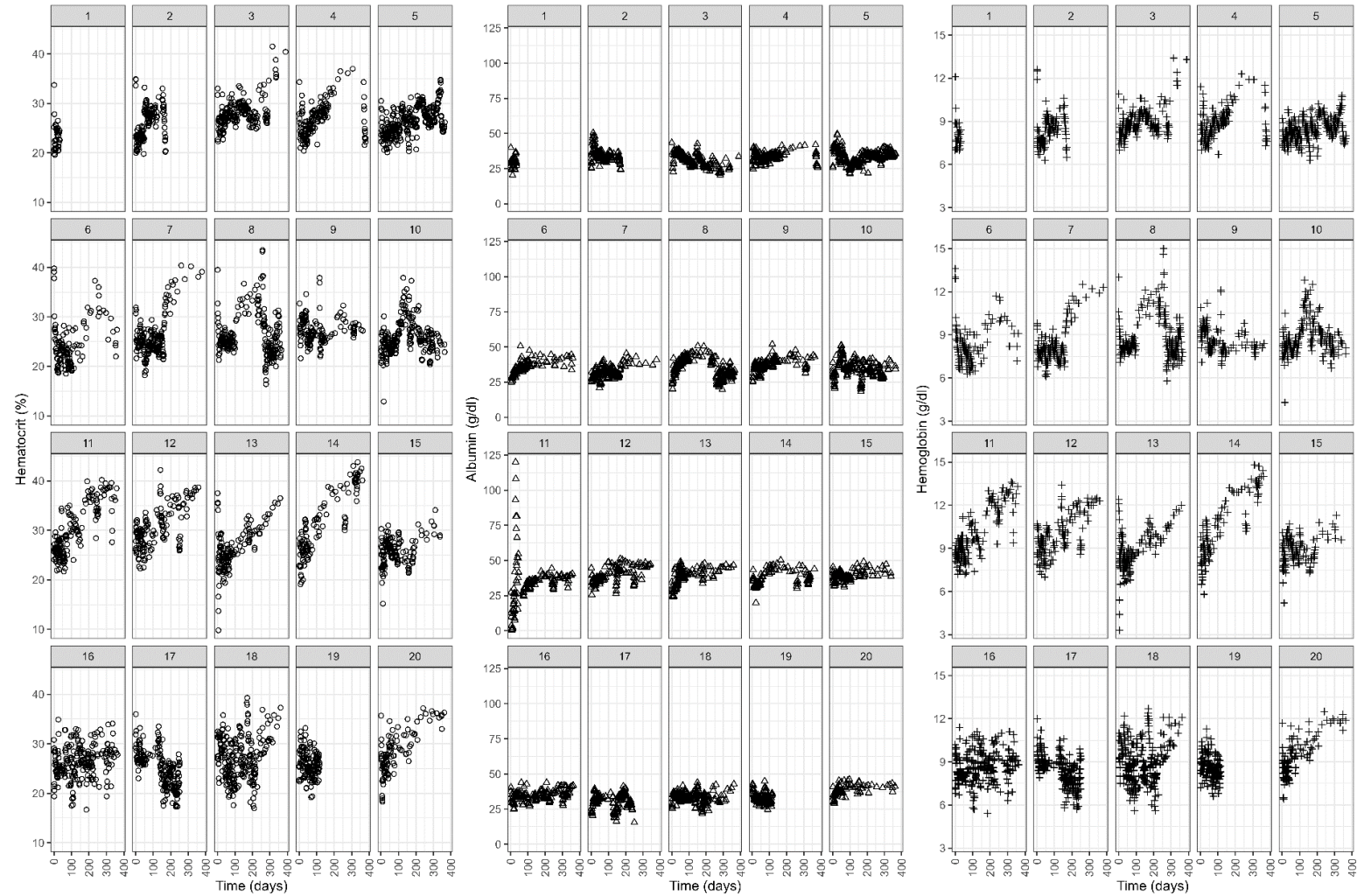


Figure S29. Longitudinal measurement of hematocrit, albumin and hemoglobin per ID.

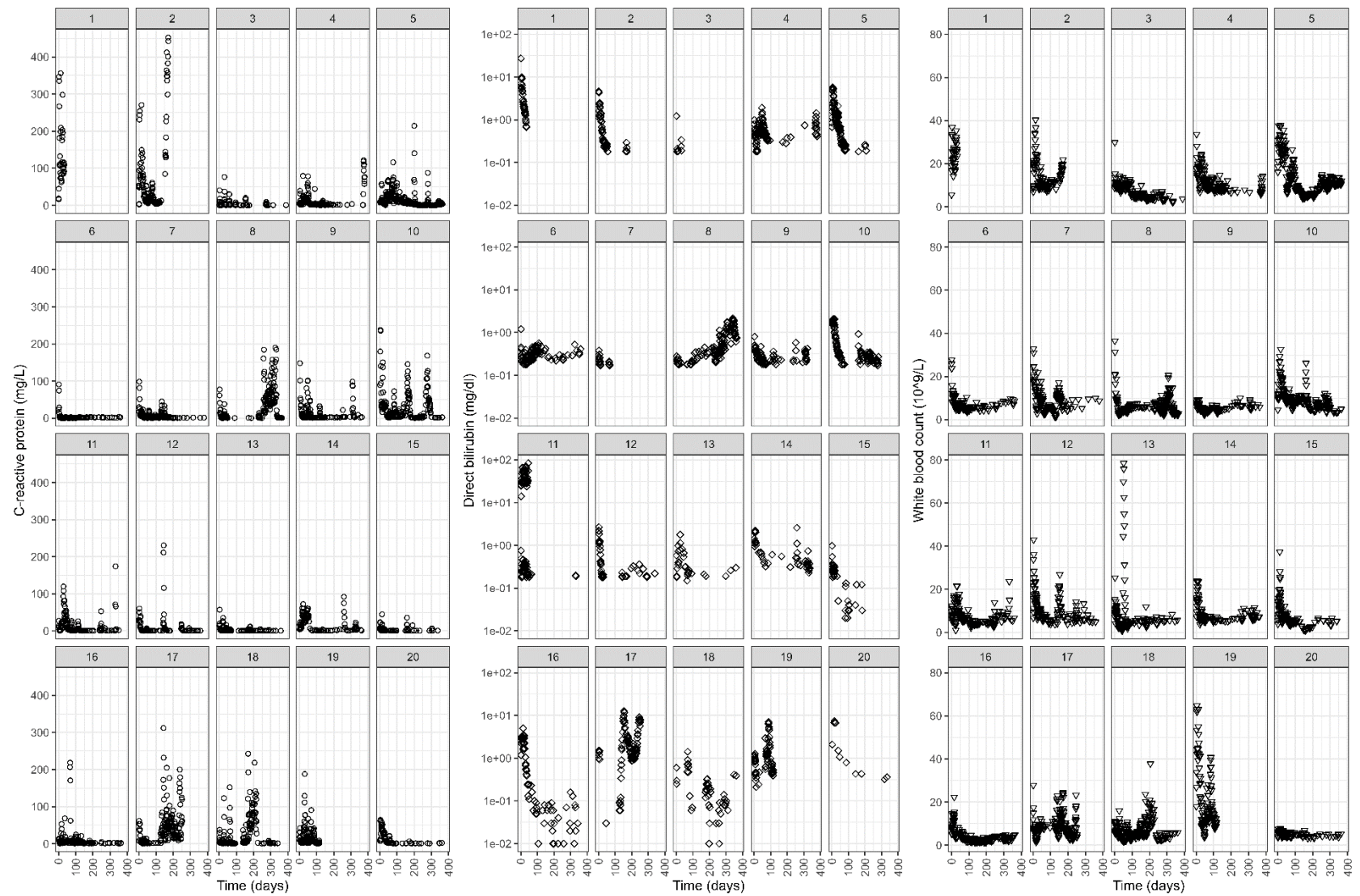


Figure S30. Longitudinal measurement of c-reactive protein, direct bilirubin, and white blood count per ID.

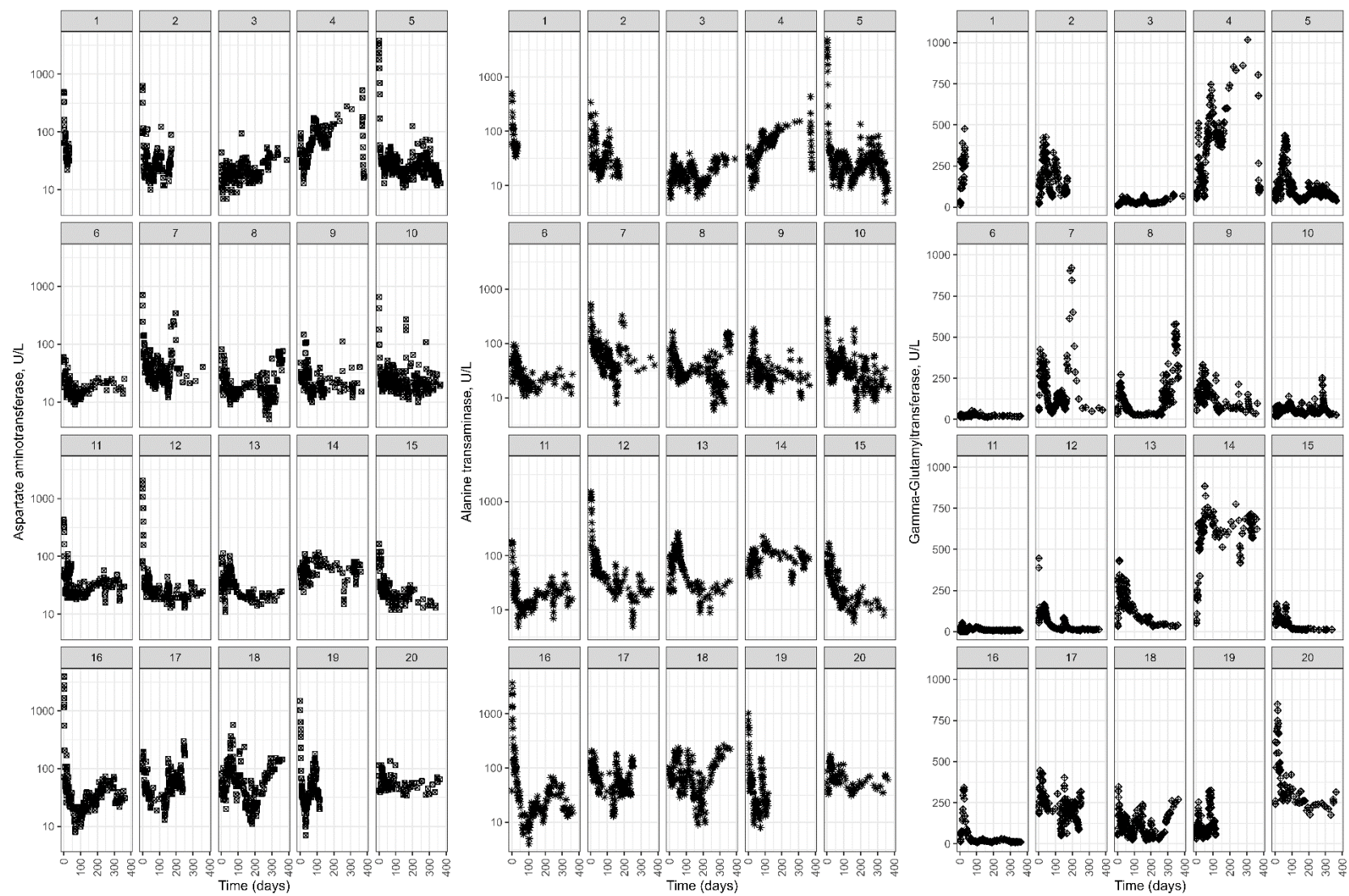


Figure S31. Longitudinal measurement of aspartate aminotransferase, alanine transaminase, gamma-glutamyltransferase per ID.

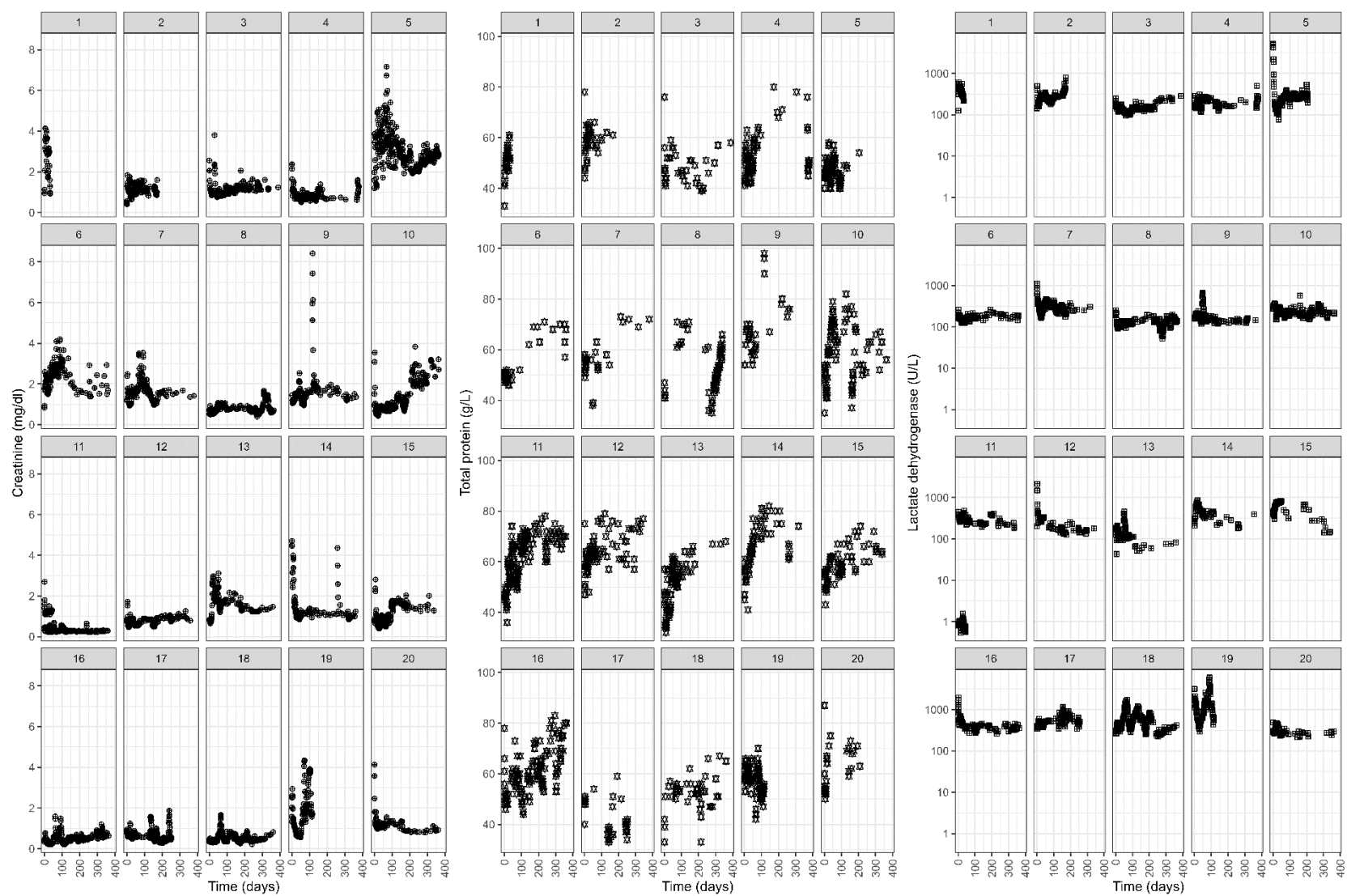


Figure S32. Longitudinal measurement of creatinine, total protein, and lactate dehydrogenase per ID.

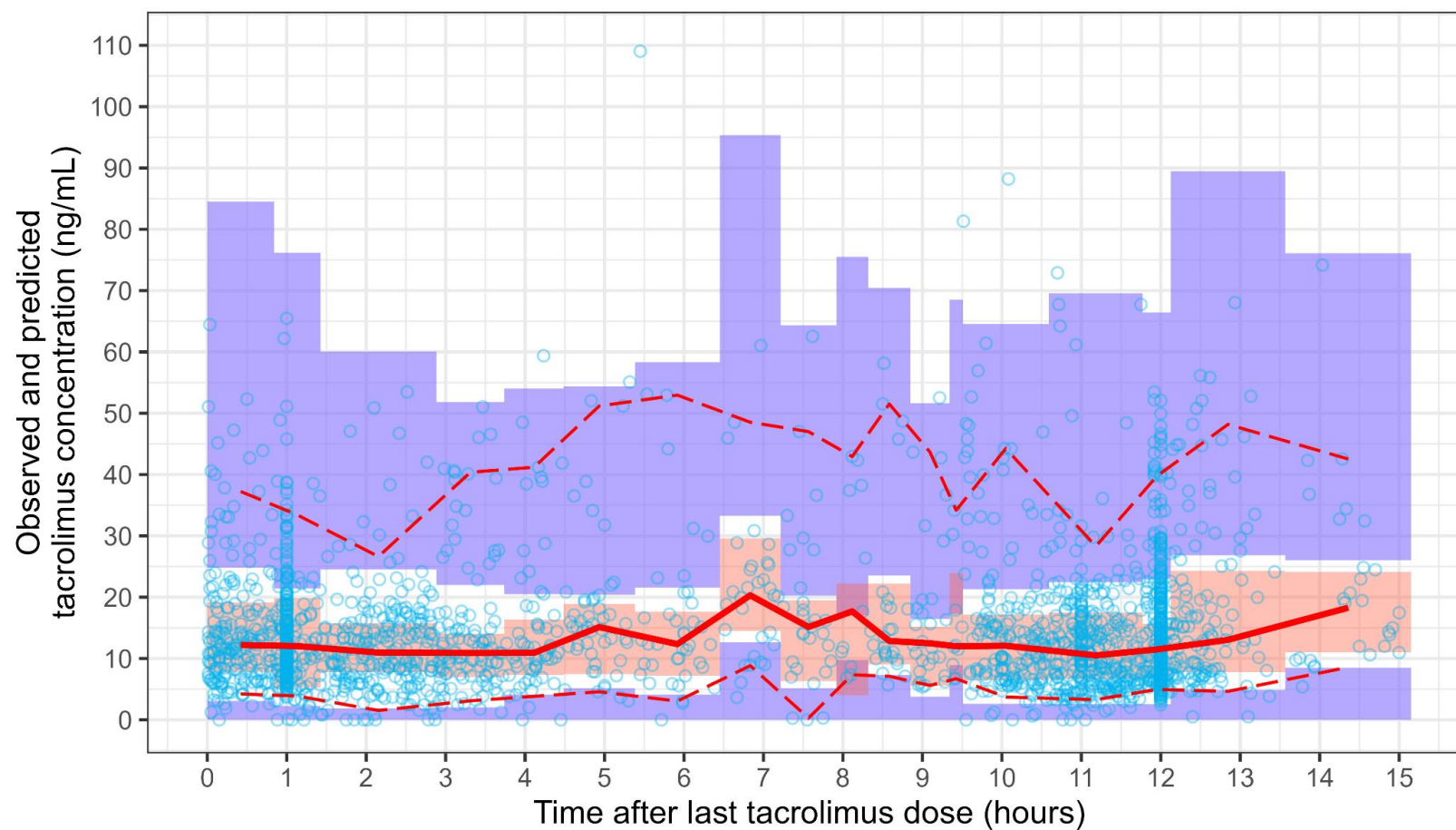


Figure S33. Visual predictive check plots of the final population pharmacokinetic model with time after dose (TAD) on the x-axis. Open circles represent prediction- and variance-corrected observed data. The solid red line is the median of the observed data. The dashed red lines are the 5th and 95th percentiles of the observed data. The red and purple shaded areas indicate the 95% confidence intervals of the median and 5th and 95th percentiles, respectively (N = 2000 simulations). Negative predictions due to the error model were set to zero.

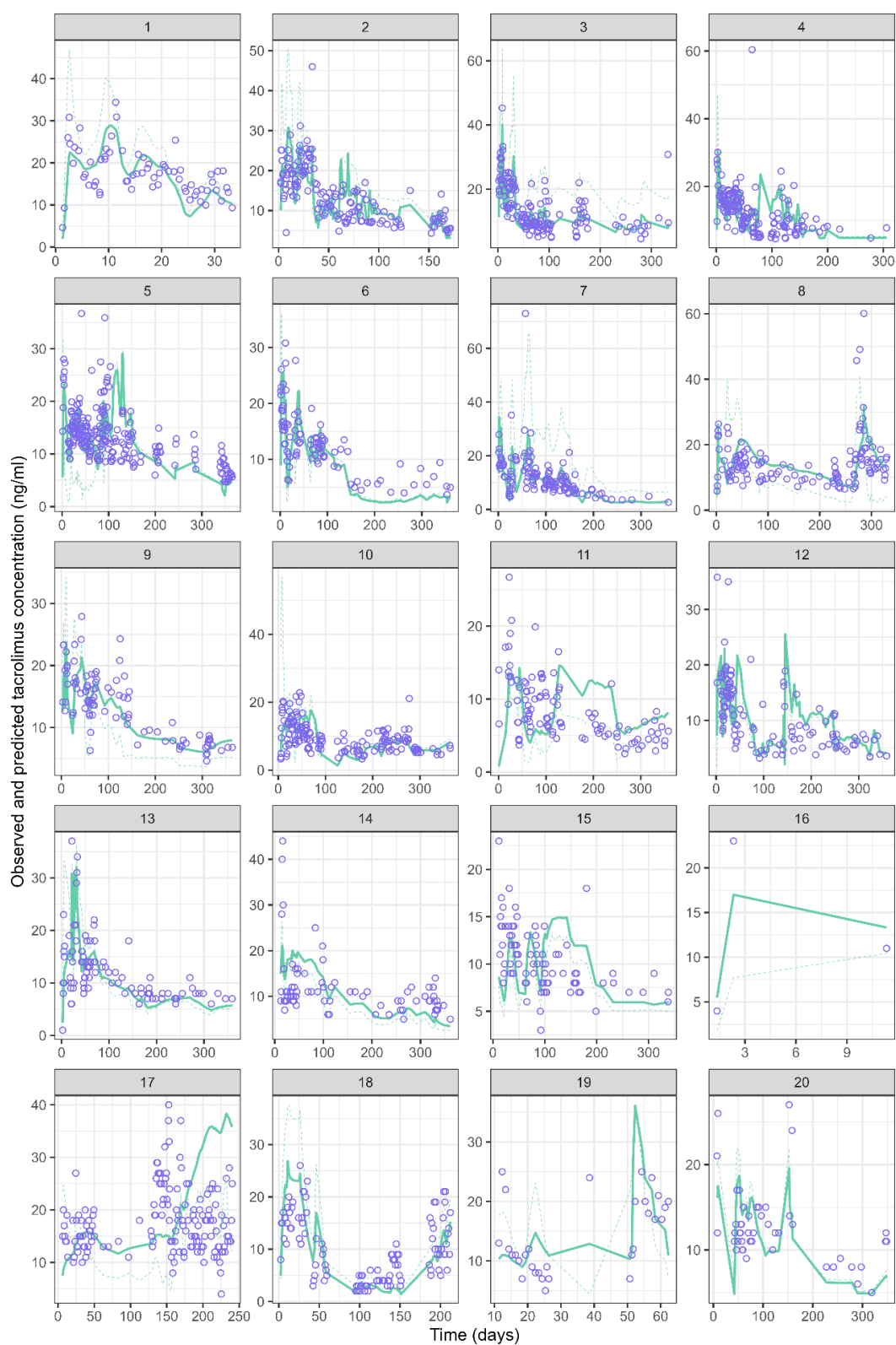


Figure S34. Individual model fit per patient. The purple open circles represent observed tacrolimus concentration. The green solid line represents the individual model prediction and the green dashed line represents the population prediction.

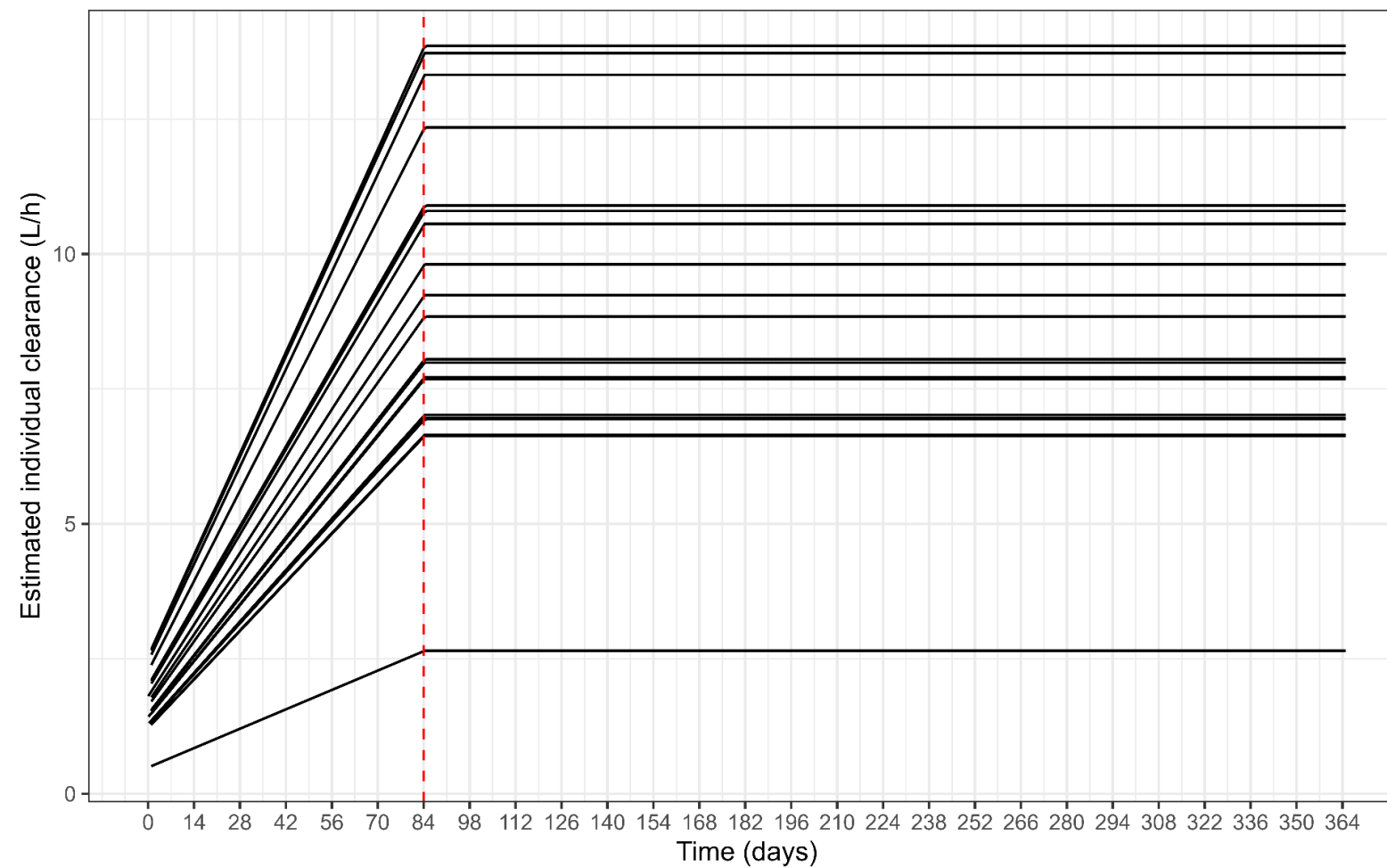


Figure S35. Individual patients' clearance over time. The red dashed line indicates the estimated inflection point of 84 days.

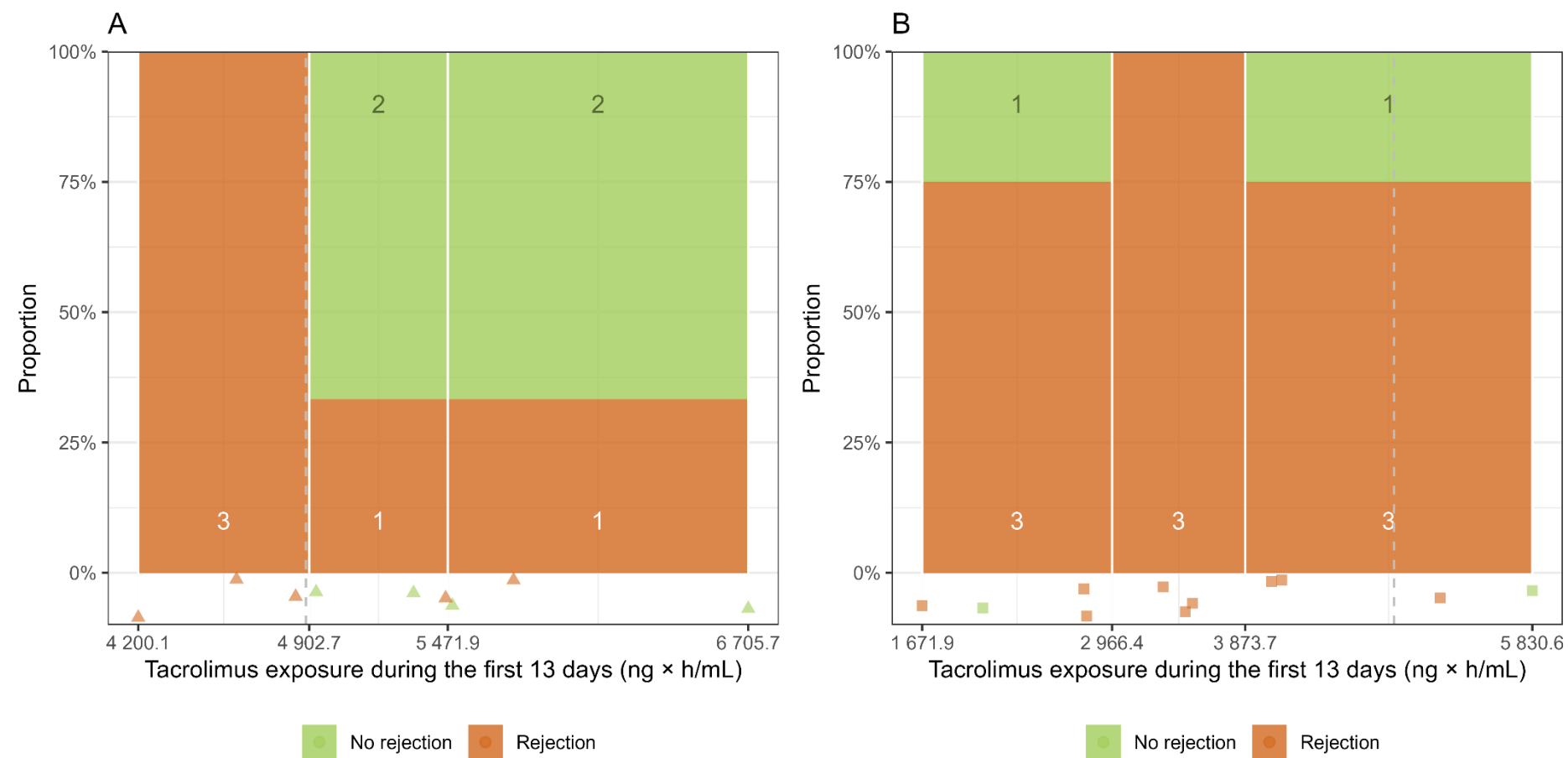


Figure S36. Model-predicted cumulative tacrolimus exposure stratified by patients receiving IV (A) and oral (B) tacrolimus initiation. The grey dashed line corresponds to the identified threshold relying on the whole cohort.

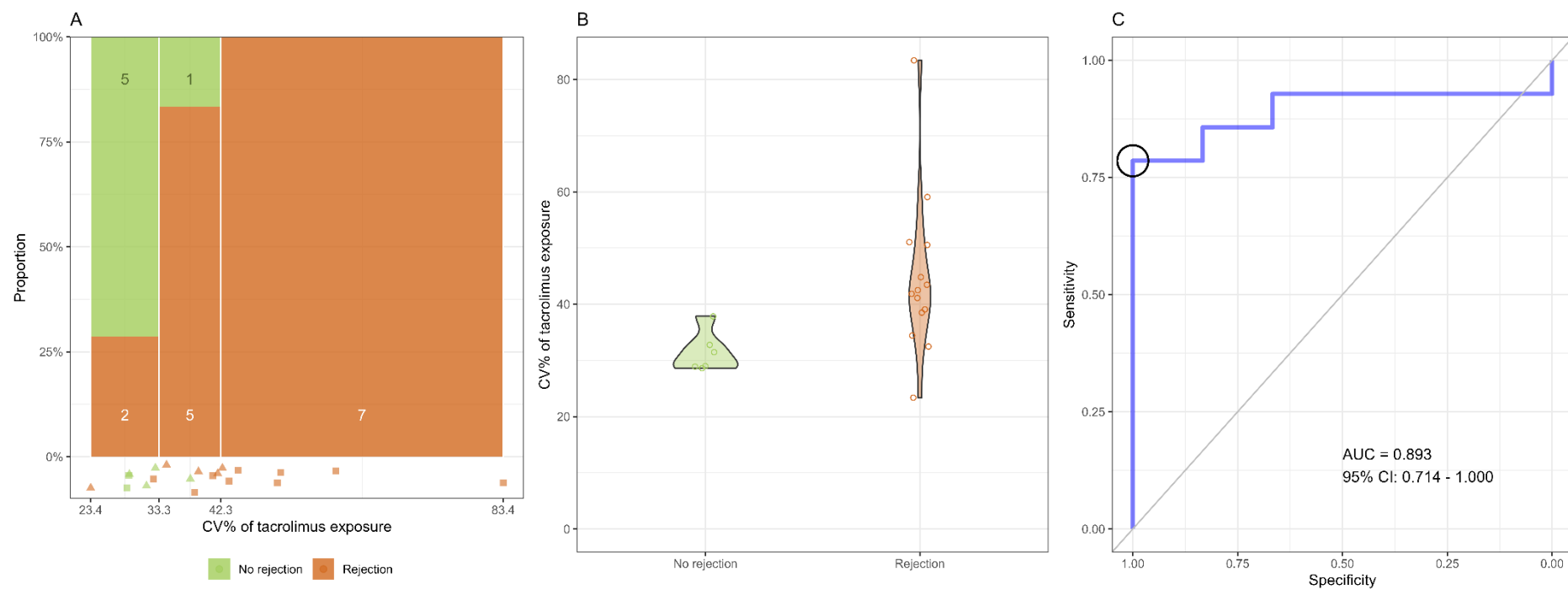


Figure S37. Coefficient of variation of model-predicted tacrolimus exposure (A) and its distribution (B) stratified by patients with and without rejection during the first year post intestinal transplantation and the corresponding receiver operating characteristic curve (C). The triangles and squares in panel (A) indicate intravenous and oral tacrolimus initiation, respectively. AUC, area under the curve; CV%, coefficient of variation in percent.

Supplemental Code Pharmacokinetics Model

\$PROBLEM

```
;; 1. run132_corr_allo
;; 2. Description: 1 CMT; IIV on CL, V1, F, KA; POD (HS) on CL; BW on V
;; x1. Author: u0155394
```

\$INPUT ID

```
TIME
POD
DV
MDV
AMT
RATE
CMT
HCT_dec
HGB_g_dl
WBC
CRP_mg_l
TP
ALB_g_dl
BILD
GAMMAGT_u_l
AST
ALT
CREA_mg_dl
BICAR_mmol_l
AGE
SEX
WT
HT
BMI
ABO
RHE
DIAB_PRE
DIAB_POST
TAD
LOQ
ASSAY_FACT
FORM_FACT
TYPE_FACT
GRAFT
TTD
```

```
$DATA CORR_DF.csv IGNORE=@ IGNORE(TAD.GT.15)
```

```
$SUBROUTINE ADVAN13 TOL=9
```

```
$MODEL NCOMP=3
```

```
COMP=(DEPOT) ; Absorption compartment
COMP=(CENTRAL, DEFOBS) ; Central compartment
COMP=(AUC)
```

\$PK

```
;;START OF COV EFFECT
```

```
PODINF = THETA(1) ; inflection point of which linear to constant CL
```

```
PODIND = 0
```

```
IF(POD >= PODINF) PODIND = 1
```

```
;;END OF COV EFFECT
```

```
; A_0(1)= 0
```

```
; A_0(2)= 0
```

```
TVF1 = THETA(2)
```

```
F1 = TVF1*EXP(ETA(1))
```

```
;absorption
```

```
TVKA = 10**THETA(3)
```

```
; TVKA = TVKA*KACOV
```

```

KA    = TVKA*EXP(ETA(2))

;elimination
TVCL = THETA(4)
TVCL = TVCL + THETA(5)*PODINF*PODIND + THETA(5)*POD*(1-PODIND)
CL    = TVCL*EXP(ETA(3))

;distribution
TVV1 = THETA(6)
TVV1 = TVV1 * (WT/55.4)**0.75
V1    = TVV1*EXP(ETA(4))

; lag time
; TVALAG = THETA(5)
; ALAG = TVALAG*EXP(ETA(5))

S2    = V1/1000 ; DV is ug/L, V is in L, AMT is in mg!
KE    = CL/V1

$DES
DADT(1) = -KA*A(1) ; Absorption compartment
DADT(2) =  KA*A(1) - KE*A(2) ; Central compartment
DADT(3) =  A(2)/S2

$ERROR
; IPRED = A(2)/V1 ; individual prediction by
NONMEM
IPRED = F
IRES  = DV-IPRED ; individual residual
SD    = SQRT(SIGMA(1,1)*(IPRED)**2 + SIGMA(2,2)) ; standard deviation
IWRES = IRES/SD ; individual weighted residual
Y      = IPRED*(1+ERR(1))+ERR(2) ; combined proportional and additive residual
error
AUC2   = A(3)

IF(IPRED.LT.0) THEN
IPRED = 0
ENDIF

IF(Y.LT.0) THEN
Y = 0
ENDIF

$THETA
(0, 84.3) ; POD on CLINF
(0, 0.269, 1) ; F
(-9, -1.88, 3) ; KA
(0, 1.59) ; CL
(0.0846) ; SLOPE INF
(0, 37.1) ; V1

$OMEGA
(0, 0.257) ; F
(0, 3.15) ; KA
(0, 0.176) ; CL
(0, 3.47) ; V1

$SIGMA
(0, 0.107) ; proportional
(0, 9.9) ; additive
$ESTIMATION METHOD=1 INTERACTION MAXEVALS=9999 PRINT=1 NSIG=3 SIGL=9 ; METHOD=1,
FOCE+1 does not work for M3
$COVARIANCE PRINT=E
$TABLE ID TIME TAD IPRED IWRES PRED Y TVCL CL V1 KA F1 ETA(1) ETA(2) ETA(3) ETA(4)
CWRES
AUC2 NOPRINT ONEHEADER FILE=sdtab132_corr_allo

```

Supplemental Code Multistate Time-to-Event Model

\$PROBLEM

```
;; 1. Based on: run1007
;; 2. Description: Base model + AUC daily CV + AUC0-13 (a)
;; x1. Author: u0155394
```

\$INPUT

```
ID
TIME
CMT
AMT
DV
EVID
AUCC72
AUCC168
AUCC312
AUCC_72_168
AUCC_168_312
AUC_daily_CV
CRP_MEDIAN
CREA_MEDIAN
HCT_MEDIAN
AST_MEDIAN
ALT_MEDIAN
AGE
WT
```

\$DATA MSM.csv IGNORE=@

\$SUBROUTINES ADVAN13 TOL=6

\$MODEL NCOMP=3 ;; 1 is ongoing treatment
 ;; 2 is rejection
 ;; 3 is death

\$PK

```
BSV = ETA(1)           ; place holder
IF(NEWIND.NE.2) DV1 = 1
IF(NEWIND.NE.2) XDV = 1
IF(NEWIND.NE.2) PDV = 1
;Sim_start: add for simulation
IF(DV.GE.1) DV1 = DV
IF(DV.GE.1) XDV = DV
;Sim_end
```

```
;;-----mean transit times in days-----
MTT12=THETA(1) ;from ongoing treatment to rejection
MTT13=THETA(2) ;from ongoing treatment to death
MTT23=THETA(3) ;from rejection to death
```

```
;;-----Weibull parameters-----
HZL12=1/MTT12
HZA12=THETA(4)
```

```
;;-----Initialize location probabilities-----
IF(NEWIND.NE.2) THEN
BIO1 = 1
BIO2 = 0
BIO3 = 0
```

ENDIF

```
;;-----
;----- "Bioavailability" factors -----
F1 = BIO1
F2 = BIO2
F3 = BIO3
```

\$DES

```
DEL = 1E-16
WEIBULL12 = HZL12*HZA12*(HZL12*(T+DEL))**(HZA12-1)
```

```

; WEIBULL15 = HZL15*HZA15*(HZL15*(T+DEL))**(HZA15-1)
;-----probability transfer constants-----
K12=WEIBULL12 * EXP(THETA(5)*(AUC_daily_CV-38.77319)) * ( 1 + (THETA(6)*(AUCC312 -
4401.2)))

K13=1/MTT13
K23=1/MTT23

;--Differential equations for the probability of each score
DADT(1) = - A(1) * (K12 + K13 ) ; ongoing tacrolimus therapy
DADT(2) = A(1) * K12 - A(2) * K23 ; rejection
DADT(3) = A(1) * K13 + A(2) * K23 ; tube dislodgement

$ERROR
P_1 = A(1) ; Probability of observing ongoing tacrolimus therapy
P_2 = A(2) ; Probability of observing rejection
P_3 = A(3) ; Probability of observing death

;-----Cumulative probabilities -----
CUP1 = A(1) ; 1
CUP2 = A(1) + A(2) ; 1 + 2
CUP3 = A(1) + A(2) + A(3) ; 1 + 2 + 3

;----- Model prediction -----
IF(DV.EQ.1.AND.EVID.EQ.0) Y = P_1
IF(DV.EQ.2.AND.EVID.EQ.0) Y = P_2
IF(DV.EQ.3.AND.EVID.EQ.0) Y = P_3

TOT = P_1 + P_2 + P_3
;-----
; Sim_start
; IF(ICALL.EQ.4) THEN
;
; IF (EVID.EQ.0) THEN
; CALL RANDOM (2,R)
; USUR=R
; ENDIF
;
; DV1=PDV
; IF(USUR.LE.CUP1.AND.EVID.EQ.0) DV1=1
; IF(USUR.GT.CUP1.AND.USUR.LE.CUP2.AND.EVID.EQ.0) DV1=2
; IF(USUR.GT.CUP2.AND.USUR.LE.CUP3.AND.EVID.EQ.0) DV1=3
;
; XDV = DV1
; IF(EVID.EQ.0) PDV=XDV
; ENDIF
;Sim_end
;----- Update Bioavailability factors -----

BIO1=0
BIO2=0
BIO3=0

IF(XDV.EQ.1) BIO1 = 1
IF(XDV.EQ.2) BIO2 = 1
IF(XDV.EQ.3) BIO3 = 1

$THETA
;-----multi-state model parameters-----
(0,200) ;MTT12 1
(0,1000) ;MTT13 2
(0,600) ;MTT23 3

(0,2) ;Shape12 4

```



```

;;; covariate effects
(-0.001) ; (AUC CV)
(-0.0016, -0.00005, 0.0003584743) ; (AUC312)

$OMEGA 0 FIX ; eta as a placeholder
; Sim_start
;$SIGMA 0 FIX
;$SIMULATION (12345) (123 UNIFORM) (123 UNIFORM) NSUB=1 ONLYSIM
$ESTIMATION METHOD=1 LIKE LAPLACE NUMERICAL SLOW PRINT=1 SIGL=9
MAXEVALS=9999 CTYPE=4 NSIG=2
$COVARIANCE MATRIX=R
;Sim_end
$TABLE ID DV1 XDV TIME AMT EVID CMT DV BIO1 BIO2 BIO3 F1 F2 F3
P_1 P_2 P_3 K12 K13 K23 Y TOT NOPRINT FILE=sdtab1101

;;;;; creating a VPC command ;;;;
vpc -samples=2000 -flip_comments -dv=XDV -levels=1,2 -seed=1234 -rplots=2 -
auto_bin=auto -dir=vpc_run1101 run1101.mod

```


GENERAL DISCUSSION AND PERSPECTIVES

Parts of the discussion are based on

Yannick Hoffert and Erwin Dreesen. Response to Dai *et al.* *Am J Gastroenterol.* 2025 Dec 1;120(12):2981-2982.

Yannick Hoffert, Anthony Buisson, Erwin Dreesen. Response to Shen *et al.* *Am J Gastroenterol.* 2026 Apr 3.

In the present work, we investigated the pharmacokinetics and pharmacodynamics of various drugs commonly used for the treatment of GI disorders. Our model-based analyses demonstrated opportunities for therapy individualization to increase PK target attainment or therapy outcomes for patients with IBD, SBS, or ITx. In almost all settings (except the chapter on PEG-related outcomes), dose individualization (covariate-based or exposure-based) is performed, indicating the inadequacy of the *one-dose-fits-all* or covariate-based dosing approach. However, despite the pressing need, guidance and algorithms on how to reach therapeutic exposure are not available.

In pharmacometrics research, dose finding and therapy individualization can be understood as an iterative *learn-and-confirm* cycle, first described by Sheiner *et al.*³⁰³ In a first step, data is being collected, modeled, followed by model simulations and ideally a prospective validation of the proposed individualized dosing scheme (**Figure 35**). Importantly, the origin of the data is not decisive. It may stem from flat-dose studies, TDM investigations, or dedicated MIPD trials. Regardless of the data source, the *learn-and-confirm* cycle represents a continuous process of learning from available data, generating hypotheses, and prospectively testing them in subsequent studies.

To counteract the *one-dose-fits-all* paradigm, decision trees and flowcharts were established for instance, for infliximab, as shown in trials like TAXIT, TAILORIX, and NOR-DRUM.^{304–306} However, they follow rather a trial-and-error approach, commonly resulting in slow and imprecise target attainment with suboptimal outcome rates. Furthermore, the lack of clinical experiences with new formulations such as SC infliximab hamper the clinical implementation.³⁰⁷

More contemporary approaches include the use of models as demonstrated for tacrolimus in trials by Størset *et al.*, Lloberas *et al.*, or Fukudo *et al.*^{74,75,165} However, this requires dedicated models in a dedicated population and often times the need for software tools to

allow an easy access for model-informed dose optimization. At the moment, there is a research void in dedicated population pharmacokinetics and exposure–response models of tacrolimus in ITx.

Lastly, we wish to pave the way for exposure-guided and model-informed dose optimization in patients with SBS, and investigate for which patients a flat dose approach might be inappropriate. The absence of suitable models impedes the clinical implementation of an MIPD approach. Besides, optimal PK (and PD targets) are yet to be established in this cohort.

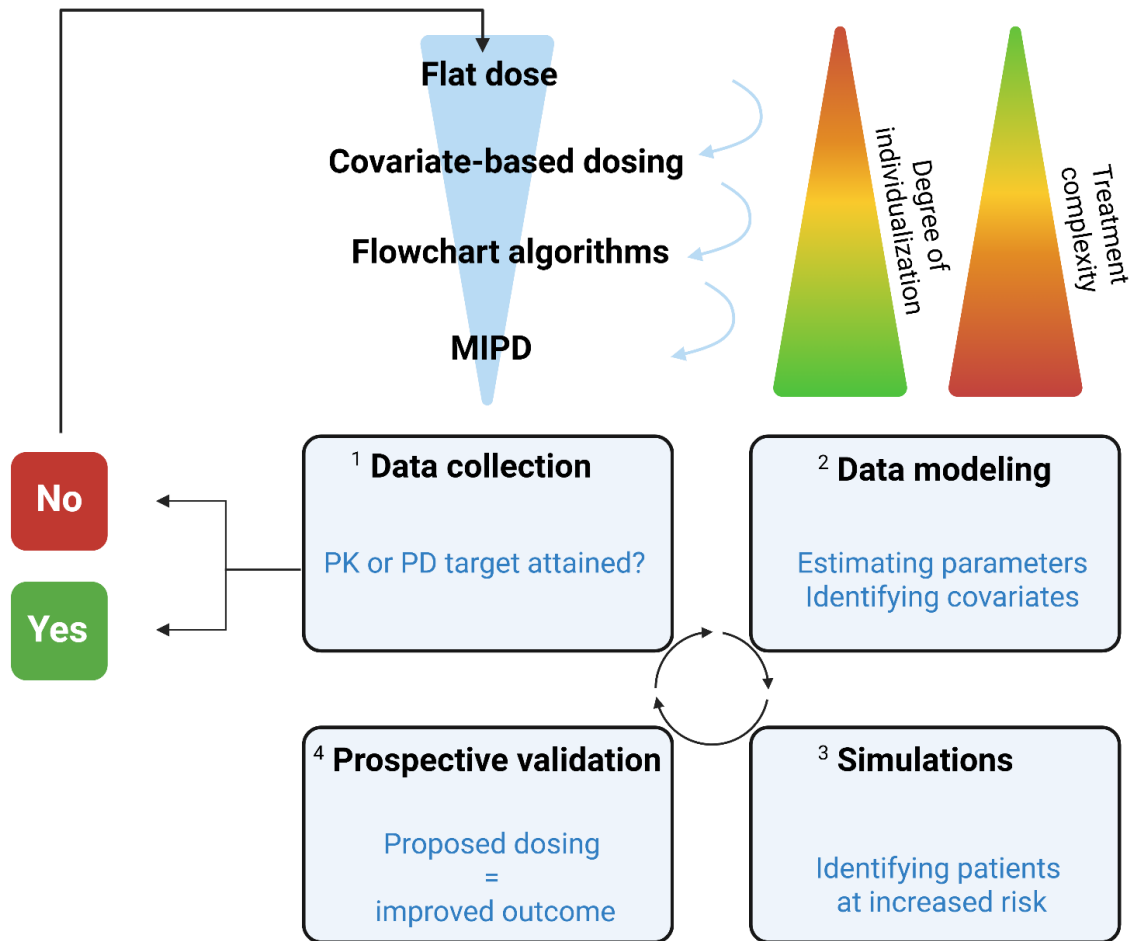


Figure 35. Four step workflow of the *learn-and-confirm* cycle proposed by Sheiner *et al.*³⁰³
Abbreviations. MIPD, model-informed precision dosing; PD, pharmacodynamics; PK, pharmacokinetics.

Dose optimization for SC infliximab

Since the EMA approval of SC infliximab in 2020, a new therapeutic option has been added to the treatment armamentarium for IBD. Although SC infliximab has demonstrated comparable clinical efficacy and safety to the IV originator, its uptake in routine clinical practice remains limited.³⁰⁸⁻³¹⁰ While IV administration is less convenient for patients, it inherently provides structured, regular clinical visits, which may facilitate TDM, reinforce adherence, and allow early detection of loss of response or adverse events. In contrast, SC infliximab is administered at home, potentially reducing the frequency of clinical contact and routine monitoring. Currently, there is a lack of evidence to identify which patient subgroups may particularly benefit from continued regular TDM and clinical follow-up while receiving SC infliximab, highlighting an important gap in existing research.

In **chapter 1** of this doctoral thesis, we conducted a predictive performance assessment aiming to predict the SC exposure of individuals taking into account previous IV exposure data. We showed that the predictive performance was poor, particularly for patients who experienced relapse. Furthermore, we developed a pharmacodynamics relapse model to predict relapse already at baseline and prior to the first SC administration.

This approach may help identify patients at high risk of relapse who could potentially benefit from more frequent clinical follow-up including TDM measurements. However, the model did not allow to identify optimized SC dosing regimens as predictors capable of mitigating the observed increased relapse risk. Consequently, it remains unclear whether patients at higher risk would benefit from SC dose intensification or whether continuation of intravenous administration would be more appropriate. Overall, our findings are in line with the DIAMOND consensus, which supports the use of TDM prior to IV-to-SC switching and recommends reactive TDM in cases of loss of response during maintenance therapy.³⁰⁷ Finally, results from dedicated trials like AMARETTO (NCT06113913) and DISCUS-IBD

(ACTRN12622001458729) are awaited to further shed light on intensified SC initiation strategies and the impact on relapse rates.

Perspectives

Our work presents a quantitative modeling approach that enables the identification of patients at increased risk of relapse prior to switching to SC infliximab. At present, it remains unclear whether these high-risk patients would benefit from SC dose intensification, as the definition of optimal drug exposure, and (more importantly) the most appropriate exposure metrics, after SC administration is still under debate.

Notwithstanding these uncertainties, we evaluated the predictive performance of a PK model with respect to predicting post-switch levels using pre-switch data. However, improving the predictive performance of PK models remains an active area of research. Potential future directions include the development of platform models that integrate multiple data sources (i.e., multi-center studies) and are continuously updated as more data becomes available. Global collaboration and data sharing will be essential.

Our models have been implemented in an open-source software tool, allowing external validation and supporting the clinical application of the proposed risk-stratification approach. Although the development of such a tool represents an important first step toward translating MIPD into clinical practice, MIPD has so far been used predominantly in research settings. By providing open access, we aim to encourage the use of our models and support their integration into electronic health records (EHRs). All our models and software tools introduced in the present work were made available on our GitLab account (<https://gitlab.kuleuven.be/pharmacometrics>).

Dose optimization of infliximab in patients with ASUC

Dosing of infliximab in patients with ASUC remains challenging, and despite the urgency of this clinical situation, available data are still limited. Through international collaboration within the spECTRUM consortium, an international group of TDM specialists, we were able to assemble multicenter data to help address this gap.³¹¹ Even though ideal infliximab exposure targets are still the subject of debate, TDM is commonly performed, and there is growing recognition and interest in infliximab MIPD particularly in ASUC.³¹²

In **chapter 2** we conducted a retrospective multicenter study aiming to describe and predict the PK and PD of infliximab in patients with ASUC. The clearance-normalized overall exposure was a significant predictor of colectomy rates, reinforcing previous findings.³¹³

Perspectives

Evidence is accumulating on infliximab clearance as a superior predictor for therapy outcomes compared to infliximab exposure, especially when doses were optimized.^{34,100,313,314} Unlike infliximab exposure, the clearance of the drug is an unmodifiable and a patient-specific parameter. However, it indicates the necessity of intensified dosing to counteract the increased CL allowing to maintain or increase infliximab exposure.^{31,312}

Currently, we are investigating the feasibility and performance of an infliximab MIPD service for patients with ASUC, and to date, three patients have been recruited [IGNITE trial; CIV-24-05-047117]. We expect one patient a month and aim for a small *n* study of 10 patients in total. Secondary endpoints of the study include effectiveness (i.e., colectomy-free survival) and treatment costs.

Our MIPD service relies on the *TDMx* software and the integrated in-house developed model-averaging algorithm (<https://tdmx.shinyapps.io/infliximab/>).³¹⁵ Furthermore, we integrated our own developed PK and PD model in our own software tool, facilitating external validation

and clinical implementation of our personalized dosing algorithm. User-friendly software tools are key for successful translation from a research setting to the clinic. In the end, integrating models directly in the EHR will ultimately facilitate access for model-based therapy individualization at the patient's bedside. At the moment, software tools for dosing support are considered as medical devices in the European Union and therefore fall under the medical device regulation (MDR). Regulatory frameworks, while requiring dedicated effort to comply with, are essential for patient safety and represent a key step toward MDR approval and broader clinical implementation.

Dose optimization of apixaban in patients with short bowel syndrome

Apixaban absorption was reduced in patients with SBS, necessitating dose adaptation to ensure and maintain exposure compared to the control cohort. While EMA recognized covariate-based dosing considering age, body weight, and serum creatinine, we provide quantitative evidence to further support dose individualization in patients with SBS.

In **chapter 3** we performed a prospective, non-randomized, cross-over study in patients with and without SBS. Furthermore, the effect of teduglutide on the recovery of apixaban absorption capacity was evaluated.

Perspectives

The popPK model developed in this chapter provides a quantitative basis showing that patients exhibit lower absorption (rate and extent) compared to non-SBS patients. Furthermore, we introduce citrulline as a biomarker for functional enterocyte mass as a significant covariate on apixaban absorption.

Model-based dosing simulations will be performed to characterize the probability of target attainment across SBS phenotypes and doses (e.g., 2.5 mg, 5 mg, 7.5 mg, and 10 mg twice daily at steady state). Such simulations may enable a “traffic-light” stratification (low/intermediate/high risk of underexposure) to guide when standard dosing is likely sufficient versus when TDM and/or dose optimization should be considered.

Within our MIDD framework, these hypotheses can then be prospectively evaluated in clinical studies following a learn–confirm cycle. Furthermore, it would be interesting to extend the prospective study to investigate higher doses for patients with SBS (e.g. 7.5 or 10 mg), trying to identify an apixaban dose effect on F. This iterative approach may ultimately support the identification of more suitable dosing regimens for patients with SBS. Additionally, it would be interesting to include more patients before and after teduglutide

administration. At the moment, we saw a trend of increased absorption capacity, yet a quantitative confirmation is pending.

Finally, clinical outcomes such as bleeding or thrombosis could be investigated in cohorts on long-term apixaban therapy to evaluate appropriate apixaban exposure in patients with SBS. Considering the rarity of SBS and scarcity of apixaban PK data in this cohort, global collaboration and data sharing will be essential to further support data-driven dosing.

Dose optimization of tacrolimus

Model-informed precision dosing of tacrolimus has shown benefits in accelerating target attainment and reducing variability in exposure.^{74,75,165,166} At present, the application of MIPD in bowel ITx is largely limited by the availability of dedicated population pharmacokinetic (popPK) models. In addition, exposure–response analyses remain challenging because competing risks, particularly death, which is common in critically ill transplant recipients, are commonly ignored, resulting in a distorted and biased exposure-response analysis.

In **chapter 4** we established a conceptual multistate modeling framework to allow outcome analysis in the presence of competing risks, commonly arising with critically ill patients such as patients on enteral nutrition or undergoing intestinal transplantation.³¹⁶ This superior framework was successfully extended and allowed a bias-free exposure-response analysis of tacrolimus in intestinal transplantation (cf. chapter 6).

In **chapter 5** we reviewed available population pharmacokinetics models and benchmark software tools and integrated tacrolimus MIPD modules. Furthermore, we developed our own software tool, which integrates four tacrolimus modules and allows the estimation of an individual patient's PK parameters, and to simulate a corresponding concentration-time curve for a hypothetical dosing scenario.

In **chapter 6** we developed a population pharmacokinetics and exposure–response model of tacrolimus in intestinal transplant recipients. We challenge the static dosing assumption during the early postoperative phase due to time-changing PK. Furthermore, we show that early exposure decreases the risk of rejection and provide a software tool, allowing individualized exposure simulations to assess an individual patient's risk of rejection

Perspectives

Looking beyond dose–exposure, we established a tacrolimus exposure–response framework. The developed model allows risk stratification by identifying patient subgroups with differing probabilities of response. Such stratification can support more informed clinical decision-making and may contribute to individualized treatment strategies.

This work builds on methodological advances, most notably the development and successful integration of a multistate framework. In clinical research, competing risks are often incorrectly handled as simple right-censored events, which can lead to biased parameter estimates. Multistate models are powerful for appropriately analyzing outcomes in critically ill patients, including patients undergoing transplantation. Although their value is increasingly acknowledged in transplantation research, we aim to further encourage their broader adoption. Currently, we are exploring opportunities to extend the developed framework to other disease areas, including oncology. Finally, in an ongoing project, we are evaluating the impact of tacrolimus exposure on longitudinal outcomes, such as kidney function. Overall, this work extends beyond PK-focused analyses and places greater emphasis on (adverse) outcomes.

Our integrated popPK and exposure–response framework represents a crucial initial step for dose optimization in intestinal recipients. At the moment, the developed popPK model is under evaluation, and its predictive performance is being compared with other popPK models identified in Chapter 5 using a model selection and averaging approach.^{59,315} In addition, data from a small number of additional ITx patients (approximately 1–3) are being collected to pave the way for external validation of the PK/PD model. This work underscores the importance of multidisciplinary and multicenter collaboration, to allow external evaluation in settings where patient numbers are exceptionally low. Data sharing should

follow the principle of “as open as possible, as closed as needed,” respecting intellectual property and regulatory frameworks.

Final reflection

As awareness grows around the value of tailoring treatments to the individuals who benefit most, personalized therapies are becoming increasingly important. With the expanding availability of healthcare data and advances in analytical and computational modeling techniques, accessible personalized pharmacotherapy is emerging as the future of medicine.

At its core, pharmacometrics embraces patient variability, with particular attention to those who deviate from the “typical” patient profile. This perspective aligns closely with the work presented in this thesis, which contributes to how we understand, quantify, and translate that variability into clinical practice.

REFERENCES

1. Peery, A. F. *et al.* Burden and Cost of Gastrointestinal, Liver, and Pancreatic Diseases in the United States: Update 2024. *Gastroenterology* **168**, 1000–1024 (2025).
2. Zhang, Z. *et al.* Global, regional, and national burden of inflammatory bowel disease in persons aged 60–89 years from 1992 to 2021. *BMC Gastroenterol* **25**, 425 (2025).
3. Torres, J., Mehandru, S., Colombel, J.-F. & Peyrin-Biroulet, L. Crohn's disease. *Lancet* **389**, 1741–1755 (2017).
4. Ungaro, R., Mehandru, S., Allen, P. B., Peyrin-Biroulet, L. & Colombel, J.-F. Ulcerative colitis. *Lancet* **389**, 1756–1770 (2017).
5. Ng, S. C. *et al.* Worldwide incidence and prevalence of inflammatory bowel disease in the 21st century: a systematic review of population-based studies. *Lancet* **390**, 2769–2778 (2017).
6. Hetta, H. F. *et al.* Gut Microbiome as a Target of Intervention in Inflammatory Bowel Disease Pathogenesis and Therapy. *Immuno* **4**, 400–425 (2024).
7. Sairenji, T., Collins, K. L. & Evans, D. V. An Update on Inflammatory Bowel Disease. *Prim Care* **44**, 673–692 (2017).
8. Guillen, B. & Atherton, N. S. Short Bowel Syndrome. in *StatPearls* (StatPearls Publishing, Treasure Island (FL), 2025).
9. Thompson, J. S. Short Bowel Syndrome and Malabsorption – Causes and Prevention. *Viszeralmedizin* **30**, 174–178 (2014).
10. Pironi, L. *et al.* Characteristics of adult patients with chronic intestinal failure due to short bowel syndrome: An international multicenter survey. *Clin Nutr ESPEN* **45**, 433–441 (2021).
11. Vanuytsel, T. Disease modifying therapy in short bowel syndrome: finally coming of age? *The Lancet Gastroenterology & Hepatology* **4**, 328–329 (2019).
12. Pironi, L. *et al.* ESPEN guidelines on chronic intestinal failure in adults. *Clin Nutr* **35**, 247–307 (2016).
13. O'Keefe, S. J. D. *et al.* Short Bowel Syndrome and Intestinal Failure: Consensus Definitions and Overview. *Clinical Gastroenterology and Hepatology* **4**, 6–10 (2006).
14. Verbiest, A., Wauters, L. & Vanuytsel, T. Enterohormone therapy for short bowel syndrome. *Current Opinion in Endocrinology, Diabetes and Obesity* **29**, 207 (2022).
15. Ceulemans, L. J. *et al.* Belgian multicenter experience with intestinal transplantation. (2015).
16. Peck, R. W. Precision Dosing: An Industry Perspective. *Clinical Pharmacology & Therapeutics* **109**, 47–50 (2021).
17. Schuck, R. N., Pacanowski, M., Kim, S., Madabushi, R. & Zineh, I. Use of Titration as a Therapeutic Individualization Strategy: An Analysis of Food and Drug Administration–Approved Drugs. *Clinical and Translational Science* **12**, 236–239 (2019).
18. Tracey, D., Klareskog, L., Sasso, E. H., Salfeld, J. G. & Tak, P. P. Tumor necrosis factor antagonist mechanisms of action: a comprehensive review. *Pharmacol Ther* **117**, 244–279 (2008).
19. Sabino, J., Verstockt, B., Vermeire, S. & Ferrante, M. New biologics and small molecules in inflammatory bowel disease: an update. *Therap Adv Gastroenterol* **12**, 1756284819853208 (2019).
20. Ben-Horin, S. & Chowers, Y. Review article: loss of response to anti-TNF treatments in Crohn's disease. *Aliment Pharmacol Ther* **33**, 987–995 (2011).
21. Papamichael, K. *et al.* Therapeutic drug monitoring of biologics in inflammatory bowel disease: unmet needs and future perspectives. *The Lancet Gastroenterology & Hepatology* **7**, 171–185 (2022).
22. Alsoud, D., Verstockt, B., Fiocchi, C. & Vermeire, S. Breaking the therapeutic ceiling in drug development in ulcerative colitis. *Lancet Gastroenterol Hepatol* **6**, 589–595 (2021).
23. Le Berre, C., Ricciuto, A., Peyrin-Biroulet, L. & Turner, D. Evolving Short- and Long-Term Goals of Management of Inflammatory Bowel Diseases: Getting It Right, Making It Last. *Gastroenterology* **162**, 1424–1438 (2022).
24. Sudhakar, P. *et al.* Holistic healthcare in inflammatory bowel disease: time for patient-centric approaches? *Gut* **72**, 192–204 (2023).
25. Peyrin-Biroulet, L. *et al.* Selecting Therapeutic Targets in Inflammatory Bowel Disease (STRIDE): Determining Therapeutic Goals for Treat-to-Target. *Am J Gastroenterol* **110**, 1324–1338 (2015).
26. Remsima | European Medicines Agency (EMA). <https://www.ema.europa.eu/en/medicines/human/EPAR/remsim> (2013).
27. Ryman, J. T. & Meibohm, B. Pharmacokinetics of Monoclonal Antibodies. *CPT Pharmacometrics Syst Pharmacol* **6**, 576–588 (2017).
28. van Schie, K. A. *et al.* Restricted immune activation and internalisation of anti-idiotypic complexes between drug and antidrug antibodies. *Ann Rheum Dis* **77**, 1471–1479 (2018).
29. Brandse, J. F. *et al.* Loss of Infliximab Into Feces Is Associated With Lack of Response to Therapy in Patients With Severe Ulcerative Colitis. *Gastroenterology* **149**, 350–355.e2 (2015).
30. Kantasiripitak, W., Wang, Z., Spriet, I., Ferrante, M. & Dreesen, E. Recent advances in clearance monitoring of monoclonal antibodies in patients with inflammatory bowel diseases. *Expert Review of Clinical Pharmacology* **14**, 1455–1466 (2021).
31. Hoffert, Y., Ferrante, M., Verstockt, B. & Dreesen, E. Intensified infliximab induction therapy for steroid-refractory acute severe ulcerative colitis. *Lancet Gastroenterol Hepatol* **10**, 18–19 (2025).
32. Steenholdt, C. *et al.* Individualised therapy is more cost-effective than dose intensification in patients with Crohn's disease who lose response to anti-TNF treatment: A randomised, controlled trial. *Gut* **63**, 919–927 (2014).
33. Egan, T. D. The drug titration paradox: something obvious finally understood. *British Journal of Anaesthesia* **128**, 900–902 (2022).

34. Wang, Z. *et al.* Infliximab and Ustekinumab Clearance Better Predict Endoscopic Outcomes Than Trough Concentrations in Crohn's Disease. *Clin Gastroenterol Hepatol* S1542-3565(25)00750-5 (2025) doi:10.1016/j.cgh.2025.08.030.
35. Byon, W., Nepal, S., Schuster, A. E., Shenker, A. & Frost, C. E. Regional Gastrointestinal Absorption of Apixaban in Healthy Subjects. *The Journal of Clinical Pharmacology* **58**, 965–971 (2018).
36. Mercer, C., Crawford, A., Shapiro, S., Allan, P. J. & Ambrose, T. Anticoagulation and antiplatelet therapy in short bowel syndrome: A systematic review. *Intest Fail* **1**, 100007 (2024).
37. Byon, W., Garonzik, S., Boyd, R. A. & Frost, C. E. Apixaban: A Clinical Pharmacokinetic and Pharmacodynamic Review. *Clinical pharmacokinetics* **58**, 1265–1279 (2019).
38. Thomson, A. W., Bonham, C. A. & Zeevi, A. Mode of action of tacrolimus (FK506): molecular and cellular mechanisms. *Ther Drug Monit* **17**, 584–591 (1995).
39. Yang, K., Zhang, M., Zhang, B., Zhang, Y. & Zhao, Q. Systematic review and meta-analysis of calcineurin inhibitors on long-term prognosis of renal transplant patients. *Transpl Immunol* **75**, 101741 (2022).
40. Grimm, M. *et al.* Superior Prevention of Acute Rejection by Tacrolimus vs. Cyclosporine in Heart Transplant Recipients—A Large European Trial. *American Journal of Transplantation* **6**, 1387–1397 (2006).
41. Pitre, T. *et al.* Tacrolimus versus cyclosporine immunosuppression in lung transplantation: a systematic review and meta-analysis. *BMJ Open Respir Res* **12**, e002672 (2025).
42. Muduma, G., Saunders, R., Odeyemi, I. & Pollock, R. F. Systematic Review and Meta-Analysis of Tacrolimus versus Cyclosporin as Primary Immunosuppression After Liver Transplant. *PLoS One* **11**, e0160421 (2016).
43. Vanlerberghe, B. *et al.* Tacrolimus drug exposure level in the first year after liver transplantation is an independent risk factor for de novo malignancy in patients transplanted for alcohol-related liver disease. *Journal of Hepatology* **77**, S781 (2022).
44. Destere, A., Premaud, A., Monchaud, C., Marquet, P. & Woillard, J.-B. Longitudinal Exposure to Tacrolimus and New-Onset Diabetes Mellitus in Renal Transplant Patients. *Therapeutic Drug Monitoring* **45**, 102–109 (2023).
45. Venkataramanan, R. *et al.* Clinical Pharmacokinetics of Tacrolimus. *Clin. Pharmacokinet.* **29**, 404–430 (1995).
46. Wallemacq, P. E. & Verbeeck, R. K. Comparative Clinical Pharmacokinetics of Tacrolimus in Paediatric and Adult Patients: *Clinical Pharmacokinetics* **40**, 283–295 (2001).
47. Jain, A. B. *et al.* Effect of hepatic dysfunction and T tube clamping on FK 506 pharmacokinetics and trough concentrations. *Transplant Proc* **22**, 57–59 (1990).
48. Bekersky, I., Dressler, D., Alak, A., Boswell, G. W. & Mekki, Q. A. Comparative tacrolimus pharmacokinetics: normal versus mildly hepatically impaired subjects. *J Clin Pharmacol* **41**, 628–635 (2001).
49. Bauer, R. J. NONMEM Tutorial Part II: Estimation Methods and Advanced Examples. *CPT Pharmacometrics Syst Pharmacol* **8**, 538–556 (2019).
50. Mould, D. R. & Upton, R. N. Basic Concepts in Population Modeling, Simulation, and Model-Based Drug Development—Part 2: Introduction to Pharmacokinetic Modeling Methods. *CPT Pharmacometrics Syst Pharmacol* **2**, e38 (2013).
51. Karlsson, M. O. & Sheiner, L. B. The importance of modeling interoccasion variability in population pharmacokinetic analyses. *Journal of Pharmacokinetics and Biopharmaceutics* **21**, 735–750 (1993).
52. Lalonde, R. L. *et al.* Comparison of different methods to evaluate population dose-response and relative potency: importance of interoccasion variability. *J Pharmacokinet Biopharm* **27**, 67–83 (1999).
53. Koehne-Voss, S., Gautier, A. & Graham, G. The impact of unmodelled interoccasion variability in bioavailability and absorption on parameter estimates in population pharmacokinetic analysis.
54. Svensson, R. J. & Jonsson, E. N. Efficient and relevant stepwise covariate model building for pharmacometrics. *CPT: Pharmacometrics & Systems Pharmacology* **11**, 1210–1222 (2022).
55. Sanghavi, K. *et al.* Covariate modeling in pharmacometrics: General points for consideration. *CPT: Pharmacometrics & Systems Pharmacology* **13**, 710–728 (2024).
56. ICH Official web site : ICH. <https://www.ich.org/news/harmonised-ich-m15-guideline-general-principles-model-informed-drug-development-adopted>.
57. Applications Using Parameter Estimates from the Individual. in *Introduction to Population Pharmacokinetic / Pharmacodynamic Analysis with Nonlinear Mixed Effects Models* 198–211 (John Wiley & Sons, Ltd, 2014). doi:10.1002/9781118784860.ch7.
58. Kantasiripitak, W. *et al.* Software Tools for Model-Informed Precision Dosing: How Well Do They Satisfy the Needs? *Front Pharmacol* **11**, 620 (2020).
59. Uster, D. W. *et al.* A Model Averaging/Selection Approach Improves the Predictive Performance of Model-Informed Precision Dosing: Vancomycin as a Case Study. *Clin. Pharmacol. Ther.* **109**, 175–183 (2021).
60. Maier, C., de Wiljes, J., Hartung, N., Kloft, C. & Huisinga, W. A continued learning approach for model-informed precision dosing: updating models in clinical practice. *CPT Pharmacometrics Syst Pharmacol* **11**, 185–198 (2022).
61. Hughes, J. H. & Keizer, R. J. A hybrid machine learning/pharmacokinetic approach outperforms maximum a posteriori Bayesian estimation by selectively flattening model priors. *CPT: Pharmacometrics & Systems Pharmacology* **10**, 1150–1160 (2021).
62. Minichmayr, I. K. *et al.* Model-informed precision dosing: State of the art and future perspectives. *Adv Drug Deliv Rev* **215**, 115421 (2024).
63. Faelens, R. *et al.* Model-Informed Precision Dosing during Infliximab Induction Therapy Reduces Variability in Exposure and Endoscopic Improvement between Patients with Ulcerative Colitis. *Pharmaceutics* **13**, 1623 (2021).
64. Dubinsky, M. C., Phan, B. L., Singh, N., Rabizadeh, S. & Mould, D. R. Pharmacokinetic Dashboard-Recommended Dosing Is Different than Standard of Care Dosing in Infliximab-Treated Pediatric IBD Patients. *AAPS J* **19**, 215–222 (2017).

65. Eser, A. *et al.* Prediction of Individual Serum Infliximab Concentrations in Inflammatory Bowel Disease by a Bayesian Dashboard System. *J Clin Pharmacol* **58**, 790–802 (2018).
66. Strik, A. S. *et al.* Efficacy of dashboard driven dosing of infliximab in inflammatory bowel disease patients; a randomized controlled trial. *Scand J Gastroenterol* **56**, 145–154 (2021).
67. Wang, Z. *et al.* S1204 Model-Informed Dosing Interval Extension of Infliximab in Patients With Inflammatory Bowel Diseases: Six Months Interim Analysis of the MODIFI Trial. *Official journal of the American College of Gastroenterology* | *ACG* **119**, S855 (2024).
68. Colman, R. J. *et al.* Model-informed Precision Dosing for Biologics Is Now Available at the Bedside for Patients With Inflammatory Bowel Disease. *Inflamm Bowel Dis* **29**, 1342–1346 (2023).
69. Eliquis | European Medicines Agency (EMA). <https://www.ema.europa.eu/en/medicines/human/EPAR/eliquis> (2018).
70. Gaspar, F. *et al.* Population pharmacokinetics of apixaban in a real-life hospitalized population from the OptimAT study. *CPT: Pharmacometrics & Systems Pharmacology* **12**, 1541–1552 (2023).
71. Cirincione, B. *et al.* Population Pharmacokinetics of Apixaban in Subjects With Nonvalvular Atrial Fibrillation. *CPT Pharmacometrics Syst Pharmacol* **7**, 728–738 (2018).
72. Hronová, K. *et al.* Pharmacokinetics of apixaban in patients with short bowel syndrome: a Bayesian re-estimation approach. *Expert Opinion on Drug Metabolism & Toxicology* **21**, 1205–1211 (2025).
73. Astellas Pharma Europe B.V. Advagraf prolonged-release hard capsules, Summary of Product Characteristics. https://www.ema.europa.eu/en/documents/product-information/advagraf-epar-product-information_en.pdf.
74. Størset, E. *et al.* Improved tacrolimus target concentration achievement using computerized dosing in renal transplant recipients – a prospective, randomized study. *Transplantation* **99**, 2158–2166 (2015).
75. Fukudo, M. *et al.* Prospective Evaluation of the Bayesian Method for Individualizing Tacrolimus Dose Early After Living-Donor Liver Transplantation. *The Journal of Clinical Pharmacology* **49**, 789–797 (2009).
76. McKeage, K. A review of CT-P13: An infliximab biosimilar. *BioDrugs* **28**, 313–321 (2014).
77. Roblin, X. *et al.* Higher serum infliximab concentrations following subcutaneous dosing are associated with deep remission in patients with inflammatory bowel disease. *Journal of Crohn's & colitis* <https://doi.org/10.1093/ecco-jcc/jjad188> (2023) doi:10.1093/ecco-jcc/jjad188.
78. Schreiber, S. *et al.* Randomized Controlled Trial: Subcutaneous vs Intravenous Infliximab CT-P13 Maintenance in Inflammatory Bowel Disease. *Gastroenterology* **160**, 2340–2353 (2021).
79. Mohamed, I., Naughton, T., Okuwobi, K., Walker, G. & Hanna, M. P652 Transitioning Inflammatory Bowel Disease patients from an intravenous to a subcutaneous infliximab protocol: a single centre experience and patient satisfaction survey. *Journal of Crohn's and Colitis* **18**, i1243 (2024).
80. Alten, R., An, Y., Kim, D.-H., Yoon, S. & Peyrin-Biroulet, L. Re-Routing Infliximab Therapy: Subcutaneous Infliximab Opens a Path Towards Greater Convenience and Clinical Benefit. *Clin Drug Investig* **42**, 477–489 (2022).
81. Harno-Tasihin, J., Siregar, L., Paajanen, M., Arkkila, P. & Punkkinen, J. Switching from intravenous to subcutaneous infliximab and vedolizumab in patients with inflammatory bowel disease: impact on trough levels, day hospital visits, and medical expenses. *Scand J Gastroenterol* **1–8** (2023) doi:10.1080/00365521.2023.2285229.
82. Buisson, A. *et al.* Effectiveness of switching from intravenous to subcutaneous infliximab in patients with inflammatory bowel diseases: the REMSWITCH study. *Clin Gastroenterol Hepatol* **21**, 2338–2346.e3 (2023).
83. Fierens, L. *et al.* Introduction of Subcutaneous Infliximab CT-P13 and Vedolizumab in Clinical Practice: A Multi-Stakeholder Position Statement Highlighting the Need for Post-Marketing Studies. *Journal of Crohn's and Colitis* **jja009** (2022) doi:10.1093/ecco-jcc/jja009.
84. Wang, Z. *et al.* Therapeutic Drug Monitoring Can Guide the Intravenous-to-Subcutaneous Switch of Infliximab and Vedolizumab: A Simulation Study. *Clin Gastroenterol Hepatol* **21**, 3188–3190 (2023).
85. Le Louedec, F., Puisset, F., Thomas, F., Chatelut, É. & White-Koning, M. Easy and reliable maximum a posteriori Bayesian estimation of pharmacokinetic parameters with the open-source R package mapbayr. *CPT: Pharmacometrics & Systems Pharmacology* **10**, 1208–1220 (2021).
86. Bernhard Lehnert. BlandAltmanLeh: Plots (Slightly Extended) Bland-Altman Plots. 0.3.1 <https://doi.org/10.32614/CRAN.package.BlandAltmanLeh> (2014).
87. Calcagno, V. glmulti: Model Selection and Multimodel Inference Made Easy. 1.0.8 <https://doi.org/10.32614/CRAN.package.glmulti> (2009).
88. Sohil, F., Sohali, M. U. & Shabbir, J. An introduction to statistical learning with applications in R: by Gareth James, Daniela Witten, Trevor Hastie, and Robert Tibshirani, New York, Springer Science and Business Media, 2013, \$41.98, eISBN: 978-1-4614-7137-7. *Statistical Theory and Related Fields* **6**, 87–87 (2022).
89. Kantasiripitak, W., Wang, Z., Spriet, I., Ferrante, M. & Dreesen, E. Recent advancements in clearance monitoring of monoclonal antibodies in patients with inflammatory bowel diseases. *Expert Rev Clin Pharmacol* **14**, 1455–1466 (2021).
90. Brandse, J. F. *et al.* Fecal Loss of Infliximab As a Cause of Lack of Response in Severe Inflammatory Bowel Disease. in *Gastroenterology* vol. 144 S-36 (8th Congress of ECCO, Vienna, Austria, February 14–16, 2013, 2013).
91. Vande Casteele, N. *et al.* Trough concentrations of infliximab guide dosing for patients with inflammatory bowel disease. *Gastroenterology* **148**, 1320–1329.e3 (2015).
92. Vande Casteele, N. *et al.* Subcutaneous Absorption Contributes to Observed Interindividual Variability in Adalimumab Serum Concentrations in Crohn's Disease: A Prospective Multicentre Study. *Journal of Crohn's & colitis* <https://doi.org/10.1093/ecco-jcc/jjz050> (2019) doi:10.1093/ecco-jcc/jjz050.
93. Hanzel, J., Bukkems, L. H., Gecse, K. B., D'Haens, G. R. & Mathôt, R. A. A. Population pharmacokinetics of subcutaneous infliximab CT-P13 in Crohn's disease and ulcerative colitis. *Aliment Pharmacol Ther* <https://doi.org/10.1111/apt.16609> (2021) doi:10.1111/apt.16609.

94. Dinesen, L. C. *et al.* The pattern and outcome of acute severe colitis. *Journal of Crohn's and Colitis* **4**, 431–437 (2010).
95. Kaplan, G. G. *et al.* Decreasing Colectomy Rates for Ulcerative Colitis: A Population-Based Time Trend Study. *American Journal of Gastroenterology* **107**, 1879–1887 (2012).
96. Narula, N. *et al.* Systematic Review and Meta-Analysis: Infliximab or Cyclosporine as Rescue Therapy in Patients With Severe Ulcerative Colitis Refractory to Steroids. *American Journal of Gastroenterology* **111**, 477–491 (2016).
97. Mould, D. & Upton, R. Basic Concepts in Population Modeling, Simulation, and Model-Based Drug Development—Part 2: Introduction to Pharmacokinetic Modeling Methods. *CPT Pharmacom & Syst Pharma* **2**, 1–14 (2013).
98. Overgaard, R., Ingwersen, S. & Tornøe, C. Establishing Good Practices for Exposure–Response Analysis of Clinical Endpoints in Drug Development. *Systems Pharmacology*.
99. Brandse, J. F. *et al.* Loss of Infliximab Into Feces Is Associated With Lack of Response to Therapy in Patients With Severe Ulcerative Colitis. *Gastroenterology* **149**, 350–355.e2 (2015).
100. Battat, R. *et al.* Baseline Clearance of Infliximab Is Associated With Requirement for Colectomy in Patients With Acute Severe Ulcerative Colitis. *Clinical Gastroenterology and Hepatology* **19**, 511–518 (2021).
101. Sands, B. E. *et al.* Infliximab in the Treatment of Severe, Steroid-Refractory Ulcerative Colitis: A Pilot Study: *Inflammatory Bowel Diseases* **7**, 83–88 (2001).
102. Seow, C. H. *et al.* Trough serum infliximab: a predictive factor of clinical outcome for infliximab treatment in acute ulcerative colitis. *Gut* **59**, 49–54 (2010).
103. Gibson, D. J. *et al.* An Accelerated Infliximab Induction Regimen Reduces the Need for Early Colectomy in Patients With Acute Severe Ulcerative Colitis. *Clinical Gastroenterology and Hepatology* **13**, 330–335.e1 (2015).
104. Choy, M. C. *et al.* Intensified versus standard dose infliximab induction therapy for steroid-refractory acute severe ulcerative colitis (PREDICT-UC): an open-label, multicentre, randomised controlled trial. *The Lancet Gastroenterology & Hepatology* **9**, 981–996 (2024).
105. Li Wai Suen, C. F. D. *et al.* Early infliximab levels and clearance predict outcomes after infliximab rescue in acute severe ulcerative colitis: Results from PREDICT-UC. *Gastroenterology* S0016-5085(25)05777–4 (2025) doi:10.1053/j.gastro.2025.07.020.
106. Truelove, S. C. & Witts, L. J. Cortisone in ulcerative colitis; final report on a therapeutic trial. *Br Med J* **2**, 1041–1048 (1955).
107. Beal, S. L. Ways to Fit a PK Model with Some Data Below the Quantification Limit. *J Pharmacokinet Pharmacodyn* **28**, 481–504 (2001).
108. Fagan, E. A. *et al.* Serum levels of C-reactive protein in Crohn's disease and ulcerative colitis. *Eur J Clin Investigation* **12**, 351–359 (1982).
109. Alsoud, D. *et al.* Best Practice for Therapeutic Drug Monitoring of Infliximab: Position Statement from the International Association of Therapeutic Drug Monitoring and Clinical Toxicology. *Therapeutic Drug Monitoring* <https://doi.org/10.1097/FTD.0000000000001204> (2024) doi:10.1097/FTD.0000000000001204.
110. Minichmayr, I. K. *et al.* Model-informed precision dosing: State of the art and future perspectives. *Adv Drug Deliv Rev* 115421 (2024) doi:10.1016/j.addr.2024.115421.
111. Geese, K. *et al.* DOP056 TDM-Based Dose-Intensification of Infliximab is not Superior to Standard Dosing in Patients with Acute Severe Ulcerative Colitis: Results from the TITRATE Study. *Journal of Crohn's and Colitis* **19**, i194–i195 (2025).
112. Goldhaber, S. Z. Risk factors for venous thromboembolism. *J Am Coll Cardiol* **56**, 1–7 (2010).
113. Tilsed, J. V. T. *et al.* ESTES guidelines: acute mesenteric ischaemia. *Eur J Trauma Emerg Surg* **42**, 253–270 (2016).
114. Santamaria, M. M. *et al.* Systematic review of drug bioavailability following gastrointestinal surgery. *Eur J Clin Pharmacol* **74**, 1531–1545 (2018).
115. Hong, W. B.-T., Tan, W. K., Law, L. S.-C., Ong, D. E.-H. & Lo, E. A.-G. Changes of Drug Pharmacokinetics in Patients with Short Bowel Syndrome: A Systematic Review. *Eur J Drug Metab Pharmacokinet* **46**, 465–478 (2021).
116. Azuz, S., Poulsen, J. L., Vinter-Jensen, L. & Olesen, A. E. Drug absorption from oral formulations in patients with short bowel syndrome: a comprehensive update of the literature. *Expert Opinion on Drug Metabolism & Toxicology* **19**, 577–600 (2023).
117. Ward, N. The impact of intestinal failure on oral drug absorption: a review. *J Gastrointest Surg* **14**, 1045–1051 (2010).
118. Severijnen, R., Bayat, N., Bakker, H., Tolboom, J. & Bongaerts, G. Enteral drug absorption in patients with short small bowel : a review. *Clin Pharmacokinet* **43**, 951–962 (2004).
119. Daoud, D. C. & Joly, F. The new place of enterohormones in intestinal failure. *Curr Opin Clin Nutr Metab Care* **23**, 344–349 (2020).
120. Connolly, S. J. *et al.* Dabigatran versus Warfarin in Patients with Atrial Fibrillation. *New England Journal of Medicine* **361**, 1139–1151 (2009).
121. Patel, M. R. *et al.* Rivaroxaban versus warfarin in nonvalvular atrial fibrillation. *N Engl J Med* **365**, 883–891 (2011).
122. Granger, C. B. *et al.* Apixaban versus Warfarin in Patients with Atrial Fibrillation. *New England Journal of Medicine* **365**, 981–992 (2011).
123. Giugliano, R. P. *et al.* Edoxaban versus Warfarin in Patients with Atrial Fibrillation. *New England Journal of Medicine* **369**, 2093–2104 (2013).
124. Ntaios, G. *et al.* Real-World Setting Comparison of Nonvitamin-K Antagonist Oral Anticoagulants Versus Vitamin-K Antagonists for Stroke Prevention in Atrial Fibrillation: A Systematic Review and Meta-Analysis. *Stroke* **48**, 2494–2503 (2017).
125. Pollak, P. T., Sun, G. R. & Kim, R. B. Personalized Anticoagulation: Guided Apixaban Dose Adjustment to Compensate for Pharmacokinetic Abnormalities Related to Short-Bowel Syndrome. *Can J Cardiol* **34**, 342.e17–342.e19 (2018).

126. Hakeam, H. A., Alkhani, M., Alyahya, Z., Alawaji, Z. & Ofori, S. Direct Acting Oral Anticoagulants Following Gastrointestinal Tract Surgery. *J Cardiovasc Pharmacol* **78**, 867–874 (2021).
127. Jeppesen, P. B. *et al.* Randomised placebo-controlled trial of teduglutide in reducing parenteral nutrition and/or intravenous fluid requirements in patients with short bowel syndrome. *Gut* **60**, 902–914 (2011).
128. Gosselin, R. C. *et al.* International Council for Standardization in Haematology (ICSH) Recommendations for Laboratory Measurement of Direct Oral Anticoagulants. *Thromb Haemost* **118**, 437–450 (2018).
129. Cummings, J. H., James, W. P. & Wiggins, H. S. Role of the colon in ileal-resection diarrhoea. *Lancet* **1**, 344–347 (1973).
130. Crenn, P., Messing, B. & Cynober, L. Citrulline as a biomarker of intestinal failure due to enterocyte mass reduction. *Clinical Nutrition* **27**, 328–339 (2008).
131. Greif, S. *et al.* Long-term results of teduglutide treatment for chronic intestinal failure – Insights from a national, multi-centric patient home-care service program. *Clinical Nutrition ESPEN* **51**, 222–230 (2022).
132. Seidner, D. L. *et al.* Increased intestinal absorption in the era of teduglutide and its impact on management strategies in patients with short bowel syndrome-associated intestinal failure. *JPEN J Parenter Enteral Nutr* **37**, 201–211 (2013).
133. Byon, W., Garonzik, S., Boyd, R. A. & Frost, C. E. Apixaban: A Clinical Pharmacokinetic and Pharmacodynamic Review. *Clin Pharmacokinet* **58**, 1265–1279 (2019).
134. Van der Linden, L., Hias, J. & Vanassche, T. The value and limitations of new oral anticoagulant plasma level assessments. *Eur Heart J Suppl* **24**, A32–A41 (2022).
135. Mundi, M. S. *et al.* Home enteral nutrition: A descriptive study. *JPEN J Parenter Enteral Nutr* **47**, 550–562 (2023).
136. Mohamed Elfadil, O., Ewy, M., Patel, J., Patel, I. & Mundi, M. S. Growing use of home enteral nutrition: a great tool in nutrition practice toolbox. *Curr Opin Clin Nutr Metab Care* **24**, 446–452 (2021).
137. Kohli, D. R. *et al.* American Society for Gastrointestinal Endoscopy guideline on gastrostomy feeding tubes: summary and recommendations. *Gastrointestinal Endoscopy* **101**, 25–35 (2025).
138. Gauderer, M. W., Ponsky, J. L. & Izant, R. J. Gastrostomy without laparotomy: a percutaneous endoscopic technique. *J Pediatr Surg* **15**, 872–875 (1980).
139. Toyama, Y. *et al.* Successful new method of extracorporeal percutaneous endoscopic gastrostomy (E-PEG). *Surg Endosc* **21**, 2034–2038 (2007).
140. Teich, N. *et al.* Usage characteristics and adverse event rates of the direct puncture and pull techniques for percutaneous endoscopic gastrostomy in patients with malignant tumors of the upper aerodigestive tract. *Endosc Int Open* **6**, E29–E35 (2018).
141. Van Dyck, E., Macken, E. J., Roth, B., Pelckmans, P. A. & Moreels, T. G. Safety of pull-type and introducer percutaneous endoscopic gastrostomy tubes in oncology patients: a retrospective analysis. *BMC Gastroenterol* **11**, 23 (2011).
142. Shigoka, H. *et al.* Comparison of modified introducer method with pull method for percutaneous endoscopic gastrostomy: prospective randomized study. *Dig Endosc* **24**, 426–431 (2012).
143. Horiuchi, A., Nakayama, Y., Tanaka, N., Fujii, H. & Kajiyama, M. Prospective randomized trial comparing the direct method using a 24 Fr bumper-button-type device with the pull method for percutaneous endoscopic gastrostomy. *Endoscopy* **40**, 722–726 (2008).
144. Köhler, G. *et al.* Comparison of 231 patients receiving either ‘pull-through’ or ‘push’ percutaneous endoscopic gastrostomy. *Surg Endosc* **29**, 170–175 (2015).
145. Bouchiba, H., Jacobs, M. A. J. M., Bouma, G. & Ramsoekh, D. Outcomes of push and pull percutaneous endoscopic gastrostomy placements in 854 patients: A single-center study. *JGH Open* **6**, 57–62 (2021).
146. Hogan, R. B., DeMarco, D. C., Hamilton, J. K., Walker, C. O. & Polter, D. E. Percutaneous endoscopic gastrostomy—to push or pull: A prospective randomized trial. *Gastrointestinal Endoscopy* **32**, 253–258 (1986).
147. Kozarek, R. A., Ball, T. J. & Ryan, J. A. When push comes to shove: a comparison between two methods of percutaneous endoscopic gastrostomy. *Am J Gastroenterol* **81**, 642–646 (1986).
148. Gkolfakis, P. *et al.* Endoscopic management of enteral tubes in adult patients – Part 2: Peri- and post-procedural management. European Society of Gastrointestinal Endoscopy (ESGE) Guideline. *Endoscopy* **53**, 178–195 (2020).
149. Kucha, P. *et al.* To push or to pull? A clinical audit on the efficacy and safety of the pull and push percutaneous endoscopic gastrostomy techniques in oncological patients. *United European Gastroenterol J* **11**, 951–959 (2023).
150. Wickham, H. The tidyverse. *Presentation* (2016).
151. Kassambara, A., Kosinski, M., Biecek, P. & Fabian, S. survminer: Drawing Survival Curves using ‘ggplot2’. (2024).
152. Therneau, T. M., until 2009), T. L. (original S.->R port and R. maintainer, Elizabeth, A. & Cynthia, C. survival: Survival Analysis. (2024).
153. Schuhmacher, L. *et al.* Complication rates of direct puncture and pull-through techniques for percutaneous endoscopic gastrostomy: Results from a large multicenter cohort. *Endoscopy International Open* **10**, E1454–E1461 (2022).
154. Friedenberg, F., Jensen, G., Gujral, N., Braitman, L. E. & Levine, G. M. Serum albumin is predictive of 30-day survival after percutaneous endoscopic gastrostomy. *JPEN J Parenter Enteral Nutr* **21**, 72–74 (1997).
155. Nair, S., Hertan, H. & Pitchumoni, C. S. Hypoalbuminemia is a poor predictor of survival after percutaneous endoscopic gastrostomy in elderly patients with dementia. *Am J Gastroenterol* **95**, 133–136 (2000).
156. Aalen, O. O. & Johansen, S. An Empirical Transition Matrix for Non-Homogeneous Markov Chains Based on Censored Observations. *Scandinavian Journal of Statistics* **5**, 141–150 (1978).
157. Reulen, H. simMSM: Simulation of Event Histories for Multi-State Models. (2022).
158. Sikma, M. A. *et al.* Pharmacokinetics and Toxicity of Tacrolimus Early After Heart and Lung Transplantation. *Am J Transplant* **15**, 2301–2313 (2015).

159. Brunet, M. *et al.* Therapeutic Drug Monitoring of Tacrolimus-Personalized Therapy: Second Consensus Report. *Therapeutic Drug Monitoring* **41**, 261–307 (2019).
160. Staatz, C. E. & Tett, S. E. Clinical Pharmacokinetics and Pharmacodynamics of Tacrolimus in Solid Organ Transplantation: *Clinical Pharmacokinetics* **43**, 623–653 (2004).
161. Vanlerberghe, B. T. K. *et al.* Tacrolimus Drug Exposure Level and Smoking Are Modifiable Risk Factors for Early De Novo Malignancy After Liver Transplantation for Alcohol-Related Liver Disease. *Transpl Int* **37**, 12055 (2024).
162. Degraeve, A. L. *et al.* Predictors of tacrolimus pharmacokinetic variability: current evidences and future perspectives. *Expert Opinion on Drug Metabolism & Toxicology* **16**, 769–782 (2020).
163. Holford, N. H. G. & Buclin, T. Safe and Effective Variability—A Criterion for Dose Individualization. *Therapeutic Drug Monitoring* **34**, 565–568 (2012).
164. Størset, E. *et al.* Improved prediction of tacrolimus concentrations early after kidney transplantation using theory-based pharmacokinetic modelling. *Brit J Clinical Pharma* **78**, 509–523 (2014).
165. Lloberas, N. *et al.* A prospective controlled, randomized clinical trial of kidney transplant recipients developed personalized tacrolimus dosing using model-based Bayesian Prediction. *Kidney Int* S0085-2538(23)00475-1 (2023) doi:10.1016/j.kint.2023.06.021.
166. Shi, B. *et al.* Genotype-guided model significantly improves accuracy of tacrolimus initial dosing after liver transplantation. *eClinicalMedicine* **55**, 101752 (2023).
167. Khong, J. *et al.* Personalized Tacrolimus Dosing After Liver Transplantation: A Randomized Clinical Trial. *medRxiv* 2023.05.26.23290604 (2023) doi:10.1101/2023.05.26.23290604.
168. Taylor, Z. L. *et al.* Tutorial on model selection and validation of model input into precision dosing software for model-informed precision dosing. *CPT: Pharmacometrics & Systems Pharmacology* **12**, 1827–1845 (2023).
169. Baron K (2023). mrgsolve: Simulate from ODE-Based Models. R package version 1.3.0.,
170. Charlton, M. *et al.* International Liver Transplantation Society Consensus Statement on Immunosuppression in Liver Transplant Recipients. *Transplantation* **102**, 727 (2018).
171. Andreu, F. *et al.* A New CYP3A5*3 and CYP3A4*22 Cluster Influencing Tacrolimus Target Concentrations: A Population Approach. *Clin Pharmacokinet* **56**, 963–975 (2017).
172. Franken, L. G. *et al.* A Population Pharmacokinetic Model of Whole-Blood and Intracellular Tacrolimus in Kidney Transplant Recipients. *Eur J Drug Metab Pharmacokinet* **47**, 523–535 (2022).
173. Andrews, L. M. *et al.* A population pharmacokinetic model to predict the individual starting dose of tacrolimus in adult renal transplant recipients. *Brit J Clinical Pharma* **85**, 601–615 (2019).
174. Lu, Y. *et al.* A population pharmacokinetic study of tacrolimus in healthy Chinese volunteers and liver transplant patients. *Acta Pharmacol Sin* **36**, 281–288 (2015).
175. Kirubakaran, R. *et al.* Adaptation of a population pharmacokinetic model to inform tacrolimus therapy in heart transplant recipients. *Brit J Clinical Pharma* **89**, 1162–1175 (2023).
176. Alqahtani, S., Alenazi, M., Alsultan, A. & Alsarhani, E. Estimation of Tacrolimus Clearance in Saudi Adult Kidney Transplant Recipients. *Saudi Journal of Kidney Diseases and Transplantation* **32**, 101 (2021).
177. Al-Kofahi, M. *et al.* Precision Dosing for Tacrolimus Using Genotypes and Clinical Factors in Kidney Transplant Recipients of European Ancestry. *J Clin Pharmacol* **61**, 1035–1044 (2021).
178. Francke, M. I. *et al.* Body composition is associated with tacrolimus pharmacokinetics in kidney transplant recipients. *Eur J Clin Pharmacol* **78**, 1273–1287 (2022).
179. Ji, E., Kim, M. G. & Oh, J. M. CYP3A5 genotype-based model to predict tacrolimus dosage in the early postoperative period after living donor liver transplantation. *TCRM Volume* **14**, 2119–2126 (2018).
180. Andreu, F. *et al.* Development of a Population PK Model of Tacrolimus for Adaptive Dosage Control in Stable Kidney Transplant Patients. *Therapeutic Drug Monitoring* **37**, 246–255 (2015).
181. Quintairos, L. *et al.* Early prognostic performance of miR155-5p monitoring for the risk of rejection: Logistic regression with a population pharmacokinetic approach in adult kidney transplant patients. *PLoS ONE* **16**, e0245880 (2021).
182. Zhang, S.-F. *et al.* Effect of drug combination on tacrolimus target dose in renal transplant patients with different CYP3A5 genotypes. *Xenobiotica* **52**, 312–321 (2022).
183. Zuo, X. *et al.* Effects of CYP3A4 and CYP3A5 polymorphisms on tacrolimus pharmacokinetics in Chinese adult renal transplant recipients: a population pharmacokinetic analysis. *Pharmacogenetics and Genomics* **23**, 251–261 (2013).
184. Zhu, L. *et al.* Effects of CYP3A5 genotypes, ABCB1 C3435T and G2677T/A polymorphism on pharmacokinetics of Tacrolimus in Chinese adult liver transplant patients. *Xenobiotica* **45**, 840–846 (2015).
185. Ling, J. *et al.* Effects of CYP3A5, ABCB1 and POR*28 polymorphisms on pharmacokinetics of tacrolimus in the early period after renal transplantation. *Xenobiotica* **50**, 1501–1509 (2020).
186. Lu, Y., Xu, L., Cui, J., Shen, S. & Li, X. Effects of Postoperative Day and NR1I2 on Tacrolimus Clearance in Chinese Liver Transplant Recipients—A Population Model Approach. *Clinical Pharm in Drug Dev* **10**, 1385–1394 (2021).
187. Chen, W. *et al.* Effects of Voriconazole Exposure on the Pharmacokinetics of Tacrolimus in Lung Transplantation Patients, Based on Therapeutic Drug Monitoring Data. *The Journal of Clinical Pharma* **62**, 1310–1320 (2022).
188. Lee, J. Y. *et al.* Factors Affecting the Apparent Clearance of Tacrolimus in Korean Adult Liver Transplant Recipients. *Pharmacotherapy* **26**, 1069–1077 (2006).
189. Press, R. R. *et al.* Explaining Variability in Tacrolimus Pharmacokinetics to Optimize Early Exposure in Adult Kidney Transplant Recipients. *Therapeutic Drug Monitoring* **31**, 187–197 (2009).
190. Sikma, M. A. *et al.* High Variability of Whole-Blood Tacrolimus Pharmacokinetics Early After Thoracic Organ Transplantation. *Eur J Drug Metab Pharmacokinet* **45**, 123–134 (2020).

191. Chen, L. *et al.* Impact of body composition on pharmacokinetics of tacrolimus in liver transplantation recipients. *Xenobiotica* **50**, 196–201 (2020).
192. Shao, J. *et al.* Impact of Donor and Recipient *CYP3A5**3 Genotype on Tacrolimus Population Pharmacokinetics in Chinese Adult Liver Transplant Recipients. *Ann Pharmacother* **54**, 652–661 (2020).
193. Fukudo, M. *et al.* Impact of MDR1 and CYP3A5 on the oral clearance of tacrolimus and tacrolimus-related renal dysfunction in adult living-donor liver transplant patients. *Pharmacogenetics and Genomics* **18**, 413–423 (2008).
194. Størset, E. *et al.* Importance of hematocrit for a tacrolimus target concentration strategy. *Eur J Clin Pharmacol* **70**, 65–77 (2014).
195. Li, D. *et al.* Population pharmacokinetics of tacrolimus and CYP3A5, MDR1 and IL-10 polymorphisms in adult liver transplant patients. *J Clin Pharm Ther* **32**, 505–515 (2007).
196. Kim, J. H. *et al.* Model based development of tacrolimus dosing algorithm considering CYP3A5 genotypes and mycophenolate mofetil drug interaction in stable kidney transplant recipients. *Sci Rep* **9**, 11740 (2019).
197. Ogasawara, K., Chitnis, S. D., Gohh, R. Y., Christians, U. & Akhlaghi, F. Multidrug Resistance-Associated Protein 2 (MRP2/ABCC2) Haplotypes Significantly Affect the Pharmacokinetics of Tacrolimus in Kidney Transplant Recipients. *Clin Pharmacokinet* **52**, 751–762 (2013).
198. Faelens, R., Luyckx, N., Kuypers, D., Bouillon, T. & Annaert, P. Predicting model-informed precision dosing: A test-case in tacrolimus dose adaptation for kidney transplant recipients. *CPT Pharmacom & Syst Pharma* **11**, 348–361 (2022).
199. Grover, A., Frassetto, L. A., Benet, L. Z. & Chakker, H. A. Pharmacokinetic Differences Corroborate Observed Low Tacrolimus Dosage in Native American Renal Transplant Patients. *Drug Metab Dispos* **39**, 2017–2019 (2011).
200. Choi, S. *et al.* Pharmacokinetic Model Based on Stochastic Simulation and Estimation for Therapeutic Drug Monitoring of Tacrolimus in Korean Adult Transplant Recipients. *Therapeutic Drug Monitoring* **44**, 729–737 (2022).
201. Allard, M. *et al.* Pharmacokinetics and Pharmacodynamics of Once-daily Prolonged-release Tacrolimus in Liver Transplant Recipients. *Clinical Therapeutics* **41**, 882–896.e3 (2019).
202. Jing, Y. *et al.* Population pharmacokinetic analysis and dosing guidelines for tacrolimus co-administration with Wuzhi capsule in Chinese renal transplant recipients. *J Clin Pharm Ther* **46**, 1117–1128 (2021).
203. Gong, Y. *et al.* Population pharmacokinetic analysis of tacrolimus in Chinese cardiac transplant recipients. *Eur J Hosp Pharm* **27**, e12–e18 (2020).
204. Rong, Y., Mayo, P., Ensom, M. H. H. & Kiang, T. K. L. Population Pharmacokinetic Analysis of Immediate-Release Oral Tacrolimus Co-administered with Mycophenolate Mofetil in Corticosteroid-Free Adult Kidney Transplant Recipients. *Eur J Drug Metab Pharmacokinet* **44**, 409–422 (2019).
205. Zahir, H. *et al.* Population Pharmacokinetic Estimation of Tacrolimus Apparent Clearance in Adult Liver Transplant Recipients: *Therapeutic Drug Monitoring* **27**, 422–430 (2005).
206. Woillard, J. *et al.* Population pharmacokinetic model and Bayesian estimator for two tacrolimus formulations – twice daily Prograf® and once daily Advagraf®. *Brit J Clinical Pharma* **71**, 391–402 (2011).
207. Li, Z. *et al.* Population Pharmacokinetic Modeling Combined With Machine Learning Approach Improved Tacrolimus Trough Concentration Prediction in Chinese Adult Liver Transplant Recipients. *The Journal of Clinical Pharma* **63**, 314–325 (2023).
208. Monchaud, C. *et al.* Population Pharmacokinetic Modelling and Design of a Bayesian Estimator for Therapeutic Drug Monitoring of Tacrolimus in Lung Transplantation: *Clinical Pharmacokinetics* **51**, 175–186 (2012).
209. Han, N. *et al.* Population Pharmacokinetic–Pharmacogenetic Model of Tacrolimus in the Early Period after Kidney Transplantation. *Basic Clin Pharma Tox* **114**, 400–406 (2014).
210. Chen, B. *et al.* Population pharmacokinetics and Bayesian estimation of tacrolimus exposure in Chinese liver transplant patients. *J Clin Pharm Ther* **42**, 679–688 (2017).
211. Benkali, K. *et al.* Population Pharmacokinetics and Bayesian Estimation of Tacrolimus Exposure in Renal Transplant Recipients on a New Once-Daily Formulation: *Clinical Pharmacokinetics* **49**, 683–692 (2010).
212. Du, Y., Song, W., Xiong, X., Ge, W. & Zhu, H. Population pharmacokinetics and dosage optimisation of tacrolimus coadministration with Wuzhi capsule in adult liver transplant patients. *Xenobiotica* **52**, 274–283 (2022).
213. Antignac, M., Barrou, B., Farinotti, R., Lechat, P. & Urien, S. Population pharmacokinetics and bioavailability of tacrolimus in kidney transplant patients. *Br J Clin Pharmacol* **0**, 070621084156051-??? (2007).
214. Cai, X. *et al.* Population pharmacokinetics and dosing regimen optimization of tacrolimus in Chinese lung transplant recipients. *European Journal of Pharmaceutical Sciences* **152**, 105448 (2020).
215. Martial, L. C. *et al.* Population pharmacokinetics and genetics of oral meltdose tacrolimus (Envarsus) in stable adult liver transplant recipients. *Brit J Clinical Pharma* **87**, 4262–4272 (2021).
216. Moes, D. J. A. R. *et al.* Population pharmacokinetics and pharmacogenetics of once daily tacrolimus formulation in stable liver transplant recipients. *Eur J Clin Pharmacol* **72**, 163–174 (2016).
217. Lu, Z., Bonate, P. & Keirns, J. Population pharmacokinetics of immediate- and prolonged-release tacrolimus formulations in liver, kidney and heart transplant recipients. *Brit J Clinical Pharma* **85**, 1692–1703 (2019).
218. Bergmann, T. K., Hennig, S., Barraclough, K. A., Isabel, N. M. & Staatz, C. E. Population Pharmacokinetics of Tacrolimus in Adult Kidney Transplant Patients: Impact of CYP3A5 Genotype on Starting Dose. *Ther Drug Monit* **36**, (2014).
219. Staatz, C. Population pharmacokinetics of tacrolimus in adult kidney transplant recipients. *Clinical Pharmacology & Therapeutics* **72**, 660–669 (2002).
220. S., F. *et al.* Population pharmacokinetics of tacrolimus in adult recipients receiving living-donor liver transplantation. *European Journal of Clinical Pharmacology* **57**, 479–484 (2001).
221. Teng, F. *et al.* Population pharmacokinetics of tacrolimus in Chinese adult liver transplant patients. *Biopharm & Drug Disp* **43**, 76–85 (2022).

222. Antignac, M. *et al.* Population pharmacokinetics of tacrolimus in full liver transplant patients: modelling of the post-operative clearance. *Eur J Clin Pharmacol* **61**, 409–416 (2005).
223. Velickovic-Radovanovic, R. *et al.* Population pharmacokinetics of tacrolimus in kidney transplant patients. *CP* **48**, 375–382 (2010).
224. Vadcharavivad, S., Praisuwan, S., Techawathanawanna, N., Treyaprasert, W. & Avihingsanon, Y. Population pharmacokinetics of tacrolimus in Thai kidney transplant patients: comparison with similar data from other populations. *J Clin Pharm Ther* **41**, 310–328 (2016).
225. Sam, W. J. *et al.* Population Pharmacokinetics of Tacrolimus in Whole Blood and Plasma in Asian Liver Transplant Patients: *Clinical Pharmacokinetics* **45**, 59–75 (2006).
226. Han, Y. *et al.* Prediction of tacrolimus dosage in the early period after heart transplantation: a population pharmacokinetic approach. *Pharmacogenomics* **20**, 21–35 (2019).
227. Han, N. *et al.* Prediction of the tacrolimus population pharmacokinetic parameters according to CYP3A5 genotype and clinical factors using NONMEM in adult kidney transplant recipients. *Eur J Clin Pharmacol* **69**, 53–63 (2013).
228. Nakagawa, J. *et al.* Relationship Between Change Rate of Tacrolimus Clearance During Continuous Intravenous Infusion and Recipient Recovery at an Early Stage After Living Donor Liver Transplantation. *Eur J Drug Metab Pharmacokinet* **45**, 619–626 (2020).
229. Oteo, I. *et al.* Tacrolimus pharmacokinetics in the early post-liver transplantation period and clinical applicability via Bayesian prediction. *Eur J Clin Pharmacol* **69**, 65–74 (2013).
230. Zhu, W. *et al.* Tacrolimus population pharmacokinetic models according to CYP3A5 / CYP3A4 / POR genotypes in Chinese Han renal transplant patients. *Pharmacogenomics* **19**, 1013–1025 (2018).
231. Benkali, K. *et al.* Tacrolimus Population Pharmacokinetic-Pharmacogenetic Analysis and Bayesian Estimation in Renal Transplant Recipients: *Clinical Pharmacokinetics* **48**, 805–816 (2009).
232. Zhang, H. J., Li, D. Y., Zhu, H. J., Fang, Y. & Liu, T. S. Tacrolimus population pharmacokinetics according to CYP3A5 genotype and clinical factors in Chinese adult kidney transplant recipients. *J Clin Pharm Ther* **42**, 425–432 (2017).
233. Campagne, O., Mager, D. E., Brazeau, D., Venuto, R. C. & Tornatore, K. M. Tacrolimus Population Pharmacokinetics and Multiple CYP3A5 Genotypes in Black and White Renal Transplant Recipients. *The Journal of Clinical Pharmacology* **58**, 1184–1195 (2018).
234. Zhang, J.-J. *et al.* The genetic polymorphisms of POR*28 and CYP3A5*3 significantly influence the pharmacokinetics of tacrolimus in Chinese renal transplant recipients. *CP* **53**, 728–736 (2015).
235. Zhang, X. *et al.* The Impact of Sulfonylureas on Tacrolimus Apparent Clearance Revealed by a Population Pharmacokinetics Analysis in Chinese Adult Liver-Transplant Patients. *Therapeutic Drug Monitoring* **34**, 126–133 (2012).
236. Zhu, L. *et al.* The Population Pharmacokinetic Models of Tacrolimus in Chinese Adult Liver Transplantation Patients. *Journal of Pharmaceutics* **2014**, 1–7 (2014).
237. Henin, E., Govoni, M., Cella, M., Laveille, C. & Piotti, G. Therapeutic Drug Monitoring Strategies for Envarsus in De Novo Kidney Transplant Patients Using Population Modelling and Simulations. *Adv Ther* **38**, 5317–5332 (2021).
238. Golubović, B. *et al.* Total plasma protein effect on tacrolimus elimination in kidney transplant patients – Population pharmacokinetic approach. *European Journal of Pharmaceutical Sciences* **52**, 34–40 (2014).
239. Sikma, M. A. *et al.* Unbound Plasma, Total Plasma, and Whole-Blood Tacrolimus Pharmacokinetics Early After Thoracic Organ Transplantation. *Clin Pharmacokinet* **59**, 771–780 (2020).
240. Staat, C. Toward better outcomes with tacrolimus therapy: Population pharmacokinetics and individualized dosage prediction in adult liver transplantation. *Liver Transplantation* **9**, 130–137 (2003).
241. Chen, L. *et al.* Wuzhi Capsule Dosage Affects Tacrolimus Elimination in Adult Kidney Transplant Recipients, as Determined by a Population Pharmacokinetics Analysis. *PGPM Volume* **14**, 1093–1106 (2021).
242. Paschier, A. *et al.* Tacrolimus population pharmacokinetics in adult heart transplant patients. *Br J Clin Pharmacol* **89**, 3584–3595 (2023).
243. Koomen, J. V. *et al.* A Joint Pharmacokinetic Model for the Simultaneous Description of Plasma and Whole Blood Tacrolimus Concentrations in Kidney and Lung Transplant Recipients. *Clin Pharmacokinet* **62**, 1117–1128 (2023).
244. Abderahmene, A. *et al.* A Population Pharmacokinetic Model to Predict the Individual Starting Dose of Tacrolimus for Tunisian Adults after Renal Transplantation. *Ther Drug Monit* <https://doi.org/10.1097/FTD.0000000000001147> (2023) doi:10.1097/FTD.0000000000001147.
245. Cheng, Y., Chen, J., Lin, X., Qiu, H. & Zhang, J. Population Pharmacokinetic Analysis for Model-Based Therapeutic Drug Monitoring of Tacrolimus in Chinese Han Heart Transplant Patients. *Eur J Drug Metab Pharmacokinet* **48**, 89–100 (2023).
246. Mohammed Ali, Z. *et al.* CYP3A5*3 and CYP3A4*22 Cluster Polymorphism Effects on LCP-Tac Tacrolimus Exposure: Population Pharmacokinetic Approach. *Pharmaceutics* **15**, 2699 (2023).
247. Li, Z.-R. *et al.* Population Pharmacokinetic Modeling Combined With Machine Learning Approach Improved Tacrolimus Trough Concentration Prediction in Chinese Adult Liver Transplant Recipients. *J Clin Pharmacol* **63**, 314–325 (2023).
248. Reséndiz-Galván, J. E. *et al.* Dosing recommendations based on population pharmacokinetics of tacrolimus in Mexican adult patients with kidney transplant. *Basic Clin Pharmacol Toxicol* **124**, 303–311 (2019).
249. Janmahasatian, S. *et al.* Quantification of lean bodyweight. *Clinical Pharmacokinetics* **44**, 1051–1065 (2005).
250. Riff, C., Debord, J., Monchaud, C., Marquet, P. & Woillard, J.-B. Population pharmacokinetic model and Bayesian estimator for 2 tacrolimus formulations in adult liver transplant patients. *Br J Clin Pharmacol* **85**, 1740–1750 (2019).

251. Woillard, J.-B., Debord, J., Monchaud, C., Saint-Marcoux, F. & Marquet, P. Population Pharmacokinetics and Bayesian Estimators for Refined Dose Adjustment of a New Tacrolimus Formulation in Kidney and Liver Transplant Patients. *Clin Pharmacokinet* **56**, 1491–1498 (2017).
252. Andrews, L. M. *et al.* A Population Pharmacokinetic Model to Predict the Individual Starting Dose of Tacrolimus Following Pediatric Renal Transplantation. *Clin Pharmacokinet* **57**, 475–489 (2018).
253. Brooks, J. T. *et al.* Population Pharmacokinetic Model Development of Tacrolimus in Pediatric and Young Adult Patients Undergoing Hematopoietic Cell Transplantation. *Front Pharmacol* **12**, 750672 (2021).
254. Jacobson, P., Ng, J., Ratanatharathorn, V., Uberti, J. & Brundage, R. C. Factors affecting the pharmacokinetics of tacrolimus (FK506) in hematopoietic cell transplant (HCT) patients. *Bone Marrow Transplant* **28**, 753–758 (2001).
255. Musuamba, F. T. *et al.* Statistical tools for dose individualization of mycophenolic acid and tacrolimus co-administered during the first month after renal transplantation. *Br J Clin Pharmacol* **75**, 1277–1288 (2013).
256. Schijvens, A. M. *et al.* Low Bioavailability of Oral Tacrolimus Suspension in Pediatric Kidney Transplant Patients. *Clin Pharmacokinet* **59**, 1483–1485 (2020).
257. Wallin, J. E., Friberg, L. E., Fasth, A. & Staatz, C. E. Population pharmacokinetics of tacrolimus in pediatric hematopoietic stem cell transplant recipients: new initial dosage suggestions and a model-based dosage adjustment tool. *Ther Drug Monit* **31**, 457–466 (2009).
258. Nanga, T. M., Doan, T. T. P., Marquet, P. & Musuamba, F. T. Toward a robust tool for pharmacokinetic-based personalization of treatment with tacrolimus in solid organ transplantation: A model-based meta-analysis approach. *Br J Clin Pharmacol* **85**, 2793–2823 (2019).
259. Wang, C.-B., Zhang, Y.-J., Zhao, M.-M. & Zhao, L. Dosage optimization of tacrolimus based on the glucocorticoid dose and pharmacogenetics in adult patients with systemic lupus erythematosus. *Int Immunopharmacol* **124**, 110866 (2023).
260. Cai, X.-J. *et al.* Prospective population pharmacokinetic study of tacrolimus in adult recipients early after liver transplantation: A comparison of Michaelis-Menten and theory-based pharmacokinetic models. *Front Pharmacol* **13**, 1031969 (2022).
261. Åsberg, A. *et al.* Inclusion of CYP3A5 genotyping in a nonparametric population model improves dosing of tacrolimus early after transplantation. *Transpl Int* **26**, 1198–1207 (2013).
262. Saint-Marcoux, F. *et al.* Pharmacokinetic study of tacrolimus in cystic fibrosis and non-cystic fibrosis lung transplant patients and design of Bayesian estimators using limited sampling strategies. *Clin Pharmacokinet* **44**, 1317–1328 (2005).
263. Saint-Marcoux, F. *et al.* Development and evaluation of a simulation procedure to take into account various assays for the Bayesian dose adjustment of tacrolimus. *Ther Drug Monit* **33**, 171–177 (2011).
264. Saint-Marcoux, F., Debord, J., Undre, N., Rousseau, A. & Marquet, P. Pharmacokinetic modeling and development of Bayesian estimators in kidney transplant patients receiving the tacrolimus once-daily formulation. *Ther Drug Monit* **32**, 129–135 (2010).
265. Ponthier, L. *et al.* Application of machine learning to predict tacrolimus exposure in liver and kidney transplant patients given the MeltDose formulation. *Eur J Clin Pharmacol* **79**, 311–319 (2023).
266. Rosen, R. D., Singh, A. & Burns, B. Trauma Organ Procurement. in *StatPearls* (StatPearls Publishing, Treasure Island (FL), 2024).
267. Holford, N., Ma, G. & Metz, D. TDM is dead. Long live TCI! *British Journal of Clinical Pharmacology* **88**, 1406–1413 (2022).
268. Wicha, S. G. *et al.* From Therapeutic Drug Monitoring to Model-Informed Precision Dosing for Antibiotics. *Clinical Pharmacology & Therapeutics* **109**, 928–941 (2021).
269. Koch, B. C. *et al.* Therapeutic Drug Monitoring of Antibiotics in Critically Ill Patients: Current Practice and Future Perspective With a Focus on Clinical Outcome. *Ther Drug Monit* <https://doi.org/10.1097/FTD.0000000000000942> (2021) doi:10.1097/FTD.0000000000000942.
270. Zwart, T. C. *et al.* Model-informed precision dosing to optimise immunosuppressive therapy in renal transplantation. *Drug Discovery Today* **26**, 2527–2546 (2021).
271. Kuypers, D. R. J. Inpatient Variability of Tacrolimus Exposure in Solid Organ Transplantation: A Novel Marker for Clinical Outcome. *Clinical Pharmacology & Therapeutics* **107**, 347–358 (2020).
272. Chen, F. *et al.* High intra-patient variability of tacrolimus within post-operative 1 month predicted worse 1-year outcomes in pediatric liver transplant recipients. *Eur J Clin Pharmacol* <https://doi.org/10.1007/s00228-024-03663-z> (2024) doi:10.1007/s00228-024-03663-z.
273. Meziyeh, S. *et al.* Tacrolimus and Mycophenolic Acid Exposure Are Associated with Biopsy-Proven Acute Rejection: A Study to Provide Evidence for Longer-Term Target Ranges. *Clinical Pharmacology & Therapeutics* **114**, 192–200 (2023).
274. Brooks, E., Tett, S. E., Isbel, N. M. & Staatz, C. E. Population Pharmacokinetic Modelling and Bayesian Estimation of Tacrolimus Exposure: Is this Clinically Useful for Dosage Prediction Yet? *Clin Pharmacokinet* **55**, 1295–1335 (2016).
275. Woillard, J.-B., Labriffe, M., Debord, J. & Marquet, P. Tacrolimus Exposure Prediction Using Machine Learning. *Clin Pharmacol Ther* **110**, 361–369 (2021).
276. Kirubakaran, R., Stocker, S. L., Hennig, S., Day, R. O. & Carland, J. E. Population Pharmacokinetic Models of Tacrolimus in Adult Transplant Recipients: A Systematic Review. *Clin Pharmacokinet* **59**, 1357–1392 (2020).
277. Abdulla, A. *et al.* Model-Informed Precision Dosing of Antibiotics in Pediatric Patients: A Narrative Review. *Front Pediatr* **9**, 624639 (2021).
278. Kantasiripitak, W. *et al.* Software Tools for Model-Informed Precision Dosing: How Well Do They Satisfy the Needs? *Frontiers in Pharmacology* **11**, 620–620 (2020).

279. Jain, A. *et al.* FK 506 in Small Bowel Transplant Recipients: Pharmacokinetics and Dosing. *Transplant Proc* **26**, 1609–1610 (1994).
280. Andres, A. M. *et al.* Graft-vs-host disease after small bowel transplantation in children. *Journal of Pediatric Surgery* **45**, 330–336 (2010).
281. Jain, A. *et al.* Intravenous, Oral Pharmacokinetics, and Oral Dosing of FK 506 in Small Bowel Transplant Patients. *Transplant Proc* **24**, 1181–1182 (1992).
282. Reyes, J. *et al.* Intestinal Transplantation under Tacrolimus Monotherapy after Perioperative Lymphoid Depletion with Rabbit Anti-Thymocyte Globulin (Thymoglobulin®). *Am J Transplant* **5**, 1430–1436 (2005).
283. Trevizol, A. P. *et al.* Intestinal and Multivisceral Transplantation Immunosuppression Protocols—Literature Review. *Transplantation Proceedings* **44**, 2445–2448 (2012).
284. Kaufman, S. S. *et al.* New Insights Into the Indications for Intestinal Transplantation: Consensus in the Year 2019. *Transplantation* **104**, 937–946 (2020).
285. Horslen, S. P., Wood, N. L., Cafarella, M. & Schnellinger, E. M. OPTN/SRTR 2021 Annual Data Report: Intestine. *Am J Transplant* **23**, S264–S299 (2023).
286. Silva, J. T. *et al.* Infectious Complications Following Small Bowel Transplantation. *American Journal of Transplantation* **16**, 951–959 (2016).
287. Yin, S. *et al.* Tacrolimus variability score outperforms coefficient of variation in predicting clinical outcomes of living kidney transplantation. *Br J Clin Pharmacol* **88**, 75–83 (2022).
288. Vanhove, T., Vermeulen, T., Annaert, P., Lerut, E. & Kuypers, D. R. J. High Intrapatient Variability of Tacrolimus Concentrations Predicts Accelerated Progression of Chronic Histologic Lesions in Renal Recipients. *Am J Transplant* **16**, 2954–2963 (2016).
289. Brunet, M. *et al.* Therapeutic Drug Monitoring of Tacrolimus—Personalized Therapy: Second Consensus Report. *Therapeutic drug monitoring* **41**, 261–307 (2019).
290. Hoffert, Y. *et al.* Model-Informed Precision Dosing of Tacrolimus: A Systematic Review of Population Pharmacokinetic Models and a Benchmark Study of Software Tools. *Clin Pharmacokinet* **63**, 1407–1421 (2024).
291. Wickham, H. *et al.* Welcome to the Tidyverse. *JOSS* **4**, 1686 (2019).
292. Shuker, N., Van Gelder, T. & Hesselink, D. A. Intra-patient variability in tacrolimus exposure: Causes, consequences for clinical management. *Transplantation Reviews* **29**, 78–84 (2015).
293. Dickmann, L. J., Patel, S. K., Rock, D. A., Wienkers, L. C. & Slatter, J. G. Effects of interleukin-6 (IL-6) and an anti-IL-6 monoclonal antibody on drug-metabolizing enzymes in human hepatocyte culture. *Drug Metab Dispos* **39**, 1415–1422 (2011).
294. Scholten, E. M. *et al.* AUC-guided dosing of tacrolimus prevents progressive systemic overexposure in renal transplant recipients. *Kidney International* **67**, 2440–2447 (2005).
295. Musuamba, F. T. *et al.* A simultaneous d-optimal designed study for population pharmacokinetic analyses of mycophenolic Acid and tacrolimus early after renal transplantation. *J Clin Pharmacol* **52**, 1833–1843 (2012).
296. Andreu, F. *et al.* A New CYP3A5*3 and CYP3A4*22 Cluster Influencing Tacrolimus Target Concentrations: A Population Approach. *Clin Pharmacokinet* **56**, 963–975 (2017).
297. Przepiorka, D. *et al.* Tacrolimus clearance is age-dependent within the pediatric population. *Bone Marrow Transplant* **26**, 601–605 (2000).
298. Kim, J. S., Aviles, D. H., Silverstein, D. M., LeBlanc, P. L. & Matti Vehaskari, V. Effect of age, ethnicity, and glucocorticoid use on tacrolimus pharmacokinetics in pediatric renal transplant patients. *Pediatric Transplantation* **9**, 162–169 (2005).
299. MacFarlane, G. D. *et al.* Therapeutic drug monitoring of tacrolimus in pediatric liver transplant patients. *Pediatric Transplantation* **5**, 119–124 (2001).
300. Jain, A. B. *et al.* Comparative study of cyclosporine and FK 506 dosage requirements in adult and pediatric orthotopic liver transplant patients. *Transplant Proc* **23**, 2763–2766 (1991).
301. Gonzales, H. M. *et al.* A comprehensive review of the impact of tacrolimus intrapatient variability on clinical outcomes in kidney transplantation. *American Journal of Transplantation* **20**, 1969–1983 (2020).
302. Schumacher, L., Leino, A. D. & Park, J. M. Tacrolimus intrapatient variability in solid organ transplantation: A multiorgan perspective. *Pharmacotherapy* **41**, 103–118 (2021).
303. Sheiner, L. B. Learning versus confirming in clinical drug development. *Clinical Pharmacology & Therapeutics* **61**, 275–291 (1997).
304. Vande Casteele, N. *et al.* Trough Concentrations of Infliximab Guide Dosing for Patients With Inflammatory Bowel Disease. *Gastroenterology* **148**, 1320–1329.e3 (2015).
305. D’Haens, G. *et al.* Increasing Infliximab Dose Based on Symptoms, Biomarkers, and Serum Drug Concentrations Does Not Increase Clinical, Endoscopic, and Corticosteroid-Free Remission in Patients With Active Luminal Crohn’s Disease. *Gastroenterology* **154**, 1343–1351.e1 (2018).
306. Syversen, S. W. *et al.* Therapeutic drug monitoring of infliximab compared to standard clinical treatment with infliximab: study protocol for a randomised, controlled, open, parallel-group, phase IV study (the NOR-DRUM study). *Trials* **21**, 13–13 (2020).
307. Fierens, L. *et al.* Introduction of Subcutaneous Infliximab CT-P13 and Vedolizumab in Clinical Practice: A Multi-Stakeholder Position Statement Highlighting the Need for Post-Marketing Studies. *Journal of Crohn’s and Colitis* **16**, 1059–1069 (2022).
308. Schreiber, S. *et al.* Randomized Controlled Trial: Subcutaneous vs Intravenous Infliximab CT-P13 Maintenance in Inflammatory Bowel Disease. *Gastroenterology* **160**, 2340–2353 (2021).
309. Hanauer, S. B. *et al.* Subcutaneous Infliximab (CT-P13 SC) as Maintenance Therapy for Inflammatory Bowel Disease: Two Randomized Phase 3 Trials (LIBERTY). *Gastroenterology* **167**, 919–933 (2024).

310. Buisson, A. *et al.* Effectiveness of switching from intravenous to subcutaneous infliximab in patients with inflammatory bowel diseases: the REMSWITCH study. *Clin Gastroenterol Hepatol* **21**, 2338-2346.e3 (2023).
311. spECTRUM | Center for Inflammatory Bowel Disease. <https://research.bidmc.org/ibdcenter/research/spECTRUM>.
312. Bourgonje, A. R. & Mian, P. Rethinking Infliximab Dosing in Acute Severe Ulcerative Colitis With Model-Informed Precision Dosing. *Gastroenterology* **0**, (2026).
313. Li Wai Suen, C. F. D. *et al.* Early infliximab levels and clearance predict outcomes after infliximab rescue in acute severe ulcerative colitis: Results from PREDICT-UC. *Gastroenterology* S0016-5085(25)05777-4 (2025) doi:10.1053/j.gastro.2025.07.020.
314. Petitcollin, A. *et al.* Pharmacokinetic Parameters of Infliximab Influence the Rate of Relapse After De-Escalation in Adults With Inflammatory Bowel Diseases. *Clinical Pharmacology and Therapeutics* **106**, 605–615 (2019).
315. Kantasiripitak, W. *et al.* Multi-model averaging improves the performance of model-guided infliximab dosing in patients with inflammatory bowel diseases. *CPT Pharmacometrics Syst Pharmacol* **11**, 1045–1059 (2022).
316. Koller, M. T., Raatz, H., Steyerberg, E. W. & Wolbers, M. Competing risks and the clinical community: irrelevance or ignorance? *Statistics in Medicine* **31**, 1089–1097 (2012).

ABSTRACT

Background Optimizing pharmacotherapy in highly vulnerable patients, such as patients with inflammatory bowel disease, short bowel syndrome, or patients undergoing intestinal transplantation, presents substantial and pressing challenges. These patients commonly experience inadequate drug exposure and therefore an increased risk of suboptimal outcomes. The absence of these special patient populations in clinical trials, together with the lack of specific guidelines that support individualized therapy optimization rather than a *one-dose-fits-all* approach, represents an important gap in current clinical practice that must be addressed.

Aims In the present work we explored the merits of population pharmacokinetics and pharmacodynamics modeling and simulation in optimizing and individualizing therapies in patients with inflammatory bowel disease, short bowel syndrome, and patients undergoing intestinal transplantation. Furthermore, we aim to explore methodological advances to enhance exposure–response analysis and support the clinical translation of our work.

Methods In *chapter 1* we use data from a prospective clinical study (IRB00013412, “CHU de Clermont Ferrand IRB #1”, number 2022-CF001) and data from 98 patients with inflammatory bowel disease to predict relapse during the intravenous-to-subcutaneous switching of infliximab. In *chapter 2* we use data from a retrospective multicenter clinical study (S64627) including 72 patients with acute severe ulcerative colitis to predict time-to-colectomy during infliximab rescue therapy. In *chapter 3* we conducted a prospective, non-randomized, controlled, cross-over pharmacokinetics study (S63950) in 70 patients with and without short bowel syndrome to assess differences in apixaban pharmacokinetics. In *chapter 4* we performed a retrospective clinical study (MP019554) investigating long-term outcomes of 305 patients undergoing percutaneous endoscopic gastrostomy placement using a multistate modeling framework. In *chapter 5*, we systematically reviewed available

tacrolimus population pharmacokinetics models in patients undergoing solid organ transplantation. In *chapter 6* we performed a retrospective study (S65900) and investigated the pharmacokinetics and pharmacodynamics of tacrolimus in 20 patients undergoing intestinal transplantation.

Results We developed population pharmacokinetics and exposure-response models for infliximab, apixaban, and tacrolimus and identified patients at risk for suboptimal exposure or therapy failure. We developed a relapse risk prediction model for patients with inflammatory bowel diseases during intravenous-to-subcutaneous switching of infliximab. An increase in infliximab clearance and baseline fecal calprotectin leads to a higher probability of relapse. For patients with acute severe ulcerative colitis, we developed a sequential population pharmacokinetics–pharmacodynamics model allowing to predict time-to-colectomy during infliximab rescue therapy. Clearance-normalized infliximab exposure was withheld as a significant predictor of colectomy rates. Our developed population pharmacokinetics model of apixaban in patients with short bowel syndrome shows a decreased rate and extent of absorption. Finally, in our tacrolimus model review, we found a plethora of models for most solid organ transplantation, but none to date for intestinal transplant recipients. In our dedicated tacrolimus population pharmacokinetics and exposure–response analysis, we found that tacrolimus clearance increased, necessitating dynamic dosing early on. Furthermore, we found that early tacrolimus exposure predicts time-to-rejection in intestinal transplantation.

Conclusions Pharmacometrics approaches have the potential to improve efficacy and safety of drugs commonly used in gastrointestinal disorders and can therefore improve clinical outcomes.

SAMENVATTING

Achtergrond Het optimaliseren van farmacotherapie bij zeer kwetsbare patiënten, zoals patiënten met inflammatoire darmziekten, het korte-darmsyndroom of patiënten die een darmtransplantatie ondergaan, brengt aanzienlijke en dringende uitdagingen met zich mee. Deze patiënten ervaren vaak een onvoldoende geneesmiddelblootstelling en lopen daardoor een verhoogd risico op suboptimale behandelresultaten. Het ontbreken van deze specifieke patiëntengroepen in klinische studies, samen met het gebrek aan richtlijnen die geïndividualiseerde therapie-optimalisatie ondersteunen in plaats van een *one-dose-fits-all* benadering, vormt een belangrijke lacune in de huidige klinische praktijk die moet worden aangepakt.

Doelstellingen In dit werk onderzochten we de waarde van populatiefarmacokinetische en farmacodynamische modellering en simulatie voor het optimaliseren en individualiseren van therapieën bij patiënten met inflammatoire darmziekten, het korte-darmsyndroom en bij patiënten die een darmtransplantatie ondergaan. Daarnaast beogen we methodologische vooruitgangen te verkennen om blootstelling-responsanalyses te verbeteren en de klinische vertaling van ons werk te ondersteunen.

Methoden In hoofdstuk 1 gebruikten we gegevens uit een prospectieve klinische studie (IRB00013412, “CHU de Clermont Ferrand IRB #1”, nummer 2022-CF001) en data van 98 patiënten met inflammatoire darmziekten om terugval te voorspellen tijdens de overstap van intraveneuze naar subcutane toediening van infliximab. In hoofdstuk 2 gebruikten we gegevens uit een retrospectieve multicentrische klinische studie (S64627) met 72 patiënten met acute ernstige colitis ulcerosa om de tijd tot colectomie te voorspellen tijdens infliximab-rescuetherapie. In hoofdstuk 3 voerden we een prospectieve, niet-gerandomiseerde, gecontroleerde cross-over farmacokinetische studie (S63950) uit bij 70 patiënten met en zonder kortedarmsyndroom om verschillen in de farmacokinetiek van apixaban te

beoordelen. In hoofdstuk 4 voerden we een retrospectieve klinische studie (MP019554) uit naar de langetermijntuitkomsten van 305 patiënten die een percutane endoscopische gastrostomie ondergingen, met behulp van een multistate-modelleringskader. In hoofdstuk 5 voerden we een systematische review uit van beschikbare populatiefarmacokinetische modellen van tacrolimus bij patiënten die een solide orgaantransplantatie ondergaan. In hoofdstuk 6 voerden we een retrospectieve studie (S65900) uit waarin we de farmacokinetiek en farmacodynamiek van tacrolimus onderzochten bij 20 patiënten die een darmtransplantatie ondergingen.

Resultaten We ontwikkelden populatiefarmacokinetische en blootstelling-responsmodellen voor infliximab, apixaban en tacrolimus en identificeerden patiënten met een verhoogd risico op suboptimale blootstelling of therapiefalen. We ontwikkelden een predictiemodel voor het risico op terugval bij patiënten met inflammatoire darmziekten tijdens de overstap van intraveneuze naar subcutane toediening van infliximab. Een toename in de klaring van infliximab en een hogere baseline fecale calprotectine leiden tot een grotere kans op terugval. Voor patiënten met acute ernstige colitis ulcerosa ontwikkelden we een sequentieel populatiefarmacokinetisch-farmacodynamisch model dat het mogelijk maakt de tijd tot colectomie tijdens infliximab-rescuetherapie te voorspellen. Klaring-genormaliseerde infliximabblootstelling werd geïdentificeerd als een significante voorspeller van het colectomierisico. Ons populatiefarmacokinetische model van apixaban bij patiënten met het kortedarmsyndroom toonde een verminderde snelheid en mate van absorptie. In onze review van tacrolimusmodellen vonden we een groot aantal modellen voor de meeste solide orgaantransplantaties, maar tot op heden geen voor ontvangers van een darmtransplantaat. In onze specifieke populatiefarmacokinetische en blootstelling-responsanalyse van tacrolimus vonden we dat de klaring van tacrolimus verhoogd is, wat vroeg na transplantatie dynamische dosisaanpassingen noodzakelijk maakt. Daarnaast

stelden we vast dat vroege blootstelling aan tacrolimus de tijd tot afstoting bij darmtransplantatie voorspelt.

Conclusies Farmacometrische benaderingen hebben het potentieel om de effectiviteit en veiligheid van geneesmiddelen die vaak worden gebruikt bij gastro-intestinale aandoeningen te verbeteren en kunnen daardoor klinische uitkomsten verbeteren.

CURRICULUM VITAE

Personal information

Yannick Hoffert
Born 15th of October 1996
Nationality: German
Rue Alfred Cluysenaar 35
1060 Bruxelles, Belgium
yannick.hoffert@kuleuven.be

Education

Aug 2022 – present	<i>Ph.D. in Pharmacometrics</i> Thematic program: Drug Design and Development. From Target to Market Department of Pharmaceutical and Pharmacological Sciences Thesis: Pharmacometrics Approaches For Therapy Individualization In Patients With Gastrointestinal Disorders Promotor: Prof. Dr. Erwin Dreesen Co-promotor: Prof. Dr. Tim Vanuytsel Prof. Dr. Bram Verstockt Prof. Dr. Jef Verbeek	Leuven, Belgium
Oct 2015 – May 2021	<i>PharmD (German eq.)</i> Pharmaceutical Sciences Freie Universität Berlin State Office for Health and Social Affairs Berlin	Berlin, Germany

Experience

Aug 2022 – present	<i>Ph.D. Researcher</i> Leuven Pharmacometrics Research Group Population PK/PD modeling to support clinical drug development. Teaching involvement in the Master's program of Pharmacometrics at KU Leuven.	Leuven, Belgium
Nov 2021 – May 2022	<i>Visiting scholar</i> Center of Pharmacometrics and Systems Pharmacology Physiology-based pharmacokinetic modeling and clinical trial simulation	Orlando, USA
May 2020 – Nov 2021	<i>Pharmacist</i> Blaue Apotheke Berlin Pharmacist on duty in a community pharmacy store (part time)	Berlin, Germany
May 2020 – Nov 2021	<i>Pharmacist</i> Vaccination Center Berlin Responsible pharmacist for batch release of the aseptic vaccine (part time)	Berlin, Germany
Nov 2019 – May 2020	<i>Pharmaceutical intern</i> Hospital pharmacy Ernst von Bergmann Klinikum Aseptic compounding and analytics department of the hospital pharmacy	Potsdam, Germany

Publications

Included as part of the doctoral thesis

Hoffert Y, Ferrante M, Verstockt B, Dreesen E. Intensified infliximab induction therapy for steroid-refractory acute severe ulcerative colitis. *Lancet Gastroenterol Hepatol*. 2025 Jan;10(1):18-19 (PMID: 39674216)

Minichmayr I K, Dreesen E, Centanni M, Wang Z, **Hoffert Y**, Friberg L E, Wicha S G. Model-informed precision dosing: State of the art and future perspectives. *Adv Drug Deliv Rev*. 2024 Dec;215:115421 (PMID: 39159868)

Hoffert Y, Wang Z, Fumery M, Nachury M, Bazoge M, Buisson A, Dreesen E. A Risk Stratification Tool for Relapse After Intravenous-To-Subcutaneous Switching of Infliximab in Patients With Inflammatory Bowel Diseases. *Am J Gastroenterol*. 2026 Jan 1;121(1):179-186 (PMID: 40192145)

Niyigena E*, **Hoffert Y***, Afif W, Pedicelli A, Roblin X, Hanžel J, Papamichael K, Kobayashi T, Elens L, Wang Z, Ferrante M, Verstockt B, Vermeire S, Vande Casteele N, Battat R, Dreesen E. *J Crohns Colitis*. 2026 Mar 10;20(3):jjag029 (PMID: 41873130)

Deleenheer B*, **Hoffert Y***, Vanassche T, Wauters L, Van der Linden L, Dreesen E, Vanuytsel T. Apixaban Absorption Is Reduced in Short Bowel Syndrome and Correlates With Citrulline: A Prospective Population Pharmacokinetic Study. [Submitted]

Hoffert Y*, Vandewalle J*, Couturier M, Wauters L, Dreesen E, Vanuytsel T. Long-term outcomes of pull and push percutaneous endoscopic gastrostomy techniques. [Submitted]

Hoffert Y, Dia N, Vanuytsel T, Vos R, Kuypers D, Van Cleemput J, Verbeek J, Dreesen E. Model-Informed Precision Dosing of Tacrolimus: A Systematic Review of Population Pharmacokinetic Models and a Benchmark Study of Software Tools. *Clin Pharmacokinet*. 2024 Oct. 63(10):1407-1421 (PMID: 39304577)

Hoffert Y*, Deleenheer B*, Boelhouwer C, Van den Eede N, Niyigena E, Ceulemans L, Elens L, Monbaliu D, Pirenne J, Wauters L, Vanuytsel T, Dreesen E. Early Tacrolimus Exposure Predicts Rejection Risk in Intestinal Transplant Recipients. [Submitted]

Hoffert Y, Dreesen E. Response to Dai *et al*. *Am J Gastroenterol*. 2025 Dec 1;120(12):2981-2982 (PMID: 40668240)

Hoffert Y, Buisson A, Dreesen E. Response to Shen *et al*. *Am J Gastroenterol*. [Accepted for publication]
[*Shared co-first authorship]

Not included in the doctoral thesis

Berckmans Y*, **Hoffert Y***, Vankerckhoven Y, Dreesen E, An Coosemans. Drug Repurposing for Targeting Myeloid-Derived Suppressor-Cell-Generated Immunosuppression in Ovarian Cancer: A Literature Review of Potential Candidates. *Pharmaceutics*. 2023.22;15(7):1792 (PMID: 37513979)

Elkayal O, **Hoffert Y**, Mertens B, Van Daele R, Lagrou K, Wauters J, Spriet I, Dreesen E. Anidulafungin exposure and population pharmacokinetics in critically ill patients with invasive candidiasis. *Infection*. 2025.53(3):1155-1165 (PMID:39641856)

Wang Z, **Hoffert Y**, Zhang W, Kantasiripitak W, Verstockt B, Sabino J, Ferrante M, Declerck P, D'Haens G, Laharie D, Vermeire S, Dreesen E. *Clin Gastroenterol Hepatol*. 2025. 7:S1542-3565(25)00750-5 (PMID: 40925517)

Morales J F, Klose M, **Hoffert Y**, Podichetty J T, Burton J, Schmidt S, Romero K, O'Doherty I, Martin F, Campbell-Thompson M, Haller M J, Atkinson M A, Kim S. Type 1 diabetes prevention clinical trial simulator: Case reports of model-informed drug development tool. *CPT*. 2024. ;13(8):1309-1316 (PMID: 38961520)

Guadagnoli L, **Hoffert Y**, Den Hond S, Dreesen E, van Ryckeghem D, Van Damme S, Zaman J, Van Oudenhove L. Do we perceive sensations inside and outside of our body differently? Perceptual, emotional, and behavioral differences between visceral and somatic sensation, discomfort, and pain. *Neurogastroenterol Motil*. 2024. 36(6):e14787 (PMID: 38523349)

Dia N, **Hoffert Y**, Edwina A E, Tournoy J, Spriet I, Dreesen E. Precision Dosing of Vancomycin: A Systematic Review of Population Pharmacokinetics Models and Comparative Evaluation of Software Tools. *Clin Pharmacokin*. [Accepted for publication] [*Shared co-first authorship]

Oral presentations at international conferences

Development and validation of a novel software tool to guide the intravenous-to-subcutaneous switching of infliximab in patients with inflammatory bowel diseases.

Hoffert et al. UEGW 2024. Vienna, Austria.

A Risk Stratification Tool for Relapse After Intravenous-to-Subcutaneous Switching of Infliximab in Patients with Inflammatory Bowel Diseases.

Hoffert et al. DDW 2025. San Diego, USA.

Risk Stratification for Relapse After Intravenous-to-Subcutaneous Infliximab Switching in Inflammatory Bowel Disease.

Hoffert et al. IATDMCT 2025. Singapore.

Personalized Infliximab Dosing For Acute Severe Ulcerative Colitis: A Multi-Center Pharmacometrics Analysis.

Niyigena and **Hoffert et al.** IATDMCT 2025. Singapore.

Poster presentations at international conferences

An automated multi-model selection algorithm to improve precision dosing of tacrolimus in liver, lung and bowel transplant recipients.

Hoffert et al. PAGE 2023. A Coruña, Spain.

A software tool for a priori model-informed dose selection of tacrolimus for solid organ transplant recipients.

Hoffert et al. RedIF. Bogota, Colombia.

Model-informed intravenous-to-subcutaneous switching of infliximab in patients with inflammatory bowel diseases.

Hoffert et al. PAGE 2024. Rome, Italy.

Clinical Validation of a Novel Software Tool to Guide the Intravenous to Subcutaneous Switching of Infliximab in Patients with Inflammatory Bowel Diseases.

Hoffert et al. ACG 2024. Philadelphia, USA.

Long-term outcomes of pull and push percutaneous endoscopic gastrostomy techniques.

Hoffert et al. DDW 2025. San Diego, USA.

Development of a personalized infliximab dosing algorithm for acute severe ulcerative colitis: results of a multi-center pharmacometrics analysis.

Niyigena and **Hoffert et al.** DDW 2025. San Diego, USA.

Development of a personalized infliximab dosing algorithm for acute severe ulcerative colitis: results of a multi-center pharmacometrics analysis.

Niyigena and **Hoffert et al.** ECCO 2025. Berlin, Germany.

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Poster of distinction. DDW 2025. San Diego, USA

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Chapter 1

- Conceptualization: Yannick Hoffert, Erwin Dreesen;
- Data curation: Yannick Hoffert, Mathurin Fumery, Maria Nachury, Maëva Bazoge, Anthony Buisson;
- Formal analysis: Yannick Hoffert;
- Funding acquisition: Yannick Hoffert, Erwin Dreesen;
- Investigation: Yannick Hoffert, Erwin Dreesen;
- Methodology: Yannick Hoffert, Erwin Dreesen;
- Project administration: Yannick Hoffert, Erwin Dreesen;
- Resources: Erwin Dreesen;
- Software: Yannick Hoffert, Zhigang Wang;
- Supervision: Erwin Dreesen;
- Validation: Zhigang Wang, Mathurin Fumery, Maria Nachury, Maëva Bazoge, Anthony Buisson, Erwin Dreesen;
- Visualization: Yannick Hoffert;
- Writing – original draft: Yannick Hoffert;
- Writing – review and editing: Yannick Hoffert, Zhigang Wang, Mathurin Fumery, Maria Nachury, Maëva Bazoge, Anthony Buisson, Erwin Dreesen.

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Chapter 2

- Conceptualization: Emmanuel Niyigena, Yannick Hoffert, Erwin Dreesen
- Data curation: Emmanuel Niyigena, Zhigang Wang
- Formal analysis: Emmanuel Niyigena
- Funding acquisition: Yannick Hoffert, Zhigang Wang, Erwin Dreesen
- Investigation: Emmanuel Niyigena
- Methodology: Emmanuel Niyigena, Yannick Hoffert
- Project administration: Erwin Dreesen
- Resources: Waqqas Afif, Alessandro Pedicelli, Xavier Roblin, Jurij Hanžel, Konstantinos Papamichael, Taku Kobayashi, Laure Elens, Marc Ferrante, Bram Verstockt, Séverine Vermeire, Niels Vande Casteele, Robert Battat, Erwin Dreesen
- Software: Emmanuel Niyigena, Yannick Hoffert
- Supervision: Erwin Dreesen
- Validation: Yannick Hoffert

- Visualization: Emmanuel Niyigena, Yannick Hoffert
- Writing – original draft: Emmanuel Niyigena, Yannick Hoffert
- Writing – review and editing: All authors

Parts of this work were presented at American College of Gastroenterology (ACG) 2024 in Philadelphia, International Association of Therapeutic Drug Monitoring and Clinical Toxicology (IATDMCT) 2025 in Singapore, Digestive Disease Week (DDW) 2025 in San Diego, European Crohn's and Colitis Organisation (ECCO) 2025 in Berlin, Population Approach Group Europe (PAGE) 2025 in Thessaloniki.

Chapter 3

Conceptualization: Barbara Deleenheer, Yannick Hoffert, Erwin Dreesen, Tim Vanuytsel;
 Data curation: Barbara Deleenheer, Yannick Hoffert;
 Formal analysis: Barbara Deleenheer, Yannick Hoffert;
 Funding acquisition: Yannick Hoffert;
 Investigation: Barbara Deleenheer, Yannick Hoffert;
 Methodology: Barbara Deleenheer, Yannick Hoffert, Erwin Dreesen, Tim Vanuytsel;
 Project administration: Tim Vanuytsel;
 Resources: Erwin Dreesen, Tim Vanuytsel;
 Supervision: Erwin Dreesen, Tim Vanuytsel;
 Validation: Thomas Vanassche, Lucas Wauters, Lorenz Van der Linden, Erwin Dreesen, Tim Vanuytsel;
 Visualization: Barbara Deleenheer, Yannick Hoffert;
 Writing – original draft: Barbara Deleenheer, Yannick Hoffert;
 Writing – review and editing: Barbara Deleenheer, Yannick Hoffert, Thomas Vanassche, Lucas Wauters, Lorenz Van der Linden, Erwin Dreesen, and Tim Vanuytsel.

Chapter 4

Conceptualization: Yannick Hoffert, Jan Vandewalle, Erwin Dreesen, and Tim Vanuytsel;
 Data curation: Yannick Hoffert, Jan Vandewalle, Maarten Couturier;
 Formal analysis: Yannick Hoffert, Jan Vandewalle;
 Funding acquisition: Yannick Hoffert;
 Investigation: Yannick Hoffert, Jan Vandewalle;
 Methodology: Yannick Hoffert and Erwin Dreesen;
 Project administration: Tim Vanuytsel;
 Resources: Erwin Dreesen, Tim Vanuytsel;
 Supervision: Erwin Dreesen, Tim Vanuytsel;
 Validation: Jan Vandewalle, Maarten Couturier, Lucas Wauters, Tim Vanuytsel;
 Visualization: Yannick Hoffert;
 Writing – original draft: Yannick Hoffert;
 Writing – review and editing: Yannick Hoffert, Jan Vandewalle, Maarten Couturier, Lucas Wauters, Erwin Dreesen, and Tim Vanuytsel.

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Chapter 5

Conceptualization: Yannick Hoffert and Erwin Dreesen;
Data curation: Yannick Hoffert;
Formal analysis: Yannick Hoffert;
Funding acquisition: Yannick Hoffert;
Investigation: Yannick Hoffert;
Methodology: Yannick Hoffert and Erwin Dreesen;
Project administration: Yannick Hoffert and Erwin Dreesen;
Resources: Erwin Dreesen;
Software: Yannick Hoffert;
Supervision: Erwin Dreesen;
Validation: Nada Dia, Tim Vanuytsel, Robin Vos, Dirk Kuypers, Johan Van Cleemput, and Jef Verbeek;
Visualization: Yannick Hoffert;
Writing – original draft: Yannick Hoffert;
Writing – review and editing: Yannick Hoffert, Nada Dia, Tim Vanuytsel, Robin Vos, Dirk Kuypers, Johan Van Cleemput, Jef Verbeek, and Erwin Dreesen.

Chapter 6

Conceptualization: Yannick Hoffert, Barbara Deleenheer, Tim Vanuytsel, Erwin Dreesen;
Data curation: Yannick Hoffert, Barbara Deleenheer, Caroline Boelhouwer, Nele Van den Eede;
Formal analysis: Yannick Hoffert;
Funding acquisition: Yannick Hoffert;
Investigation: Yannick Hoffert;
Methodology: Yannick Hoffert, Erwin Dreesen;
Project administration: Erwin Dreesen;
Resources: Tim Vanuytsel, Erwin Dreesen;
Supervision: Tim Vanuytsel, Erwin Dreesen;
Validation: Barbara Deleenheer, Caroline Boelhouwer, Nele Van den Eede, Emmanuel Niyigena, Laurens Ceulemans, Laure Elens, Diethard Monbaliu, Jacques Pirenne, Lucas Wauters, Tim Vanuytsel;
Visualization: Yannick Hoffert;
Writing – original draft: Yannick Hoffert;
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Personal contribution

For **chapter 1, 4, 5, 6** the author of the thesis manuscript fully contributed to the design of the studies, the generation and assembly of the data, the analysis and interpretation of the data, and wrote the manuscript.

For **chapter 2** the author of the thesis manuscript assisted Emmanuel Niyigena, a volunteer researcher under his supervision, with the conceptualization of the work, provided hands-on modelling advice and guidance in data analysis and interpretation, and assisted in writing the manuscript.

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Conflict of interest

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Use of generative AI

I did not use generative AI assistance tools during the research/writing process of my thesis, except for mere language assistance. The text/code/ images in this thesis are my own (unless otherwise specified) and generative AI has only been used in accordance with the KU Leuven guidelines and appropriate references have been added. I have reviewed and edited the content as needed and I take full responsibility for the content of the thesis.